Modern Physics

basics

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CONTENTS

This material is part of the **basics-books project**. It is also available as a .pdf document.

Please check out the Github repo of the project, basics-book project.

- Special Relativity
  - *Special Relativity*
  - *Special Relativity - Notes*
  - *Inertial reference frames and Lorentz's transformations*
  - *Electromagnetism*
  - *"Constant" acceleration observer and Rindler coordinates*
- General Relativity
  - *General Relativity*
  - *General Relativity - Notes*
- Statistical Mechanics
  - *Statistical Physics*
  - *Liouville theorem and Boltzmann equation*
  - *Statistical Physics - Notes*
  - *Statistical Physics - Statistics Miscellanea*
  - *Ising model*
  - *Entropy and arrow of time*
- Quantum Mechanics
  - *Quantum Mechanics*
  - *Quantum Mechanics - Notes*
  - *Wave quantum mechanics*
  - *Angular Momentum*
  - *Density Operator*
  - *Decoherence*
  - *Interaction picture, perturbation theory and Einstein coefficients*
  - *Historical Approach to Quantum Mechanics*
  - *Atomic models*
  - *Relativistic Quantum Mechanics*
  - *Many-body systems*
  - *Quantization of the electromagnetic field*
  - *Introduction to Quantum Field Theory*
  - *Analtycal Mechanics - Notes for Quantum Mechanics*
- Quantum Field Theory
  - *Introduction to QFT*
- Solid State Physics

– *Introduction to Solid State Physics*
– *Notes about Solid State Physics*
– *Introduction to Semiconductors*

Part I

Special Relativity

CHAPTER

ONE

SPECIAL RELATIVITY

Principles. (1) invariant nature of physical laws, (2) c as the maximum speed of propagation of information in space. Every inertial observer measures the same speed of light.

Lorentz's transformation as a special coordinate transformations for space-time description as seen by two inertial observers in relative motion with uniform velocity. Physical laws governing a physical process have the same expressions if seen by two inertial observers. In a homogeneous time-space the expression of equations of physics doesn't depend on translation of origin of time-space coordinates; in a isotropic space the expression of equations of physics doesn't depend on rotations space rotation; as two inertial observers may write the same expressions of the physical equations using their own coordinates, the expression of equations of physics doesn't depend on Lorentz's transformations.

- standard configuration: derivation of Lorentz's transformations from principles and symmetry considerations
- composition of transformations and general Lorentz's transformation

Some consequences and examples:

- **Finite speed of propagation and loss of simultaneity.**
- **Speed of light and causality.** Causality follows from the principle that c is the maximum speed of information: all the observers perceive causes before consequences.
- **Length contraction and time dilation**

Mechanics. Point kinematics: 4-velocity and 4-acceleration; dynamics of free particle: 4-momentum, energy-momentum relation, rest energy; dynamics of particles in external force filed: which field? For particles in EM field, Lorentz's force; Lagrangian approach

Electromagnetism. Special relativity and equations of electromagnetism. From classical electromagnetism to electromagnetism in special relativity theory; EM potentials, gauge condition, 4-current and EM field tensor; EM field equations; motion of a point charge in an EM field; energy balance equations. Relativity of electromagnetism: low-speed relativity for classical electromagnetism. Lagrangian approach for the equations of motion of charges in an EM field, and for the EM field equations.

CHAPTER

TWO

SPECIAL RELATIVITY - NOTES

An event is determined by spatio-temporal information together, $t, \vec{r}$. Absolute nature of physics needs vector algebra and calculus formalism

$$\mathbf{X} = ct\mathbf{E}_0 + \vec{r} = ct\mathbf{E}_0 + x^1\mathbf{E}_1 + x^2\mathbf{E}_2 + x^3\mathbf{E}_3 = X^\alpha \mathbf{E}_\alpha \ ,$$

having used Cartesian coordiantes for the space coordinate.

Geometry of time-space in special relativity. Minkowski metric reads

$$g_{\alpha\beta} = \mathbf{E}_\alpha \cdot \mathbf{E}_\beta = \text{diag}\{1, -1, -1, -1\}$$

Reciprocal basis

The reciprocal basis reads $\mathbf{E}_\alpha \cdot \mathbf{E}^\beta = \delta^\beta_\alpha$, $\mathbf{E}_\alpha = g_{\alpha\beta}\mathbf{E}^\beta$, s.t. the elementary interval between two events can be written as

$$d\mathbf{X} = dX^\alpha \, \mathbf{E}_\alpha = \underbrace{dX^\alpha \, g_{\alpha\beta}}_{=dX_\beta} \, \mathbf{E}^\beta = dX_\beta \, \mathbf{E}^\beta \ ,$$

having used Cartesian coordinates,

$$\begin{array}{llll} X^0 = ct & X^1 = x & X^2 = y & X^3 = z \\ X_0 = ct & X_1 = -x & X_2 = -y & X_3 = -z \end{array}$$

Elementary interval between two events

$$ds^2 = d\mathbf{X} \cdot d\mathbf{X} = (dX_\alpha \mathbf{E}^\alpha) \cdot \left(dX^\beta \mathbf{E}_\beta\right) = c^2\, dt^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 = c^2\, dt^2 - |d\vec{r}|^2$$

2.1 Kinematics of point

Proper time. For a co-moving observer, $d\vec{r}' = \vec{0}$, and t' is commonly indicated with τ, and its differential is invariant itself, being the product of a constant (c is a universal constant in special relativity) and an invariant quantity.

$$ds^2 = c^2 dt'^2 - |d\vec{r}'|^2 = c^2 d\tau^2 \ .$$

Given the invariant nature of ds,

$$ds^2 = c^2\, d\tau^2 = c^2\, dt^2 - |d\vec{r}|^2 = c^2\, dt^2 \left[1 - \frac{1}{c^2}\frac{|d\vec{r}|^2}{dt^2}\right] = c^2\, dt^2 \left[1 - \frac{|\vec{v}|^2}{c^2}\right]$$

and thus

$$ds = c\,d\tau = \gamma^{-1}(v/c)\,c\,dt\ ,$$

with $\gamma(w) = \frac{1}{\sqrt{1-w^2}}$.

Note: ds is invariant **todo** prove it. And/or add a section about the role of invariance.

4-Velocity. Given the parametric representation of an event in space-time as a function of its proper time, $\mathbf{X}(\tau)$ or coordinate s, $\mathbf{X}(s)$ the derivative w.r.t. this parameter is defined as the 4-velocity of the event in space time. Using Cartesian coordinates inducing constant and uniform basis $\mathbf{E}_\alpha$, as a function of the observer time t, ct, $x^i(t)$, and the transformation of coordinates $t(\tau)$, with differential $dt = \frac{1}{\gamma}\,d\tau$

$$\mathbf{U}(\tau) := \mathbf{X}'(\tau) = \frac{d}{d\tau}\left(X^\alpha(\tau)\mathbf{E}_\alpha\right) = \frac{dt}{d\tau}(ct\mathbf{E}_0 + x^i(t)\mathbf{E}_i) = \gamma(v/c)\left(c\mathbf{E}_0 + \dot{x}^i(t)\mathbf{E}_i\right) = \gamma(v/c)\left(c\mathbf{E}_0 + \vec{v}\right)$$

or

$$\mathbf{U}(s) := \mathbf{X}'(s) = \frac{dt}{ds}\frac{d}{dt}\mathbf{X}(t) = \cdots = \gamma(v/c)\left(\mathbf{E}_0 + \frac{\vec{v}}{c}\right)\ .$$

Note: Using s as the parameter, $\mathbf{U}$ is non-dimensional, and has pseudo-norm = 1,

$$\mathbf{U}(s)\cdot\mathbf{U}(s) = \gamma^2\underbrace{\left(1 - \frac{|\vec{v}|^2}{c^2}\right)}_{=\gamma^{-2}} = 1\ .$$

Using τ as the parameter, $\mathbf{U}$ has physical dimension of a velocity and pseudo-norm = c.

4-acceleration $\mathbf{X}''(\tau)$ or $\mathbf{X}''(s)$, **todo**

2.2 Dynamics of a point mass

4-momentum

$$\mathbf{P} = m\mathbf{U}$$

Using Cartesian coordinates and τ as independent variable,

$$\mathbf{P} = m\mathbf{U} = m\frac{d\mathbf{X}}{d\tau} = m\gamma(c, \vec{v})\ .$$

Low-speed limit - classical mechanics

The spatial component is γ times the 3-dimensional momentum $\vec{p} = m\vec{v}$; the time component reads

$$P^0 = m\gamma(w)c\ ,$$

and for small ratio $w := \frac{v}{c}$ it can be expanded in Taylor series around $w = 0$ as

$$\gamma(w) \sim \gamma(0) + w\,\gamma'(0) + \frac{1}{2}\,w^2\gamma''(0) + o(w^2)\ ,$$

with

$$\begin{aligned}
\gamma(w)|_{w=0} &= \left.\frac{1}{\sqrt{1-w^2}}\right|_{w=0} = 1 \\
\gamma'(w)|_{w=0} &= \left.-\frac{1}{2}(1-w^2)^{-\frac{3}{2}}(-2w)\right|_{w=0} = w(1-w^2)^{-\frac{3}{2}} = 0 \\
\gamma''(w)|_{w=0} &= \left.\left((1-w^2)^{-\frac{3}{2}} + w\left(-\frac{3}{2}\right)(1-w^2)^{-\frac{5}{2}}(-2w)\right)\right|_{w=0} = \\
&= \left.\left((1-w^2)^{-\frac{3}{2}} + 3w^2(1-w^2)^{-\frac{5}{2}}\right)\right|_{w=0} = 1
\end{aligned}$$

and thus

$$\gamma(w) = 1 + \frac{1}{2}w^2 + o(w^2)$$

and

$$\gamma(v/c)\, m\, c \sim m\, c\left(1 + \frac{v^2}{c^2}\right) = \frac{1}{c}\left(mc^2 + \frac{1}{2}m|\vec{v}|^2\right)$$

Energy-momentum relation

Recognizing energy ($E = \gamma mc^2$) and 3-momentum ($\vec{p} = m_3\vec{v}$, with $m_3 := \gamma m$)[1], the 4-momentum can be written as

$$\mathbf{P} = m\mathbf{U} = \gamma m\left(1, \frac{\vec{v}}{c}\right) =: \frac{1}{c}\left(\frac{E}{c}, \vec{p}\right)$$

Its pseudo-norm reads

$$m^2 = \mathbf{P}\cdot\mathbf{P} = \frac{1}{c^4}\left(E^2 - c^2|\vec{p}|^2\right)$$

and thus the relation between E, $\vec{p}$, m and c,

$$E^2 = m^2c^4 + c^2|\vec{p}|^2\ ,$$

from which, for $\vec{v} = \vec{0} \to \vec{p} = \vec{0}$,

$$E^2 = m^2c^4\ ,$$

and keeping only the solution with positive energy (**todo** *reference to Dirac's equation and anti-matter?*)

$$E = mc^2\ .$$

4-force and dynamical equation for a point mass. The dynamical equation of a point mass subject to a 4-force $\mathbf{f}$ reads

$$\frac{d\mathbf{P}}{d\tau} = \mathbf{f}\ . \tag{2.1}$$

Examples:

- **Free particle** is subject to no net force, $\mathbf{f}$,
- Point charge q in EM field is subject to **Lorentz force**, $\mathbf{f} = q\mathbf{F}\cdot\mathbf{U}$, being $\mathbf{F}$ the EM field tensor, (2.4),

$$\mathbf{F} = \nabla\mathbf{A} - (\nabla\mathbf{A})^T\ .$$

[1] This definitions naturally arise in Lagrange approach.

2.2.1 Lagrangian approach

Here variational approach to the motion of point particle in a force field $\mathbf{f}$ is derived from the weak form of the dynamical equation (2.1). In particular, the derivation is performed for a particle moving in a known EM field. The motion of a free particle immediately follows if the EM field is zero.

$$m\frac{d\mathbf{U}}{d\tau} = q\mathbf{F}\cdot\mathbf{U}$$

The equation of motion can be manipulated to get Lagrange equations

$$\begin{aligned}
\mathbf{0} &= m\frac{d\mathbf{U}}{d\tau} - q\nabla\mathbf{A}\cdot\mathbf{U} + q\left(\nabla\mathbf{A}\right)^T\cdot\mathbf{U} = \\
&= \frac{d}{d\tau}\left[m\mathbf{U} + q\mathbf{A}(\mathbf{X}(\tau))\right] - \nabla\left(q\mathbf{A}\cdot\mathbf{U}\right) = (1) \\
&= \frac{d}{d\tau}\left[\mathbf{E}^\alpha\frac{\partial}{\partial U^\alpha}\left(mc\sqrt{\mathbf{U}\cdot\mathbf{U}} + q\mathbf{A}(\mathbf{X}(\tau))\cdot\mathbf{U}\right)\right] - \mathbf{E}^\alpha\frac{\partial}{\partial X^\alpha}\left(mc\sqrt{\mathbf{U}\cdot\mathbf{U}} + q\mathbf{A}\cdot\mathbf{U}\right) = (2) \\
&= \mathbf{E}^\alpha\left[\frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial X'^\alpha} - \frac{\partial\mathcal{L}}{\partial X^\alpha}\right] = \\
&= \frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\mathbf{X}'} - \frac{\partial\mathcal{L}}{\partial\mathbf{X}}
\end{aligned} \tag{2.2}$$

since (1) $\mathbf{U} = \mathbf{E}^\alpha U_\alpha = c\mathbf{E}^\alpha\frac{\partial}{\partial U^\alpha}\sqrt{U^\beta U_\beta}$, see (2.3), and $\sqrt{\mathbf{U}\cdot\mathbf{U}}$ independent from $\mathbf{X}$, and equal to c^2 and (2) with the definition of the **Lagrangian function**

$$\mathcal{L}(\mathbf{X}'(\tau),\mathbf{X}(\tau),\tau) := -mc\sqrt{\mathbf{X}'(\tau)\cdot\mathbf{X}'(\tau)} - q\mathbf{A}(\mathbf{X}(\tau))\cdot\mathbf{X}'(\tau)$$

Proof of the relation used in (1)

$$\begin{aligned}
\mathbf{E}^\alpha\frac{\partial}{\partial U^\alpha}\sqrt{U^\beta U_\beta} &= \mathbf{E}^\alpha\frac{\partial}{\partial U^\alpha}\sqrt{g_{\beta\gamma}U^\beta U^\gamma} = \\
&= \mathbf{E}^\alpha\frac{1}{2\sqrt{U^\beta U_\beta}}\left(g_{\beta\gamma}\delta^\beta_\alpha U^\gamma + g_{\beta\gamma}U^\beta\delta^\gamma_\alpha\right) = \\
&= \mathbf{E}^\alpha\frac{1}{2\sqrt{U^\beta U_\beta}}\left(2U_\alpha\right) = \\
&= \mathbf{E}^\alpha\frac{U_\alpha}{\sqrt{U^\beta U_\beta}} = \\
&= \frac{\mathbf{U}}{\sqrt{\mathbf{U}\cdot\mathbf{U}}} = \frac{\mathbf{U}}{c}\,.
\end{aligned} \tag{2.3}$$

since $\mathbf{U}\cdot\mathbf{U} = c^2$ and the components of the metric tensor are independent from the velocity.

Weak form of the problem is derived from the Lagrange equation (2.2), with dot product with arbitrary 4-vector $\mathbf{W}$,

$$0 = \mathbf{W}\cdot\left[\frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\mathbf{X}'} - \frac{\partial\mathcal{L}}{\partial\mathbf{X}}\right]\,.$$

and integrating over an interval of proper time and using integration by parts,

$$\begin{aligned}
0 &= \int_{\tau=\tau_0}^{\tau_1}\mathbf{W}\cdot\left[\frac{d}{d\tau}\frac{\partial\mathcal{L}}{\partial\mathbf{X}'} - \frac{\partial\mathcal{L}}{\partial\mathbf{X}}\right]d\tau = \\
&= \left[\mathbf{W}(\tau)\cdot\frac{\partial\mathcal{L}}{\partial\mathbf{X}'}(\mathbf{X}'(\tau),\mathbf{X}(\tau),\tau)\right]\Bigg|_{\tau_0}^{\tau_1} - \int_{\tau=\tau_0}^{\tau_1}\left[\mathbf{W}'\cdot\frac{\partial\mathcal{L}}{\partial\mathbf{X}'} + \mathbf{W}\cdot\frac{\partial\mathcal{L}}{\partial\mathbf{X}}\right]d\tau\,.
\end{aligned}$$

With the choice of the test function $\mathbf{W}(\tau)$ as the variation of the position of the particle, $\mathbf{W} = \delta\mathbf{X}$ just to remember its variational nature, with given values at τ_0, τ_1 and thus $\mathbf{W}(\tau_0) = \mathbf{W}(\tau_1) = 0$, the weak form of the dynamical equations is thus recast as the stationary condition of an action functional in a **variational principle**,

$$0 = -\delta S = -\delta \int_{\tau=\tau_0}^{\tau_1} \mathcal{L}\left(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau\right)\, d\tau\ .$$

todo *Repeat the process using generalized coordinates $q^k(\tau)$, $\mathbf{X}(q^k(\tau), \tau)$…*

Lagrange equations and variational principle with generalized coordinates

Choosing the test function as it's usually done in deriving Lagrange formulation of mechanics from the strong formulation, after writing the position and its velocity of the particle as a function of generalized coordinates q^k,

$$\begin{aligned}
&\mathbf{X}\left(q^k(\tau), \tau\right) \\
&\mathbf{U}\left(q^k(\tau), \tau\right) = \frac{d\mathbf{X}}{d\tau} = q^{k'}(\tau)\ \underbrace{\frac{\partial \mathbf{X}}{\partial q^k}}_{=\frac{\partial \mathbf{X}'}{\partial q^{k'}}} + \frac{\partial \mathbf{X}}{\partial \tau}
\end{aligned}$$

With the particular choice of the test function, $\mathbf{W} = w^k(\tau)\dfrac{\partial \mathbf{X}}{\partial q^k}(q^l(\tau), \tau)$

Variational principle as a function of time. Using the relation between proper time τ and the observer time t, $d\tau = \gamma^{-1}\, dt = \sqrt{1 - \frac{|\vec{v}|^2}{c^2}}dt$,

$$\begin{aligned}
S &= \int_{\tau=\tau_0}^{\tau_1} \mathcal{L}\left(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau\right)\, d\tau = \\
&= \int_{t=t_0}^{t_1} \mathcal{L}\left(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau\right)\, \gamma^{-1}\, dt = \\
&= \int_{t=t_0}^{t_1} \sqrt{1 - \frac{|\vec{v}|^2}{c^2}}\left(-mc^2 - \gamma q\phi(\vec{r}, t) + \gamma q\vec{a}(\vec{r}, t)\cdot\vec{v}(t)\right)\, dt = \\
&= \int_{t=t_0}^{t_1} \left(-\sqrt{1 - \frac{|\vec{v}|^2}{c^2}}mc^2 - q\phi(\vec{r}, t) + q\vec{a}(\vec{r}, t)\cdot\vec{v}(t)\right)\, dt = \\
&= \int_{t=t_0}^{t_1} L\left(\vec{v}(t), \vec{r}(t), t\right)\, dt
\end{aligned}$$

Three-dimensional momentum can be defined as the partial derivative of the Lagrangian function L w.r.t. the velocity $\vec{v}$,

$$\vec{p} := \frac{\partial L}{\partial \vec{v}} = \gamma\frac{\vec{v}}{c^2}mc^2 + q\vec{a} = \gamma m\vec{v} + q\vec{a}$$

The Hamiltoninan function reads

$$\begin{aligned}
H &= \vec{v}\cdot\frac{\partial L}{\partial \vec{v}} - L = \\
&= \gamma m|\vec{v}|^2 + q\vec{a}\cdot\vec{v} + \gamma^{-1}mc^2 + q\phi - q\vec{a}\cdot\vec{v} = \\
&= \gamma m|\vec{v}|^2\left[1 + \left(1 - \frac{|\vec{v}|^2}{c^2}\right)\frac{c^2}{|\vec{v}|^2}\right] + q\phi = \\
&= \gamma mc^2 + q\phi\ .
\end{aligned}$$

The expressions of the Hamiltonian function (divided by the light speed) and the three-dimensional momentum can be recognized to be respectively the time and space components of 4-momentum,

$$\mathbf{P} = m\mathbf{U} + q\mathbf{A} \ ,$$

remembering the expression of the contravariant components of the 4-velocity and the 4-EM potential,

$$[\mathbf{P}]^{\alpha} = \left(\gamma mc + q\frac{\phi}{c},\ \gamma m\vec{v} + q\vec{a}\right) = \left(\frac{H}{c},\ \vec{p}\right)$$

2.3 Electromagnetism

2.3.1 Classical electromagnetic theory

Maxwell equations

Maxwell equations read

$$\begin{cases} \nabla \cdot \vec{d} = \rho_f \\ \nabla \times \vec{e} + \partial_t \vec{b} = \vec{0} \\ \nabla \cdot \vec{b} = 0 \\ \nabla \times \vec{h} - \partial_t \vec{d} = \vec{j}_f \end{cases}$$

or in vacuum, with $\rho_f = \rho$, $\vec{j} = \vec{j}_f$, $\vec{d} = \varepsilon_0 \vec{e}$, $\vec{b} = \mu_0 \vec{h}$

$$\begin{cases} \nabla \cdot \vec{e} = \frac{\rho}{\varepsilon_0} \\ \nabla \times \vec{e} + \partial_t \vec{b} = \vec{0} \\ \nabla \cdot \vec{b} = 0 \\ \nabla \times \vec{b} - \mu_0 \varepsilon_0 \partial_t \vec{e} = \mu_0 \vec{j} \end{cases}$$

Electromagnetic potentials

The electromagnetic field can be written in terms of the electromagnetic potentials

$$\begin{cases} \vec{b} = \nabla \times \vec{a} \\ \vec{e} = -\partial_t \vec{a} - \nabla \varphi \end{cases}$$

Lorentz force

A particle in motion in a electromagnetic field is subject to Lorentz force. In classical electromagnetism, the expression of Lorentz force reads

$$\vec{F} = q\left(\vec{e} - \vec{b} \times \vec{v}\right) \ ,$$

whose power is

$$\vec{v} \cdot \vec{F} = \vec{v} \cdot q\left(\vec{e} - \vec{b} \times \vec{v}\right) = q\vec{v} \cdot \vec{e} \ .$$

2.3.2 Electromagnetic potential

$$\begin{cases} \vec{b} = \nabla \times \vec{a} \\ \vec{e} = -\partial_t \vec{a} - \nabla \varphi \end{cases}$$

$$\mathbf{A} = \mathbf{E}_\alpha A^\alpha = \frac{\varphi}{c}\mathbf{E}_0 + \vec{a}$$

$$\boldsymbol{\nabla}\mathbf{A} = \left(\mathbf{E}^\alpha \frac{\partial}{\partial X^\alpha}\right)\left(A^\beta \mathbf{E}_\beta\right) = \frac{\partial A^\beta}{\partial X^\alpha}\mathbf{E}^\alpha \otimes \mathbf{E}_\beta = g^{\alpha\gamma}\frac{\partial A^\beta}{\partial X^\alpha}\mathbf{E}_\gamma \otimes \mathbf{E}_\beta \, .$$

whose components may be collected in a 2-dimensional array (first index for rows, second index for columns),

$$(\nabla\mathbf{A})_\alpha{}^\beta = \frac{\partial A^\beta}{\partial X^\alpha} = \begin{bmatrix} c^{-2}\partial_t\varphi & c^{-1}\partial_x\varphi & c^{-1}\partial_y\varphi & c^{-1}\partial_z\varphi \\ c^{-1}\partial_t a_x & \partial_x a_x & \partial_y a_x & \partial_z a_x \\ c^{-1}\partial_t a_y & \partial_x a_y & \partial_y a_y & \partial_z a_y \\ c^{-1}\partial_t a_z & \partial_x a_z & \partial_y a_z & \partial_z a_z \end{bmatrix}$$

or covariant-covariant coomponents,

$$(\nabla\mathbf{A})_{\alpha\beta} = \frac{\partial A_\beta}{\partial X^\alpha} = \frac{\partial A^\gamma}{\partial X^\alpha} g_{\gamma\beta} = \begin{bmatrix} c^{-2}\partial_t\varphi & -c^{-1}\partial_x\varphi & -c^{-1}\partial_y\varphi & -c^{-1}\partial_z\varphi \\ c^{-1}\partial_t a_x & -\partial_x a_x & -\partial_y a_x & -\partial_z a_x \\ c^{-1}\partial_t a_y & -\partial_x a_y & -\partial_y a_y & -\partial_z a_y \\ c^{-1}\partial_t a_z & -\partial_x a_z & -\partial_y a_z & -\partial_z a_z \end{bmatrix}$$

or contravariant-contravariant coomponents,

$$(\nabla\mathbf{A})^{\alpha\beta} = \frac{\partial A^\beta}{\partial X_\alpha} = g_{\alpha\gamma}\frac{\partial A^\beta}{\partial X^\gamma} = \begin{bmatrix} c^{-2}\partial_t\varphi & c^{-1}\partial_x\varphi & c^{-1}\partial_y\varphi & c^{-1}\partial_z\varphi \\ -c^{-1}\partial_t a_x & -\partial_x a_x & -\partial_y a_x & -\partial_z a_x \\ -c^{-1}\partial_t a_y & -\partial_x a_y & -\partial_y a_y & -\partial_z a_y \\ -c^{-1}\partial_t a_z & -\partial_x a_z & -\partial_y a_z & -\partial_z a_z \end{bmatrix}$$

The electromagnetic field tensor is defined as the anti-symmetric part of the gradient of the 4-electromagnetic potential,

$$\mathbf{F} = \left[\boldsymbol{\nabla}\mathbf{A} - (\boldsymbol{\nabla}\mathbf{A})^T\right] \tag{2.4}$$

whose components may be collected in a 2-dimensional array (first index for rows, second index for columns),

$$F^{\alpha\beta} = \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & \underline{b}_\times \end{bmatrix} \quad , \quad F_{\alpha\beta} = \begin{bmatrix} 0 & \frac{\underline{e}^T}{c} \\ -\frac{\underline{e}}{c} & \underline{b}_\times \end{bmatrix}$$

$$F_\alpha{}^\beta = \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ -\frac{\underline{e}}{c} & -\underline{b}_\times \end{bmatrix} \quad , \quad F^\alpha{}_\beta = \begin{bmatrix} 0 & \frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & -\underline{b}_\times \end{bmatrix}$$

2.3.3 Electromagnetic field and electromagnetic field equations

The pair of Maxwell equations

$$\begin{cases} \rho_f = \nabla \cdot \vec{d} \\ \vec{j}_f = -\partial_t \vec{d} + \nabla \times \vec{h} \end{cases}$$

can be re-written in 4-formalism, using 4-gradient in Cartesian coordinates

$$\boldsymbol{\nabla} = \mathbf{E}^\alpha \frac{\partial}{\partial X^\alpha} = \mathbf{E}_0 \frac{\partial}{c\partial t} + \mathbf{E}_i \frac{\partial}{\partial x^i} = \mathbf{E}_0 \frac{\partial}{c\partial t} + \nabla \, ,$$

and the definition of the 4-current density vector

$$\mathbf{J} = J^{\alpha}\mathbf{E}_{\alpha} = c\rho\,\mathbf{E}_0 + \vec{j}$$

so that

$$c\rho\mathbf{E}_0 + \vec{j} = \boldsymbol{\nabla}\cdot\mathbf{F} = \boldsymbol{\nabla}\cdot\left[(0\,\mathbf{E}_0 + c\vec{d})\otimes\mathbf{E}_0 + (-\mathbf{E}_0 c\vec{d} + \vec{h}_{\times})\right] ,$$

with the displacement field tensor,

$$\mathbf{D} = D^{\alpha\beta}\,\mathbf{E}_{\alpha}\,\mathbf{E}_{\beta} ,$$

with components (rows for the first index, columns for the second index)

$$D^{\alpha\beta} = \begin{bmatrix} 0 & -cd_x & -cd_y & -cd_z \\ cd_x & 0 & -h_z & h_y \\ cd_y & h_z & 0 & -h_x \\ cd_z & -h_y & h_x & 0 \end{bmatrix} = \begin{bmatrix} 0 & -c\underline{d}^T \\ c\underline{d} & \underline{h}_{\times} \end{bmatrix} .$$

The pair of Maxwell equations

$$\begin{cases} \nabla\cdot\vec{b} = 0 \\ \partial_t\vec{b} + \nabla\times\vec{e} = \vec{0} \end{cases}$$

can be re-written in 4-formalism as

$$0 = \partial_{\mu}F_{\eta\xi} + \partial_{\eta}F_{\xi\mu} + \partial_{\xi}F_{\mu\eta}$$

Among these $64 = 4^3$ equations, there are only 4 independent equations.

- If 2 indices are the same, the corresponding equation is the identity $0 = 0$. As an example, if $\mu = \eta$

$$0 = \partial_{\mu}F_{\mu\xi} + \partial_{\mu}\underbrace{F_{\xi\mu}}_{-F_{\mu\xi}} + \partial_{\xi}\underbrace{F_{\mu\mu}}_{=0} = 0 ,$$

  thus only combinations with different indices may provide some information.

- Given the ordered set of indices (μ,η,ξ), switching a pair of indices provides the same equation. As an example, switching μ and η

$$\begin{aligned} 0 &= \partial_{\eta}F_{\mu\xi} + \partial_{\mu}F_{\xi\eta} + \partial_{\xi}F_{\eta\mu} = \\ &= \partial_{\eta}(-F_{\xi\mu}) + \partial_{\mu}(-F_{\eta\xi}) + \partial_{\xi}(-F_{\mu\eta}) . \end{aligned}$$

- Thus, only 4 combination of different indices, without taking order into account, provide independent information

$$\begin{aligned} (1,2,3): &\quad 0 = \partial_1F_{23} + \partial_2F_{31} + \partial_3F_{12} = \partial_x(-b_x) + \partial_y(-b_y) + \partial_z(-b_z) \\ (2,3,0): &\quad 0 = \partial_2F_{30} + \partial_3F_{02} + \partial_0F_{23} = \partial_y\left(-\frac{e_z}{c}\right) + \partial_z\left(\frac{e_y}{c}\right) + \partial_{ct}(-b_x) \\ (3,0,1): &\quad 0 = \partial_3F_{01} + \partial_0F_{13} + \partial_1F_{30} = \partial_z\left(\frac{e_x}{c}\right) + \partial_{ct}(-b_y) + \partial_x\left(-\frac{e_z}{c}\right) \\ (0,1,2): &\quad 0 = \partial_0F_{12} + \partial_1F_{20} + \partial_2F_{01} = \partial_{ct}(-b_z) + \partial_x\left(-\frac{e_y}{c}\right) + \partial_y\left(\frac{e_x}{c}\right) \end{aligned}$$

  i.e.

$$\begin{aligned} (1,2,3): &\quad 0 = -\nabla\cdot\vec{b} \\ (2,3,0): &\quad 0 = -\frac{1}{c}\left[\partial_t b_x + (\partial_y e_z - \partial_z e_y)\right] \\ (3,0,1): &\quad 0 = -\frac{1}{c}\left[\partial_t b_y + (\partial_z e_x - \partial_x e_z)\right] \\ (0,1,2): &\quad 0 = -\frac{1}{c}\left[\partial_t b_z + (\partial_x e_y - \partial_y e_x)\right] \end{aligned}$$

i.e.

$$\begin{cases} 0 = \nabla \cdot \vec{b} \\ \vec{0} = \partial_t \vec{b} + \nabla \times \vec{e} \end{cases}$$

2.3.4 Point particle in electromagnetic field

Lorentz 4-force acting on a point charge of electric charge charge q reads

$$\mathbf{f} = \mathbf{F} \cdot \mathbf{J} = q\,\mathbf{F} \cdot \mathbf{U}\,.$$

so that the dynamical equation reads

$$m\,\mathbf{X}'' = q\,\mathbf{F} \cdot \mathbf{X}'$$

2.3.5 Energy balance

Continuity

$$\nabla \cdot \mathbf{J} = 0$$

Maxwell's equations

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \mathbf{J} \\ \nabla \cdot (\boldsymbol{\varepsilon} : \mathbf{F}) &= \mathbf{0} \end{aligned}$$

Energy and momentum balance equation of the EM field using 3-dimensional formalism,

$$\begin{aligned} \frac{\partial u}{\partial t} + \nabla \cdot \vec{s} &= -\vec{e} \cdot \vec{j}^f \\ \frac{\partial \vec{s}}{\partial t} + \nabla \cdot \left\{ c^2 \left[\frac{1}{2} \left(\vec{d} \cdot \vec{e} + \vec{h} \cdot \vec{b} \right) \mathbb{I} - \left(\vec{d} \otimes \vec{e} + \vec{h} \otimes \vec{b} \right) \right] \right\} &= -c^2 (\rho^f \vec{e} - \vec{b} \times \vec{j}^f) \\ \frac{1}{c} \frac{\partial u}{\partial t} + \nabla \cdot \left(\frac{\vec{s}}{c} \right) &= -\frac{\vec{e}}{c} \cdot \vec{j}^f \\ \frac{1}{c} \frac{\partial}{\partial t} \frac{\vec{s}}{c} + \nabla \cdot \left[\frac{1}{2} \left(\vec{d} \cdot \vec{e} + \vec{h} \cdot \vec{b} \right) \mathbb{I} - \left(\vec{d} \otimes \vec{e} + \vec{h} \otimes \vec{b} \right) \right] &= -\rho^f c \frac{\vec{e}}{c} + \vec{b} \times \vec{j}^f \end{aligned} \tag{2.5}$$

The governing equations of energy and momentum of the EM field can be recast in 4-dimensional formalism,

$$\boldsymbol{\nabla} \cdot \mathbf{T} = -\mathbf{F} \cdot \mathbf{J}$$

with the **energy-momentum-stress tensor**, **T**,

$$\begin{aligned} T^{\alpha\beta} &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} - F^{\alpha\nu} D^{\beta}{}_{\nu} = \\ &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} - F^{\alpha\nu} g^{\beta\mu} D_{\mu\nu} = \\ &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} + F^{\alpha\nu} D_{\nu\mu} g^{\mu\beta} \end{aligned} \tag{2.6}$$

Term $\mathbf{g}\,\mathbf{F}:\mathbf{D}$

$$\mathbf{F}:\mathbf{D}=F_{\mu\nu}D^{\mu\nu}=\begin{bmatrix}0 & \frac{\underline{e}^T}{c}\\ -\frac{\underline{e}}{c} & \underline{b}_\times\end{bmatrix}:\begin{bmatrix}0 & -c\underline{d}^T\\ c\underline{d} & \underline{h}_\times\end{bmatrix}=-2\underline{e}^T\underline{d}+2\underline{b}^T\underline{h}=2\left(-\vec{e}\cdot\vec{d}+\vec{b}\cdot\vec{h}\right)\,. \tag{2.7}$$

Term $F^{\alpha\nu}D^{\beta}{}_{\nu}$

$$F^{\alpha\nu}D^{\beta}{}_{\nu}=\begin{bmatrix}0 & -\frac{\underline{e}^T}{c}\\ \frac{\underline{e}}{c} & \underline{b}_\times\end{bmatrix}\begin{bmatrix}0 & c\underline{d}^T\\ c\underline{d} & -\underline{h}_\times^T\end{bmatrix}=\begin{bmatrix}-\underline{e}^T\underline{d} & \frac{(\underline{h}\times\underline{e})^T}{c}\\ c\underline{b}\times\underline{d} & \underline{e}\underline{d}^T+\underline{b}_\times\underline{h}_\times\end{bmatrix} \tag{2.8}$$

being

$$\underline{e}^T\underline{h}_\times^T=e_i\varepsilon_{kji}h_j=\left(\vec{h}\times\vec{e}\right)_k$$

$$\left[\underline{b}_\times\underline{h}_\times\right]_{im}=\varepsilon_{ijk}b_j\varepsilon_{klm}h_l=\left(\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}\right)b_jh_l=h_ib_m-b_lh_l\delta_{im}=\underline{h}\underline{b}^T-\underline{b}^T\underline{h}\underline{\underline{I}}_3\,.$$

Energy-stress tensor

Putting together the expression (2.7) of term $\frac{1}{4}g^{\alpha\beta}F_{\mu\nu}D^{\mu\nu}$ and (2.8) of term $F^{\alpha\nu}D^{\beta}{}_{\nu}$, with $\underline{d}=\varepsilon\underline{e}$, $\underline{b}=\mu\underline{h}$ (**todo** do energy balance for all charges and currents and spearating contributions of free and bound charges and currents)

$$\begin{aligned}T^{\alpha\beta}&=\frac{1}{4}\begin{bmatrix}2(-\underline{e}^T\underline{d}+\underline{b}^T\underline{h}) & \underline{0}^T\\ \underline{0} & -2(-\underline{e}^T\underline{d}+\underline{b}^T\underline{h})\underline{\underline{I}}\end{bmatrix}-\begin{bmatrix}-\underline{e}^T\underline{d} & \frac{(\underline{e}\times\underline{h})^T}{c}\\ c\underline{b}\times\underline{d} & \underline{e}\underline{d}^T+\underline{b}_\times\underline{h}_\times\end{bmatrix}=\\ &=\begin{bmatrix}\frac{1}{2}\left(\underline{e}^T\underline{d}+\underline{b}^T\underline{h}\right) & \frac{(\underline{e}\times\underline{h})^T}{c}\\ \frac{\underline{e}\times\underline{h}}{c} & \frac{1}{2}(\underline{e}^T\underline{d}+\underline{b}^T\underline{h})\underline{\underline{I}}_3-\underline{e}\underline{d}^T-\underline{b}\underline{h}^T\end{bmatrix}=\\ &=\begin{bmatrix}u & \frac{\vec{s}}{c}\\ \frac{\vec{s}}{c} & u\mathbb{I}-\vec{e}\otimes\vec{d}-\vec{b}\otimes\vec{h}\end{bmatrix}\end{aligned}$$

Energy-stress equation in 4-dimensional formalism

Energy and momentum balance equations (2.5) can be recast using 4-dimensional formalism,

$$\begin{aligned}\nabla\cdot\mathbf{T}&=-\mathbf{F}\cdot\mathbf{J}\\ \frac{\partial T^{\nu\mu}}{\partial X^\nu}&=-F^{\mu\nu}J_\nu\end{aligned}$$

as

$$F^{\mu\nu}J_\nu=\begin{bmatrix}0 & -\frac{\underline{e}^T}{c}\\ \frac{\underline{e}}{c} & \underline{b}_\times\end{bmatrix}\begin{bmatrix}\rho c\\ -\underline{j}\end{bmatrix}=\begin{bmatrix}\frac{1}{c}\underline{e}^T\underline{j}\\ \rho\underline{e}-\underline{b}_\times\underline{j}\end{bmatrix}$$

$$\begin{aligned}0:\quad &\frac{\partial T^{\nu 0}}{\partial X^\nu}=-F^{0\nu}J_\nu \quad \frac{1}{c}\frac{\partial}{\partial t}u+\frac{\partial}{\partial x_k}\frac{s_k}{c}=-\frac{1}{c}e_kj_k\\ 1:3:\quad &\frac{\partial T^{\nu i}}{\partial X^\nu}=-F^{i\nu}J_\nu \quad \frac{1}{c}\frac{\partial}{\partial t}\frac{s_i}{c}+\frac{\partial\sigma_{ki}}{\partial x_k}=-\rho e_i+\varepsilon_{ijk}b_jj_k\end{aligned}$$

Relativity under Lorentz's transformation. Components of the description of two inertial observers in relative motion, with aligned Cartesian space coordinates, are related by Lorentz's transformation

Energy-momentum-stress tensor

todo *Complete*

Contravariant (Cartesian in space) components of the energy-momentu-stress tensor can be collected in a symmetric matrix

$$\mathbf{T}' = \mathbf{\Lambda}\mathbf{T}\mathbf{\Lambda}^T$$

$$\mathbf{T} = \begin{bmatrix} u & \frac{\mathbf{s}^T}{c} \\ \frac{\mathbf{s}}{c} & \boldsymbol{\sigma} \end{bmatrix}$$

$$\mathbf{\Lambda} = \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} = \gamma \begin{bmatrix} 1 & -\mathbf{v}^T \\ -\mathbf{v} & \hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} + \begin{bmatrix} 0 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{I} - \hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{P} \end{bmatrix}$$

$$\begin{aligned} \mathbf{T}' &= \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{P} \end{bmatrix} \begin{bmatrix} u & \frac{\mathbf{s}^T}{c} \\ \frac{\mathbf{s}}{c} & \boldsymbol{\sigma} \end{bmatrix} \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{P} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma u - \frac{\gamma}{c}\mathbf{s}^T\mathbf{v} & \frac{\gamma}{c}\mathbf{s}^T - \gamma\mathbf{v}^T\boldsymbol{\sigma} \\ -\gamma u\mathbf{v} + \mathbf{P}\frac{\mathbf{s}}{c} & -\frac{\gamma}{c}\mathbf{v}\mathbf{s}^T + \mathbf{P}\boldsymbol{\sigma} \end{bmatrix} \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{P} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma^2 u - 2\frac{\gamma^2}{c}\mathbf{s}^T\mathbf{v} + \gamma^2\mathbf{v}^T\boldsymbol{\sigma}\mathbf{v} & -\gamma^2 u\mathbf{v}^T + \frac{\gamma^2}{c}\mathbf{s}^T\mathbf{v}\mathbf{v}^T + \frac{\gamma}{c}\mathbf{s}^T\mathbf{P} - \gamma\mathbf{v}^T\boldsymbol{\sigma}\mathbf{P} \\ -\gamma^2 u\mathbf{v} + \frac{\gamma}{c}\mathbf{P}\mathbf{s} + \frac{\gamma^2}{c}\mathbf{v}\mathbf{s}^T\mathbf{v} - \gamma\mathbf{P}\boldsymbol{\sigma}\mathbf{v} & \gamma^2 u\mathbf{v}\mathbf{v}^T - \frac{\gamma}{c}\left(\mathbf{P}\mathbf{s}\mathbf{v}^T + \mathbf{v}\mathbf{s}^T\mathbf{P}\right) + \mathbf{P}\boldsymbol{\sigma}\mathbf{P} \end{bmatrix} \end{aligned}$$

CHAPTER

THREE

INERTIAL REFERENCE FRAMES AND LORENTZ'S TRANSFORMATIONS

Equations of physics as seen by inertial reference frames (**todo** *but what's an inertial reference frame?*) have the same expressions. Change of observer can be related to change of coordinates (and choice of basis vectors, in special relativity) used to describe a physical phenomenon, that is invariant to this change for its very nature: reality doesn't change if observed from different point of view (if observations don't interact with reality itself, at least). Proper mathematical tools for dealing with invariance are vectors and tensors in general.

Using 2 different sets of coordinates associated with constant and uniform basis vectors $\mathbf{E}_\alpha$, $\mathbf{E}'_\beta$ a vector can be written as

$$\mathbf{V} = V^\alpha \mathbf{E}_\alpha = V'^\beta \mathbf{E}'_\beta \, .$$

Lorentz transformation describe the change of description between two inertial reference frames.

3.1 Lorentz's transformations

3.1.1 Lorentz's transformations in standard configuration

Standard configuration is defined for two observers with Cartesian bases for the space components, with the axes aligned, and with relative motion along x, x' axes.

…**todo** *derivation…see material for high school: Special Relativity:Relativity and Lorentz's transformations*

$$\begin{aligned} ct' &= \gamma(ct - vx) \\ x' &= \gamma(-vct + x) \\ y' &= y \\ z' &= z \end{aligned} \, ,$$

with $\gamma := \dfrac{1}{\sqrt{1-v^2}}$, so that the coordinates transformations can be written using matrix of change of coordinates

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}$$

$$X'^\alpha = \Lambda^\alpha_\beta X^\beta \, .$$

This is the same transformation of the components of all the vectors of the 4-dimensional space as seen by two different inertial observers

3.1.2 Lorentz's transformation in general configuration

Beside change of origin of the coordinates, general transformation can be derived composing Lorentz's transformations in stanrdard configuration (the only one derived so far), and rotations of the space coordinates.

As an example, the general expression of Lorentz's transformation between two inertial reference frames with general relative (space) velocity $\vec{v}$ - and aligned axes - can be derived introducing two intermediate inertial reference frames: reference $Otxyz$ and $O't'x'y'z'$ are the two refence frames we'd like to link with a Lorentz transformation; $O_a t_a x_a y_a z_a$ is a reference frame at rest w.r.t. $Otxyz$ with the x_a axis with the same direction of the relative (space) velocity $\vec{v}$, $O_b t_b x_b y_a z_b$ is in standard configuration with O_a. The transformation of coordinates reads

$$\begin{aligned}
\begin{bmatrix} x'_0 \\ \mathbf{r}' \end{bmatrix} &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} x_{b,0} \\ \mathbf{x}_b \end{bmatrix} = \\
&= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} x_{a,0} \\ \mathbf{x}_a \end{bmatrix} = \\
&= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \left(\begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma - 1 & & \\ & & 0 & \\ & & & 0 \end{bmatrix} + \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \right) \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix},
\end{aligned}$$

and the two contributions give

$$\begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma - 1 & & \\ & & 0 & \\ & & & 0 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v \mathbf{R}_{1,:} \\ -\gamma v & (\gamma - 1)\mathbf{R}_{1,:} \\ \mathbf{0}_{2,1} & \mathbf{0}_{2,4} \end{bmatrix} =$$

$$= \begin{bmatrix} \gamma & -\gamma v \mathbf{R}_{1,:} \\ -\gamma v \mathbf{R}^T_{1,:} & (\gamma - 1)\mathbf{R}^T_{1,:}\mathbf{R}_{1,:} \end{bmatrix} =$$

$$= \begin{bmatrix} \gamma & -\gamma R_{11} v & -\gamma R_{12} v & -\gamma R_{13} v \\ -\gamma R_{11} v & (\gamma - 1) R_{11} R_{11} & (\gamma - 1) R_{11} R_{12} & (\gamma - 1) R_{11} R_{13} \\ -\gamma R_{12} v & (\gamma - 1) R_{12} R_{11} & (\gamma - 1) R_{12} R_{12} & (\gamma - 1) R_{12} R_{13} \\ -\gamma R_{13} v & (\gamma - 1) R_{13} R_{11} & (\gamma - 1) R_{13} R_{12} & (\gamma - 1) R_{13} R_{13} \end{bmatrix}.$$

$$\begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} 0 & \\ & \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} = \begin{bmatrix} 0 & \\ & \mathbf{I}_3 \end{bmatrix}$$

Here, components $(R_{11}v, R_{12}v, R_{13}v)$ are the components of velocity $\vec{v}$ as seen by observer O, and thus the matrix for coordinate transformation reads

$$\begin{aligned}
\mathbf{\Lambda} &= \begin{bmatrix} 0 & \\ & \mathbf{I}_3 \end{bmatrix} + \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & (\gamma - 1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} = \\
&= \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{I}_3 + (\gamma - 1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix}
\end{aligned} \tag{3.1}$$

and the transformation of coordinates can be written as

$$\begin{cases}
ct' &= \gamma(ct - \vec{v} \cdot \vec{r}) \\
\vec{r}' &= -\gamma ct \vec{v} + (\gamma - 1)\, \hat{v} \otimes \hat{v} \cdot \vec{r} + \vec{r} = \\
&= -\gamma ct \vec{v} + \underbrace{[\mathbb{I} - \hat{v} \otimes \hat{v}]}_{\mathbb{P}_{\perp \hat{v}}} \cdot \vec{r} + \gamma \underbrace{\hat{v} \otimes \hat{v}}_{\mathbb{P}_{\parallel \hat{v}}} \cdot \vec{r}\,,
\end{cases} \tag{3.2}$$

having introduced the two orthogonal projectors in directions parallel and orthogonal to the direction of the velocity, $\hat{v}$.

Example (Standard configuration, as a special case.)

If $\vec{v} = v\hat{x}$, Lorentz's transformation for two inertial observers in starndard configuration is retrieved from the general expression of Lorentz's transformation

$$\begin{cases} ct' &= \gamma(ct - vx) \\ \vec{r}' &= -\gamma vct\hat{x} + y\hat{y} + z\hat{z} + \gamma vx\hat{x} \end{cases}$$

3.2 Transformation of vector components and vectors of the bases

$$\mathbf{V} = V^{\alpha}\mathbf{E}_{\alpha} = V'^{\beta}\mathbf{E}'_{\beta}$$
$$V'^{\beta} = \Lambda^{\beta}_{\alpha}V^{\alpha} \,,$$

being α and β the indice of columns and rows of matrix $\mathbf{\Lambda}$ of change of coordinates of the general Lorentz's transformation (3.1). In order to keep invariance, vectors of the bases transform with the transpose transformation, namely

$$\mathbf{V} = V^{\alpha}\mathbf{E}_{\alpha} = V'^{\beta}\mathbf{E}'_{\beta} = V^{\alpha}\underbrace{\Lambda^{\beta}_{\alpha}\mathbf{E}'_{\beta}}_{\mathbf{E}_{\alpha}}$$

$$\mathbf{E}_{\alpha} = \Lambda^{\beta}_{\alpha}\mathbf{E}'_{\beta} \,. \tag{3.3}$$
$$\left[\Lambda^{-1}\right]^{\alpha}_{\phi}\mathbf{E}_{\alpha} = \underbrace{[\Lambda^{-1}]^{\alpha}_{\phi}\Lambda^{\beta}_{\alpha}}_{=\delta^{\beta}_{\phi}}\mathbf{E}'_{\beta}$$

3.3 Transformation of tensor components

$$\mathbf{D} = D^{\alpha\beta}\mathbf{E}_{\alpha}\mathbf{E}_{\beta} = D'^{\phi\eta}\mathbf{E}'_{\phi}\mathbf{E}'_{\eta} =$$

and using the rule of transformation of vectors of the bases (3.3)

$$D'^{\phi\eta} = D^{\alpha\beta}\Lambda^{\phi}_{\alpha}\Lambda^{\eta}_{\beta} \,,$$

or using matrix notation (**todo** *avoid abuse of notation! Use underline for arrays of coomponents, bold for vectors and tensors*)

$$\mathbf{D}' = \mathbf{\Lambda}\mathbf{D}\mathbf{\Lambda}^{T}$$

Example (Electromagnetic field tensor)

Contravariant components of the electromagnetic field tensor

$$\begin{aligned} \begin{bmatrix} 0 & -c\mathbf{d}'^{T} \\ c\mathbf{d}' & \mathbf{h}'_{\times} \end{bmatrix} &= \begin{bmatrix} \gamma & -\gamma\mathbf{v}^{T} \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^{T} \end{bmatrix} \begin{bmatrix} 0 & -c\mathbf{d}^{T} \\ c\mathbf{d} & \mathbf{h}_{\times} \end{bmatrix} \begin{bmatrix} \gamma & -\gamma\mathbf{v}^{T} \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^{T} \end{bmatrix}^{T} = \\ &= \begin{bmatrix} \gamma & -\gamma\mathbf{v}^{T} \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^{T} \end{bmatrix} \begin{bmatrix} \gamma cvd_v & -c\mathbf{d}^{T} - (\gamma-1)cd_v\hat{\mathbf{v}}^{T} \\ \gamma c\mathbf{d} - \gamma\mathbf{h}_{\times}\mathbf{v} & -\gamma c\mathbf{d}\mathbf{v}^{T} + \mathbf{h}_{\times} + (\gamma-1)\mathbf{h}_{\times}\hat{\mathbf{v}}\hat{\mathbf{v}}^{T} \end{bmatrix} = \\ &= \begin{bmatrix} 0 & \text{anti-sym} \\ c\mathbf{d}' & \mathbf{h}'_{\times} \end{bmatrix} = \end{aligned}$$

with

$$\begin{aligned} c\mathbf{d}' &= -\gamma^2 cv^2\hat{\mathbf{v}}d_v + \gamma c\mathbf{d} - \gamma\mathbf{h}_\times\mathbf{v} + \gamma(\gamma-1)c\hat{\mathbf{v}}d_v = \\ &= c\gamma^2\hat{\mathbf{v}}d_v\underbrace{(1-v^2)}_{=\gamma^{-2}} + \gamma c\mathbf{d} - \gamma\mathbf{h}\times\mathbf{v} - \gamma c\hat{\mathbf{v}}d_v = \\ &= \gamma c\mathbf{d} - \gamma\mathbf{h}\times\mathbf{v} + (1-\gamma)c\hat{\mathbf{v}}\hat{\mathbf{v}}\cdot\mathbf{d}\,, \end{aligned}$$

and

$$\begin{aligned} \mathbf{h}'_\times &= && (1)\\ &= \gamma cv\hat{\mathbf{v}}\mathbf{d}^T + \gamma(\gamma-1)cvd_v\hat{\mathbf{v}}\hat{\mathbf{v}}^T + \\ &- \gamma cv\mathbf{d}\hat{\mathbf{v}} + \mathbf{h}_\times + (\gamma-1)(\mathbf{h}\times\hat{\mathbf{v}})\hat{\mathbf{v}}^T - \gamma(\gamma-1)cvd_v\hat{\mathbf{v}}\hat{\mathbf{v}}^T + (\gamma-1)\hat{\mathbf{v}}\mathbf{v}^T\mathbf{h}_\times + \mathbf{0} = && (2)\\ &= \gamma cv\left(\hat{\mathbf{v}}\mathbf{d}^T - \mathbf{d}\hat{\mathbf{v}}^T\right) + \mathbf{h}_\times + (\gamma-1)\left((\mathbf{h}_\times\hat{\mathbf{v}})\,\hat{\mathbf{v}}^T - \hat{\mathbf{v}}\,(\mathbf{h}_\times\hat{\mathbf{v}})^T\right) = && (3)\\ &= \gamma cv\left(\mathbf{d}\times\hat{\mathbf{v}}\right)_\times + \mathbf{h}_\times + (\gamma-1)\left(\hat{\mathbf{v}}\times(\mathbf{h}\times\hat{\mathbf{v}})\right)_\times \end{aligned}$$

having (1) recognized that $\hat{\mathbf{v}}\hat{\mathbf{v}}^T\mathbf{h}_\times\hat{\mathbf{v}}\hat{\mathbf{v}}^T = \hat{\mathbf{v}}(\hat{\mathbf{v}}\cdot(\mathbf{h}\times\hat{\mathbf{v}}))\hat{\mathbf{v}} = \mathbf{0}$, (2) $\{\mathbf{v}^T\mathbf{h}_\times\}_k = v_i\varepsilon_{ijk}h_j = -\{\mathbf{h}\times\hat{\mathbf{v}}\}_k$ and (3)

$$\begin{aligned} \left[(\mathbf{a}_\times\mathbf{b})_\times\right]_{ij} &= \varepsilon_{ikj}\varepsilon_{klm}a_l b_m = \\ &= \left(\delta_{jl}\delta_{im} - \delta_{jm}\delta_{il}\right)a_l b_m = \\ &= a_j b_i - a_i b_j = \left[\mathbf{b}\mathbf{a}^T - \mathbf{a}\mathbf{b}^T\right]_{ij}\,. \end{aligned}$$

Thus

$$\begin{aligned} \mathbf{h}' &= \mathbf{h} - \gamma c\mathbf{v}\times\mathbf{d} + (1-\gamma)(\mathbf{h}\times\hat{\mathbf{v}})\times\hat{\mathbf{v}} = \\ &= \mathbf{h} - \gamma c\mathbf{v}\times\mathbf{d} + (1-\gamma)\left(\hat{\mathbf{v}}\hat{\mathbf{v}}\cdot\mathbf{h} - \mathbf{h}\right) = \\ &= \gamma\mathbf{h} - \gamma c\mathbf{v}\times\mathbf{d} + (1-\gamma)\hat{\mathbf{v}}\hat{\mathbf{v}}\cdot\mathbf{h}\,. \end{aligned}$$

Going back from non dimensional velocity to dimensional velocity $c\mathbf{v}\to\mathbf{v}$,

$$\begin{aligned} \mathbf{d}' &= \gamma\left(\mathbf{d} - \frac{\mathbf{h}\times\mathbf{v}}{c^2}\right) + (1-\gamma)\hat{\mathbf{v}}\,\hat{\mathbf{v}}\cdot\mathbf{d} \\ \mathbf{h}' &= \gamma\left(\mathbf{h} - \mathbf{v}\times\mathbf{d}\right) + (1-\gamma)\hat{\mathbf{v}}\,\hat{\mathbf{v}}\cdot\mathbf{h} \end{aligned}$$

Repeating the same process for the electromagnetic field tensor

$$[\mathbf{F}]^{\alpha\beta} = \begin{bmatrix} 0 & -\dfrac{\mathbf{e}^T}{c} \\ \dfrac{\mathbf{e}}{c} & \mathbf{b}_\times \end{bmatrix}$$

$$\begin{aligned} \mathbf{e}' &= \gamma\left(\mathbf{e} - \mathbf{b}\times\mathbf{v}\right) + (1-\gamma)\hat{\mathbf{v}}\,\hat{\mathbf{v}}\cdot\mathbf{e} \\ \mathbf{b}' &= \gamma\left(\mathbf{b} - \frac{\mathbf{v}\times\mathbf{e}}{c^2}\right) + (1-\gamma)\hat{\mathbf{v}}\,\hat{\mathbf{v}}\cdot\mathbf{b} \end{aligned}$$

Example (4-Current)

$$\mathbf{J}' = \mathbf{\Lambda}\mathbf{J}$$

$$\begin{aligned} c\rho' &= \gamma c\rho - \gamma v\mathbf{j} \\ \mathbf{j}' &= -\gamma vc\rho + \gamma\mathbf{j} \end{aligned}$$

Example (Constitutive equations)

Linear isotropic medium

$$\mathbf{d} = \varepsilon_0 \mathbf{e} + \mathbf{p}$$
$$\mathbf{b} = \mu_0 \mathbf{h} + \mu_0 \mathbf{m}$$

with $c^2 = \dfrac{1}{\varepsilon_0 \mu_0}$

$$c\mathbf{d} = \varepsilon_0 c^2 \frac{\mathbf{e}}{c} + c\mathbf{p} = \frac{1}{\mu_0}\frac{\mathbf{e}}{c} + c\mathbf{p}$$
$$\mathbf{h} = \frac{1}{\mu_0}\mathbf{b} - \mathbf{m}$$

the constitutive equations can be written in 4-dimensional form as

$$\mathbf{D} = \frac{1}{\mu_0}\mathbf{F} + \mathbf{P}$$

$$\begin{bmatrix} 0 & -c\mathbf{d}^T \\ c\mathbf{d} & \mathbf{h}_\times \end{bmatrix} = \frac{1}{\mu_0}\begin{bmatrix} 0 & -\mathbf{e}^T/c \\ \mathbf{e}/c & \mathbf{b}_\times \end{bmatrix} + \begin{bmatrix} 0 & -c\mathbf{p}^T \\ c\mathbf{p} & -\mathbf{m}_\times \end{bmatrix}$$

CHAPTER

FOUR

ELECTROMAGNETISM

From 3-dimensional to 4-dimensional formalism. In this section the 4-dimensional formalism is introduced, and the 4-dimensional version of governing equations of electromagnetism that naturally suits special relativity theory is derived starting from the governing equations of classical electromagnetism.

Definition of physical quantities

- 4-potential vector, $\mathbf{A} = \frac{\phi}{c}\mathbf{E}_0 + a^i\mathbf{E}_i$
- 4-current vector, $\mathbf{J} = c\rho\mathbf{E}_0 + j^i\mathbf{E}_i$
- EM field tensor,

$$\mathbf{F} = -\mathbf{E}_0 \otimes \mathbf{E}_i \frac{e_i}{c} + \frac{e_i}{c}\mathbf{E}_i \otimes \mathbf{E}_0 + \varepsilon_{ijk} b_j \mathbf{E}_i \otimes \mathbf{E}_k$$
$$\mathbf{D} = -\mathbf{E}_0 \otimes \mathbf{E}_i c d_i + d_i \mathbf{E}_i \otimes \mathbf{E}_0 + \varepsilon_{ijk} h_j \mathbf{E}_i \otimes \mathbf{E}_k$$

  or

$$[F^{\alpha\beta}] = \begin{bmatrix} 0 & -\mathbf{e}^T/c \\ \mathbf{e}/c & \mathbf{b}_\times \end{bmatrix} \quad , \quad [D^{\alpha\beta}] = \begin{bmatrix} 0 & -c\mathbf{d}^T \\ c\mathbf{d} & \mathbf{h}_\times \end{bmatrix}$$

- 4-momentum-energy density tensor, $\mathbf{T}$

Definitions, relations and equations

- Lorentz's gauge, $\nabla \cdot \mathbf{A} = 0$
- EM field tensor, $\mathbf{F} = \nabla\mathbf{A} - (\nabla\mathbf{A})^T$
- 4-current continuity equation, $\nabla \cdot \mathbf{J} = 0$
- Maxwell's equations

$$\boldsymbol{\nabla} \cdot \mathbf{D} = \mathbf{J}^f$$
$$\nabla \cdot (\boldsymbol{\epsilon} : \mathbf{F}) = \mathbf{0}$$

- Constitutive equations

$$\mathbf{D} = \frac{1}{\mu_0}\mathbf{F} + \mathbf{P}$$

- Wave equation for the pontential (what about wave equations for the EM field?)

$$\nabla \cdot \nabla\mathbf{A} = \mu_0 \mathbf{J}$$

- Energy balance equation

$$\nabla \cdot \mathbf{T} = -\mathbf{F} \cdot \mathbf{J}$$

- Equation of motion of a point charge in a EM field,

$$m\mathbf{X}'' = q\mathbf{F} \cdot \mathbf{X}'$$

Einstein's special relativity and Lorentz's transformations for the EM quantities. After the equations are derived, Lorentz's transformations are used to discuss special relativity. Low-speed relations, used in classical electromagnetism for systems with charateristic speed $v \ll c$, are derived.

Lagrangian approach to electromagnetism in special relativity. Weak form of the equations are derived, and the Lagrangian approach to electromagnetism in special relativity is discussed: both field equations and dynamical equations of charges moving in an electromagnetic field are re-derived with a Lagrangian approach.

CHAPTER

FIVE

"CONSTANT" ACCELERATION OBSERVER AND RINDLER COORDINATES

todo Check the hand-written notes into .md converted by Gemini below

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This section covers the kinematics of uniformly accelerated motion in Special Relativity, the construction of the reference frame of an accelerated observer, the derivation of Rindler coordinates, and their geometric properties (metric, Christoffel symbols, geodesics, zero Riemann curvature tensor, and the Newtonian/classical limit).

5.1 3+1 and 4D Kinematics of an Accelerated Point

5.1.1 4-Vector Definitions in an Inertial Frame

Consider an inertial reference frame with Cartesian basis vectors $\{\underline{E}_\mu\}$ such that $\underline{E}_\mu \cdot \underline{E}_\nu = \eta_{\mu\nu} = \text{diag}(1,-1,-1,-1)$.

- **4-position:**

$$\underline{X}(\tau) = X^\mu(\tau)\underline{E}_\mu = ct\underline{E}_0 + x_i\underline{E}_i$$

where τ is the proper time defined by:

$$c^2 d\tau^2 = c^2 dt^2 - |d\vec{x}|^2 = c^2\left(1 - \frac{|\vec{v}|^2}{c^2}\right)dt^2 \implies d\tau = \gamma^{-1}dt$$

with $\gamma = \left(1 - \frac{|\vec{v}|^2}{c^2}\right)^{-1/2}$.

- **4-velocity:**

$$\underline{U} = \frac{d\underline{X}}{d\tau} = \frac{dt}{d\tau}\frac{d\underline{X}}{dt} = c\gamma\underline{E}_0 + \gamma v_i\underline{E}_i$$

which satisfies $\underline{U} \cdot \underline{U} = c^2$.

- **4-acceleration:**

$$\underline{A} = \frac{d\underline{U}}{d\tau} = \gamma\frac{d\underline{U}}{dt}$$

Expanding component-wise:

$$\frac{d}{dt}(\gamma c) = c\frac{d\gamma}{dt} = c\left(-\frac{1}{2}\right)\left(1 - \frac{|\vec{v}|^2}{c^2}\right)^{-3/2}\left(-2\vec{v}\cdot\frac{d\vec{v}}{dt}\right)\frac{1}{c^2} = \frac{\gamma^3}{c}(\vec{v}\cdot\vec{a})$$

$$\frac{d}{dt}(\gamma v_i) = \gamma^3 \frac{1}{c^2}(\vec{v}\cdot\vec{a})v_i + \gamma a_i$$

Combining these gives:

$$\underline{A} = \left(\frac{\gamma^4}{c}(\vec{v}\cdot\vec{a}),\ \gamma^2\vec{a} + \frac{\gamma^4}{c^2}\vec{v}(\vec{v}\cdot\vec{a})\right)$$

Using the longitudinal/transverse decomposition of acceleration:

$$\underline{A} = \gamma^4 \left\{ \frac{\vec{v}\cdot\vec{a}}{c},\ \frac{\vec{v}}{c}\left(\frac{\vec{v}}{c}\cdot\vec{a}\right) + \left(1 - \frac{|\vec{v}|^2}{c^2}\right)\vec{a}_\perp \right\}$$

5.2 Uniformly Accelerated Trajectory

Consider motion along one spatial direction (x) with constant proper acceleration magnitude α_p such that $\underline{A}\cdot\underline{A} = -\alpha_p^2$. Assuming initial conditions at $\tau = 0$:

$$(ct, x)|_{\tau=0} = (0, x_{P0}), \quad v(0) = 0$$

The solution to the equations of motion yields hyperbolic motion:

$$ct(\tau) = \frac{c^2}{\alpha_p}\sinh\left(\frac{\alpha_p\tau}{c}\right)$$

$$x(\tau) = \frac{c^2}{\alpha_p}\cosh\left(\frac{\alpha_p\tau}{c}\right) - \frac{c^2}{\alpha_p} + x_{P0}$$

Choosing $x_{P0} = \frac{c^2}{\alpha_p}$, this simplifies to:

$$\begin{cases} ct = \frac{c^2}{\alpha_p}\sinh\left(\frac{\alpha_p\tau}{c}\right) \\ x = \frac{c^2}{\alpha_p}\cosh\left(\frac{\alpha_p\tau}{c}\right) \end{cases}$$

5.3 Construction of the Accelerated Observer Frame & Rindler Coordinates

An accelerated observer P builds a local coordinate system using:

1. **Proper time standard (τ):** Measuring time along their worldline using their own clock.
2. **Orthogonal local frame:** Constructing an instantaneous rest frame basis $\{\underline{e}_0(\tau), \underline{e}_1(\tau)\}$:

$$\underline{e}_0(\tau) = \frac{\underline{U}}{c} = \cosh\left(\frac{g\tau}{c}\right)\underline{E}_0 + \sinh\left(\frac{g\tau}{c}\right)\underline{E}_1$$

$$\underline{e}_1(\tau) = \sinh\left(\frac{g\tau}{c}\right)\underline{E}_0 + \cosh\left(\frac{g\tau}{c}\right)\underline{E}_1$$

where $g = \alpha_p$. Note that $\underline{e}_0\cdot\underline{e}_0 = 1$, $\underline{e}_1\cdot\underline{e}_1 = -1$, and $\underline{e}_0\cdot\underline{e}_1 = 0$.

5.3.1 Coordinate Assignment for Event Q

An event Q at spatial distance z along the observer's instantaneous spatial axis $\underline{e}_1(\tau)$ has the spacetime position:

$$\underline{X}_Q(\tau, z) = \underline{X}_P(\tau) + z\,\underline{e}_1(\tau)$$

Expanding in the Minkowski background frame $\{\underline{E}_0, \underline{E}_1\}$:

$$\underline{X}_Q = \underline{E}_0 \left(\frac{c^2}{g} \sinh \frac{g\tau}{c} + z \sinh \frac{g\tau}{c} \right) + \underline{E}_1 \left(\frac{c^2}{g} \cosh \frac{g\tau}{c} + z \cosh \frac{g\tau}{c} \right)$$

$$\underline{X}_Q = \left(\frac{c^2}{g} + z \right) \sinh \left(\frac{g\tau}{c} \right) \underline{E}_0 + \left(\frac{c^2}{g} + z \right) \cosh \left(\frac{g\tau}{c} \right) \underline{E}_1$$

5.3.2 Definition of Rindler Coordinates $(\eta, \bar{\xi})$

Define the transformed coordinates:

$$\bar{\xi} = \frac{c^2}{g} + z, \quad \eta = \frac{g\tau}{c}$$

This maps Minkowski coordinates (ct, x) to Rindler coordinates $(\eta, \bar{\xi})$:

$$\begin{cases} ct = \bar{\xi} \sinh \eta \\ x = \bar{\xi} \cosh \eta \end{cases}$$

Curves of constant spatial coordinate $\bar{\xi} = \bar{\xi}_k$ represent hyperbolas ($x^2 - c^2t^2 = \bar{\xi}_k^2$) corresponding to observers maintaining constant acceleration $\alpha = c^2/\bar{\xi}_k$.

5.4 Metric Tensor in Rindler Coordinates

Differentials of the coordinate transformation are given by:

$$cdt = \cosh \eta \, d\bar{\xi} + \bar{\xi} \sinh \eta \, d\eta$$

$$dx = \sinh \eta \, d\bar{\xi} + \bar{\xi} \cosh \eta \, d\eta$$

Substituting these into the flat Minkowski line element $ds^2 = c^2dt^2 - dx^2$:

$$ds^2 = (\cosh \eta \, d\bar{\xi} + \bar{\xi} \sinh \eta \, d\eta)^2 - (\sinh \eta \, d\bar{\xi} + \bar{\xi} \cosh \eta \, d\eta)^2$$

$$ds^2 = \bar{\xi}^2 d\eta^2 - d\bar{\xi}^2$$

Alternatively, writing the metric in terms of proper time τ and local spatial coordinate z:

$$ds^2 = \left(1 + \frac{gz}{c^2}\right)^2 c^2 d\tau^2 - dz^2$$

5.4.1 Natural Basis Vectors and Dual Metric Components

The natural basis vectors $\underline{b}_\eta = \partial_\eta \underline{X}$ and $\underline{b}_{\bar{\xi}} = \partial_{\bar{\xi}} \underline{X}$ are:

$$\underline{b}_\eta = \bar{\xi} \cosh\eta \, \underline{E}_0 + \bar{\xi} \sinh\eta \, \underline{E}_1$$

$$\underline{b}_{\bar{\xi}} = \sinh\eta \, \underline{E}_0 + \cosh\eta \, \underline{E}_1$$

The metric components $g_{\mu\nu} = \underline{b}_\mu \cdot \underline{b}_\nu$ are:

$$g_{\eta\eta} = \bar{\xi}^2, \quad g_{\bar{\xi}\bar{\xi}} = -1, \quad g_{\eta\bar{\xi}} = 0$$

$$\mathbf{g} = \begin{pmatrix} \bar{\xi}^2 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{g}^{-1} = \begin{pmatrix} \frac{1}{\bar{\xi}^2} & 0 \\ 0 & -1 \end{pmatrix}$$

5.5 Christoffel Symbols & Riemann Tensor

5.5.1 Derivatives of Basis Vectors

Calculating derivatives of the natural basis vectors with respect to coordinates $(\bar{\xi}, \eta)$:

$$\partial_{\bar{\xi}} \underline{b}_{\bar{\xi}} = 0$$

$$\partial_\eta \underline{b}_{\bar{\xi}} = \cosh\eta \, \underline{E}_0 + \sinh\eta \, \underline{E}_1 = \frac{1}{\bar{\xi}} \underline{b}_\eta$$

$$\partial_\eta \underline{b}_\eta = \bar{\xi} \sinh\eta \, \underline{E}_0 + \bar{\xi} \cosh\eta \, \underline{E}_1 = \bar{\xi} \, \underline{b}_{\bar{\xi}}$$

5.5.2 Non-Zero Christoffel Symbols

From $\partial_\mu \underline{b}_\nu = \Gamma^\lambda_{\mu\nu} \underline{b}_\lambda$, the non-zero Christoffel symbols are:

$$\Gamma^\eta_{\eta\bar{\xi}} = \Gamma^\eta_{\bar{\xi}\eta} = \frac{1}{\bar{\xi}}$$

$$\Gamma^{\bar{\xi}}_{\eta\eta} = \bar{\xi}$$

5.5.3 Curvature (Zero Riemann Tensor)

Calculating the Riemann curvature tensor components using $R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}$:

$$R^{\bar{\xi}}{}_{\eta\bar{\xi}\eta} = \partial_{\bar{\xi}} \Gamma^{\bar{\xi}}_{\eta\eta} - \partial_\eta \Gamma^{\bar{\xi}}_{\bar{\xi}\eta} + \Gamma^{\bar{\xi}}_{\bar{\xi}\lambda}\Gamma^\lambda_{\eta\eta} - \Gamma^{\bar{\xi}}_{\eta\lambda}\Gamma^\lambda_{\bar{\xi}\eta}$$

$$R^{\bar{\xi}}{}_{\eta\bar{\xi}\eta} = \frac{\partial}{\partial\bar{\xi}}(\bar{\xi}) - 0 + 0 - \left(\Gamma^{\bar{\xi}}_{\eta\eta}\Gamma^\eta_{\bar{\xi}\eta}\right) = 1 - \left(\bar{\xi} \cdot \frac{1}{\bar{\xi}}\right) = 0$$

Thus, all components of the Riemann tensor vanish ($R^\rho{}_{\sigma\mu\nu} = 0$), confirming that the Rindler frame describes **flat spacetime** in non-inertial coordinates.

5.6 Geodesics in Rindler Coordinates

The geodesic equations are given by:

$$\ddot{x}^\mu + \Gamma^\mu_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta = 0$$

where dots denote derivatives with respect to an affine parameter (e.g., proper time s). Writing explicit components for $\bar{\xi}$ and η:

$$\begin{cases} \ddot{\bar{\xi}} + \Gamma^{\bar{\xi}}_{\eta\eta}\dot{\eta}^2 = 0 \implies \ddot{\bar{\xi}} + \bar{\xi}\dot{\eta}^2 = 0 \\ \ddot{\eta} + 2\Gamma^{\eta}_{\bar{\xi}\eta}\dot{\bar{\xi}}\dot{\eta} = 0 \implies \ddot{\eta} + \frac{2}{\bar{\xi}}\dot{\bar{\xi}}\dot{\eta} = 0 \end{cases}$$

5.7 Classical / Newtonian Limit

For low speeds ($v \ll c$) and small spatial extents around the observer line ($z \ll c^2/g$):

1. **Coordinate expansion:**

$$\bar{\xi} = \frac{c^2}{g} + z \implies 1 + \frac{gz}{c^2} \approx 1$$

$$\tau = \frac{c}{g}\eta \implies \eta \ll 1$$

2. **Metric limit:**

$$ds^2 = \left(1 + \frac{gz}{c^2}\right)^2 c^2 d\tau^2 - dz^2 \approx \left(1 + \frac{2gz}{c^2}\right) c^2 d\tau^2 - dz^2$$

   Matching with the weak static field limit metric $g_{00} = 1 + \frac{2\Phi}{c^2}$, we identify the fictitious Newtonian gravitational potential:

$$\Phi(z) = gz$$

3. **Geodesic / Acceleration Equation:** Substituting $\bar{\xi} = \frac{c^2}{g} + z$ into the spatial geodesic equation with $\dot{\eta} \approx \frac{g}{c}\dot{t}$:

$$\ddot{z} + \left(\frac{c^2}{g} + z\right)\left(\frac{g}{c}\right)^2 = 0 \implies \ddot{z} + g + \frac{g^2 z}{c^2} = 0$$

   Taking $c \to \infty$:

$$\ddot{z} + g = 0 \implies \ddot{z} = -g$$

This recovers the expected non-relativistic inertial force (apparent constant downward gravitational field $-g$) experienced in a uniformly accelerating reference frame.

Part II

General Relativity

CHAPTER

SIX

GENERAL RELATIVITY

CHAPTER

SEVEN

GENERAL RELATIVITY - NOTES

- ***Differential geometry***: metrics, derivatives, curvatures. This could be a short version of the notes about Differential Geometry and Tensor Calclulus
- **Equivalence principle** and the free fall as the natural state of motion: the physics in a freely falling laboratory is locally undistinguishable from physics in a gravity-free inertial frame in special relativity.
- ***Einstein field equation***. EFE is derived as the simplest linear relation between Ricci tensor - a second-order curvature tensor - and the energy momentum tensor, satisfying the energy-momentum conservation relation. *Linearization of Einstein equation* is performed to show the finite velocity of propagation: in the limit of weak gravitation, EFE can be recast as a linear wave equation with characteristic speed c.
- Examples:
  - Schwarzschild metrics
    - orbit of a planet, and precession of perihelion
    - trajectory of a zero-mass particle, and bending of rays of light

7.1 Differential Geometry

References

- Math basics: **todo** check and complete the chapters about **tensor calculus** and **differential geometry**

Basics

Let $\mathbf{X}(q^k)$ a parametrization of points in a n-dimensional space, $k = 1 : n$.

Natural basis, $\mathbf{b}_k := \frac{\partial \mathbf{X}}{\partial q^k}$. Vectors of the natural basis can be used to write vectors and tensor fields as a linear combination of them, or in their components w.r.t. that basis

$$\mathbf{v} = v^k \mathbf{b}_k \quad , \quad \mathbf{A} = A^{kl} \mathbf{b}_k \otimes \mathbf{b}_l \ .$$

Metric tensor, The covariant compoenents are $\mathbf{b}_k \cdot \mathbf{b}_l =: g_{kl}$. Some properties: $g_k^{\ l} = \delta_k^l$, $g_{ab} g^{bc} = \delta_a^c$.

Contravariant basis, $\{\mathbf{b}^k\}$, s.t. $\mathbf{b}_k \cdot \mathbf{b}^l = \delta_k^l$. It's easy to prove that $\mathbf{b}_k = g_{kl} \mathbf{b}^l$ (just scalar product with $\mathbf{b}^j \ldots$), and the law for raising or lower indices holds, $A^{ij} = g^{ik} A_k^{\ j}, \ldots$

Derivatives of the basis vectors. The components of the derivative $\frac{\partial \mathbf{b}_i}{\partial q^k} = \Gamma_{ik}^l \mathbf{b}_l$ are defined as the Christoffel symbols of the second type. As these are second-order derivatives, for Schwartz theorem about mixed partial derivatives, the

symmetry $\Gamma^k_{ij} = \Gamma^k_{ji}$ immediately follows. The derivative of the vectors of the contravariant basis follows from

$$\begin{aligned} 0 &= \partial_k \left(\mathbf{b}_l \cdot \mathbf{b}^m \right) = \\ &= \partial_k \mathbf{b}_l \cdot \mathbf{b}^m + \mathbf{b}_l \cdot \partial_k \mathbf{b}^m = \\ &= \Gamma^n_{kl} \underbrace{\mathbf{b}_n \cdot \mathbf{b}^m}_{=\delta^m_n} + \mathbf{b}_l \cdot \partial_k \mathbf{b}^m \ , \end{aligned}$$

and thus $\partial_k \mathbf{b}^m = -\Gamma^m_{kl} \mathbf{b}^l$.

Derivatives of the metric tensor. Using the definition of the covariant components of the metric tensor and the derivatives of the vectors of the natural basis, it's easy to prove

$$\partial_c g_{ab} = \Gamma^d_{ac} g_{db} + \Gamma^d_{bc} g_{ad}$$

Using the property $g_{ab} g^{bc} = \delta_{a^c}$, from its derivatives ∂_d,

$$\partial_d g_{ab} g^{bc} + g_{ab} \partial_d g^{bc} = 0 \ ,$$

it follows that $\partial_d g^{ec} = -g^{ea} \partial_d g_{ab} g^{bc}$.

Relations between the Christoffel symbols and the derivatives of the metric tensor.

$$\begin{aligned} \partial_c g_{ab} &= \Gamma^d_{ac} g_{db} + \Gamma^d_{bc} g_{ad} \\ \partial_a g_{bc} &= \Gamma^d_{ba} g_{dc} + \Gamma^d_{ca} g_{bd} \\ \partial_b g_{ca} &= \Gamma^d_{cb} g_{da} + \Gamma^d_{ab} g_{cd} \end{aligned}$$

so that, summing the first 2 equations and subtracting the last one,

$$\Gamma^b_{ac} = \frac{1}{2} g^{bd} \left(\partial_c g_{ad} + \partial_a g_{dc} - \partial_d g_{ac} \right) \tag{7.1}$$

7.1.1 Gradient, covariant derivative and directional derivative

Gradient of a scalar field

$$\nabla f = \mathbf{b}^k \partial_k f \ .$$

Gradient of a vector field

$$\nabla \mathbf{v} = \mathbf{b}^k \mathbf{b}_i \left(\partial_k v^i + \Gamma^i_{kl} v^l \right) \ .$$

Proof

$$\begin{aligned} \nabla \mathbf{v} &= \mathbf{b}^k \frac{\partial}{\partial q^k} \left(v^i \mathbf{b}_i \right) = \\ &= \mathbf{b}^k \mathbf{b}_i \partial_k v^i + \mathbf{b}^k \mathbf{b}_l \Gamma^l_{ik} v^i = \\ &= \mathbf{b}^k \mathbf{b}_i \left(\partial_k v^i + \Gamma^i_{kl} v^l \right) \ . \end{aligned}$$

Gradient of a 2-nd order tensor field

$$\nabla \mathbf{A} = \mathbf{b}^k \mathbf{b}_i \mathbf{b}_j \left(\partial_k A^{ij} + \Gamma^i_{kl} A^{lj} + \Gamma^j_{kl} A^{il} \right) = \mathbf{b}^k \mathbf{b}_i \mathbf{b}_j \nabla_k A^{ij}.$$

Proof

…

7.1.2 Curvature Tensor

todo. Meaning of this definition

…

Definition through the action on an arbitrary vector field, whose components are

$$\{\mathbf{v}\cdot\mathbf{R}\}_{\sigma\eta\nu} = v_\xi R^\xi{}_{\sigma\eta\nu} = \left(\nabla_\eta\nabla_\nu - \nabla_\nu\nabla_\eta\right) v_\sigma$$

First term

$$\begin{aligned}
\nabla\nabla\mathbf{v} &= \mathbf{b}^\eta\partial_\eta\left[\mathbf{b}^\nu\partial_\nu\left(v_\sigma\mathbf{b}^\sigma\right)\right] = \\
&= \mathbf{b}^\eta\partial_\eta\left[\mathbf{b}^\nu\mathbf{b}^\sigma\left(\partial_\nu v_\sigma - \Gamma^\xi_{\nu\sigma}v_\xi\right)\right] = \\
&= \mathbf{b}^\eta\mathbf{b}^\nu\mathbf{b}^\sigma\left[-\Gamma^\mu_{\eta\nu}(\dots)_{\mu\sigma} - \Gamma^\mu_{\eta\sigma}(\dots)_{\nu\mu} + \partial_{\eta\nu}v_\sigma - \partial_\eta\left(\Gamma^\xi_{\nu\sigma}v_\xi\right)\right] = \\
&= \mathbf{b}^\eta\mathbf{b}^\nu\mathbf{b}^\sigma\left[-\Gamma^\mu_{\eta\nu}\left(\partial_\mu v_\sigma - \Gamma^\xi_{\mu\sigma}v_\xi\right) - \Gamma^\mu_{\eta\sigma}\left(\partial_\nu v_\mu - \Gamma^\xi_{\mu\nu}v_\xi\right) + \partial_{\eta\nu}v_\sigma - \partial_\eta\Gamma^\xi_{\nu\sigma}v_\xi - \Gamma^\xi_{\nu\sigma}\partial_\eta v_\xi\right] = \\
&= \mathbf{b}^\eta\mathbf{b}^\nu\mathbf{b}^\sigma\nabla_{\eta\nu}v_\sigma\,.
\end{aligned}$$

$\nabla_{\sigma\eta}v_\nu - \nabla_{\eta\sigma}v_\nu$

Exploiting symmetry properties of Christoffel symbols,

todo!!! change indices to match the definition of the curvature tensor

$$\begin{aligned}
\nabla_{\sigma\eta}v_\nu - \nabla_{\eta\sigma}v_\nu = &-\underbrace{\Gamma^\xi_{\sigma\eta}\partial_\xi v_\nu}_{1} + \underbrace{\Gamma^\mu_{\sigma\eta}\Gamma^\xi_{\mu\nu}v_\xi}_{2} - \underbrace{\Gamma^\xi_{\sigma\nu}\partial_\eta v_\xi}_{3} + \Gamma^\mu_{\sigma\nu}\Gamma^\xi_{\mu\eta}v_\xi + \underbrace{\partial_{\sigma\eta}v_\nu}_{4} - \partial_\sigma\Gamma^\xi_{\eta\nu}v_\xi - \underbrace{\Gamma^\xi_{\eta\nu}\partial_\sigma v_\xi}_{5} + \\
&- \left[-\underbrace{\Gamma^\xi_{\eta\sigma}\partial_\xi v_\nu}_{1} + \underbrace{\Gamma^\mu_{\eta\sigma}\Gamma^\xi_{\mu\nu}v_\xi}_{2} - \underbrace{\Gamma^\xi_{\eta\nu}\partial_\sigma v_\xi}_{5} + \Gamma^\mu_{\eta\nu}\Gamma^\xi_{\mu\sigma}v_\xi + \underbrace{\partial_{\eta\sigma}v_\nu}_{4} - \partial_\eta\Gamma^\xi_{\sigma\nu}v_\xi - \underbrace{\Gamma^\xi_{\sigma\nu}\partial_\eta v_\xi}_{3}\right] = \\
= &\left[\partial_\eta\Gamma^\xi_{\sigma\nu} - \partial_\sigma\Gamma^\xi_{\eta\nu} + \Gamma^\mu_{\sigma\nu}\Gamma^\xi_{\mu\eta} - \Gamma^\mu_{\eta\nu}\Gamma^\xi_{\mu\sigma}\right]v_\xi\,.
\end{aligned}$$

Thus, it follows that

$$R^\xi{}_{\sigma\eta\nu} = \partial_\eta\Gamma^\xi_{\sigma\nu} - \partial_\nu\Gamma^\xi_{\eta\sigma} + \Gamma^\mu_{\sigma\nu}\Gamma^\xi_{\mu\eta} - \Gamma^\mu_{\eta\sigma}\Gamma^\xi_{\mu\nu}$$

$$R_{\phi\sigma\eta\nu} = g_{\phi\xi}R^\xi{}_{\sigma\eta\nu} = g_{\phi\xi}\left(\partial_\eta\Gamma^\xi_{\sigma\nu} - \partial_\nu\Gamma^\xi_{\eta\sigma} + \Gamma^\mu_{\sigma\nu}\Gamma^\xi_{\mu\eta} - \Gamma^\mu_{\eta\sigma}\Gamma^\xi_{\mu\nu}\right)$$

Ricci's tensor

todo Meaning

Ricci's tensor is defined as the contraction of the first and third indices of the curvature tensor with the metric tensor,

$$R_{\sigma\nu} := g^{\alpha\beta} R_{\alpha\sigma\beta\nu} = \underbrace{g^{\alpha\beta} g_{\alpha\gamma}}_{\delta^{\beta}_{\gamma}} R^{\gamma}{}_{\sigma\beta\nu} = R^{\mu}{}_{\sigma\mu\nu}$$

$$R_{\sigma\nu} = \partial_{\xi}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\xi\sigma} + \Gamma^{\mu}_{\sigma\nu}\Gamma^{\xi}_{\mu\xi} - \Gamma^{\mu}_{\xi\nu}\Gamma^{\xi}_{\mu\sigma}$$

Curvature scalar

todo Meaning

$$R := g^{\sigma\nu} R_{\sigma\nu} = R^{\sigma}{}_{\sigma} \, .$$

$$R = g^{\sigma\nu} R_{\sigma\nu} = g^{\sigma\nu} \left(\partial_{\xi}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\xi\sigma} + \Gamma^{\mu}_{\sigma\nu}\Gamma^{\xi}_{\mu\xi} - \Gamma^{\mu}_{\xi\nu}\Gamma^{\xi}_{\mu\sigma} \right)$$

Properties

- $R_{abcd} = -R_{abdc}$

Proof

By direct inspection of the expression of the components of the curvature tensor,

$$R_{\phi\sigma\eta\nu} = g_{\phi\xi} R^{\xi}{}_{\sigma\eta\nu} = g_{\phi\xi} \left(\partial_{\eta}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\eta\sigma} + \Gamma^{\mu}_{\sigma\nu}\Gamma^{\xi}_{\mu\eta} - \Gamma^{\mu}_{\eta\sigma}\Gamma^{\xi}_{\mu\nu} \right)$$

and the symmetry of the Christoffel symbols, $\Gamma^{c}_{ab} = \Gamma^{c}_{ba}$.

- $R_{abcd} = -R_{bacd}$
- $R_{abcd} + R_{acdb} + R_{adbc} = 0$
- $R_{abcd} = R_{cdab}$
- $R_{abcd;e} + R_{abde;c} + R_{abec;d} = 0$ (Second Bianchi identity)

Proof

$$\begin{aligned} R_{\phi\sigma\eta\nu;\chi} &= \left(g_{\phi\xi} R^{\xi}{}_{\sigma\eta\nu} \right)_{;\chi} = \\ &= g_{\phi\xi} \left(\partial_{\eta}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\eta\sigma} + \Gamma^{\mu}_{\sigma\nu}\Gamma^{\xi}_{\mu\eta} - \Gamma^{\mu}_{\eta\sigma}\Gamma^{\xi}_{\mu\nu} \right)_{;\chi} = \\ &= \end{aligned}$$

todo

Divergence of Ricci's tensor

$$\nabla_\mu R^\mu{}_\nu = \frac{1}{2}\nabla_\nu R \, .$$

Proof

$$R_{\sigma\nu} = R^\xi{}_{\sigma\xi\nu} = \partial_\xi \Gamma^\xi_{\sigma\nu} - \partial_\sigma \Gamma^\xi_{\xi\nu} + \Gamma^\mu_{\sigma\nu}\Gamma^\xi_{\mu\xi} - \Gamma^\mu_{\xi\nu}\Gamma^\xi_{\mu\sigma}$$
$$R^\mu{}_\nu = g^{\mu\sigma} R_{\sigma\nu}$$
$$\nabla_\mu R^\mu{}_\nu = \partial_\mu R^\mu{}_\nu + \Gamma^\mu_{\mu\sigma} R^\sigma{}_\nu - \Gamma^\mu_{\sigma\nu} R^\sigma{}_\mu =$$
$$=$$

7.2 Einstein's equation

Idea:

1. Try with **the simplest relation** between momentum-energy tensor T and a second order tensor representing the geometry of space, i.e. **proportionality** with Ricci's curvature tensor R

   $$\mathsf{R} = \kappa \mathsf{T} \, .$$

   Remark. This relationship doesn't agree with local energy-momentum balance $\nabla \cdot \mathsf{T} = \mathbf{0}$, as $\nabla \cdot \mathsf{R} \neq \mathbf{0}$,

   $$\nabla_\mu R^\mu{}_\nu = \frac{1}{2}\nabla_\nu R$$
   $$\nabla_\mu R^{\mu\sigma} = \nabla_\mu \left(g^{\sigma\nu} R^\mu{}_\nu\right) = g^{\sigma\nu}\nabla_\mu R^\mu{}_\nu = \frac{1}{2} g^{\sigma\nu}\nabla_\nu R = \nabla_\mu \left(\frac{1}{2} g^{\sigma\mu} R\right),$$

   as the covariant derivative of the metric components of the metric tensor are identically zero. Moving the extreme terms of this equality on the same side of the equal sign, $\nabla_\mu \left(R^{\mu\sigma} - \frac{1}{2} g^{\mu\sigma} R\right) = 0$.

   #### Covariant derivative of the metric tensor

   $$\nabla_\nu g_{\mu\eta} = \partial_\nu g_{\mu\eta} - \Gamma^\sigma_{\nu\mu} g_{\sigma\eta} - \Gamma^\sigma_{\nu\eta} g_{\mu\sigma} = 0 \, ,$$

   as $\partial_\nu g_{\mu\eta} = \Gamma^\sigma_{\nu\mu} g_{\sigma\eta} + \Gamma^\sigma_{\nu\eta} g_{\mu\sigma}$ (see the derivative of the components of the metric tensor in *Differential geometry for general relativity*.

2. Correct the relation, replacing Ricci's tensor with its divergence-free part

   $$R^{\mu\sigma} - \frac{1}{2} g^{\mu\sigma} R = \kappa T^{\mu\sigma} \, , \tag{7.2}$$

   or

   $$\mathsf{R} - \frac{1}{2}\mathsf{g} R = \kappa \mathsf{T} \, .$$

3. Check against the classical limit, to find:

   - if Newton gravitation is the classical limit of general relativity

- the values of the constants involved in the model, κ

Under some assumptions:

- **weak gravitation** for the linearization of the metric tensor $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ around Minkowski flat time-space (here $\eta_{\mu\nu}$ describing Minkowski metrics).
- **pressure-less mass distribution at rest**
- **mass** distribution "at rest" (w.r.t. the "quasi inertial observer")

7.2.1 Alternative expression of Einstein's equations

Let's evaluate the trace of R, i.e. $R = R^{\sigma}{}_{\sigma}$ as a function of the trace of T.

$$\begin{aligned} R &:= R^{\sigma}{}_{\sigma} = \\ &= g_{\sigma\varphi} R^{\varphi\sigma} = \\ &= g_{\sigma\varphi} \left(\frac{1}{2} g^{\varphi\sigma} R + \kappa T^{\varphi\sigma} \right) = 2R + \kappa T \end{aligned}$$

and thus the relation $R = -\kappa T$ holds between the trace of Ricci and energy-momentum tensors (here $g_{\sigma\varphi} g^{\sigma\varphi} = \delta^{\sigma}_{\sigma} = 4$ in the 4-dimensional time-space). Thus, Einstein equation can be recast as

$$\mathsf{R} = \kappa \left(\mathsf{T} - \frac{1}{2} \mathsf{g}\, T \right) . \tag{7.3}$$

7.2.2 Classical limit

1. Linearization of the metric tensor. $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, and the inverse (linearized) relation gives $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$

2. Linearization of Christoffel symbols.

$$\Gamma^{i}_{00} = -\frac{1}{2} \partial_i h_{00}$$

in the static limit, $\partial_0 \equiv 0$, for the spatial components, $i = 1 : 3$.

3. Linearization of the geodesics equation.

$$\ddot{q}^{\mu} + \Gamma^{\mu}_{\nu\sigma} \dot{q}^{\nu} \dot{q}^{\mu} = 0$$

For $\mu = i = 1 : 3$, as the $\tau \sim t$ in the slow-regime limit, and $q^0 \sim ct$ and $q^i = x_i$,

$$\ddot{x}_i = -c^2 \Gamma^{i}_{00} = \frac{c^2}{2} \partial_i h_{00} \, .$$

This equation must be compared with the dynamical equation of the Newtonian mechanics, $\ddot{\vec{r}} = -\nabla\Phi$. In order to get the same equation, $\partial_i h_{00} = -\frac{2}{c^2} \partial_i \Phi$, and thus - except for an arbitrary constant (irrelevant) - $h_{00} = -\frac{2}{c^2} \Phi$.

4. Linearized Einstein equation, with energy-momentum tensor (of the pressure-less medium) $T^{\mu\nu} = \rho c^2 \delta^{\mu}_{0} \delta^{\nu}_{0}$, so that its trace reads $T = \rho c^2$. Thus the linearized 00 component of Einstein equation reads

$$R_{00} = \kappa \left(\rho c^2 - \frac{1}{2} \rho c^2 \right) = \kappa \frac{1}{2} \rho c^2$$

5. Linearized relation between curvature tensor and the linearized metrics.

$$R_{00} = \partial_i \Gamma^{i}_{00} = -\frac{1}{2} \partial_{ii} h_{00} = \frac{1}{c^2} \nabla^2_{\vec{r}} \Phi$$

6. Comparison of the two expressions of R_{00}.

$$\frac{\kappa}{2}\rho c^2 = \frac{1}{c^2}\nabla^2_{\vec{r}}\Phi \ ,$$

or rearranging,

$$\nabla^2\Phi = \frac{\kappa c^4}{2}\rho \ .$$

By direct comparison with the Poisson equation for Newtonian gravitation, $\nabla^2\Phi = 4\pi G\rho$, it follows that

$$\kappa = \frac{8\pi G}{c^4} \ .$$

7.3 Linearization of Einstein's equation

In this section, linearization of Einstein field equation (7.2)

$$R^{\mu\sigma} - \frac{1}{2}g^{\mu\sigma}R = \kappa T^{\mu\sigma} \ ,$$

is performed in the limit of weak gravitational field, inducing a metric that only slightly deviates from Minkowski metric $\eta_{\mu\nu} = \text{diag}\{1, -1, -1, -1\}$,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \ .$$

Using *harmonic gauge condition*, the linearized Einstein field equation reduces to a wave equation for $\overline{h}^{\mu\sigma} := h^{\mu\sigma} - \frac{1}{2}\eta^{\mu\sigma}h$,

$$\Box\overline{h}^{\mu\sigma} = -2\kappa T^{\mu\sigma} \ . \tag{7.4}$$

First the inverse of the metric and Christoffel symbols are linearized, and then they're used to find the linearized expression of Riemann tensor, Ricci tensor, and scalar curvature, to find the expression of the linearized Einstein equation.

Linearized Christoffel symbols

Christoffel symbols can be written in terms of the metric tensor and its derivatives with (7.1)

$$\Gamma^b_{ac} = \frac{1}{2}g^{bd}\left(\partial_c g_{ad} + \partial_a g_{dc} - \partial_d g_{ac}\right) \ .$$

The derivatives are first-order contributions, thus linearization only involves zero-order term of g^{bd}. Thus linearized Christoffel symbols are

$$\Gamma^b_{ac} \sim \frac{1}{2}\eta^{bd}\left(\partial_c h_{ad} + \partial_a h_{dc} - \partial_d h_{ac}\right) \ .$$

Linearized inverse metric

Linearization of the inverse metric

$$\delta^\xi_\mu = g_{\mu\nu}g^{\nu\xi} = (\eta_{\mu\nu} + h_{\mu\nu})(\eta^{\nu\xi} + h^{\nu\xi}) = \delta^\xi_\mu + h_{\mu\nu}\eta^{\nu\xi} + \eta_{\mu\nu}h^{\nu\xi} + h_{\mu\nu}h^{\nu\xi} \ ,$$

implies $0 = h_{\mu\nu}\eta^{\nu\xi} + \eta_{\mu\nu}h^{\nu\xi}$, so that the linear correction to the Minkowski metric is

$$h^{\mu\nu} = -h_{\xi\chi}\eta^{\mu\xi}\eta^{\nu\chi} \ ,$$

and the linearized inverse metric is

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu} = \eta^{\mu\nu} - \eta^{\mu\xi}\eta^{\nu\chi}h_{\xi\chi} \ .$$

Linearization of Riemann tensor

Riemann tensor,

$$
\begin{aligned}
R^{\xi}{}_{\sigma\chi\nu} &= \partial_{\chi}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\chi\sigma} + \Gamma^{\mu}_{\sigma\nu}\Gamma^{\xi}_{\mu\chi} - \Gamma^{\mu}_{\chi\sigma}\Gamma^{\xi}_{\mu\nu} = \\
&\sim \partial_{\chi}\Gamma^{\xi}_{\sigma\nu} - \partial_{\nu}\Gamma^{\xi}_{\chi\sigma} = \\
&= \frac{1}{2}\partial_{\chi}\left\{\eta^{\xi\phi}\left(\partial_{\nu}h_{\sigma\phi} + \partial_{\sigma}h_{\phi\nu} - \partial_{\phi}h_{\sigma\nu}\right)\right\} - \frac{1}{2}\partial_{\nu}\left\{\eta^{\xi\phi}\left(\partial_{\chi}h_{\sigma\phi} + \partial_{\sigma}h_{\phi\chi} - \partial_{\phi}h_{\sigma\chi}\right)\right\} = \\
&= \frac{1}{2}\eta^{\xi\phi}\left(\partial_{\chi\sigma}h_{\phi\nu} - \partial_{\chi\phi}h_{\sigma\nu} - \partial_{\nu\sigma}h_{\phi\chi} + \partial_{\nu\phi}h_{\sigma\chi}\right) \ .
\end{aligned}
$$

Linearization of Ricci tensor

Ricci tensor, $R_{\sigma\nu} = R^{\mu}{}_{\sigma\mu\nu}$

$$
R_{\sigma\nu} = R^{\xi}{}_{\sigma\xi\nu} \sim \frac{1}{2}\eta^{\xi\phi}\left(\partial_{\xi\sigma}h_{\phi\nu} - \partial_{\xi\phi}h_{\sigma\nu} - \partial_{\nu\sigma}h_{\phi\xi} + \partial_{\nu\phi}h_{\sigma\xi}\right) = R^{(1)}_{\sigma\nu} \ ,
$$

and the contravariant components read

$$
R^{\mu\nu(1)} = \eta^{\mu\chi}\eta^{\nu\xi}R^{(1)}_{\chi\xi} \ . \tag{7.5}
$$

or, explicitly

$$
\begin{aligned}
R^{\mu\chi(1)} &= \eta^{\mu\sigma}\eta^{\chi\nu}R^{(1)}_{\sigma\nu} = \\
&= \eta^{\mu\sigma}\eta^{\chi\nu}\left\{\frac{1}{2}\eta^{\xi\phi}\left(\partial_{\xi\sigma}h_{\phi\nu} - \partial_{\xi\phi}h_{\sigma\nu} - \partial_{\nu\sigma}h_{\phi\xi} + \partial_{\nu\phi}h_{\sigma\xi}\right)\right\} = \\
&= \frac{1}{2}\left\{\partial^{\mu}\partial_{\xi}h^{\xi\chi} - \Box h^{\mu\chi} - \partial^{\mu\chi}h + \partial^{\chi}\partial_{\phi}h^{\phi\mu}\right\} \ .
\end{aligned}
$$

Linearization of the term $-\frac{1}{2}g^{\mu\sigma}R$

$$
\begin{aligned}
R &:= g^{\sigma\nu}R_{\sigma\nu} = \\
&\sim \frac{1}{2}\eta^{\sigma\nu}\eta^{\xi\phi}\left(\partial_{\xi\sigma}h_{\phi\nu} - \partial_{\xi\phi}h_{\sigma\nu} - \partial_{\nu\sigma}h_{\phi\xi} + \partial_{\nu\phi}h_{\sigma\xi}\right) = \\
&= \partial^{\nu\phi}h_{\nu\phi} - \Box h = R^{(1)} \ ,
\end{aligned}
$$

with

$$
\begin{aligned}
\eta^{\sigma\nu}\partial_{\sigma} &= \partial^{\nu} \\
\eta^{\xi\phi}\partial_{\xi\phi} &= \Box \\
\eta^{\sigma\nu}h_{\sigma\nu} &=: h
\end{aligned}
$$

Linearization of the term $-\frac{1}{2}g^{\mu\sigma}R$ only involves the constant contribution of the metric, i.e.

$$
-\frac{1}{2}g^{\mu\sigma}R \sim -\frac{1}{2}\eta^{\mu\sigma}R^{(1)} \tag{7.6}
$$

Linearized Einstein field equation

Putting together (7.5) and (7.6), the linearized Einstein equation becomes

$$\begin{aligned}\kappa T^{\mu\sigma} &= R^{\mu\sigma} - \frac{1}{2}g^{\mu\sigma}R = \\ &\sim R^{\mu\sigma(1)} - \frac{1}{2}\eta^{\mu\sigma}R^{(1)} = \\ &= \frac{1}{2}\left\{\partial^\mu\partial_\xi h^{\xi\sigma} - \Box h^{\mu\sigma} - \partial^{\mu\sigma}h + \partial^\sigma\partial_\phi h^{\phi\mu}\right\} - \frac{1}{2}\eta^{\mu\sigma}\left\{\partial^{\nu\phi}h_{\nu\phi} - \Box h\right\} = \\ &= \frac{1}{2}\left\{\partial^\mu\partial_\xi h^{\xi\sigma} - \Box h^{\mu\sigma} - \partial^{\mu\sigma}h + \partial^\sigma\partial_\phi h^{\phi\mu} - \eta^{\mu\sigma}\partial^{\nu\phi}h_{\nu\phi} + \eta^{\mu\sigma}\Box h\right\} =\end{aligned}$$

Now, defining

$$\overline{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h\,,$$

$$\overline{h} = \eta^{\mu\nu}\overline{h}_{\mu\nu} = \eta^{\mu\nu}\left(h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h\right) = h - 2h = -h\,,$$

so that

$$h_{\mu\nu} = \overline{h}_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}h = \overline{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\overline{h}$$

Thus, exploiting the the expression (7.7) in the box, the equation becomes

$$\begin{aligned}\kappa T^{\mu\sigma} &= \frac{1}{2}\Box\left(-h^{\mu\sigma} + \frac{1}{2}\eta^{\mu\sigma}h + \frac{1}{2}\eta^{\mu\sigma}h\right) + \frac{1}{2}\left\{\partial^\mu\partial_\xi h^{\xi\sigma} - \partial^{\mu\sigma}h + \partial^\sigma\partial_\phi h^{\phi\mu} - \eta^{\mu\sigma}\partial^{\nu\phi}h_{\nu\phi}\right\} = \\ &= \frac{1}{2}\Box\left(-h^{\mu\sigma} + \frac{1}{2}\eta^{\mu\sigma}h + \frac{1}{2}\eta^{\mu\sigma}h\right) + \frac{1}{2}\left\{\partial^\mu\partial_\xi\overline{h}^{\xi\sigma} + \partial^\sigma\partial_\xi\overline{h}^{\xi\mu} - \eta^{\mu\sigma}\partial_{\nu\xi}\overline{h}^{\nu\xi} + \frac{1}{2}\eta^{\mu\sigma}\Box\overline{h}\right\} \\ &= -\frac{1}{2}\Box\overline{h}^{\mu\sigma} + \frac{1}{2}\left\{\partial^\mu\partial_\xi\overline{h}^{\xi\sigma} + \partial^\sigma\partial_\xi\overline{h}^{\xi\mu} - \eta^{\mu\sigma}\partial_{\nu\xi}\overline{h}^{\nu\xi}\right\}\,.\end{aligned}$$

Prescribing harmonic gauge condition (de Donder), $\partial_\xi\overline{h}^{\xi\mu} = 0$, linearized Einstein equation becomes a wave equation for $\overline{h}^{\mu\sigma}$,

$$\Box\overline{h}^{\mu\sigma} = -2\kappa T^{\mu\sigma}\,.$$

$\partial^\mu\partial_\xi h^{\xi\sigma} - \partial^{\mu\sigma}h + \partial^\sigma\partial_\phi h^{\phi\mu} - \eta^{\mu\sigma}\partial^{\nu\phi}h_{\nu\phi}$

$$\begin{aligned}&\partial^\mu\partial_\xi h^{\xi\sigma} - \partial^{\mu\sigma}h + \partial^\sigma\partial_\xi h^{\xi\mu} - \eta^{\mu\sigma}\partial^{\nu\xi}h_{\nu\xi} = \\ &= \partial^\mu\partial_\xi\overline{h}^{\xi\sigma} - \underbrace{\frac{1}{2}\eta^{\xi\sigma}\partial^\mu\partial_\xi\overline{h}}_{} + \underbrace{\partial^{\mu\sigma}\overline{h}}_{} + \partial^\sigma\partial_\xi\overline{h}^{\xi\mu} - \underbrace{\frac{1}{2}\eta^{\xi\mu}\partial^\sigma\partial_\xi\overline{h}}_{} - \eta^{\mu\sigma}\partial_{\nu\xi}\overline{h}^{\nu\xi} + \underbrace{\frac{1}{2}\eta^{\mu\sigma}\eta^{\nu\xi}\partial_{\nu\xi}\overline{h}}_{\frac{1}{2}\eta^{\mu\sigma}\Box\overline{h}} = \\ &= \partial^\mu\partial_\xi\overline{h}^{\xi\sigma} + \partial^\sigma\partial_\xi\overline{h}^{\xi\mu} - \eta^{\mu\sigma}\partial_{\nu\xi}\overline{h}^{\nu\xi} + \frac{1}{2}\eta^{\mu\sigma}\Box\overline{h}\,.\end{aligned} \tag{7.7}$$

7.4 Properties of Einstein field equation

10 independent components, due to symmetries

Only 4 independent components, after removing arbitrary choice of coordinates

7.4.1 Properties of linearized Einstein field equation

7.5 Local inertial reference frame and equivalence principle

Goal. Starting from *generic* coordinates q^μ on spacetime, build coordinates x^α around an event P in which the metric is Minkowskian up to second-order terms. Then show:

1. what is arbitrary in this construction (and what is not),
2. that "$\Gamma'(P) = 0$" and "$\partial g'(P) = 0$" are the same condition,
3. that the first deviation from flatness is the Riemann tensor, which no coordinate choice can remove.

This is the mathematical content of the **equivalence principle**.

Notation, and signature

Conventions. Signature $\eta_{\alpha\beta} = \text{diag}(+1, -1, -1, -1)$, so $c^2\, d\tau^2 = d\mathbf{X} \cdot d\mathbf{X}$. Nothing below depends on the signature. Riemann tensor:

$R^\gamma{}_{\alpha\beta\delta} = \partial_\beta \Gamma^\gamma{}_{\delta\alpha} - \partial_\delta \Gamma^\gamma{}_{\beta\alpha} + \Gamma^\gamma{}_{\beta\lambda}\Gamma^\lambda{}_{\delta\alpha} - \Gamma^\gamma{}_{\delta\lambda}\Gamma^\lambda{}_{\beta\alpha}$.

7.5.1 Setting and notation

Events in space time are parametrized with a set of coordinates $\{q^\mu\}_{\mu=0:3}$ as $\mathbf{X}(q^\mu)$. The event P has coordinates q^μ_P. The choice of the coordinates $\{q^\mu\}$ induces

- Coordinate natural basis $\mathbf{b}_\mu = \partial_\mu \mathbf{X}$ and metrics $g_{\mu\nu} = \mathbf{b}_\mu \cdot \mathbf{b}_\nu$, and the reciprocal basis $\mathbf{b}^\mu \cdot \mathbf{b}_\nu = \delta^\mu_\nu$.
- Connection - Christoffel symbols, $\Gamma^\lambda_{\mu\nu}$, defined as the components in the natural basis of the derivative of the vectors of the natural basis themselves, i.e.

$$\frac{\partial^2 \mathbf{X}}{\partial q^\mu q^\nu} = \partial_{q^\nu} \mathbf{b}_\mu = \partial_{q^\mu} \mathbf{b}_\nu = \Gamma^\lambda_{\mu\nu} \mathbf{b}_\lambda \ .$$

  Exploiting the orthogonality of the vectors of a basis and the vectors of its reciprocal basis,

$$\Gamma^\lambda_{\mu\nu} = \mathbf{b}^\lambda \cdot \partial_\mu \mathbf{b}_\nu, \qquad \Gamma_{\lambda,\mu\nu} := g_{\lambda\sigma}\Gamma^\sigma{}_{\mu\nu} = \mathbf{b}_\lambda \cdot \partial_\mu \mathbf{b}_\nu.$$

  It is symmetric in $\mu\nu$ because $\partial_\mu \mathbf{b}_\nu = \partial_\nu \mathbf{b}_\mu$.

7.5.2 Local vector basis

Definition of a local basis $\mathbf{e}_\mu$, s.t. $\mathbf{e}_\mu \cdot \mathbf{e}_\nu = \eta_{\mu\nu}$.

…

Transformation of coordinates

…

7.5.3 Local set of coordinates

Let the vectors $\mathbf{e}_\mu$ of the local basis the derivatives of the events of the spacetime w.r.t. a set of local coordinates $\{x^\mu\}_{\mu=0:3}$, identifying the event P with $x^\mu_P = 0$

$$\mathbf{e}_\mu = \left.\frac{\partial \mathbf{X}}{\partial x^\mu}\right|_{x^\nu=0} .$$

A polynomial expansion of the coordinates q^μ as a function of the local coordinates x^α around P ($x^\alpha = 0$ at P) reads

$$q^\mu = q^\mu_P + \tilde{e}^\mu{}_\alpha x^\alpha + \tfrac{1}{2} A^\mu{}_{\alpha\beta} x^\alpha x^\beta + \tfrac{1}{6} B^\mu{}_{\alpha\beta\gamma} x^\alpha x^\beta x^\gamma + O(x^4), \tag{7.8}$$

where A and B are totally symmetric in their lower indices. The inverse of $\tilde{e}$ is $e^\alpha{}_\mu$:

$$e^\alpha{}_\mu \, \tilde{e}^\mu{}_\beta = \delta^\alpha_\beta, \qquad \tilde{e}^\mu{}_\alpha \, e^\alpha{}_\nu = \delta^\mu_\nu.$$

7.5.4 First order: the tetrad

The vectors $\mathbf{e}_\alpha = \partial_\alpha \mathbf{X}|_P = \tilde{e}^\mu{}_\alpha \, \mathbf{b}_\mu$ form a **local basis** (tetrad) at P. Requiring $g'_{\alpha\beta}(P) = \eta_{\alpha\beta}$ gives

$$g_{\mu\nu}(P) \, \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta = \eta_{\alpha\beta} \quad \Longleftrightarrow \quad g_{\mu\nu}(P) = e^\alpha{}_\mu \, e^\beta{}_\nu \, \eta_{\alpha\beta}, \tag{7.9}$$

and the reciprocal basis is $\mathbf{e}^\alpha = e^\alpha{}_\mu \, \mathbf{b}^\mu$ with $\mathbf{e}^\alpha \cdot \mathbf{e}_\beta = \delta^\alpha_\beta$.

How many free parameters?

- **Counting.** $\tilde{e}^\mu{}_\alpha$ has 16 components, and the symmetric equation (7.9) gives 10 conditions. So $16 - 10 = \mathbf{6}$ parameters remain: the Lorentz group $SO(1,3)$ (3 boosts + 3 rotations).
- **Counting along an observer's worldline.** Let the observer have worldline $\mathbf{X}_P(\tau)$ and 4-velocity

  $$\mathbf{u}_P = \frac{d\mathbf{X}_P}{d\tau} = \frac{dq^\mu}{d\tau}\mathbf{b}_\mu, \qquad \mathbf{u}_P \cdot \mathbf{u}_P = c^2.$$

  Choose the time leg

  $$\mathbf{e}_0 = \frac{\mathbf{u}_P}{c}, \qquad \mathbf{e}_0 \cdot \mathbf{e}_0 = 1.$$

  Then $\mathbf{e}_0$ is **fully fixed by the worldline**. The other three vectors must satisfy $\mathbf{e}_k \cdot \mathbf{e}_0 = 0$ and $\mathbf{e}_k \cdot \mathbf{e}_l = -\delta_{kl}$, i.e. they form an orthonormal triad of the spacelike 3-plane orthogonal to $\mathbf{u}_P$ (Gram–Schmidt always works there). The remaining freedom is the choice of this triad, and counting it as

  $$\underbrace{3}_{\text{choice of } \mathbf{u}_P \text{ (boost)}} + \underbrace{2}_{\mathbf{e}_1} + \underbrace{1}_{\mathbf{e}_2} + \underbrace{0}_{\mathbf{e}_3} = 6.$$

  Choosing $\mathbf{u}_P$ ties the frame to the observer. The 3 remaining parameters ($SO(3)$) are the orientation of the spatial axes. Different $\mathbf{u}_P$ give different, equally valid, local inertial frames related by boosts.

Result of the first order

$g'_{\alpha\beta}(P) = \eta_{\alpha\beta}$. The residual freedom is exactly the Lorentz group: **6 parameters**.

Christoffel symbols and coordinate transformations

... (7.10)

7.5.5 Second order, $\Gamma^{(x)\gamma}_{\alpha\beta} = 0$, and $\partial_\gamma g^{(x)}_{\alpha\beta} = 0$

The requirement that $\Gamma^{(x)\gamma}_{\alpha\beta} = 0$ fixes the value of the coefficients $A^\lambda_{\alpha\beta}$ in the expansion (7.8). From the transformation law (7.10), using $\partial_\alpha q^\mu|_P = \tilde{e}^\mu{}_\alpha$ and $\partial_\alpha\partial_\beta q^\lambda|_P = A^\lambda{}_{\alpha\beta}$:

$$\Gamma'^\gamma{}_{\alpha\beta}(P) = e^\gamma{}_\lambda \Big[\Gamma^\lambda{}_{\mu\nu}(P)\, \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta + A^\lambda{}_{\alpha\beta}\Big].$$

Setting this to zero:

$$\boxed{A^\lambda{}_{\alpha\beta} = -\Gamma^\lambda{}_{\mu\nu}(P)\, \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta} \tag{7.11}$$

The right side is symmetric in $\alpha\beta$, as required. The solution is **unique**: no freedom is left at second order. So the quadratic approximation of the coordinate map is

$$q^\mu = q^\mu_P + \tilde{e}^\mu{}_\alpha x^\alpha - \tfrac{1}{2}\,\Gamma^\mu{}_{\nu\sigma}(P)\, \tilde{e}^\nu{}_\alpha \tilde{e}^\sigma{}_\beta\, x^\alpha x^\beta + O(x^3).$$

The same A removes $\partial g'$

Expand `eq-g-transf` to first order in x. The ingredients are

$$\partial_\alpha q^\mu = \tilde{e}^\mu{}_\alpha + A^\mu{}_{\alpha\sigma} x^\sigma + O(x^2), \qquad g_{\mu\nu}(q(x)) = g_{\mu\nu}(P) + \partial_\rho g_{\mu\nu}\, \tilde{e}^\rho{}_\sigma x^\sigma + O(x^2).$$

Using (7.9) for the zeroth-order term:

$$g'_{\alpha\beta}(x) = \eta_{\alpha\beta} + x^\sigma \Big[\partial_\rho g_{\mu\nu}\, \tilde{e}^\rho{}_\sigma \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta + g_{\mu\nu} A^\mu{}_{\alpha\sigma} \tilde{e}^\nu{}_\beta + g_{\mu\nu} \tilde{e}^\mu{}_\alpha A^\nu{}_{\beta\sigma}\Big] + O(x^2).$$

Insert (7.11) and use $g_{\mu\nu}\Gamma^\mu{}_{\kappa\rho} = \Gamma_{\nu,\kappa\rho}$ (and the symmetry in the last two indices):

$$g'_{\alpha\beta}(x) = \eta_{\alpha\beta} + x^\sigma\, \tilde{e}^\rho{}_\sigma \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta \Big[\partial_\rho g_{\mu\nu} - \Gamma_{\mu,\rho\nu} - \Gamma_{\nu,\rho\mu}\Big] + O(x^2).$$

The bracket vanishes identically by metric compatibility `eq-compat`. Therefore

$$\boxed{g'_{\alpha\beta}(x) = \eta_{\alpha\beta} + O(x^2)} \qquad \Longleftrightarrow \qquad \Gamma'(P) = 0 \iff \partial_\gamma g'_{\alpha\beta}(P) = 0. \tag{7.12}$$

Note: The two statements "$\Gamma'(P) = 0$" and "$\partial g'(P) = 0$" are equivalent *because* of `eq-compat`. This is the precise sense in which gravity is "transformed away" at a point: $g_{\mu\nu}$ and its first derivatives, which are what the gravitational field is made of in the Newtonian limit, can both be set to their Minkowski values.

The term $A\,A\,x^\sigma x^\xi$ is not trustworthy

The product of the two A terms is $O(x^2)$. At that order one also needs B (from $\partial_\alpha q^\mu$) and $\partial\partial g$ (from the expansion of $g_{\mu\nu}(q(x))$). A consistent expansion must therefore be organised by order, which is done in the next section.

7.5.6 Third order in $q^\mu(x^\alpha)$

The cubic coefficients $B^\lambda{}_{\alpha\beta\gamma}$ are totally symmetric. Differentiate (7.10) once more and evaluate at P, where $\Gamma'(P)=0$:

$$T^\gamma{}_{\alpha\beta\delta} := \partial_\delta \Gamma'^\gamma{}_{\alpha\beta}(P) = e^\gamma{}_\lambda B^\lambda{}_{\alpha\beta\delta} + C^\gamma{}_{\alpha\beta\delta}, \tag{7.13}$$

where C is built only from $\tilde{e}$, A, $\Gamma(P)$ and $\partial\Gamma(P)$ (it does not depend on B). The derivative of the prefactor $\partial x^\gamma/\partial q^\lambda$ does not contribute, because it multiplies the bracket of (7.10), which vanishes at P. The only place B enters is through $\partial_\delta\partial_\alpha\partial_\beta q^\lambda = B^\lambda{}_{\alpha\beta\delta}$. T is symmetric in $\alpha\beta$.

3.1 Decomposition of T

Define the totally symmetric part

$$S^\gamma{}_{\alpha\beta\delta} = \tfrac{1}{3}(T^\gamma{}_{\alpha\beta\delta} + T^\gamma{}_{\beta\delta\alpha} + T^\gamma{}_{\delta\alpha\beta}).$$

The Riemann tensor at P (where the $\Gamma\Gamma$ terms vanish) is

$$R'^\gamma{}_{\alpha\beta\delta} = T^\gamma{}_{\delta\alpha\beta} - T^\gamma{}_{\beta\alpha\delta}.$$

Symmetrising in $(\alpha\beta)$ and using the symmetry of T in its first two lower indices:

$$R'^\gamma{}_{\alpha\beta\delta} + R'^\gamma{}_{\beta\alpha\delta} = (T^\gamma{}_{\alpha\delta\beta} + T^\gamma{}_{\beta\delta\alpha}) - 2\,T^\gamma{}_{\alpha\beta\delta} = 3S^\gamma{}_{\alpha\beta\delta} - 3\,T^\gamma{}_{\alpha\beta\delta}.$$

Hence, in any coordinate system with $\Gamma'(P)=0$,

$$\boxed{\partial_\delta \Gamma'^\gamma{}_{\alpha\beta}(P) = S^\gamma{}_{\alpha\beta\delta} - \tfrac{1}{3}(R'^\gamma{}_{\alpha\beta\delta} + R'^\gamma{}_{\beta\alpha\delta})} \tag{7.14}$$

- The **totally symmetric part** S is pure gauge: by (7.13) it can be shifted at will through B.
- The **rest** is fixed by the Riemann tensor, which transforms as a tensor and therefore cannot be changed by any coordinate choice:

$$R'^\gamma{}_{\alpha\beta\delta}(P) = e^\gamma{}_\lambda\, \tilde{e}^\mu{}_\alpha \tilde{e}^\nu{}_\beta \tilde{e}^\kappa{}_\delta\, R^\lambda{}_{\mu\nu\kappa}(P).$$

Riemann normal coordinates

Choose

$$B^\lambda{}_{\alpha\beta\delta} = -\tilde{e}^\lambda{}_\gamma\, C^\gamma{}_{(\alpha\beta\delta)}$$

with a total symmetrisation of C. Then $S=0$. The freedom in B is **entirely used up**. This defines **Riemann normal coordinates**, in which

$$\partial_\delta \Gamma'^\gamma{}_{\alpha\beta}(P) = -\tfrac{1}{3}(R^\gamma{}_{\alpha\beta\delta} + R^\gamma{}_{\beta\alpha\delta}).$$

The metric to second order

Differentiate the metric-compatibility relation `eq-compat` (written in the primed coordinates) once more and evaluate at P, where $\Gamma' = 0$, so that derivatives of g' drop out:

$$\partial_\delta \partial_\gamma g'_{\alpha\beta}(P) = \eta_{\alpha\lambda} T^{\lambda}{}_{\gamma\beta\delta} + \eta_{\beta\lambda} T^{\lambda}{}_{\gamma\alpha\delta} = T_{\alpha\gamma\beta\delta} + T_{\beta\gamma\alpha\delta}.$$

Insert $T_{\alpha\beta\gamma\delta} = -\frac{1}{3}(R_{\alpha\beta\gamma\delta} + R_{\alpha\gamma\beta\delta})$:

$$\partial_\gamma \partial_\delta g'_{\alpha\beta} = -\tfrac{1}{3}\Big[R_{\alpha\gamma\beta\delta} + R_{\alpha\beta\gamma\delta} + R_{\beta\gamma\alpha\delta} + R_{\beta\alpha\gamma\delta}\Big].$$

The terms $R_{\alpha\beta\gamma\delta} + R_{\beta\alpha\gamma\delta}$ cancel by antisymmetry in the first pair, and $R_{\beta\gamma\alpha\delta} = R_{\alpha\delta\beta\gamma}$ by pair symmetry. So

$$\partial_\gamma \partial_\delta g'_{\alpha\beta}(P) = -\tfrac{1}{3}(R_{\alpha\gamma\beta\delta} + R_{\alpha\delta\beta\gamma}),$$

and the Taylor expansion of the metric is

$$\boxed{g'_{\alpha\beta}(x) = \eta_{\alpha\beta} - \tfrac{1}{3}\, R_{\alpha\gamma\beta\delta}(P)\, x^\gamma x^\delta + O(x^3)} \tag{7.15}$$

Counting check

- Second derivatives $\partial_\gamma \partial_\delta g'_{\alpha\beta}$: $10 \times 10 = 100$ numbers.
- Cubic coefficients $B^{\lambda}{}_{\alpha\beta\delta}$: $4 \times 20 = 80$ numbers, all used to set $S = 0$.
- Left over: $100 - 80 =$ **20**, exactly the number of independent components of the Riemann tensor in four dimensions.

Summary: what each order does

Order of $q(x)$	Coefficient	What it fixes	Freedom left
1	$\tilde{e}^{\mu}{}_{\alpha}$	$g'(P) = \eta$	6 (Lorentz group)
2	$A^{\mu}{}_{\alpha\beta} = -\Gamma^{\mu}{}_{\nu\sigma}\tilde{e}^{\nu}{}_{\alpha}\tilde{e}^{\sigma}{}_{\beta}$	$\Gamma'(P) = 0 \iff \partial g'(P) = 0$	none (unique)
3	$B^{\mu}{}_{\alpha\beta\gamma}$	totally symmetric part S of $\partial\Gamma'(P)$	all of S (80 numbers). The remaining 20 are R and are **not removable**.

Each order of the coordinate map controls the same order of the metric: the first removes $g - \eta$, the second removes ∂g, and the third can only rearrange $\partial\partial g$, never eliminate it.

7.5.7 The equivalence principle

In the coordinates x^α built above:

1. **Free fall is inertial motion at P.** The geodesic equation

$$\frac{d^2 x^\alpha}{d\tau^2} = -\Gamma'^{\alpha}{}_{\beta\gamma}\frac{dx^\beta}{d\tau}\frac{dx^\gamma}{d\tau}$$

has vanishing right-hand side at P: free particles move in straight lines, with corrections that grow linearly in x through $\partial\Gamma' \sim R$.

2. **Non-gravitational physics reduces to special relativity at** P (*minimal coupling*). Replacing $\eta \to g$ and $\partial \to \nabla$, for example

$$\nabla_\mu T^{\mu\nu} = \partial_\mu T^{\mu\nu} + \Gamma^\mu{}_{\mu\lambda} T^{\lambda\nu} + \Gamma^\nu{}_{\mu\lambda} T^{\mu\lambda} \xrightarrow{P} \partial_\mu T^{\mu\nu},$$

so all local laws take their special-relativistic form, with relative corrections of order $|R|\, L^2$ in a region of size L.

3. **What remains is tidal.** The residual gravitational information is the Riemann tensor, as seen in (7.15) and in the **geodesic deviation** of two neighbouring free-falling particles with separation ξ^μ and 4-velocity u^μ:

$$\frac{D^2 \xi^\mu}{d\tau^2} = R^\mu{}_{\nu\rho\sigma}\, u^\nu u^\rho\, \xi^\sigma.$$

Hence "local" means: the region is small enough that $|R|\, L^2$ is below the precision of the measurement. The equivalence principle is the statement that the metric and its first derivatives can be made to coincide with those of Minkowski at P, but the second derivatives cannot.

7.5.8 Example: uniformly accelerated frame in free space - Rindler coordinates in special relativity

See *"Constant" acceleration observer and Rindler coordinates*.

In Rindler coordinates $(T = ct, \xi)$ with a the proper acceleration of the observer at $\xi = 0$,

$$ds^2 = \left(1 + \frac{a\xi}{c^2}\right)^2 dT^2 - d\xi^2 - dy^2 - dz^2.$$

The Riemann tensor is identically zero (flat spacetime), but the connection is not:

$$\Gamma^\xi{}_{TT} = \frac{a}{c^2}\left(1 + \frac{a\xi}{c^2}\right) \xrightarrow{\xi=0} \frac{a}{c^2}.$$

Take $\tilde{e} = \mathbb{1}$ at P (the observer's own frame). Then by (7.11),

$$A^\xi{}_{TT} = -\frac{a}{c^2} \qquad \Longrightarrow \qquad \xi = x^\xi - \frac{a}{2c^2}\,(x^T)^2 + O(x^3).$$

This is exactly the transformation to the freely falling (inertial) frame: a free particle at rest in x has $\xi = \text{const} - \frac{a}{2c^2} T^2$, i.e. it "falls" with acceleration $-a$ as seen by the accelerated observer. This is the equivalence between a uniform gravitational field and an accelerated frame: both are removed at P by A, and in flat spacetime nothing remains at third order since $R = 0$.

In a genuine gravitational field (e.g. a static observer in Schwarzschild) the same quadratic map removes the local acceleration, but the third-order term leaves the tidal tensor $R \neq 0$, which distinguishes the two situations.

7.5.9 Extension along a worldline (Fermi normal coordinates)

If the frame is carried along the whole worldline with the tetrad transported (parallel transport for a geodesic, Fermi–Walker transport otherwise), the same construction gives coordinates in which, for a geodesic observer with a non-rotating triad,

$$g_{00} = 1 - R_{0i0j}\, x^i x^j + \dots, \qquad g_{0i} = -\tfrac{2}{3} R_{0jik}\, x^j x^k + \dots, \qquad g_{ij} = -\delta_{ij} - \tfrac{1}{3} R_{ikjl}\, x^k x^l + \dots$$

For an accelerated or rotating observer the leading terms also contain the acceleration a_i and the angular velocity ω:

$$g_{00} = (1 + a_i x^i)^2 - (\omega \times \mathbf{x})^2 + \dots$$

In flat spacetime with $a = \text{const}$ and $\omega = 0$ this is exactly the Rindler metric of Section 6.

7.6 Examples

Schwarzschild metric. First Schwarzschild metric is derived, and some properties are discussed (Schwarzschild radius, event horizon,…). Then, the orbit of a light but non-zero mass object in a Schwarzschild metric is derived, and the **precession of perihelion** is discussed (precession of Mercury perihelion was used as a test for GR). Eventually, the trajectory of a zero-mass particle in a Schwarzschild metric is derived, and the **deflection of a light ray** is discussed (Eddington, 1919. Experimental measurements of the deflection of light during a solar eclipse).

- *Schwarzschild metric*
- *Orbits in Schwarzschild spacetime*
- *Deflection of a ray of light*

7.6.1 Schwarzschild metrics

Spatially spherically symmetric and **static vacuum** solution of the Einstein equation, representing the exterior gravitational field of a **non-rotating**, **uncharged** massive body.

Schwarzschild metrics gives

$$ds^2 = -\left(1-\frac{r_s}{r}\right)c^2dt^2 + \left(1-\frac{r_s}{r}\right)^{-1}dr^2 + r^2d\theta^2 + r^2\sin^2\theta\,d\phi^2\,. \tag{7.16}$$

Vacuum equation

Without mass density and electric charge and current, $\mathsf{T} = 0$. From the expression of (7.3), it immediately follows, $\mathsf{R} = 0$.

Metric Ansatz & Symmetry Assumptions

The four coordinates used to parametrize the space-time are (t, r, θ, ϕ). The two physical symmetry constraints in Schwarzschild solution of EFE are

1. **Spherical Symmetry:** the spatial geometry is invariant under rotation, s.t. the angular part of the metric is given by the standard round metric on S^2, $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$
2. **Staticity:** the metric coefficients are independent of the time coordinate t ($\partial_t g_{\mu u} = 0$); the further assumption that the line element is invariant under time reversal eliminates cross-terms such as $dt\,dr$.

Under these assumptions, the general line element can be parameterized using two undetermined functions of radius, $A(r)$ and $B(r)$:

$$ds^2 = -e^{2A(r)}c^2dt^2 + e^{2B(r)}dr^2 + r^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)$$

Remark. Using exponential parameterizations $e^{2A(r)}$ and $e^{2B(r)}$ ensures that $g_{tt} < 0$ and $g_{rr} > 0$ outside any horizon.
todo *Add some remark/examples about the role of time, and its consequence on metrics*

From the choosen parametrization, the non-zero covariant components of the metric tensors are

$$\begin{aligned} g_{tt} &= -e^{-2A(r)}c^2 \\ g_{rr} &= e^{2B(r)} \\ g_{\theta\theta} &= r^2 \\ g_{\phi\phi} &= r^2\sin^2\theta \end{aligned}$$

As the metric tensor is diagonal, it's contravariant components are just the inverse of its covariant components.

Christoffel Symbols

The Christoffel symbols of the second kind are computed via the expression (7.1),

$$\Gamma^b_{ac} = \frac{1}{2} g^{bd} \left(\partial_c g_{ad} + \partial_a g_{dc} - \partial_d g_{ac} \right) \ .$$

Writing the derivative w.r.t. r as $\partial_r A(r) = A'(r)$

- $\Gamma^t_{tr} = \Gamma^t_{rt} = \frac{1}{2} g^{td} \left(\partial_r g_{td} + \partial_t g_{dr} - \partial_d g_{tr} \right) = -\frac{1}{2} e^{2A} c^{-2} \cdot (-2A') e^{-2A} c^2 = A'$
- $\Gamma^r_{tt} = A' e^{2(A-B)} c^2$
- $\Gamma^r_{rr} = B'$
- $\Gamma^r_{\theta\theta} = -r e^{-2B}$
- $\Gamma^r_{\phi\phi} = -r \sin^2 \theta \, e^{-2B}$
- $\Gamma^\theta_{r\theta} = \Gamma^\theta_{\theta r} = \frac{1}{r}$
- $\Gamma^\theta_{\phi\phi} = -\sin\theta \cos\theta$
- $\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = \frac{1}{r}$
- $\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta$

Components of the Ricci Tensor

The Ricci tensor is defined by contracting the Riemann curvature tensor:

$$R_{\mu u} = \partial_\lambda \Gamma^\lambda_{\mu u} - \partial_u \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\mu u} \Gamma^\sigma_{\lambda\sigma} - \Gamma^\sigma_{\mu\lambda} \Gamma^\lambda_{u\sigma}$$

Evaluating the non-zero independent components yields:

- **Time-Time Component (R_{tt})**

$$R_{tt} = e^{2(A-B)} c^2 \left[A'' + (A')^2 - A' B' + \frac{2A'}{r} \right]$$

- **Radial-Radial Component (R_{rr})**

$$R_{rr} = -A'' - (A')^2 + A' B' + \frac{2B'}{r}$$

- **Angular Component ($R_{\theta\theta}$)**

$$R_{\theta\theta} = 1 - e^{-2B} \left[1 + r(A' - B') \right]$$

Remarks.

- The $R_{\phi\phi}$ component yields $R_{\phi\phi} = R_{\theta\theta} \sin^2 \theta$, providing no independent equation).
- The non-diagonal components are identically zero (**todo** *prove it!*)

Solving the Differential Equations

Combining R_{tt} and R_{rr} (linear combination $e^{-2(A-B)}\frac{R_{tt}}{c^2} + R_{rr} = 0$) immediately gives

$$A'(r) = -B'(r) \, .$$

and integrating with respect to r,

$$A(r) + B(r) = C = \text{const.}$$

Boundary conditions at infinity. To satisfy the boundary condition of asymptotic flatness—that spacetime approaches flat Minkowski space as $r \to \infty$, then

$$\begin{aligned} g_{tt} &= -e^{2A(r)} \to -1 \quad & A(r) \to 0 \\ g_{rr} &= e^{2B(r)} \to 1 \quad & B(r) \to 0 \end{aligned}$$

and thus $C = 0$, and $A(r) = -B(r)$.

Solving for $A(r)$ via $R_{\theta\theta}$. Substituting $B' = -A'$ and $e^{-2B} = e^{2A}$ into $R_{\theta\theta} = 0$:

$$\begin{aligned} 0 = R_{\theta\theta} &= \\ &= 1 - e^{2A}\left(1 + 2rA'\right) = \\ &= 1 - \frac{d}{dr}\left(re^{2A}\right) \, , \end{aligned}$$

and thus, integrating in r,

$$re^{2A(r)} = r + C \, .$$

or

$$e^{2A(r)} = 1 + \frac{C}{r}$$

where C is a constant of integration. Since $e^{2B(r)} = e^{-2A(r)}$, it follows $e^{2B(r)} = \left(1 + \frac{C}{r}\right)^{-1}$.

Determining the Integration Constant C (The Newtonian Limit)

To determine C, we examine the metric in the weak-field, low-velocity limit ($r \to \infty$). In this limit, general relativity recovers **Newtonian gravity**, as shown in *classical limit*,

$$g_{tt} = \eta_{tt} + h_{tt} \approx -1 - \frac{2}{c^2}\Phi \, ,$$

where $\Phi(r) = -\frac{GM}{r}$ is the Newtonian gravitational potential of a central mass M. Equating the two expression of the coefficient g_{tt}

$$g_{tt} = -e^{2A(r)} = -1 - \frac{C}{r} = -1 + \frac{2}{c^2}\frac{GM}{r}$$

and thus

$$C = -\frac{2GM}{c^2} \, .$$

Defining the **Schwarzschild radius** $r_s := \frac{2GM}{c^2}$, the expression of **Schwarzschild metric** becomes

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2 + r^2d\theta^2 + r^2\sin^2\theta\, d\phi^2 \, .$$

Special Case: Complete Vacuum ($M = 0$)

If the spacetime contains no mass ($M = 0$), the Minkowski flat spacetime follows

$$ds^2 = -c^2dt^2 + dr^2 + r^2d\theta^2 + r^2\sin^2\theta\, d\phi^2 = -c^2dt^2 + |d\vec{r}|^2 \,,$$

or, using the common transformation between spherical and Cartesian space coordinates,

$$\begin{cases} t = t \\ x = r\sin\theta\cos\phi \\ y = r\sin\theta\sin\phi \\ z = r\cos\theta \,, \end{cases}$$

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2 \,.$$

7.6.2 Planetary orbits in Schwarzschild spacetime

Schwarzschild metrics (7.16) provides the metrics induced by a massive object in the vacuum spacetime around. The motion of a light object approximately doesn't affect the massive object generating the steady Schwarzschild metrics. The light object moves following the geodesics. Let the motion occurs in the equatorial plane, i.e. $\theta = \frac{\pi}{2}$, and $d\theta = 0$. Thus

$$-c^2d\tau^2 = ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1}dr^2 + r^2\,d\phi^2 \,.$$

Geodesics. With $A(r) = \frac{1}{2}\ln\left(1 - \frac{r_s}{r}\right)$, $A'(r) = \frac{1}{2}\frac{1}{1-\frac{r_s}{r}}\frac{r_s}{r^2}$

$$0 = \ddot{q}^k + \Gamma^k_{ab}\dot{q}^a\dot{q}^b$$

$$
\begin{aligned}
t: \qquad 0 &= \ddot{t} + \Gamma^t_{ab}\dot{q}^a\dot{q}^b = \\
&= \ddot{t} + 2\Gamma^t_{tr}\dot{t}\dot{r} = \\
&= \ddot{t} + \frac{1}{1-\frac{r_s}{r}}\frac{r_s}{r^2}\dot{t}\dot{r} \\
0 &= \left(1-\frac{r_s}{r}\right)^{-1}\left[\left(1-\frac{r_s}{r}\right)\ddot{t} + \frac{r_s}{r^2}\dot{t}\dot{r}\right] = \\
0 &= \left(1-\frac{r_s}{r}\right)^{-1}\frac{d}{d\tau}\left[\left(1-\frac{r_s}{r}\right)\dot{t}\right] \quad \rightarrow \quad \left(1-\frac{r_s}{r}\right)\dot{t} =: \frac{E}{c} = \text{const.} \\
r: \qquad 0 &= \ddot{r} + \Gamma^r_{ab}\dot{q}^a\dot{q}^b = \\
&= \ddot{r} + \Gamma^r_{tt}\dot{t}^2 + \Gamma^r_{rr}\dot{r}^2 + \Gamma^r_{\theta\theta}\dot{\theta}^2 + \Gamma^r_{\phi\phi}\dot{\phi}^2 = \qquad (\dot{\theta}=0) \\
&= \ddot{r} + \Gamma^r_{tt}\dot{t}^2 + \Gamma^r_{rr}\dot{r}^2 + \Gamma^r_{\phi\phi}\dot{\phi}^2 = \\
&= \ddot{r} + A'e^{4A}c^2\dot{t}^2 - A'(r)\dot{r}^2 - r\sin^2\theta e^{2A}\dot{\phi}^2 = \\
&= \ddot{r} + A'e^{4A}c^2\dot{t}^2 - A'(r)\dot{r}^2 - r\theta e^{2A}\dot{\phi}^2 = \\
\phi: \qquad 0 &= \ddot{\phi} + \Gamma^\phi_{ab}\dot{q}^a\dot{q}^b \\
&= \ddot{\phi} + 2\Gamma^\phi_{r\phi}\dot{r}\dot{\phi} + 2\Gamma^\phi_{\theta\phi}\dot{\theta}\dot{\phi} = \qquad (\dot{\theta}=0) \\
&= \ddot{\phi} + 2\Gamma^\phi_{r\phi}\dot{r}\dot{\phi} = \\
&= \ddot{\phi} + 2\frac{1}{r}\dot{r}\dot{\phi}\,, \\
0 &= \frac{1}{r^2}\frac{d}{d\tau}\left(r^2\dot{\phi}\right) \quad \rightarrow \quad r^2(\tau)\dot{\phi}(\tau) =: L = \text{const.} \\
\theta: \qquad 0 &= \ddot{\theta} + \Gamma^\theta_{ab}\dot{q}^a\dot{q}^b \\
&= \ddot{\theta} + 2\Gamma^\theta_{r\phi}\dot{\theta}\dot{\phi} + \Gamma^\theta_{\phi\phi}\dot{\phi}^2 = \qquad (\dot{\theta}=0) \\
&= \ddot{\theta} - \sin\theta\cos\theta\Gamma^\theta_{\phi\phi}\dot{\phi}^2 = \\
&= \ddot{\theta}\,.
\end{aligned}
$$

Replacing the expression for $\dot{t}$ and $\dot{\phi}$ into the relation $u^\mu u_\mu = -c^2$ (from $ds^2/d\tau^2$)

$$
\begin{aligned}
-c^2 &= -\left(1-\frac{r_s}{r}\right)c^2\dot{t}^2 + \left(1-\frac{r_s}{r}\right)^{-1}\dot{r}^2 + r^2\dot{\phi}^2 = \\
&= -()^{-1}E^2 + ()^{-1}\dot{r}^2 + \frac{L}{r^2}
\end{aligned}
$$

and thus

$$
\dot{r}^2 = E^2 - c^2\left(1-\frac{r_s}{r}\right)\left(1+\frac{L^2}{r^2c^2}\right)\,.
$$

Now, changing variable $u := \frac{1}{r}$ and using the relation $\phi(\tau)$ to write

$$
\dot{r} = \frac{dr}{d\tau} = \frac{d\phi}{d\tau}\frac{d}{d\phi} = \frac{L}{r^2}\frac{dr}{d\phi} = Lu^2\left(-\frac{1}{u^2}\right)u'(\phi) = -Lu'(\phi)\,,
$$

the equation becomes

$$
L^2u'^2 = E^2 - c^2(1-r_su)\left(1+\frac{L^2}{c^2}u^2\right)\,.
$$

Differentiating both sides by ϕ

$$
0 = L^2 2u'u'' - c^2r_su'\left(1+\frac{L^2}{c^2}u^2\right) + c^2(1-r_su)2\frac{L^2}{c^2}uu'
$$

or dividing by $2L^2u'$,

$$0 = u'' - \frac{c^2 r_s}{2L^2} - \frac{3}{2} r_s u^2 + u$$

or, with $r_s = \frac{2GM}{c^2}$

$$u'' + u = \frac{c^2 r_s}{2L^2} + \frac{3}{2} r_s u^2 = \frac{GM}{L^2} + \frac{3GM}{c^2} u^2 \, .$$

Solution via perturbation method. As $\frac{3GM}{c^2} \to 0$, the equation becomes the equation of the Kepler orbits.

Let the solution be $u(\phi) = u_0(\phi) + u_1(\phi)$, with $u_0(\phi)$ the solution of the equations of classical mechanics.

$$u_0(\phi) = \frac{GM}{L^2}(1 + e\cos\phi) \, .$$

As

$$0 = \underbrace{u_0'' + u_0 - \frac{GM}{L^2}}_{=0} + u_1'' + u_1 - \frac{3GM}{c^2}(u_0 + u_1)^2 \, ,$$

the perturbation $u_1(\phi)$ (of order $\frac{GM}{c^2}$ must satisfy (approximately, truncation) the equation

$$\begin{aligned}
u_1'' + u_1 &\approx \frac{3GM}{c^2} u_0^2(\phi) = \\
&= \frac{3(GM)^3}{c^2 L^4} \left(1 + 2e\cos\phi + e^2\cos^2\phi\right) = \qquad\qquad (\cos 2x = 2\cos^2 x - 1) = \\
&= 3\left(\frac{(GM)^{3/2}}{cL^2}\right)^2 \left(1 + 2e\cos\phi + \frac{e^2}{2}\left(1 + \cos(2\phi)\right)\right) = \\
&= 3\left(\frac{(GM)^{3/2}}{cL^2}\right)^2 \left(1 + \frac{e}{2} + 2e\cos\phi + \frac{e^2}{2}\cos(2\phi)\right) \, ,
\end{aligned}$$

whose solution undergoes resonance, as the forcing $2e\cos\phi$ has the same frequency as the natural frequency of the system.The general solution reads

$$u_1(\phi) = A\cos\phi + B\sin\phi + 3\left(\frac{(GM)^{3/2}}{cL^2}\right)^2 \left[\left(1 + \frac{e}{2}\right) + e\phi\sin\phi + \frac{e^2}{6}\cos 2\phi\right]$$

Only the term $\propto \phi\sin\phi$ is unbounded. Combining this term with the unperturbed solution gives

$$u(\phi) \approx \frac{GM}{L^2}\left\{(1 + e\cos\phi) + 3\left(\frac{GM}{cL}\right)^2 e\phi\sin\phi\right\} \, ,$$

and using small angle approximation $\cos(\alpha + x) = \cos\alpha\cos x - \sin\alpha\sin x \approx \cos x - \alpha\sin x$, for $\alpha \sim 0$,

$$\begin{aligned}
u(\phi) &\approx \frac{GM}{L^2}\left\{1 + e\cos\phi + \underbrace{3\left(\frac{GM}{cL}\right)^2 \phi}_{\text{"}-\alpha\text{"}} e\sin\phi\right\} = \\
&\approx \frac{GM}{L^2}\left\{1 + e\cos\left(\phi - \left(\frac{GM}{cL}\right)^2 \phi\right)\right\} = \\
&= \frac{GM}{L^2}\left\{1 + e\cos\left[\left(\phi - 3\left(\frac{GM}{cL}\right)^2\right)\phi\right]\right\} \, .
\end{aligned}$$

As $u := \frac{1}{r}$, the perihelion occurs for $u(\phi_{perih}) = \max_\phi u(\phi)$, i.e. for

$$\left(1 - 3\left(\frac{GM}{cL}\right)^2\right)\phi_{perih,n} = n2\pi \quad , \quad n \in \mathbb{Z} \, .$$

The **precession of the perihelion** can be evaluated as the difference between the angular coordinate of two successive revolutions (adding a $2pi$ to the first angle, mimicing an orbit with no precession) i.e.

$$\begin{aligned}\Delta\phi &= \phi_{perih,n+1} - (\phi_{perih,n} + 2\pi) = \\ &= \frac{2\pi}{1 - 3\left(\frac{GM}{cL}\right)^2} - 2\pi = \\ &\approx 2\pi\left(1 + 3\left(\frac{GM}{cL}\right)^2 - 1\right) = 6\pi\left(\frac{GM}{cL}\right)^2 \, .\end{aligned}$$

General solution

The general solution has the expression

$$u_1(\phi) = \underbrace{A\cos\phi + B\sin\phi}_{u_{1,homo}(\phi)} + \underbrace{C + D\phi\cos\phi + E\phi\sin\phi + F\cos(2\phi)}_{u_{1,part}(\phi)} \, .$$

The coefficients of the particular solution are evaluated by coefficient matching

$$\begin{aligned}(\phi\cos\phi)' &= \cos\phi - \phi\sin\phi \\ (\phi\cos\phi)'' &= -2\sin\phi - \phi\cos\phi \\ (\phi\sin\phi)' &= \sin\phi + \phi\cos\phi \\ (\phi\sin\phi)'' &= 2\cos\phi - \phi\sin\phi \\ (\cos(2\phi))'' &= -4\cos(2\phi)\end{aligned}$$

then

$$\begin{aligned}\cos\phi: &\quad 0 = A - A - 2E + 2\alpha e \\ \sin\phi: &\quad 0 = B - B + 2D \\ \phi\cos\phi: &\quad 0 = D - D \\ \phi\sin\phi: &\quad 0 = E - E \\ 1: &\quad 0 = -C + \alpha\left(1 + \frac{e}{2}\right) \\ \cos 2\phi: &\quad 0 = 4F - F + \alpha\frac{e^2}{2} \, ,\end{aligned}$$

so that

$$\begin{aligned}E &= \alpha e \\ D &= 0 \\ C &= \alpha\left(1 + \frac{e}{2}\right) \\ F &= -\frac{1}{6}\alpha e^2 \, .\end{aligned}$$

7.6.3 Deflection of a ray of light, and Eddington measurements (1919)

Newton's theory of universal gravitation is fundamentally incompatible with special relativity: in Newtonian mechanics, gravitational interactions act instantaneously across space, violating the relativistic principle that no physical signal or interaction can propagate faster than the speed of light c.

In this section, the equation of the trajectory of zero-mass object in Schwarzschild metrics is derived here, using perturbation method for weak gravitational fields. The equations of general relativity imply that a ray of light is deflected by a body with mass M by an angle

$$\delta \simeq 4\frac{GM}{Rc^2} \,,$$

being R the minimum distance of the ray from the center of the massive body. This results is compared is twice the deflection predicted by the laws of classical mechanics using Newton gravitation, $\delta^{\text{Newton}} \simeq 2\frac{GM}{Rc^2}$. **Eddington experiment** during the total solar eclipse of the 1919 measured the amount of deflection of light rays passing close to the sun, confirming that general gravitation has better predictions and matches the experimental measurements, providing the first experimental tests of Einstein theory of general relativity as a new and better theory of gravitation.

Trajectory of zero-mass objects in Schwarzschild metrics

Light particles travel at the speed of light c. Thus the elementary increment of proper interval (or proper time τ) is zero,

$$0 = ds^2 = g_{ij}\frac{dq^i}{d\lambda}\frac{dq^j}{d\lambda} \,.$$

Using Schwarzschild metric (7.16)

$$ds^2 = -\left(1-\frac{r_s}{r}\right)c^2dt^2 + \left(1-\frac{r_s}{r}\right)^{-1}dr^2 + r^2d\theta^2 + r^2\sin^2\theta\, d\phi^2 \,.$$

A similar procedure as the one followed to compute *orbits of non-zero mass light objects in a Schwarzschild metric* is followed. The main and fundamental difference is in the use of ds. For non-zero mass objects the relation $ds^2 = c^2d\tau^2 = g_{\mu\nu}dq^\mu dq^\nu$ holds, and thus $c^2 = g_{\mu\nu}\dot{q}^\mu\dot{q}^\nu$, with $\dot{q}^\mu = \frac{dq^\mu}{d\tau}$. For zero mass objects, it's not possible to use the proper time as a parameters (as it's constant and thus $d\tau = 0$); using an arbitrary parameter λ, it follows that $0 = \frac{ds^2}{d\lambda^2} = g_{\mu\nu}\frac{dq^\mu}{d\lambda}\frac{dq^\nu}{d\lambda}$.

Let's choose a planar trajectory with prescribed $\theta = \frac{\pi}{2}$ (or with prescribed $\phi = \overline{\phi}$). The equations of the geodesics are

$$\begin{aligned}
0 &= \ddot{t} + \frac{1}{1-\frac{r_s}{r}}\frac{r_s}{r}\dot{t}\,\dot{r} \\
0 &= \ddot{r} + A'e^{4A}c^2\dot{t}^2 - A'(r)\dot{r}^2 - r\theta e^{2A}\dot{\phi}^2 \\
0 &= \ddot{\phi} + \frac{2}{r}\dot{r}\,\dot{\phi} \\
0 &= \ddot{\theta} \,.
\end{aligned}$$

From the first and the third equations, the conserved quantities follow

$$\begin{aligned}
\frac{E}{c} &= \left(1-\frac{r_s}{r}\right)\dot{t} = \text{const.} \\
L &= r^2\dot{\phi} = \text{const.}
\end{aligned}$$

Replacing these relations in the constraint $ds = 0$,

$$\begin{aligned}
0 &= -\left(1-\frac{r_s}{r}\right)c^2\dot{t}^2 + \left(1-\frac{r_s}{r}\right)^{-1}\dot{r}^2 + r^2\,\dot{\phi}^2 = \\
&= -E^2\left(1-\frac{r_s}{r}\right)^{-1} + \left(1-\frac{r_s}{r}\right)^{-1}\dot{r}^2 + \frac{L^2}{r^2} \,,
\end{aligned}$$

and thus the differential equation for $r(\lambda)$ becomes

$$0 = \dot{r}^2 - E^2 + \left(1 - \frac{r_s}{r}\right)\frac{L^2}{r^2} \ .$$

Now, changing variable $u = \frac{1}{r}$, and exploiting a relation $r(\phi(\lambda))$

$$\dot{r} = \frac{d\phi}{d\lambda}\frac{dr}{d\phi} = \frac{L}{r^2}\frac{dr}{d\phi} = Lu^2\left(-\frac{1}{u^2}\right)u'(\phi) = -Lu'(\phi) \ .$$

the equation becomes

$$0 = L^2\left(u'(\phi)\right)^2 - E^2 + (1 - r_s u)L^2u^2 \ .$$

Differentiating both sides by ϕ and dividing by $2L^2u'(\phi)$, a second order equation for $u(\phi)$ appears

$$u''(\phi) + u(\phi) = \frac{3}{2}r_s u^2(\phi) = \frac{3GM}{c^2}u^2 \ .$$

Solution via perturbation method. As $\frac{3GM}{c^2} \to 0$, the equation becomes $u''(\phi) + u(\phi) = 0$, whose solution is $u_0(\phi) = A\cos(\phi) + B\sin(\phi)$, i.e.

$$r_0(\phi) = \frac{1}{A\cos(\phi) + B\sin(\phi)} \ .$$

The problem is independent from an offset in ϕ. Setting $\lim_{\phi\to\dots} r(\phi) = +\infty$, then $B = 0$, and. **todo** *Set this condition,* $\overline{\phi} = \frac{\pi}{2}$*? Which condition sets* $A = 1R$*? Minimum distance*

$$u_0(\phi) = \frac{1}{R}\cos(\phi) \ ,$$

i.e. the equation of a straight line with minimum distance R from the origin of the coordinates, i.e. the trajectory of a body in free-space. Let the solution be $u(\phi) = u_0(\phi) + u_1(\phi)$. An approximate equation for u_1 is obtained introducing this decomposition in the original equation,

$$\begin{aligned}
0 &= u''(\phi) + u(\phi) - \frac{3}{2}r_s u^2(\phi) = \\
&= u_0''(\phi) + u_1''(\phi) + u_0(\phi) + u_1(\phi) - \frac{3}{2}r_s(u_0(\phi) + u_1(\phi))^2 = \\
&= \underbrace{u_0''(\phi) + u_0(\phi)}_{=0} + u_1''(\phi) + u_1(\phi) - \frac{3}{2}r_s u_0^2(\phi) - 3r_s u_0(\phi)u_1(\phi) - \frac{3}{2}r_s u_1^2(\phi) = \\
&= u_1''(\phi) + u_1(\phi) - \frac{3}{2}r_s u_0^2(\phi)u_1(\phi) = \\
&\sim u_1''(\phi) + u_1(\phi) - \frac{3}{2}\frac{r_s}{R^2}\cos^2(\phi) \ .
\end{aligned}$$

With $\cos^2(\phi) = \frac{1}{2} + \frac{1}{2}\cos(2\phi)$, the approximate equation for $u_1(\phi)$ becomes

$$u_1''(\phi) + u_1(\phi) = \frac{3}{4}\frac{r_s}{R^2}\left[1 + \cos(2\phi)\right] \ .$$

The solution is

$$u_1(\phi) = A\cos(\phi) + B\sin(\phi) + \frac{3}{4}\frac{r_s}{R^2}\left[1 - \frac{1}{3}\cos(2\phi)\right] \ .$$

Details

$$\frac{d}{dx}\cos(2x) = -2\sin(2x)$$
$$\frac{d^2}{dx^2}\cos(2x) = -4\cos(2x)$$
$$(-4+1)\,D = \frac{3}{4}\frac{r_s}{R^2}\,,$$

thus $D = -\frac{1}{4}\frac{r_s}{R^2}$.

Putting together the zero- and first-order contributions, the approximate solution is

$$u(\phi) = A\cos(\phi) + B\sin(\phi) + \frac{3}{2}\frac{GM}{R^2c^2}\left(1 - \frac{1}{3}\cos(2\phi)\right)\,.$$

Assuming symmetric trajectory w.r.t. $\theta = 0$, then $B = 0$. Setting the other integration constant by the condition $u(0) = \frac{1}{R}$, then

$$\frac{1}{R} = A + \frac{GM}{R^2c^2} \quad \rightarrow \quad A = \frac{1}{R}\left(1 - \frac{GM}{Rc^2}\right)$$

so that

$$r(\phi) = \frac{R}{\left(1 - \frac{GM}{Rc^2}\right)\cos(\phi) + \frac{3}{2}\frac{GM}{Rc^2}\left(1 - \frac{1}{3}\cos(2\phi)\right)}\,.$$

Extremes of the trajectory

Limit angles for $\phi = \frac{\pi}{2} - \frac{\delta}{2}$, or $\phi = -\left(\frac{\pi}{2} - \frac{\delta}{2}\right)$ with "small" $\delta > 0$.

$$\cos\left(\frac{\pi}{2} - \frac{\delta}{2}\right) = \sin\frac{\delta}{2} \sim \frac{\delta}{2} - \frac{(\delta/2)^3}{3!}$$

$$\begin{aligned}
1 - \frac{1}{3}\cos\left[2\left(\frac{\pi}{2} - \frac{\delta}{2}\right)\right] &= 1 - \frac{1}{3}\cos(\pi - \delta) = \\
&= 1 + \frac{1}{3}\cos(\delta) = \\
&\sim 1 + \frac{1}{3}\left(1 - \frac{\delta^2}{2!}\right) = \\
&= \frac{4}{3} - \frac{1}{12}\delta^2
\end{aligned}$$

An approximation of the denominator for "small" δ thus becomes

$$\text{den}(\delta) = (1-\gamma)\,\frac{\delta}{2} + \frac{3}{2}\gamma\left(\frac{4}{3} - \frac{1}{12}\delta^2\right) = (1-\gamma)\,\frac{\delta}{2} + 2\gamma - \frac{1}{8}\gamma\delta^2\,,$$

and setting it equal to zero, neglecting the second-degree contribution

$$\delta = \frac{4\gamma}{1-\gamma} \sim 4\gamma = 4\frac{GM}{Rc^2}\,. \tag{7.17}$$

Comparison with the laws of classical mechanics - Newton gravitation

See: Physics - HS: Classical Mechanics: Dynamics: Central Motions

With the laws of classical mechanics, the equation of the trajectory of central motions with force $\vec{F} = c\frac{\vec{r}}{|\vec{r}|^3}$,

$$r(\phi) = \frac{\frac{L^2}{m|c|}}{-\text{sign}(c) + \sqrt{1 + 2\frac{EL^2}{m|c|^2}}\cos(\phi - \phi_0)} ,$$

being m the mass of the light body, L the constant angular momentum, and E the constant energy of the light body, and ϕ_0 a reference angle to describe trajectories with different orientation: without loss of generallity, let's set it equal to zero, $\phi_0 = 0$. For gravitation, $c = GMm$, so that $\text{sign}(c) = 1$.

Open trajectories have $E \geq 0$, with parabolic orbits for $E = 0$, and hyperbolic orbits for $E > 0$. The minimum distance between the bodies occurs when the denominator is maximum, and thus for $\cos(\phi) = 1$, i.e. $\phi = 0$.

$$R = \frac{\frac{L^2}{GMm^2}}{-1 + \sqrt{1 + 2\frac{E}{m}\frac{L^2}{G^2M^2m^2}}} .$$

Hyperbolic orbits havei asymptotes corresponding to limit angles ϕ_∞ and $-\phi_\infty$, where the denominator becomes zero. The total deflection of the trajectory is the angle δ (better, $\pi - \delta$) between the asymptotes. The limit angles satisfy the condition

$$\cos\phi_\infty = \frac{1}{\sqrt{1 + 2\frac{E}{m}\frac{L^2}{G^2M^2m^2}}} .$$

The total delection is $\delta = \pi - 2\phi_\infty$. For small deviations of the trajectory, the angles at infinity only slightly differs from $\pm\frac{\pi}{2}$. Writing $\phi_\infty = \frac{\pi}{2} - \frac{\delta}{2}$, then $\cos(\phi_\infty) = \sin\left(\frac{\delta}{2}\right)$. As $\delta \to 0$, then $\cos\phi_\infty \to 0$. Since

$$\cos\phi_\infty = \frac{1}{\sqrt{1+\varepsilon}} = \sin\left(\frac{\delta}{2}\right) \sim \frac{\delta}{2} ,$$

it follows that the parameter $\varepsilon = 2\frac{E}{m}\left(\frac{L}{GMm}\right)^2$ is "large", and the approximation of $\cos\phi_\infty$ holds

$$\cos\phi_\infty = \frac{1}{\sqrt{1+\varepsilon}} \sim \frac{1}{\sqrt{\varepsilon}} .$$

If this approximation holds, the minimum distance of the trajectory from the massive body approximately is

$$R \sim \frac{GM}{2}\frac{m}{E}\sqrt{\varepsilon} \quad \to \quad \frac{1}{\sqrt{\varepsilon}} \sim \frac{GMm}{2ER} .$$

Then, the total delection for a body travelling close to the speed of light - assuming that laws of classical mechanics still hold, and that kinetic energy prevails over potential energy - $E \sim \frac{1}{2}mc^2$,

$$\delta \sim \frac{GMm}{ER} \sim \frac{2GM}{Rc^2} . \tag{7.18}$$

Part III

Statistical Mechanics

CHAPTER

EIGHT

STATISTICAL PHYSICS

CHAPTER

NINE

LIOUVILLE THEOREM AND BOLTZMANN EQUATION

9.1 Liouville theorem

Theorem 9.1.1 (Liouville's theorem)

Let $f(\mathbf{r}_i, \mathbf{v}_i, t)$ the **multi-particle** probability density function. Liouville's equations reads

$$0 = \frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v},i} f \, .$$

Let $f(\mathbf{r}_i, \mathbf{v}_i, t)$ be the N-particle distribution of a system as a function of the $2dN$ coordinates and components of the velocity. Probability is normalized to 1

$$1 = \int_{\overline{\Omega}} f(\mathbf{r}_i, \mathbf{v}_i, t) d\mathbf{r}_i d\mathbf{v}_i \, .$$

The governing equations of each particle read

$$\begin{cases} \dot{\mathbf{r}}_i = \mathbf{v}_i \\ \dot{\mathbf{v}}_i = \frac{\mathbf{F}_i}{m} \end{cases} .$$

A governing equation for the probability P_Ω of the system to be in a state belonging to an arbitrary "material" volume Ω_t in the state space reads

$$\begin{aligned} 0 = \frac{d}{dt} P_\Omega = \frac{d}{dt} \int_{\Omega_t} f = \\ = \int_{\Omega_t} \frac{\partial f}{\partial t} + \oint_{\partial \Omega_t} \left(\hat{\mathbf{n}}_{\mathbf{r},i} \cdot \mathbf{v}_i + \hat{\mathbf{n}}_{\mathbf{v},i} \cdot \mathbf{a}_i \right) f = \\ = \int_{\Omega_t} \left\{ \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i} \cdot (\mathbf{v}_i f) + \nabla_{\mathbf{v},i} \cdot (\mathbf{a}_i f) \right\} \, , \end{aligned}$$

having used Reynolds' transport theorem in the state space, applied the multi-dimensional version of the divergence theorem. As this balance holds for any arbitrary volume Ω_t, the differential form follows

$$0 = \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i} \cdot (\mathbf{v}_i f) + \nabla_{\mathbf{v},i} \cdot (\mathbf{a}_i f) \, .$$

Now, using Hamilton equations in absence of non-conservative forces, in terms of generalized coordinates and generalized momenta

$$\begin{cases} \dot{\mathbf{q}}^i = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \\ \dot{\mathbf{p}}_i = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}^i} \end{cases} ,$$

or in terms of positions and velocities

$$\begin{cases} \mathbf{v}_i = \dot{\mathbf{r}}_i = \frac{1}{m}\nabla_{\mathbf{v}_i}\mathcal{H} \\ \mathbf{a}_i = \dot{\mathbf{v}}_i = -\frac{1}{m}\nabla_{\mathbf{r}_i}\mathcal{H} \end{cases},$$

it's possible to find the "convective form" of the Liouville's theorem

$$\begin{aligned} 0 &= \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i}\cdot(\mathbf{v}_i f) + \nabla_{\mathbf{v},i}\cdot(\mathbf{a}_i f) = \\ &= \frac{\partial f}{\partial t} + f\underbrace{\left(\nabla_{\mathbf{r},i}\cdot\mathbf{v}_i + \nabla_{\mathbf{v},i}\cdot\mathbf{a}_i\right)}_{=0} + \mathbf{v}_i\cdot\nabla_{\mathbf{r},i}f + \mathbf{a}_i\cdot\nabla_{\mathbf{v},i}f \,, \end{aligned}$$

i.e.

$$\begin{aligned} 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i\cdot\nabla_{\mathbf{r},i}f + \mathbf{a}_i\cdot\nabla_{\mathbf{v},i}f \\ 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i\cdot\nabla_{\mathbf{r},i}f + \frac{\mathbf{F}_i}{m}\cdot\nabla_{\mathbf{v},i}f \\ 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i\cdot\nabla_{\mathbf{r},i}f + \frac{\mathbf{F}_i^{ext}}{m}\cdot\nabla_{\mathbf{v},i}f + \sum_{j\neq i}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v},i}f \,. \end{aligned}$$

9.2 Boltzmann equation

Indistinguishable particles. *State counting…*

1-particle probability distribution function. Integrating all but one particle (marginalizing over all the variables expect for $\mathbf{r}_1$, $\mathbf{v}_1$),

$$f_1(\mathbf{r}_1,\mathbf{v}_1,t) = \int_{\mathbf{r}_{k=2:N}}\int_{\mathbf{v}_{k=2:N}} f(\mathbf{r}_i,\mathbf{v}_i,t)d\mathbf{r}_k d\mathbf{v}_k \,,$$

s-particle probability distribution function.

$$f_s(\mathbf{r}_{j=1:s},\mathbf{v}_{j=1:s},t) = \int_{\mathbf{r}_{k=s+1:N}}\int_{\mathbf{v}_{k=s+1:N}} f(\mathbf{r}_i,\mathbf{v}_i,t)d\mathbf{r}_k d\mathbf{v}_k \,,$$

Time derivative.

$$\begin{aligned} \partial_t f_s &= \partial_t \int_{\mathbf{r}_{k=s+1:N}}\int_{\mathbf{v}_{k=s+1:N}} f(\mathbf{r}_i,\mathbf{v}_i,t)d\mathbf{r}_k d\mathbf{v}_k = \\ &= -\int_{\mathbf{r}_{k=s+1:N}}\int_{\mathbf{v}_{k=s+1:N}} \left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f + \frac{\mathbf{F}_i^{ext}}{m}\cdot\nabla_{\mathbf{v}_i}f + \sum_{j\neq i}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f\right\} d\mathbf{r}_k d\mathbf{v}_k = \end{aligned}$$

Using Hamilton's equation to get a divergence, and assuming zero probability at infinity (impossible to be at infinite distance, impossible to have infinite velocity. Sounds likely),

Details

$$\int_{\mathbf{r}_k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f+\frac{\mathbf{F}_i}{m}\cdot\nabla_{\mathbf{v}_i}f\right\}=\int_{\mathbf{r}_k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f+\frac{\mathbf{F}_i}{m}\cdot\nabla_{\mathbf{v}_i}f+f\left(\nabla_{\mathbf{r}_i}\cdot\mathbf{v}_i+\nabla_{\mathbf{v}_i}\cdot\frac{\mathbf{F}_i}{m}\right)\right\}=$$

$$=\int_{\mathbf{r}_k}\left\{\nabla_{\mathbf{r}_i}\cdot(f\mathbf{v}_i)+\nabla_{\mathbf{v}_i}\cdot\left(f\frac{\mathbf{F}_i}{m}\right)\right\}=$$

$$=\sum_{i\neq k}\int_{\mathbf{r}_k}\left\{\nabla_{\mathbf{r}_i}\cdot(f\mathbf{v}_i)+\nabla_{\mathbf{v}_i}\cdot\left(f\frac{\mathbf{F}_i}{m}\right)\right\}=$$

$$=\sum_{i\neq k}\int_{\mathbf{r}_k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f+\frac{\mathbf{F}_i}{m}\cdot\nabla_{\mathbf{v}_i}f\right\}=$$

$$=\sum_{i\neq k}\int_{\mathbf{r}_k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f+\left(\frac{\mathbf{F}_i^{ext}}{m}+\sum_{j\in K}\frac{\mathbf{F}_{ij}}{m}+\sum_{j\notin K}\frac{\mathbf{F}_{ij}}{m}\right)\cdot\nabla_{\mathbf{v}_i}f\right\}=$$

$$=\sum_{i\neq k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}\int_{\mathbf{r}_k}f+\frac{\mathbf{F}_i^{ext}}{m}\cdot\nabla_{\mathbf{v}_i}\int_{\mathbf{r}_k}f+\sum_{j\notin K}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}\int_{\mathbf{r}_k}f\right\}+\sum_{i\neq k}\sum_{j\in K}\int_{\mathbf{r}_k}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f=$$

$$=\sum_{i\neq k}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f_s+\frac{\mathbf{F}_i^{ext}}{m}\cdot\nabla_{\mathbf{v}_i}f_s+\sum_{j\notin K}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f_s\right\}+\sum_{i\neq k}\sum_{j\in K}\int_{\mathbf{r}_k}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f=$$

$$0=\partial_t f_s+\sum_{i\notin K}\left\{\mathbf{v}_i\cdot\nabla_{\mathbf{r}_i}f_s+\frac{\mathbf{F}_i^{ext}}{m}\cdot\nabla_{\mathbf{v}_i}f_s+\sum_{j\notin K}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f_s\right\}+\sum_{i\notin K}\sum_{j\in K}\int_{\mathbf{x}_k}\frac{\mathbf{F}_{ij}}{m}\cdot\nabla_{\mathbf{v}_i}f\,.$$

with $K=s+1:N$.

1-particle equation. With $K=2:N$,

$$0=\partial_t f_1+\mathbf{v}_1\cdot\nabla_{\mathbf{r}_1}f_1+\frac{\mathbf{F}_1^{ext}}{m}\cdot\nabla_{\mathbf{v}_1}f_1+\sum_{j=2:N}\int_{\mathbf{x}_k,k=2:N}\frac{\mathbf{F}_{1j}}{m}\cdot\nabla_{\mathbf{v}_1}f\,d\mathbf{r}_k d\mathbf{v}_k\,.$$

Details

Force $\mathbf{F}_{1j}$ only depends on variables of particle 1 and j. Therefore,

$$\sum_{j=2:N}\int_{\mathbf{r}_k,k=2:N}\frac{\mathbf{F}_{1j}}{m}\cdot\nabla_{\mathbf{v}_1}f\,d\mathbf{r}_k d\mathbf{v}_k=\sum_{j=2:N}\int_{\mathbf{x}_j}\frac{\mathbf{F}_{1j}}{m}\cdot\int_{\mathbf{r}_k,k=2:N,k\neq j}\nabla_{\mathbf{v}_1}f\,d\mathbf{r}_k d\mathbf{v}_k=$$

$$=\sum_{j=2:N}\int_{\mathbf{x}_j}\frac{\mathbf{F}_{1j}}{m}\cdot\nabla_{\mathbf{v}_1}\int_{\mathbf{x}_k,k=2:N,k\neq j}f\,d\mathbf{r}_k d\mathbf{v}_k=$$

$$=\sum_{j=2:N}\int_{\mathbf{x}_j}\frac{\mathbf{F}_{1j}}{m}\cdot\nabla_{\mathbf{v}_1}f_2^{K/\{1,j\}}(\mathbf{r}_1,\mathbf{v}_1,\mathbf{r}_j,\mathbf{v}_j,t)d\mathbf{r}_j d\mathbf{v}_j\,.$$

For identical particles, the summation becomes $(N-1)\int_{\mathbf{x}_2}\frac{\mathbf{F}_{12}}{m}\cdot\nabla_{\mathbf{v}_1}f_2(\mathbf{x}_1,\mathbf{x}_2,t)$

Collision integral. Assumptions

Details

When a collision occurs, the dynamics of a particle is not smooth anymore. Thus, is it possible to use partial derivative in time? Force is impulsive. Thus, either function in the sense of distribution and/or integration in time is required

Trash

- only 2-particle collision. Good assumption for "dilute systems".
- a collision occurs when 2 particles are in the same position (i.e. the dimension of the particles is negligible), $\mathbf{r}_2 = \mathbf{r}_1$

$$\propto \int \Gamma_{\mathbf{p}_1',\mathbf{p}_2' \to \mathbf{p}_1,\mathbf{p}_2} f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{p}_1', \mathbf{p}_2') - \Gamma_{\mathbf{p}_1,\mathbf{p}_2 \to \mathbf{p}_1',\mathbf{p}_2'} f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{p}_1, \mathbf{p}_2)$$

with

- $\Gamma_{\mathbf{p}_1',\mathbf{p}_2' \to \mathbf{p}_1,\mathbf{p}_2}$ representing the probability of starting from $\mathbf{p}_i'$ before the collision and ending with momenta $\mathbf{p}_i$
  - this probability is determined by the analysis of 2-particle collision dynamics
  - this probability is "scaled" by the probability of being in the state before the collision $f(\mathbf{r}_1, \mathbf{r}_2 = \mathbf{r}_1, \mathbf{p}_1', \mathbf{p}_2')$

Molecular chaos (Stosszahlansatz). Before and after the collision

$$f_2(\mathbf{v}_1, \mathbf{v}_2, t) \sim f_1(\mathbf{v}_1)\, f(\mathbf{v}_2, t)$$

9.3 H-theorem

Starting from Boltzmann equation for the 1-particle probability density function, with the molecular chaos assumption,

$$D_t f_1 = \int d\mathbf{v}_2 \int d\Omega \sigma(\Omega) |\mathbf{v}_1 - \mathbf{v}_2| \left(f_1' f_2' - f_1 f_2\right)$$

Functional $H(t)$ is defined as

$$H(t) = \int f(\mathbf{v}, t) \ln f(\mathbf{v}, t) d\mathbf{v}$$

Taking time derivative (integration domain is unbounded. Use divergence theorem and boundary conditions to prove that flux contributions are zero)

$$d_t H(t) = \int \partial_t \left(f \ln f\right) d\mathbf{v} = \int \partial_t f \left(1 + \ln f\right) d\mathbf{v}\,.$$

Details

todo *include convective terms of Boltzmann equation*

todo *prove each step*

$$
\begin{aligned}
d_t H(t) &= \int \partial_t f \,(1+\ln f)\, d\mathbf{v} = \\
&= \int\int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f_1' f_2' - f_1 f_2)\,(1+\ln f_1)\,\sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
&= \frac{1}{2}\int\int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f_1' f_2' - f_1 f_2)\,(\ln f_1 + \ln f_2)\,\sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
&= \frac{1}{4}\int\int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f_1' f_2' - f_1 f_2)\,(\ln f_1 + \ln f_2 - \ln f_1' - \ln f_2')\,\sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
&= \frac{1}{4}\int\int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f_1' f_2' - f_1 f_2) \ln \frac{f_1 f_2}{f_1' f_2'} \sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
&\leq 0 \,,
\end{aligned}
$$

as

$$
(a-b)\ln\frac{b}{a} \leq 0 \,,
$$

since, if $a > b$ the first factor is positive and the second negative, and viceversa, and $a = b$ implies equality.

CHAPTER

TEN

STATISTICAL PHYSICS - NOTES

10.1 Ensembles

10.2 Microcanonical ensemble

10.3 Canonical ensemble

10.4 Macrocanonical ensemble

10.5 Statistics

Each of the N components of the system is in an **energy level** i. Energy level i has g_i sublevels with the same energy level.

- energy levels, E_i of each component
- occupation number N_i of level i
- **Central role of energy.** In a system macroscopically at rest, the energy of a system is the only macroscopic meaningful non-zero mechanical quantity, constant for closed and isolated systems
- **Principle of maximum uncertainty**, **maximum entropy**, **minimum information**: given a measurement of a macroscopic variable V, describing the macrostate of the system, the feasible un-observed/able microstates of the system are the microstates consistent with it: there's usually a sharp maximum of in the probability density of the micrsotates.

Given a macrostate, what's the number of ways $W(N_i; g_i)$ to get a consistent microstate? Once the expression is found, constrained optimization follows: optimization w.r.t. N_i is usually performed in the limit of $N_i \to +\infty$ (why in Fermi-Dirac distribution, obeying Pauli exclusion principle?), with the values of the macroscopic variables as constraints usually treated with Lagrange multiplier.

10.5.1 Boltzmann

Statistics of distinguishible components. Let N_i the number of elementary components in the energy level i with energy E_i. Let the number of elementary components $N = \sum_i N_i$ and the total energy of the system $E = \sum_i N_i E_i$ be given.

10.5.2 Bose-Einstein

Statistics of undistinguishable components that can be in the same (sub)level. Given the number of elementary components $\sum_i N_i = N$ and the energy $\sum_i N_i E_i = E$,

$$W_{BE,i} = \frac{(N_i + g_i - 1)!}{N_i!(g_i - 1)!} \quad , \quad W_{BE} = \prod_i W_{BE,i} \, . \tag{10.1}$$

Counting microstates

todo write page *Combinatorics* and add link

Most likely microstate. Instead of maximizing (10.1), the objective function is $\ln W_{BE}$, after using Stirling approximation in the limit of large N_i and g_i, $N_i! \sim \left(\frac{N_i}{e}\right)^{N_i}$. The approximate occupation number of one of the G_i sublevels of the ith level of the most likely microstate is

$$n_i := \frac{N_i}{g_i} = \frac{1}{e^{\alpha + \beta E_i} - 1} \, .$$

Optimization

$$\begin{aligned}
J(N_i, \alpha, \beta) &= \ln W_{BE} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) = \\
&= \sum_i \{ \ln(N_i + g_i - 1)! - \ln N_i! - \ln(g_i - 1)! \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) \simeq \\
&\simeq \sum_i \{ (N_i + g_i - 1) \ln(N_i + g_i - 1) - N_i \ln N_i - (g_i - 1) \ln(g_i - 1) + N_i + g_i - 1 - N_i - (g_i - 1) \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) \\
&= \sum_i \{ (N_i + g_i - 1) \ln(N_i + g_i - 1) - N_i \ln N_i - (g_i - 1) \ln(g_i - 1) \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right)
\end{aligned}$$

Using $\partial_n (n + a) \ln(n + a) = \ln(n + a) + 1$,

$$0 = \partial_{N_k} J \simeq \{ \ln(N_k + g_k - 1) - \ln N_k \} - \alpha - \beta E_k \, ,$$

and thus

$$\ln \frac{N_k + g_k - 1}{N_k} = \alpha + \beta E_k \, ,$$

$$\frac{N_k + g_k - 1}{N_i} = e^{\alpha + \beta E_k}$$

$$N_k = \frac{g_k - 1}{e^{\alpha + \beta E_k} - 1} \simeq \frac{g_k}{e^{\alpha + \beta E_k} - 1} \, ,$$

Thus, in the limit of $g_k \gg 1$, the occupation number of the k level is

$$N_k = \frac{G_k}{e^{\alpha+\beta E_k} - 1} \, ,$$

and the average occupation number of one of the g_k sublevels in the k level is

$$n_k := \frac{N_k}{G_k} = \frac{1}{e^{\alpha+\beta E_k} - 1}$$

Meaning of α, β

Example 10.5.1 (Black-body radiation: Planck, Wien, and Stefan-Boltzmann laws)

Planck's law. Energy density w.r.t. frequency

$$u_f(f,T) = \frac{8\pi h f^3}{c^3} \frac{1}{e^{\frac{hf}{k_B T}} - 1}$$

Planck's law in a cubic box

Planck's law uses:

- relation between pulsation and wave vector, or frequency and wave number and the speed of light c for light waves

$$c = \frac{\omega}{|\vec{k}|} = \lambda f$$

$$f = \frac{\omega}{2\pi} = \frac{c|\vec{k}|}{2\pi}$$

- Planck assumption that the minimum non-zero energy of a mode with frequency f is $E = hf$, and all the possible values of the energy of the mode is

$$E_m = mhf \quad , \quad m \in \mathbb{N} \, .$$

Taking a cubic box with sides $L_x = L_y = L_z = L$, the possibile modes have (**todo** *why? Which boundary condition? Periodic? Some physical? Just fictitious discretization?*) in each direction wave-lengths $\lambda_n = \frac{L}{|\vec{n}|} = \frac{2\pi}{|\vec{k}|}$,

$$\vec{k} = \frac{2\pi}{L}\vec{n} \, .$$

Mode density in $\vec{n}$-domain is 2 mode per each volume of unit length (2 polarization), and thus the number of modes dN in an elementary volume is

$$dN = 2 \, d^3\vec{n} \, ,$$

Changing variables, it's possible to find the mode density w.r.t. wave vector $\vec{k}$,

$$dN = 2 \, d^3\vec{n} = 2 \, \frac{L^3}{(2\pi)^3} \, d^3\vec{k} \, ,$$

or with its absolute value, exploiting the isotropy of the density function - and writing the elementary volume using "spherical coordinates" $d^3\vec{k} = 4\pi \left|\vec{k}\right|^2 d\left|\vec{k}\right|$,

$$\begin{aligned} dN &= \frac{V}{(2\pi)^3} 8\pi \left|\vec{k}\right|^2 d\left|\vec{k}\right| = \\ &= \frac{V}{(2\pi)^3} 8\pi \frac{8\pi^3}{c^3} f^2 df = \\ &= V\frac{8\pi}{c^3} f^2 df =: Vg(f)df \ . \end{aligned}$$

Average energy of a mode

Using Boltzmann distribution (**why?**) for the energy distribution in a single mode,

$$P(E_r) = \frac{e^{-\beta E_r}}{Z} \ ,$$

with $E_r = rhf$, and the partition function

$$Z = \sum_s e^{-\beta E_s} = \sum_s e^{-\beta hfs} = \frac{1}{1-e^{-\beta hf}} \ .$$

The average energy of the mode reads

$$\begin{aligned} \langle E \rangle &= \sum_r E_r P(E_r) = \\ &= \sum_r rhf \frac{e^{-\beta hfr}}{Z} = \\ &= hf(1-e^{-\beta hf}) \sum_r re^{-\beta hfr} = \\ &= hf(1-e^{-\beta hf}) \frac{e^{-\beta hf}}{(1-e^{-\beta hf})^2} = \\ &= \frac{hf}{e^{\beta hf}-1} \ . \end{aligned}$$

Putting together the mode number density and the average energy of a mode, the energy density per unit volume, per frequency reads

$$\begin{aligned} u(f,T) &= \langle E \rangle (f)\, g(f) = \\ &= \frac{hf}{e^{\beta hf}-1} \frac{8\pi}{c^3} f^2 = \\ &= \frac{8\pi hf^3}{c^3} \frac{1}{e^{\beta hf}-1} \ . \end{aligned}$$

Property of the series

$$\sum_{n=0}^{+\infty} nx^n = \frac{x}{(1-x)^2}$$

Proof. If the series is convergent (is this the required condition?)

$$\frac{d}{dx} \sum_{n=0}^{+\infty} x^n = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}$$

$$\frac{d}{dx}\sum_{n=0}^{+\infty} x^n = \sum_{n=0}^{+\infty} n x^{n-1}$$

$$x\frac{d}{dx}\sum_{n=0}^{+\infty} x^n = \sum_{n=0}^{+\infty} n x^n = \frac{x}{(1-x)^2}$$

Sperctral radiance, B_f, so that an infinitesimal amount of power radiated by a surface ... is $dP = B_f(f,T)\cos\theta\, dA\, d\Omega\, df$

$$B_f(f,T) = \frac{2hf^3}{c^2}\frac{1}{e^{\frac{hf}{k_BT}}-1}\,.$$

This expression is obtained[1] assuming homogeneous radiation from a small hole cut into a wall of the box. Only half of the energy radiates through the hole - so factor $\frac{1}{2}$ in front of the energy density - through a solid angle 2π - and thus this process give the same result as a radiation of all the energy density in all the space directions, just providing the same factor $\frac{1}{4\pi}$. The flux of energy "has velocity" c and thus

$$B_f(f,T) = \frac{1}{4\pi}u_f(f,T)c\,.$$

Wien's law. Wien's law tells that the frequency f^* corresponding to the maximum of the spectral radiance of a black-body radiation described by Planck's law is proportional to its temperature.

From direct evaluation of the derivative of the spectral radiance as a function of f,

$$\partial_f B_f(f,T) = \frac{2h}{c^2}\left[3f^2\frac{1}{e^{\frac{hf}{k_BT}}-1} + f^3\left(-\frac{\frac{h}{k_BT}e^{\frac{hf}{k_BT}}}{\left(e^{\frac{hf}{k_BT}}-1\right)^2}\right)\right] =$$

$$= \frac{2hf^2e^{\frac{hf}{k_BT}}}{c^2\left(e^{\frac{hf}{k_BT}}-1\right)^2}\left[3\left(1-e^{-\frac{hf}{k_BT}}\right) - \frac{hf}{k_BT}\right]\,.$$

Now, if $\partial_f B_f(f,T) = 0$ the frequency is either $f = 0$, or the solution of the nonlinear algebraic equation

$$0 = 3\left(1-e^{-\frac{hf}{k_BT}}\right) - \frac{hf}{k_BT}\,.$$

Defining $x := \frac{hf}{k_BT}$, this equation becomes

$$0 = 3(1-e^x) - x\,,$$

whose solution $x^* \approx 2.82$ can be easily evaluated with an iterative method (or expressed in term of the Lambert's function W, so loved at Stanford and on Youtube: they'd probaly like to look at tabulated values, or pose). Once the solution x^* of this non-dimensional equation is found, the frequency where maximum energy density occurs reads

$$f^* = \frac{k_BT}{h}x^* \simeq 2.82\frac{k_B}{h}T\,.$$

[1] Derivation of Planck's Law.

Stefan-Boltzmann law.

$$\begin{aligned}\frac{P}{A} &= \int B_f(f,T)\cos\phi\, df\, d\Omega = \\ &= \int_{f=0}^{+\infty}\int_{\phi=0}^{\frac{\pi}{2}}\int_{\theta=0}^{2\pi} B_f(f,T)\cos\phi\sin\phi\, df\, d\phi\, d\theta = \\ &= \pi\int_{f=0}^{+\infty} B_f(f,T)\, df = \\ &= \frac{2\pi h}{c^2}\int_{f=0}^{+\infty}\frac{f^3}{e^{\frac{hf}{k_BT}}-1}\, df = \\ &= \frac{2\pi h}{c^2}\left(\frac{k_BT}{h}\right)^4\int_{u=0}^{+\infty}\frac{u^3}{e^u-1}\, du\ .\end{aligned}$$

The value of the integral is $\frac{\pi^4}{15}$ and thus

$$\frac{P}{A} = \sigma T^4 \qquad , \qquad \sigma = \frac{2\pi^5 k_B^4}{15c^2h^3}\ .$$

Example 10.5.2 (Energy density and radiance)

Radiance. The radiance $L_{e,\Omega}$ of a surface is the flux of energy per unit solid angle, per unit projected area of the source.

Spectral radiance in frequency is the radiance per unit frequency, $L_{e,\Omega,f} = \frac{\partial L_{e,\Omega}}{\partial f}$.

10.5.3 Fermi-Dirac

Statistics of undistinguishable components that can't be in the same (sub)level, obeying to the Pauli exclusion principle. Given the number of elementary components $\sum_i N_i = N$ and the energy $\sum_i N_i E_i = E$,

$$W_{FD,i} = \frac{G_i!}{(G_i-N_i)!N_i!} \qquad , \qquad W_{FD} = \prod_i W_{FD,i}\ . \tag{10.2}$$

Counting microstates

todo write page *Combinatorics* and add link

Most likely microstate. The approximate occupation number of the ith level of the most likely microstate is

$$n_i := \frac{N_i}{G_i} = \frac{1}{1+e^{\alpha+\beta E_i}}\ .$$

Optimization

$$J(N_i,\alpha,\beta) = \ln W_{FD} + \alpha\left(N - \sum_i N_i\right) + \beta\left(E - \sum_i N_i E_i\right) =$$
$$= \sum_i \{\ln G_i! - \ln(G_i - N_i)! - \ln N_i!\} + \alpha\left(N - \sum_i N_i\right) + \beta\left(E - \sum_i N_i E_i\right) =$$
$$= \sum_i \{G_i \ln G_i - (G_i - N_i)\ln(G_i - N_i) - N_i \ln N_i\} + \alpha\left(N - \sum_i N_i\right) + \beta\left(E - \sum_i N_i E_i\right) =$$

Using $\partial_n (n+a)\ln(n+a) = \ln(n+a) + 1$,

$$0 = \partial_{N_k} J \simeq \{\ln(G_k - N_k) - \ln N_k\} - \alpha - \beta E_k \,,$$

and thus

$$\ln \frac{G_k - N_k}{N_k} = \alpha + \beta E_k \,,$$

$$\frac{G_k}{N_k} - 1 = e^{\alpha + \beta E_k}$$

The occupation number of the k level is

$$N_k = \frac{G_k}{1 + e^{\alpha + \beta E_k}} \,.$$

The average occupation of the G_k sublevels of the k level is

$$n_k := \frac{N_k}{G_k} = \frac{1}{1 + e^{\alpha + \beta E_k}} \,.$$

Meaning of α, β

CHAPTER

ELEVEN

STATISTICAL PHYSICS - STATISTICS MISCELLANEA

Information content and Entropy

Given a discrete random variable X with probability mass function $p_X(x)$, the self-information (**todo** *what about mutual information of random variables?*) is defined as the opposite of the logaritm of the mass function $p_X(x)$,

$$I_X(x) := -\ln\left(p_X(x)\right) \ .$$

Information content of indenpendent random variables is additive. Since $p_{X,Y}(x,y) = p_X(x)p_Y(y)$,

$$I_{X,Y}(x,y) = -\ln\left(p_{X,Y}(x,y)\right) = -\ln\left(p_X(x)p_Y(y)\right) = -\ln p_X(x) - \ln p_Y(y) \ .$$

Shannon entropy. Shannon entropy of a discrete random variable X is defined as the expected value of the information content,

$$H(X) := \mathbb{E}[I_X(X)] = \sum p_X(x) I_X(x) = -\sum p_X \ln p_X(x) \ .$$

Gibbs entropy. Gibbs entropy was defined by J.W.Gibbs in 1878,

$$S = -k_B \sum_i p_i \ln p_i \ .$$

Additivity holds for independent random variables.

Boltzmann entropy. Boltmann entropy holds for uniform distributions over Ω possible states, $p_i = \frac{1}{\Omega}$. Gibbs' entropy of this uniform distribution becomes

$$S = -k_B \Omega \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_B \ln \Omega \ .$$

Entropy in Quantum Mechanics. todo

Boltzmann distribution

Given a set of discrete states with probability p_i, and the average measure as "macroscopic quantity" $E = \sum_i p_i E_i$, Boltzann distribution maximizes the entropy (**todo** *Link to min info, max uncertainty*)

$$S = -k_B \sum_i p_i \ln p_i \ .$$

The distribution follows from the constrained optimization

$$\widetilde{S} = S - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - E \right)$$

$$0 = \partial_\alpha \widetilde{S} = -\sum_i p_i - 1$$
$$0 = \partial_\beta \widetilde{S} = -\sum_i p_i E_i - E$$
$$0 = \partial_{p_k} \widetilde{S} = -k_B \left(\ln p_k + 1\right) - \alpha - \beta E_k$$

and thus

$$p_k = e^{-1-\frac{\alpha}{k_B}-\frac{\beta}{k_B}E_k} = e^{-\left(1+\frac{\alpha}{k_B}\right)} e^{-\frac{\beta}{k_B}E_k} = C e^{-\frac{\beta}{k_B}E_k} \ ,$$

and the normalization constant C is determined by normalization condition

$$1 = \sum_k p_k = C \sum_k e^{-\frac{\beta E_k}{k_B}}$$

The inverse $Z = C^{-1}$ is defined as the **partition function**,

$$Z = C^{-1} = \sum_k e^{-\frac{\beta E_k}{k_B}} \ ,$$

and the probability distribution becomes

$$p_k = \frac{e^{-\frac{\beta E_k}{k_B}}}{Z} = \frac{e^{-\frac{\beta E_k}{k_B}}}{\sum_i e^{-\frac{\beta E_i}{k_B}}} \ .$$

Properties.

$$\frac{p_k}{p_i} = e^{-\frac{\beta}{k_B}(E_k - E_i)} \ .$$

Thermodynamics. Comparison of statistics and classical thermodynamics

First principle of classical thermodynamics (for a monocomponent gas with no electric charge,…) reads

$$T\, dS = dE + P\, dV$$

Entropy for Boltzmann distribution reads

$$\begin{aligned} S &= -k_B \sum_i p_i \ln p_i = \\ &= -k_B \sum_i \left[p_i \left(-\frac{\beta E_i}{k_B} - \ln Z \right) \right] = \\ &= \beta \langle E \rangle + k_B \ln Z \end{aligned}$$

From classical thermodyamics, temperature T can be defined as the partial derivative of the entropy of a system w.r.t. its internal energy keeping constant all the other independent variables,

$$\begin{aligned}
\frac{1}{T} &= \left(\frac{\partial S}{\partial E}\right)\Big|_X = \\
&= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{\partial \ln Z}{\partial E} = \\
&= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \frac{\partial Z}{\partial E} = \\
&= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \frac{\partial Z}{\partial \beta} \frac{\partial \beta}{\partial E} = \\
&= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \left(-\sum_i \frac{E_i}{k_B} e^{-\frac{\beta E_i}{k_B}} \right) \frac{\partial \beta}{\partial E} = \\
&= \frac{\partial \beta}{\partial E} E + \beta - \left(\sum_i E_i p_i \right) \frac{\partial \beta}{\partial E} = \\
&= \frac{\partial \beta}{\partial E} E + \beta - E \frac{\partial \beta}{\partial E} = \beta \,.
\end{aligned}$$

todo

- write the derivative above clearly in terms of composite functions
- microscopical/statistical approach to the first principle of thermodynamics

$$dE = d\left(\sum_i p_i E_i\right) = \sum_i E_i \, dp_i + \sum_i p_i dE_i$$

CHAPTER

TWELVE

ISING MODEL

- Introduction, and some notes about analytical results and numerical methods. Ising model on regular lattices
- **todo** *Notebook implementing methods and showing some examples*

References.

- https://rajeshrinet.github.io/blog/2014/ising-model/#:~:text=Flip the spin of a,Repeat 2-4
- https://mattbierbaum.github.io/ising.js/

12.1 Ising model - Notes

12.1.1 Introduction

12.1.2 Time evolution

Metropolis-Hastings

Metropolis-Hastings algorithm is designed to generate a collection of states according to a desired distribution $p(s)$, using a **Markov process** with a unique **stationary distribution** $\pi(s) = p(s)$.

Algorithm

- Initialize state s_0 with energy E_0
- while **not** terminal condition:
  - flip the spin of a randomly chosen site, to get the possible state $s_{\widetilde{n}}$
  - calculate the change in energy of the state $\Delta\widetilde{E}_n := \widetilde{E}_n - E_{n-1}$
  - update state:
    - if $\Delta\widetilde{E}_n < 0$ accept the change, $s_n = s_{\widetilde{n}}$
    - if $\Delta\widetilde{E}_n > 0$ accept the change with probability $\exp\left(-\frac{\Delta\widetilde{E}_n}{T}\right)$

Starting from the **detailed balance** (sufficient but not necessary condition for the existence of a (unique?) stationary state), $P(s'|s)p(s) = P(s|s')p(s')$,

$$\frac{P(s'|s)}{P(s|s')} = \frac{p(s')}{p(s)} ,$$

each transition is splitted in **two sub-steps**: proposal and acceptance-rejection steps. Transition is written as $P(s'|s) = g(s'|s)A(s',s)$ being $g(s'|s)$ the proposal conditional probability, and $A(s',s)$ the acceptance probability. Acceptance probability is choosen in a way to satisfy the conditions of time-reversal and ergodicity. Metropolis choice reads

$$A(s',s) = \min\left(1, \frac{p(s')}{p(s)}\frac{g(s|s')}{g(s'|s)}\right) .$$

todo *check if Metropolis choice fulfills the necessary conditions.*

Example: Boltmann distribution with uniform proposal. Let a system have Boltzmann stationary distribution, $p(s) \propto \exp(-\beta E(s))$, and uniform proposal conditional probability $g(s'|s) = \frac{1}{N}$, with N the number of accessible states from the state s.

Example 12.1.1 (Ising model on a 2D lattice with 1 flip per update)

As an example, for an Ising model on a 2D lattice with dimension $m \times n$ with proposal update with 1 flip $N = mn$.

Example 12.1.2 (Ising model on a 2D lattice with k flips per update)

todo *Is a probability built with independent successive flips ok? Target states at each flip have the same probability, s.t. the probability of reaching steps s_k after k flips starting from s_0 is not uniform.*

Let $g(s|s') = g(s'|s)$. The acceptance probability of the transition between different states $s' \neq s$ reads

$$A(s',s) = \min\left(1, e^{-\beta(E(s')-E(s))}\right) ,$$

and the corresponding element of transition matrix is

$$P(s'|s) = \frac{1}{N}\min\left(1, e^{-\beta(E(s')-E(s))}\right) .$$

The *diagonal elements* of the probability transition matrix - representing no state transition - are evaluated as the probability that makes $\sum_{s'} P(s'|s) = P(s|s) + \sum_{s' \in T(s)} P(s'|s) = 1$, i.e.

$$P(s|s) = \frac{1}{N} \sum_{\substack{s' \in T(s) \\ E(s') > E(s)}} \left(1 - e^{-\beta(E(s')-E(s))}\right) .$$

$P(s|s)$

$$\begin{aligned}
P(s|s) &= 1 - \sum_{s' \in T(s)} P(s'|s) = \\
&= 1 - \sum_{s' \in T(s)} \frac{1}{N}\min\left(1, e^{-\beta(E(s')-E(s))}\right) = \\
&= \frac{1}{N}\sum_{s' \in T(s)} \left[1 - \min\left(1, e^{-\beta(E(s')-E(s))}\right)\right] = \\
&= \frac{1}{N}\sum_{s' \in T(s)} \max\left(0, 1 - e^{-\beta(E(s')-E(s))}\right) = \\
&= \frac{1}{N} \sum_{\substack{s' \in T(s) \\ E(s') > E(s)}} \left(1 - e^{-\beta(E(s')-E(s))}\right) .
\end{aligned}$$

It's easy to prove that this probability functions are related by the detailed balance equations,

$$\frac{P(s'|s)}{P(s|s')} = \frac{\min\left(1, e^{-\beta(E(s')-E(s))}\right)}{\min\left(1, e^{-\beta(E(s)-E(s'))}\right)} = e^{-\beta(E(s')-E(s))} = \frac{e^{-\beta E(s')}}{e^{-\beta E(s)}} = \frac{p(s')}{p(s)} \,.$$

12.2 Markov processes

Markov Process. Let S a finite set of state s that can be indexed with integer indices i, P the transition probability between two states at consecutive time-steps, $P_{ss'} = P(s'|s)$, and p_0 the initial probability. A Markox process is uniquely defined by the triple

$$\langle S, P, p_0 \rangle \,.$$

If the Markov process is **time-homogeneous**, state transition matrix P is independent from the time-step.

12.2.1 State transition matrix

Given the finite set of states with integer indices, their probabilities at time n can be collected in the $|S|$-dimensional vector $\mathbf{p}_n$. Transition from time $n-1$ to n can be represented as

$$p_n(s') = \sum_s P(s'|s) p_{n-1}(s) = \sum_s P_{ss'} p_{n-1}(s) \,,$$

or using matrix formalism

$$\mathbf{p}_n^T = \mathbf{p}_{n-1}^T \mathbf{P} \,.$$

As $\mathbf{p}_n$ and $\mathbf{P}$ represents probability, their elements are non-negative. As the ith element of $\mathbf{p}_n$ represents the probability of the system of being in state i at time n, it follows

$$\sum_i \{\mathbf{p}_n\}_i = \sum_i p_n(i) = 1 \,.$$

As the element $P_{ij} = P(i|j)$ represents the conditional probability of reaching j from state i in one step, it follows

$$\sum_j \{\mathbf{P}\}_{ij} = \sum_j P(j|i) = 1 \,.$$

12.2.2 Properties

Eigenvalue $s = 1$, and stationary state π

The sum of the elements of each row of $\mathbf{P}$ is equal to 1, as $\sum_{s'} P_{ss'} = \sum_{s'} P(s'|s) = 1$. Equivalently, state transition matrix $\mathbf{P}$ has an eigenvalue $\lambda = 1$, with the uniform right eigenvector $\mathbf{1}$, i.e. $\mathbf{P1} = \mathbf{1}$.

The corresponding left eigenvector is a stationary state π,

$$\pi^T \mathbf{P} = \pi^T \,.$$

Spectrum of the transition matrix and asymptotic behavior of the system

todo *Are all the transition matrices diagonalizable?*

Assuming the transition matrix can be diagonalized, the evolution of the system from the initial probability $\mathbf{p}_0$ can be easily represented. The initial probabbility is written as $\mathbf{p}_0 = c_k \mathbf{w}_k$, being $\mathbf{w}_k$ the left eigenvectors of the transition matrix, and

$$\mathbf{p}_n^T = \mathbf{p}_{n-1}^T \mathbf{P} = \mathbf{p}_0^T \underbrace{\mathbf{P} \dots \mathbf{P}}_{n \text{ times}} = a_k \underbrace{\mathbf{w}_k^T \mathbf{P}}_{=\lambda_k \mathbf{w}_k^T} \dots \mathbf{P} = a_k \lambda_k^n \mathbf{w}_k^T \ .$$

As $n \to +\infty$ only the contributions of the eigenvectors with $|\lambda_k| = 1$ survive. If there's only one of these eigenvalues, $\lambda_1 = 1$, the system converges to $\mathbf{1}$. If there are many eigenvalues with $\lambda_i = 1$, there system converges with probability a_i to those stationary states, being a_i the component of the initial state of the system along the ith eigenvector,

$$\mathbf{p}_n \sim a_k \mathbf{w}_k \ .$$

Components a_i can be evaluated as the dot product between the **dual basis** $\{\mathbf{v}_i\}_i$ w.r.t. the left eigenvectors and the initial state $\mathbf{p}_0$, as

$$\mathbf{p}_0^T \mathbf{v}_i = a_k \underbrace{\mathbf{w}_k^T \mathbf{v}_i}_{=\delta_{ik}, \text{ def. of dual basis}} = a_i \ .$$

Existence of eigenvalues $|s_j| = 1$, $s_j = e^{i\theta_j}$ makes limit cycles possible as

$$\mathbf{p}_n \sim \mathbf{w}_j e^{in\theta_j} + \text{c.c.} = 2\left(\text{re}\{\mathbf{w}_j\} \cos\theta_j - \text{im}\{\mathbf{w}_j\} \cos\theta_j\right) \ .$$

todo *Discuss the possibility (?) of negative components. What's a negative probability?*

Detailed balance

By definition, a **reversible Markov chain** has a positive stationary distribution π that satisfies the detailed balance equations

$$\pi_i P_{ij} = \pi_j P_{ji} \ ,$$

i.e. the flow between each pair of states i, j is individaully balanced. Reversibility is a sufficient - even not necessary - condition for the existence of a stationary state.

Let $\pi_i \neq 0$ for $\forall i$, it's possible to define a symmetric matrix via a similarity transformation, as

$$\begin{aligned} \sqrt{\pi_i}\sqrt{\pi_i} P_{ij} &= \sqrt{\pi_j}\sqrt{\pi_j} P_{ji} \\ \sqrt{\pi_i} P_{ij} \frac{1}{\sqrt{\pi_j}} &= \sqrt{\pi_j} P_{ji} \frac{1}{\sqrt{\pi_i}} \\ S_{ij} &= S_{ji} \ , \end{aligned}$$

with $\mathbf{S} = \mathbf{D}\mathbf{P}\mathbf{D}^{-1}$, with $\mathbf{D} = \text{diag}\{\sqrt{\pi_i}\}$. As the matrices $\mathbf{P}$ and $\mathbf{S}$ are related via a similarity transformation, they have the same eigenvalues

todo

- *Discuss the meaning of "reveresible"*

Ergodicity. …**todo** *start from https://en.wikipedia.org/wiki/Markov_chain#Ergodicity*

12.3 Ising model - Numerics

Some references:

- https://farside.ph.utexas.edu/teaching/329/lectures/node110.html
- O.Rojas, Spectral mechanism and nearly reducible transfer matrices for pseudotransitions in one-dimensional systems, https://arxiv.org/html/2511.18109v2
- L.Onsager, 1944, https://arxiv.org/html/2412.07328v1 *really useful?*

```
"""
Ising model on regular 2D lattice with periodic boundaries

The Hamiltonian, H, is used to define the probability of the (micro)states of the␣
 ↪system
following a Boltzmann probability distribution p_i ~ e^( - E_i / T ) = e^( - beta * E_
 ↪i ).
The expression of the Hamiltonian reads: H = .5 * J_{ij} s_i s_j + M p_i s_i

First with low-dimensional lattice:
Then increasing the dimensions:

Samples states from the distribution p_i using Metropolis-Hastings method

"""

import numpy as np
```

12.3.1 $m \times n$ lattice

```
#> Useful functions for ising model on 2D lattice
class IsingLattice():
    def __init__(self, m=3, n=3, J=1., M=0., T=1.):
        """  """
        self.m = m
        self.n = n
        self.J = J
        self.M = M
        self.T = T

    def initialize(self,):
        """  """
        set_random_state()
        # Other initializations? Compatible with {-1, +1} spin values

    def set_random_state(self,):
        """  """
        self.state = 2 * np.random.randint(0,2, size=(self.m, self.n)) - 1
        self.E0 = self.compute_total_energy()

    def flip_indices(self,):
        """  """
        return np.random.randint(0, self.m), np.random.randint(0, self.n)
```

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```
    def flip_loc(self, i,j):
        """  """
        self.state[i,j] *= -1

    def flip(self,):
        """  """
        self.state[self.flip_indices()] *= -1

    def compute_local_energy(self, i,j):
        """
        Energy contribution due to [i,j] element of state reads
          E_local[i,j] = - J s[i,j] \sum_{(k,l)\in B(i,j)} s[k,l] - M s[i,j]
        Flipping the state, s[i,j] -> -s[i,j], gives a energy difference
        E_local_flip[i,j] - E_local[i,j] = 2 ( J s[i,j] \sum_{} s[k,l] + M s[i,j] ) =
                                         = - 2 E_local[i,j]
        """
        neighbor_sum = self.state[(i+1)%self.m, j] + self.state[(i-1)%self.m, j] + \
                       self.state[i, (j+1)%self.n] + self.state[i, (j-1)%self.n]
        return - self.state[i,j] * ( self.J * neighbor_sum + self.M )

    def compute_total_energy(self,):
        # For efficiency, use vector operations and avoid nested for loops
        # Just shift up and left, to avoid double counting in an efficient way
        up = np.roll(self.state, shift=1, axis=0)
        left = np.roll(self.state, shift=1, axis=1)

        # Energy = -J * sum(spin * neighbors)
        return - self.J * np.sum(self.state * (up + left)) - self.M * np.sum(self.
↪state)

    def compute_total_magnetization(self,):
        """  """
        return np.sum(isi.state)

    def compute_avg_magnetization(self,):
        """  """
        return np.mean(isi.state)

    def metropolis_hastings_run(self, n_steps=100, output_states=False):
        """   """
        E = self.compute_total_energy()
        Es = [ E ]
        Ms = [ self.compute_total_magnetization() ]
        states = []

        if output_states: states += [ self.state.copy() ]

        i_step = 0
        for i_step in range(n_steps):
            #> 1 step of Metropolis-Hastings algorithm
            dE = self.metropolis_hastings_step()    # Updating state with flip
            E += dE                                 # Update system energy

            #> Update output arrays
            Es += [ E ]
            Ms += [ self.compute_total_magnetization() ]
```

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```
            if output_states:
                states += [ self.state.copy() ]    # <-- ! Memory. Add some sampling␣
↪period,
                                                   # or dumping method. Don't store␣
↪the state
                                                   # at all the timesteps.
            #> Update step index
            i_step += 1

        return Es, Ms, states

    def metropolis_hastings_step(self,):
        """  """
        #> Proposal step
        i_flip, j_flip = self.flip_indices()                 # random sampling of the␣
↪flip indices
        dE = - 2 * self.compute_local_energy(i_flip, j_flip) # evaluate delta energy

        if dE < 0:   # accept
            self.flip_loc(i_flip, j_flip)
        else:
            p_flip = np.exp( -dE / self.T )

            if np.random.random() < p_flip:        # flip
                self.flip_loc(i_flip, j_flip)
            else:                                  # reject flip proposal
                dE = 0

        return dE
```

```
#> Parameters
m, n = 32,32   # 16, 16
J, M = 1., 0.  # 1.
Tc = 2.26918   # 2.26918 * J/k = T_critical
T = Tc * 1.

#> Ising Lattice Model
isi = IsingLattice(m=m, n=n, T=T, M=M, J=J)
isi.set_random_state()

print(f"Lattice dimensions: {m},{n}")
print( "Initial conditions:")
print(f"Energy       : {isi.E0}")
print(f"N. of spin up: {np.sum(isi.state[isi.state > 0])}")
print(f"Magnetization: {np.sum(isi.state)}")

#> Run Metropolis-Hastings algorithm
Es, Ms, states = isi.metropolis_hastings_run(n_steps=int(1e6), output_states=True)
```

```
Lattice dimensions: 32,32
Initial conditions:
Energy       : -24.0
N. of spin up: 516
Magnetization: 8
```

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(2,1, figsize=(5,5))
ax[0].plot(Es);  ax[0].set_ylabel('E');  ax[0].grid()
ax[1].plot(Ms);  ax[1].set_ylabel('M');  ax[1].grid()
ax[1].set_xlabel('step')

nrows_plots = 6
ncols_plots = 5
n_plots = nrows_plots * ncols_plots
i_plot = 0

#> Indices of all the flipping time steps
indices = np.where(np.diff(Es) != 0)[0] + 1
n_flips = len(indices)

d_indices = np.max([n_flips // n_plots, 1])
indices_plot = indices[d_indices * np.arange(n_plots)]
indices_plot[n_plots - 2] = len(Es)-1

# print(d_indices)
# print(np.arange(n_plots))
# print(indicesthere
# print(indices_plot)

fig, ax = plt.subplots(nrows_plots, ncols_plots, figsize=(3*ncols_plots, 3*nrows_
↪plots))

ax[0,0].imshow(states[0])
ax[0,0].set_title(f"it: {indices_plot[i_plot]}")
i_plot += 1

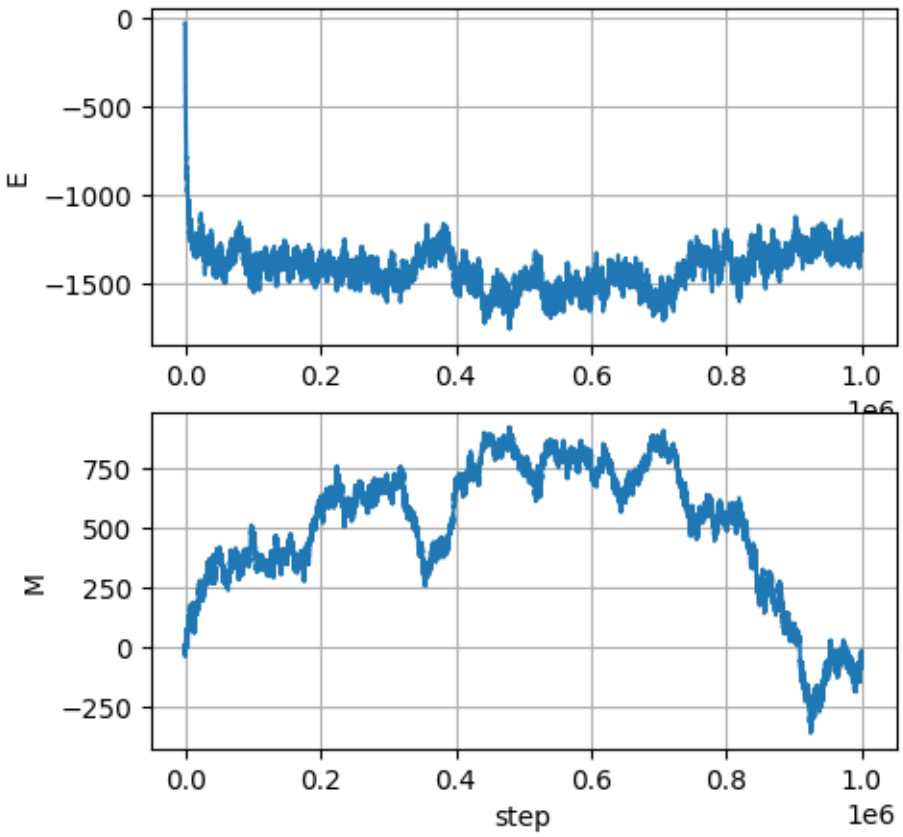
while i_plot < n_plots and i_plot < len(indices):

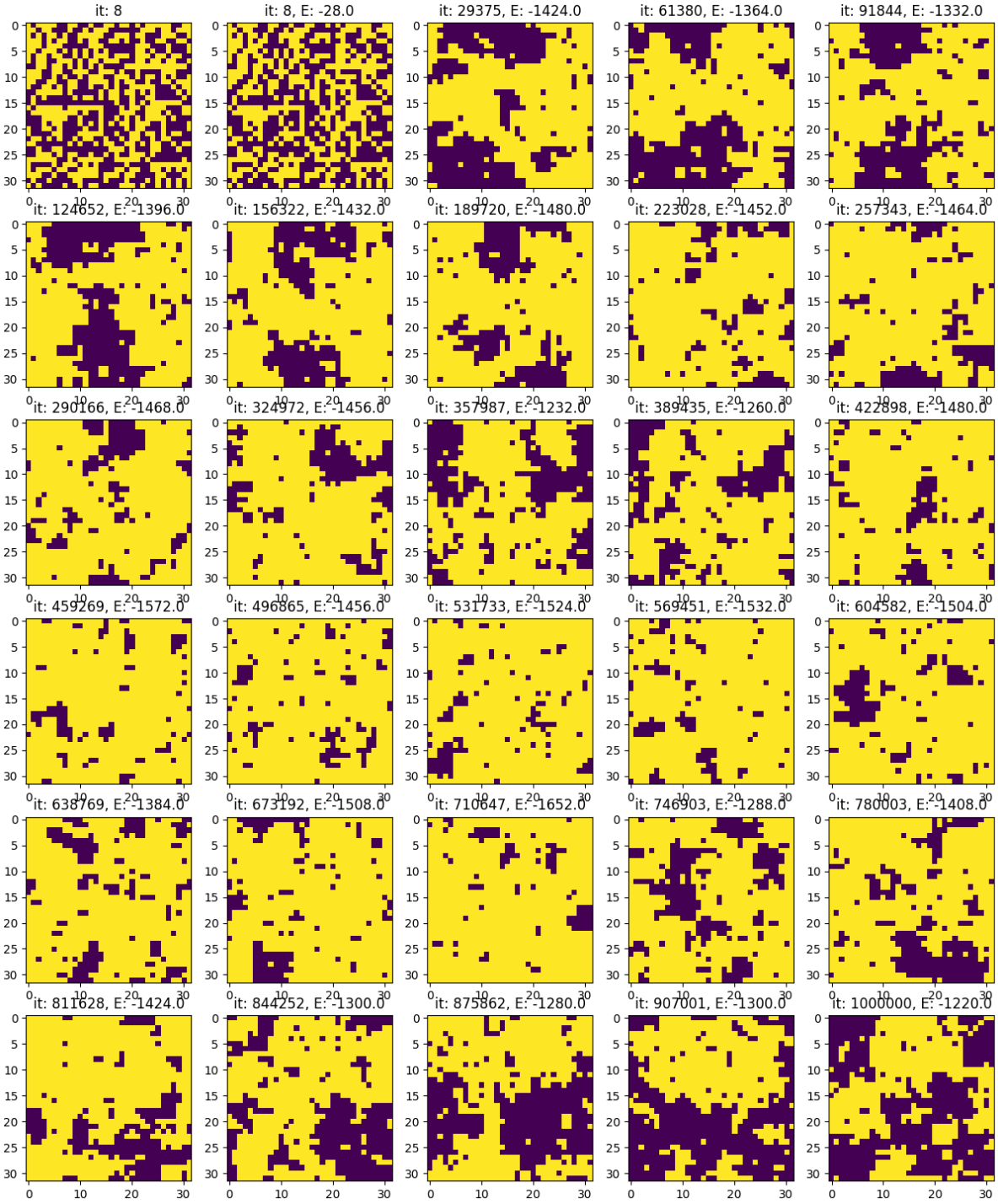
    row_plot = i_plot // ncols_plots
    col_plot = i_plot - row_plot * ncols_plots

    ax[row_plot, col_plot].imshow(states[indices_plot[i_plot-1]], vmin=-1, vmax=1)
    ax[row_plot, col_plot].set_title(f"it: {indices_plot[i_plot-1]}, E: {Es[indices_
↪plot[i_plot-1]]}")

    i_plot += 1

plt.show()
```


0
−500
−1000
−1500
E
0.0
0.2
0.4
0.6
0.8
1.0
1e6
750
500
250
0
−250
M
0.0
0.2
0.4
0.6
0.8
1.0
step
1e6




```
from scipy.special import logsumexp

#> Sample the stationary probability density function
burn_in = 5000                              # From visual inspection
Es_stationary = np.array(Es[burn_in:])      # Discard transient evolution

#> Evaluate P(E) from time history
```

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```
energy_levels, counts = np.unique(Es_stationary, return_counts=True)
pE = counts / np.sum(counts)

ln_g_tilde = np.log(pE) + energy_levels / isi.T

# N = 2 ** (isi.m, isi.n)
ln_N = isi.m * isi.m * np.log(2)

ln_Z = ln_N - logsumexp(ln_g_tilde)
ln_g = ln_Z + ln_g_tilde

#> P(M) probability distribution of magnetization, from time history
Ms_stationary = np.array(Ms[burn_in:])    # Discard transient evolution
mag_levels, mag_counts = np.unique(Ms_stationary, return_counts=True)
pM = mag_counts / np.sum(mag_counts)
```

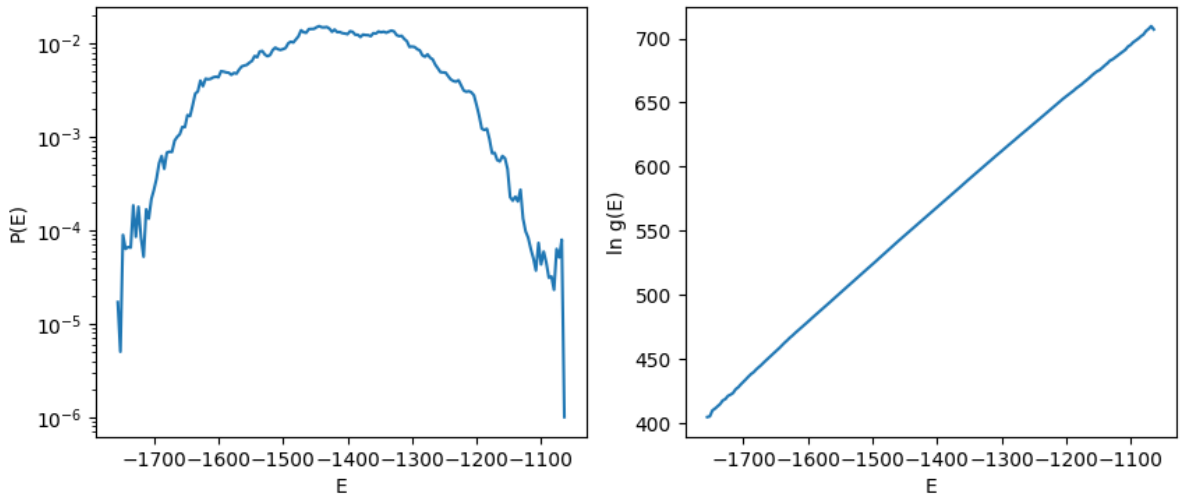
```
fig, ax = plt.subplots(1,2, figsize=(10,4))
ax[0].semilogy(energy_levels, pE)
ax[1].plot(energy_levels, ln_g)

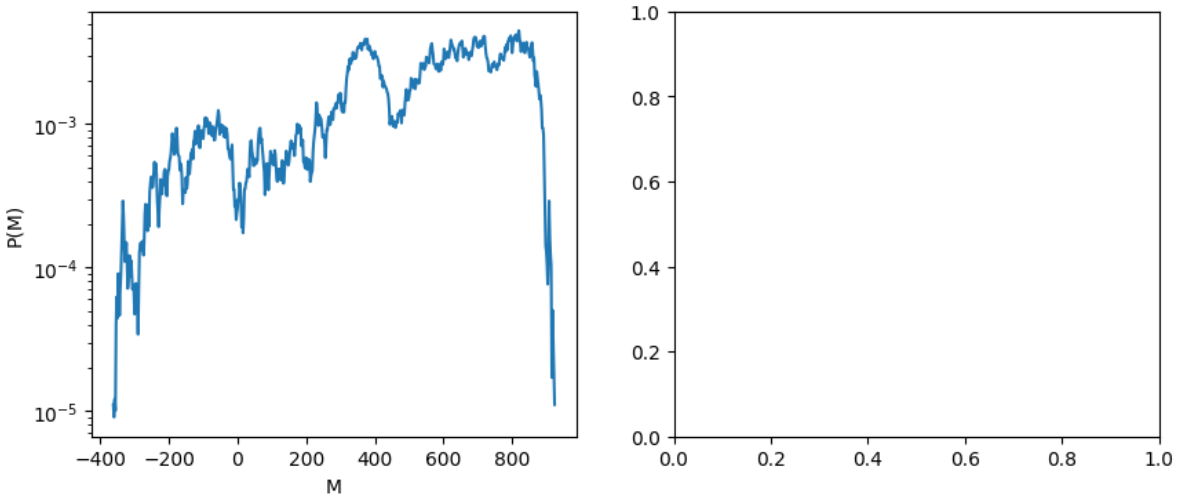
ax[0].set_xlabel('E');  ax[0].set_ylabel('P(E)')
ax[1].set_xlabel('E');  ax[1].set_ylabel('ln g(E)')

# ---
fig, ax = plt.subplots(1,2, figsize=(10,4))
ax[0].semilogy(mag_levels, pM)
# ax[1].plot(energy_levels, ln_g)

ax[0].set_xlabel('M');  ax[0].set_ylabel('P(M)')
# ax[1].set_xlabel('E');  ax[1].set_ylabel('ln g(E)')
```

```
Text(0, 0.5, 'P(M)')
```






```
"""
Old script using histrograms

#> Sample the stationary probability density function
burn_in = 1000                          # From visual inspection
Es_stationary = np.array(Es[burn_in:])   # Discard transient evolution

fig, ax = plt.subplots(1,1, figsize=(5,5))
ax.hist(Es_stationary, density=True)
ax.grid()
ax.set_xlabel('E')
# ax.plot(Es_stationary, np.exp(Es_stationary/isi.T), color='red')

# P(E) = g(E) * exp(-beta*E) / Z
# Z is the partition function, that works as the normalization factor
# g(E) is the number of degenerate states with energy = E
# Thus:
# - P(E) can be evaluated from samples
# - exp(-beta*E) is known for every value E
# - g(E) / Z can be evaluated as P(E) / exp(-beta*E) for every value E
#
# Z = sum_E ( g(E) * exp(-beta*E) )
# ...is it possible to find Z and thus g(E)?

"""
```

```
"\nOld script using histrograms\n\n#> Sample the stationary probability density␣
↪function\nburn_in = 1000                          # From visual inspection\nEs_
↪stationary = np.array(Es[burn_in:])   # Discard transient evolution\n\nfig, ax =␣
↪plt.subplots(1,1, figsize=(5,5))\nax.hist(Es_stationary, density=True)\nax.
↪grid()\nax.set_xlabel('E')\n# ax.plot(Es_stationary, np.exp(Es_stationary/isi.T),
↪ color='red')\n\n# P(E) = g(E) * exp(-beta*E) / Z\n# Z is the partition function,
↪ that works as the normalization factor\n# g(E) is the number of degenerate␣
↪states with energy = E\n# Thus:\n# - P(E) can be evaluated from samples\n# -␣
↪exp(-beta*E) is known for every value E\n# - g(E) / Z can be evaluated as P(E) /␣
↪exp(-beta*E) for every value E\n#\n# Z = sum_E ( g(E) * exp(-beta*E) )\n# ...is␣
↪it possible to find Z and thus g(E)?\n\n"
```

The relation between state probability $P(E)$ - i.e. probability of getting a microstate with energy E -, degeneracy number

$g(E)$ - i.e. the number of microstates with energy E - and the Boltzmann distribution $\propto e^{-\beta E}$ reads

$$P(E) = \frac{g(E)e^{-\beta E}}{Z} = \frac{g(E)e^{-\beta E}}{\sum_{E'} g(E')e^{-\beta E'}} \, .$$

Once the value E of energy levels is known, the exponential is known. Probability $P(E)$ is sampled from experiments. Thus

$$\frac{g(E)}{Z} = P(E)e^{\beta E} \, .$$

If the total number of microstate is not known, $g(E)$ can be evaluated up to a multiplicative constant.

If the total number of microstates is known, $\sum_E g(E) = N$, it's possible to compute all the values of $g(E)$ and Z. In this case, a linear system reads

$$\begin{cases} g(E) = ZP(E)e^{\beta E} \\ \\ \displaystyle\sum_{E'} g(E') = N \, , \end{cases}$$

and thus

$$Z = \frac{N}{\sum_{E'} P(E')e^{\beta E'}} \quad , \quad g(E) = \frac{NP(E)e^{\beta E}}{\sum_{E'} P(E')e^{\beta E'}} \, .$$

Log to avoid numerical exceptions.

$$\ln g(E) = \ln Z + \ln P(E) + \beta E$$

In order to evaluate Z,

$$Z = \sum_E g(E)e^{-\beta E} = \sum_E e^{\ln g(E) - \beta E} \, ,$$

and avoid overflow, the largest contribution is factored using the relation

$$\ln \sum_i e^{x_i} = \ln \left(e^{x_{max}} \sum_i e^{-(x_{max} - x_i)} \right) = \ln e^{x_{max}} + \ln \sum_i e^{-(x_{max} - x_i)} = x_{max} + \ln \sum_i e^{-(x_{max} - x_i)} \, .$$

The largest exponential corresponds to energy value E_0 at the maximum of $P(E)$ as

$$\begin{aligned} E_0 &= \text{argmax}_E \left(\ln g(E) - \beta E \right) = \\ &= \text{argmax}_E \left(\ln Z + \ln P(E) \right) = \\ &= \text{argmax}_E \ln P(E) = \\ &= \text{argmax}_E P(E) \, . \end{aligned}$$

Then the value of the partition function becomes

$$Z = \sum_E e^{\ln g(E) - \beta E} = \ln g(E_0) - \beta E_0 + \sum_E e^{-\Delta(\ln g(E) - \beta E)} \, ,$$

begin $\Delta f = f(E) - f(E_0)$, and E_0

If N is known, $N = \sum_E g(E)$

CHAPTER

THIRTEEN

ENTROPY AND ARROW OF TIME

Microscopic view vs. macroscopic view. Usually we don't know it accurately/we don't care to know it, and summarize it with a macroscopic view: in this description some information is lost

Usually:

- Microscopic symmetry, with uniform transition matrix
- Many microstates correspond to the same macrostate
- As the system evolves, its macrostate may evolve towards a stationary state, corrsponding with the macrostate with the most microstates. This evolution has a preferred direction - defining an **arrow of time** -, the one with **non-decreasing** expected **entropy**. This evolution can also be described in terms of **increasing ignorance**
- A macroscopic equilibrium corresponds to a stationary distribution of the macrostates with quite sharp maximum. While the macroscopic view looks stationary, the system is still evolving at the microscopic level. The amplitude of oscillations around an equilibrium decreases as $\sim \frac{1}{\sqrt{N}}$

13.1 Box occupation example

Remark. In this example, the microscopic dynamics of the system is not deterministic.

A box with two rooms contains N chambers. This balls are numbered so that they are distingushable if you can see their labels.

Macrostate definition. A macrostate of the system is defined here by the number of balls in each room, (N_L, N_R). As the total number of balls is given and constant, the number of balls in one chamber, e.g. N_L in the left chamber, is enough to determine the macrostate being $N_R = N - N_L$.

Microstate counting. The number of microstates producing the macrostate N_L is the number of combinations, C_{N,N_L}

$$\Omega(N_L) = C_{N,N_L} = \binom{N}{N_L} = \frac{N!}{(N-N_L)!N_L!} \, .$$

Entropy of a macrostate. …

$$S(n) = \ln \binom{N}{n} \, .$$

Probability transition between macrostates. Being the probability uniform for any transition of microstates, the probability of transition between macrostates reads

$$\begin{aligned} P_-(n) &:= P(n \to n-1) = \frac{n}{N} \\ P_+(n) &:= P(n \to n+1) = \frac{N-n}{N} \end{aligned} \tag{13.1}$$

Expected change of entropy.

$$\begin{aligned}
\mathbb{E}\left[\Delta S(n)\right] &= \sum_i P_i(n)\Delta S_i(n) = \\
&= P_-(n)\left(S(n-1)-S(n)\right) + P_+(n)\left(S(n+1)-S(n)\right) = \\
&= \frac{n}{N}\ln\frac{\binom{N}{n-1}}{\binom{N}{n}} + \frac{N-n}{N}\ln\frac{\binom{N}{n+1}}{\binom{N}{n}} = \\
&= \frac{n}{N}\ln\frac{N!}{(N-n+1)!(n-1)!}\frac{(N-n)!n!}{N!} + \frac{N-n}{N}\ln\frac{N!}{(N-n-1)!(n+1)!}\frac{(N-n)!n!}{N!} = \\
&= \frac{n}{N}\ln\frac{(N-n)!n!}{(N-n+1)!(n-1)!} + \frac{N-n}{N}\ln\frac{(N-n)!n!}{(N-n-1)!(n+1)!} = \\
&= \ln\frac{(N-n)!n!}{(N-n-1)!(n+1)!} + \frac{n}{N}\ln\frac{(N-n-1)!(n+1)!}{(N-n+1)!(n-1)!} = \\
&= \ln\frac{(N-n)}{n+1} + \frac{n}{N}\ln\frac{n(n+1)}{(N-n)(N-n+1)} = \\
&= \ldots
\end{aligned}$$

todo *keep going with the algebra without the limit of large number of particles...*

In the limit $N \gg 1, n \gg 1, N-n \gg 1$,

$$\begin{aligned}
\mathbb{E}\left[\Delta S(n)\right] &\simeq \ln\frac{N-n}{n} + \frac{n}{N}\ln\frac{n^2}{(N-n)^2} = \\
&= \ln\frac{N-n}{n} - \frac{2n}{N}\ln\frac{N-n}{n} = \\
&= \left(1-2\frac{n}{N}\right)\ln\frac{1-\frac{n}{N}}{\frac{n}{N}} \,.
\end{aligned}$$

Both factors change sign when $\frac{n}{N} = \frac{1}{2}$. For $\frac{n}{N} < \frac{1}{2}$ both factors are positive, for $\frac{n}{N} > \frac{1}{2}$ both factors are negative. Thus, the expected variation of entropy of this process is non-negative for every value of $n \in [0, N]$ and thus for every transition,

$$\mathbb{E}\left[\Delta S(n)\right] \geq 0 \,, \quad \forall n \in [0, N] \,.$$

Markov process of macrostates. The macrostates and the transition between them can be model as a stationary Markov process. The probability of the system of being in (macro)state n at time t is $p_n(t)$, and using the transition probability $P(n|m)$ from state m at $t-1$ to state n at t, the following holds

$$p_n(t) = \sum_m P(n|m)p_m(t-1) \,,$$

or using matrix formalism, introducing transition probability matrix $\{\mathbf{P}\}_{mn} := P(n|m)$,

$$\mathbf{p}(t) = \mathbf{p}(t-1)\mathbf{P} \,.$$

The transition probabilities (13.1) define the elements of the transition probability matrix $\mathbf{P}$,

$$\mathbf{P} = \begin{bmatrix}
\cdot & 1 & & & & & \\
\frac{1}{N} & \cdot & 1-\frac{1}{N} & & & & \\
 & \frac{2}{N} & \cdot & 1-\frac{2}{N} & & & \\
 & & \ldots & \cdot & \ldots & & \\
 & & & 1-\frac{2}{N} & \cdot & \frac{2}{N} & \\
 & & & & 1-\frac{1}{N} & \cdot & \frac{1}{N} \\
 & & & & & 1 & \cdot
\end{bmatrix} .$$

Stationary distribution. The stationary distribution $\mathbf{p}^*$ is invariant under a transition, i.e. it satisfies $\mathbf{p}^* = \mathbf{p}^*\mathbf{P}$. The stationary distribution of the process is the Binomial distribution

$$p_n^* = \frac{1}{2^N}\binom{N}{n} .$$

Proof

This can be proved by direct computation. The first element reads

$$p_0^* = \frac{1}{N}p_1^* ,$$

or

$$\frac{1}{2^N}\cdot 1 = \frac{1}{N}\frac{1}{2^N}\frac{1}{N} .$$

The nth element ($n \in (1, N-1)$) reads

$$p_n^* = \left(1 - \frac{n-1}{N}\right)p_{n-1}^* + \frac{n+1}{N}p_{n+1}^* .$$

or

$$\begin{aligned}
\frac{1}{2^N}\frac{N!}{(N-n)!n!} &= \left(1-\frac{n-1}{N}\right)\frac{1}{2^N}\frac{N!}{(N-n+1)!(n-1)!} + \frac{n+1}{N}\frac{1}{2^N}\frac{N!}{(N-n-1)!(n+1)!} = \\
&= \frac{1}{2^N}\left[\frac{N-n+1}{N}\frac{N!}{(N-n+1)!(n-1)!} + \frac{n+1}{N}\frac{N!}{(N-n-1)!(n+1)!}\right] = \\
&= \frac{1}{2^N}\frac{N!}{(N-n)!n!}\left[\frac{n}{N} + \frac{N-n}{N}\right] = \\
&= \frac{1}{2^N}\frac{N!}{(N-n)!n!} .
\end{aligned}$$

Statistics of the stochastic process. …

Statistics of the stationary distribution.

- Average $\mu = \frac{N}{2}$
- Variance $\sigma^2 = \frac{N}{4}$
- Standard deviation $\sigma = \frac{\sqrt{N}}{2}$

Thus the relative fluctuation around the average can be estimated with the ratio of the standard deviation and the average value,

$$\frac{\sigma}{\mu} = \frac{1}{\sqrt{N}} ,$$

so, the relative fluctuation decreases as N increases.

N	Relative fluctuation
10^2	10^{-1}
10^4	10^{-2}
10^6	10^{-3}
10^8	10^{-4}

Average of a binomial distribution

$$\begin{aligned}\mathbb{E}[n] &= \sum_{n=0}^{N} n\, p_n^* = \\ &= \frac{1}{2^N}\sum_{n=0}^{N} n \binom{N}{n} = \\ &= \frac{1}{2^N}\sum_{n=1}^{N} N \binom{N-1}{n-1} = \\ &= \frac{1}{2^N} N \sum_{n=0}^{N-1} \binom{N-1}{n} = \\ &= \frac{1}{2^N} N 2^{N-1} = \frac{N}{2}\,.\end{aligned}$$

Variance of a binomial distribution

$$\sigma_n^2 = \mathbb{E}[n^2] - \left(\mathbb{E}[n]\right)^2$$

Then,

$$\mathbb{E}[n^2] = \mathbb{E}[n(n-1)] + \mathbb{E}[n]\,,$$

where the first term follows from a similar procedure as the one used for the average, using the identity $n(n-1)\binom{N}{n} = N(N-1)\binom{N-2}{n-2}$,

$$\mathbb{E}[n(n-1)] = \frac{N(N-1)}{4}\,.$$

Thus the variance reads

$$\sigma_n^2 = \frac{N(N-1)}{4} + \frac{N}{2} - \frac{N^2}{4} = \frac{N}{4}\,.$$

Markov process of microstates. A microstate is defined by the id of the box of each of the N balls. One-step transition switch the index of one ball. If the transition probability between these microstates is uniform it has probability $\frac{1}{N}$. Not, let's consider two states i, j. If these states are adjacent, the probabilities $P^{micro}(i \to j) = P^{micro}(j \to i) = \frac{1}{N}$, and zero otherwise. Thus, the transition probability matrix $\mathbf{P}^{micro}$ between microstates is symmetric. Since the **transition matrix is symmetric**, the **stationary distribution is uniform**. This is immediately proved using the normalization condition $\mathbf{P1} = \mathbf{1}$, and the definition of a stationary distribution, $\mathbf{p}^{*T} = \mathbf{p}^{*T}\mathbf{P}$, and the definition of symmetric transition matrix $\mathbf{P} = \mathbf{P}^T$. The transpose of the definition of the stationary distribution, $\mathbf{p}^* = \mathbf{P}\mathbf{p}^*$, compared with the normalization condition, provides the relation $\mathbf{p}^* = \alpha\mathbf{1}$, and normalization condition $1 = |\mathbf{p}^*|_1 = \alpha N$, and $\alpha = \frac{1}{N}$. The stationary distribution for the symmetric Markov process is $\mathbf{p}^* = \frac{1}{N}\mathbf{1}$.

Combinatorics naturally links the distribution of microstates and the distribution of the macrostates.

H-Theorem for symmetric Markov chains. A discrete version of the Boltzmann's H-Theorem exists for symmetric Markov process, assessing that the entropy $H(\mathbf{p}(t)) := -\sum_k p_k(t) \ln p_k(t)$ is non-decreasing, i.e.

$$H(\mathbf{p}(t+1)) \geq H(\mathbf{p}(t))\,.$$

Proof

Remark. A generic transition probability matrix $\mathbf{P}$ satisfies the normalization condition $\sum_j P_{ij} = 1$, for every i (row sum). For a symmetric probability matrix $\mathbf{P} = \mathbf{P}^T$, also the column sum holds $\sum_i P_{ij} = 1$.

Remark. The function $f(x) = x \ln x$ is convex, i.e. $f''(x) > 0$, for $x > 0$.

$$f'(x) = \ln x + 1$$
$$f''(x) = \frac{1}{x} .$$

$$H(t+1) = -\sum_k p_k(t+1) \ln p_k(t+1) =$$
$$= -\sum_k f(p_k(t+1)) =$$
$$= -\sum_k f\left(\sum_i p_i(t) P_{ik}\right) .$$

Jensen's Inequality. As f is convex, and $\sum_i P_{ik} = 1$, with $P_{ik} \in (0,1)$,

$$f\left(\sum_i p_i P_{ik}\right) \leq \sum_i P_{ik} f(p_i) .$$

Using Jensen's inequality in the expression of $H(t+1)$,

$$H(t+1) \geq -\sum_k \sum_i P_{ik} f(p_i) =$$
$$= -\sum_i f(p_i) \underbrace{\sum_k P_{ik}}_{=1} = H(t) ,$$

i.e. $H(t+1) \geq H(t)$.

Part IV

Quantum Mechanics

CHAPTER

FOURTEEN

QUANTUM MECHANICS

- Principles and postulates
  - statistics and measurements outcomes (Heisenberg built its matrix mechanics only on observables…)
  - CCR
- angluar momentum, spin, and atom

14.1 Mathematical tools for quantum mechanics

Definition 14.1.1 (Operator)

Definition 14.1.2 (Adjoint operator)

Given an operator $\hat{A} : U \to V$, its adjoint operator $\hat{A}^* : V \to U$ is the operator s.t.

$$(\mathbf{v},\ \hat{A}\mathbf{u})_V = (\mathbf{u}, \hat{A}^*\mathbf{v})_U$$

holds for $\forall \mathbf{u} \in U,\ \mathbf{v} \in V$.

Definition 14.1.3 (Hermitian (self-adjoint) operator)

The operator $\hat{A} : U \to U$ is a self-adjoint operator if

$$\hat{A}^* = \hat{A}\,.$$

Self-adjoint operators have real eigenvalues, and orthogonal eigenvectors (at least those associated to different eigenvalues; those associated with the same eigenvalues can be used to build an orthogonal set of vectors with orthogonalization process).

14.2 Postulates of Quantum Mechanics

- …
- Canonical Commutation Relation (CCR) *and Canonical Anti-Commutation Relation…*
- …

14.3 Non-relativistic Mechanics

14.3.1 Wave function

The state of a system is described by a wave function $|\Psi\rangle$

todo

- properties: domain, image,…
- unitary $1 = \langle\Psi|\Psi\rangle = |\Psi|^2$, for statistical interpretation of $|\Psi|^2$ as a density probability function

14.3.2 Operators and Observables

Physical **observable** quantities are represented by *Hermitian operators*. Possible outcomes of measurement are the eigenvalues of the operator

Given $\hat{A}$ and the set of its eigenvectors $\{|A_i\rangle\}_i$ (**todo** *continuous or discrete spectrum…, need to treat this difference quite in details*), with associated eigenvalues $\{a_i\}_i$

$$\hat{A}|A_i\rangle = a_i|A_i\rangle$$

$$|\Psi\rangle = |A_i\rangle\langle A_i|\Psi\rangle = |A_i\rangle\Psi_i^A$$

$$\langle A_j|\Psi\rangle = \langle A_j|A_i\rangle\langle A_i|\Psi\rangle = \Psi_j^A$$

and thus

$$\Psi_j^A = \langle A_j|\Psi\rangle$$
$$\Psi_j^{A*} = \langle\Psi|A_j\rangle$$

- identity operator $\sum_i |A_i\rangle\langle A_i| = \mathbb{1}$, since

$$\sum_i |A_i\rangle\langle A_i|\Psi\rangle = \sum_i |A_i\rangle\langle A_i|\Psi_j^A A_j\rangle = \sum_i |A_i\rangle\delta_{ij}\Psi_j^A = \sum_i |A_i\rangle\Psi_i^A = |\Psi\rangle$$

- Normalization:

$$1 = \langle\Psi|\Psi\rangle = \Psi_j^{A*}\underbrace{\langle A_j|A_i\rangle}_{\delta_{ij}}\Psi_i^A = \sum_i \left|\Psi_i^A\right|^2$$

  with $|\Psi_i^A|^2$ that can be interpreted as the probability of finding the system in state $|\Psi_i^a\rangle$

- Expected value of the physical quantity in the a state $|\Psi\rangle$, with possible values a_i with probability $|\Psi_i^A|^2$

$$\begin{aligned}\bar{A}_\Psi &= \sum_i a_i |\Psi_i^A|^2 = \\ &= \sum_i a_i \Psi_i^{A*} \Psi_i^A = \\ &= \sum_i a_i \langle\Psi|A_i\rangle\langle A_i|\Psi\rangle = \\ &= \langle\Psi| \left(\sum_i a_i |A_i\rangle\langle A_i| \right) |\Psi\rangle = \\ &= \langle\Psi|\hat{A}|\Psi\rangle =\end{aligned}$$

since an operator $\hat{A}$ can be written as a function of its eigenvalues and eigenvectors

$$\begin{aligned}\left(\sum_i a_i |A_i\rangle\langle A_i| \right) \Psi\rangle = \left(\sum_i a_i |A_i\rangle\langle A_i| \right) c_k |A_k\rangle = \\ = \sum_i a_i |A_i\rangle c_i = \\ = \sum_i \hat{A}|A_i\rangle c_i = \\ = \hat{A} \sum_i |A_i\rangle c_i = \hat{A}|\Psi\rangle \ .\end{aligned}$$

14.3.3 Space and Momentum Representation

Space Representation

Position operator $\hat{\mathbf{r}}$ has eigenvalues $\mathbf{r}$ identifying the possible measurements of the position

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle \ ,$$

being $\mathbf{r}$ the result of the measurement (position in space, mathematically it could be a vector), and $|\mathbf{r}\rangle$ the state function corresponding to the measurement $\mathbf{r}$ of the position.

- Result of measurement, $\mathbf{r}$, is a position in space. As an example, it could be a point in an Euclidean space $P \in E^n$. It could be written using properties of Dirac's delta "function"

$$\mathbf{r} = \int_{\mathbf{r}'} \delta(\mathbf{r}' - \mathbf{r})\, \mathbf{r}' d\mathbf{r}'$$

- Projection of wave function over eigenstates of position operator

$$\begin{aligned}\langle\mathbf{r}|\Psi\rangle(t) = \Psi(\mathbf{r}, t) = \int_{\mathbf{r}'} \delta(\mathbf{r} - \mathbf{r}')\Psi(\mathbf{r}', t) d\mathbf{r}' = \\ = \int_{\mathbf{r}'} \langle\mathbf{r}|\mathbf{r}'\rangle\Psi(\mathbf{r}', t) d\mathbf{r}' = \\ = \int_{\mathbf{r}'} \langle\mathbf{r}|\mathbf{r}'\rangle\langle\mathbf{r}'|\Psi\rangle(t) d\mathbf{r}' = \\ = \langle\mathbf{r}| \underbrace{\left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle\mathbf{r}'| d\mathbf{r}' \right)}_{=\hat{\mathbf{1}}} |\Psi\rangle(t) \ .\end{aligned}$$

- having used orthogonality (**todo** *why? provide definition and examples of operators with continuous spectrum*)

$$\langle \mathbf{r}' | \mathbf{r} \rangle = \delta(\mathbf{r}' - \mathbf{r})$$

- Expansion of a state function $|\Psi\rangle(t)$ over the basis of the position operator

$$|\Psi\rangle(t) = \hat{\mathbf{1}} |\Psi\rangle(t) = \left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle \mathbf{r}' d\mathbf{r}' \right) |\Psi\rangle(t) = \int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle \mathbf{r}' | \Psi\rangle(t) \, d\mathbf{r}' \ .$$

- Unitariety and probability density

$$\begin{aligned} 1 = \langle \Psi | \Psi \rangle(t) &= \langle \Psi | \left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle \mathbf{r}' d\mathbf{r}' \right) |\Psi\rangle \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle \, d\mathbf{r}' \\ &= \int_{\mathbf{r}'} \Psi^*(\mathbf{r}', t) \Psi(\mathbf{r}', t) \, d\mathbf{r}' \\ &= \int_{\mathbf{r}'} |\Psi(\mathbf{r}', t)|^2 \ d\mathbf{r}' \end{aligned}$$

  and thus $|\Psi(\mathbf{r}, t)|^2$ can be interpreted as the **probability density function** of measuring position of the system equal to $\mathbf{r}'$.

- Average value of the operator

$$\begin{aligned} \bar{\mathbf{r}} = \langle \Psi | \hat{\mathbf{r}} | \Psi \rangle &= \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | d\mathbf{r}' \ |\hat{\mathbf{r}}| \ \int_{\mathbf{r}''} |\mathbf{r}''\rangle \langle \mathbf{r}'' | \Psi \rangle \, d\mathbf{r}'' \\ &= \int_{\mathbf{r}'} \int_{\mathbf{r}''} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | \hat{\mathbf{r}} | \mathbf{r}'' \rangle \langle \mathbf{r}'' | \Psi \rangle \, d\mathbf{r}' d\mathbf{r}'' = \\ &= \int_{\mathbf{r}'} \int_{\mathbf{r}''} \langle \Psi | \mathbf{r}' \rangle \ \underbrace{\langle \mathbf{r}' | \mathbf{r}'' \rangle}_{=\delta(\mathbf{r}' - \mathbf{r}'')} \ \mathbf{r}'' \langle \mathbf{r}'' | \Psi \rangle \, d\mathbf{r}' d\mathbf{r}'' = \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \mathbf{r}' \langle \mathbf{r}' | \Psi \rangle \, d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} \Psi^*(\mathbf{r}', t) \ \mathbf{r}' \ \Psi(\mathbf{r}', t) \, d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} |\Psi(\mathbf{r}', t)|^2 \ \mathbf{r}' \, d\mathbf{r}' \ . \end{aligned}$$

Momentum Representation

Momentum operator as the limit of…**todo** *prove the expression of the momentum operator as the limit of the generator of translation*

$$\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{r} | \tag{14.1}$$

- Spectrum

$$\hat{\mathbf{p}} | \mathbf{p} \rangle = \mathbf{p} | \mathbf{p} \rangle$$

$$\langle \mathbf{r} | \hat{\mathbf{p}} | \mathbf{p} \rangle = -i\hbar \nabla \langle \mathbf{r} | \mathbf{p} \rangle = \mathbf{p} \langle \mathbf{r} | \mathbf{p} \rangle$$

and thus the eigenvectors in space base $\mathbf{p}(\mathbf{r}) = \langle \mathbf{r}|\mathbf{p}\rangle$ are the solution of the differential equation

$$-i\hbar\nabla\mathbf{p}(\mathbf{r}) = \mathbf{p}\mathbf{p}(\mathbf{r}) \ ,$$

that in Cartesian coordinates reads

$$-i\hbar\partial_j p_k(\mathbf{r}) = p_j p_k(\mathbf{r})$$

$$p_k(\mathbf{r}) = p_{k,0} \exp\left[i\frac{p_j}{\hbar}r_j\right]$$

or

$$\langle \mathbf{r}|\mathbf{p}\rangle = \mathbf{p}(\mathbf{r}) = \mathbf{p}_0 \exp\left[i\frac{\mathbf{p}\cdot\mathbf{r}}{\hbar}\right]$$

todo

- normalization factor $\frac{1}{(2\pi)^{\frac{3}{2}}}$

$$\mathcal{F}\{\delta(x)\}(k) = \int_{-\infty}^{\infty} \delta(x) e^{-ikx}\, dx = 1$$

- Fourier transform and inverse Fourier transform: definitions and proofs (link to a math section)
- representation in basis of wave vector operator $\hat{\mathbf{k}}$, $\hat{\mathbf{p}} = \hbar\hat{\mathbf{k}}$

From position to momentum representation

Momentum and wave vector, $\mathbf{p} = \hbar\mathbf{k}$

$$\begin{aligned}
\langle \mathbf{p}|\Psi\rangle &= \langle \mathbf{p}| \int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' = \\
&= \int_{\mathbf{r}'} \langle \mathbf{p}|\mathbf{r}'\rangle\langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' = \\
&= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp\left[i\frac{\mathbf{p}\cdot\mathbf{r}}{\hbar}\right] \langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' =
\end{aligned}$$

Relation between position and wave-number representation can be represented with a Fourier transform

$$\begin{aligned}
\langle \mathbf{k}|\Psi\rangle &= \langle \mathbf{k}| \int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' = \\
&= \int_{\mathbf{r}'} \langle \mathbf{k}|\mathbf{r}'\rangle\langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' = \\
&= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp\left[i\mathbf{k}\cdot\mathbf{r}'\right] \langle \mathbf{r}'|\Psi\rangle d\mathbf{r}' = \\
&= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp\left[i\mathbf{k}\cdot\mathbf{r}'\right] \Psi(\mathbf{r}') d\mathbf{r}' = \\
&= \mathcal{F}\{\Psi(\mathbf{r})\}(\mathbf{k})
\end{aligned}$$

14.3.4 Schrodinger Equation

$$i\hbar \frac{d}{dt}|\Psi\rangle = \hat{H}|\Psi\rangle$$

being $\hat{H}$ the Hamiltonian operator and $|\Psi\rangle$ the wave function, as a function of time t as an independent variable.

Stationary States

Eigenspace of the Hamiltonian operator

$$\hat{H}|\Psi_k\rangle = E_k|\Psi_k\rangle \ ,$$

with E_k possible values of energy measurements. *If no eigenstates with the same eigenvalue exists, then…otherwise… Without external influence* **todo** *be more detailed!*, energy values and eigenstates of the systems are constant in time.

Thus, exapnding the state of the system $|\Psi\rangle$ over the stationary states gives $|\Psi_k\rangle$, $|\Psi\rangle = |\Psi_k\rangle c_k(t)$, and inserting in Schrodinger equation

$$i\hbar \dot{c}_k|\Psi_k\rangle = c_k E_k|\Psi_k\rangle$$

and exploiting orthogonality of eigenstates, a diagonal system for the amplitudes of stationary states ariese,

$$i\hbar \dot{c}_k = c_k E_k \ .$$

whose solution reads

$$c_k(t) = c_{k,0} \exp\left[-i\frac{E_k}{\hbar}t\right]$$

Thus the state of the system evolves like a superposition of monochromatic waves with frequencies $\omega_k = \frac{E_k}{\hbar}$,

$$|\Psi\rangle = |\Psi_k\rangle c_k(t) = |\Psi_k\rangle c_{k,0} \exp\left[-i\frac{E_k}{\hbar}t\right] \ .$$

$$\begin{aligned}
\frac{d}{dt}\bar{A} &= \frac{d}{dt}\left(\langle\Psi|\hat{A}|\Psi\rangle\right) = \\
&= \frac{d}{dt}\langle\Psi|\hat{A}|\Psi\rangle + \langle\Psi|\frac{d\hat{A}}{dt}|\Psi\rangle + \langle\Psi|\hat{A}\frac{d}{dt}|\Psi\rangle = \\
&= \langle\Psi|\frac{d\hat{A}}{dt}|\Psi\rangle + \frac{i}{\hbar}\langle\Psi|\hat{H}\hat{A}|\Psi\rangle - \frac{i}{\hbar}\langle\Psi|\hat{A}\hat{H}|\Psi\rangle = \\
&= \langle\Psi|\left(\frac{i}{\hbar}[\hat{H},\hat{A}] + \frac{d\hat{A}}{dt}\right)|\Psi\rangle \ .
\end{aligned}$$

Pictures

- Schrodinger
- Heisenberg
- Interaction

Schrodinger

If $\hat{H}$ not function of time

$$|\Psi\rangle(t) = \exp\left[-i\frac{\hat{H}}{\hbar}(t-t_0)\right]|\Psi\rangle(t_0) = \hat{U}(t,t_0)|\Psi\rangle(t_0)$$
$$\bar{A} = \langle\Psi|\hat{A}|\Psi\rangle = \langle\Psi_0|\hat{U}^*(t,t_0)\hat{A}\hat{U}(t,t_0)|\Psi_0\rangle$$

Heisenberg

While Schrodinger picture works with (usually) time-independent operators and time-dependent wave functions, Heisenberg picture works with time-dependent operators and time-independent wave functions. The state of a quantum system at two different times is connected by a unitary operator U_{t,t_0},

$$|\Psi_t\rangle = U_{t,t_0}|\Psi_{t_0}\rangle$$

Expected value. The expected value of the physical quantity represented by the Hermitian operator $\hat{A}$ reads

$$\begin{aligned}\langle\hat{A}\rangle &= \langle\Psi_t|\hat{A}|\Psi_t\rangle = \\ &= \langle\Psi_{t_0}|U^\dagger_{t,t_0}\hat{A}U_{t,t_0}|\Psi_{t_0}\rangle = \\ &= \langle\Psi_{t_0}|\hat{A}_H(t)|\Psi_{t_0}\rangle\ ,\end{aligned}$$

having defined the counterpart in Heisenberg picture, $\hat{A}_H(t) = U^\dagger_{t,t_0}\hat{A}U_{t,t_0}$ of the operator $\hat{A}$ in Schrodinger picture.

Hamiltonian operator in Heisenberg picture.

$$\hat{H}_H = U^\dagger\hat{H}U\ .$$

If the Hamiltonian operator is **time-independent in Schrodinger picture**, the evolution operator can be written as $U_t = \exp\left(-i\frac{\hat{H}}{\hbar}t\right)$. In this case, U and $\hat{H}$ commute, and the Hamiltonian operator in Heisenberg picture has the same expression as the Hamiltonian operator in Schrodinger equation,

$$\hat{H}_H = U^\dagger\hat{H}U = \quad (\quad \text{if } \partial_t\hat{H} = 0 \quad) \quad = U^\dagger U\hat{H} = \hat{H}\ .$$

Time derivative of the operator $\hat{A}_H(t)$.

$$\begin{aligned}\frac{d}{dt}\hat{A}_H(t) &= \frac{d}{dt}\left(U^\dagger_{t,t_0}\hat{A}U_{t,t_0}\right) = \\ &= \partial_t U^\dagger_{t,t_0}\hat{A}U_{t,t_0} + U^\dagger_{t,t_0}\partial_t\hat{A}U_{t,t_0} + U^\dagger_{t,t_0}\hat{A}\partial_t U_{t,t_0} = \\ &= -\frac{1}{i\hbar}U^\dagger_{t,t_0}\hat{H}\hat{A}U_{t,t_0} + U^\dagger_{t,t_0}\partial_t\hat{A}U_{t,t_0} + \frac{1}{i\hbar}U^\dagger_{t,t_0}\hat{A}\hat{H}U_{t,t_0} = \\ &= -\frac{1}{i\hbar}U^\dagger_{t,t_0}\hat{H}U_{t,t_0}U^\dagger_{t,t_0}\hat{A}U_{t,t_0} + U^\dagger_{t,t_0}\partial_t\hat{A}U_{t,t_0} + \frac{1}{i\hbar}U^\dagger_{t,t_0}\hat{A}U_{t,t_0}U^\dagger_{t,t_0}\hat{H}U_{t,t_0} = \\ &= -\frac{1}{i\hbar}\left[\hat{H}_H,\hat{A}_H\right] + \left(\partial_t\hat{A}\right)_H\ ,\end{aligned} \tag{14.2}$$

as the relation (16.2) gives $\partial_t U_{t,t_0}U^\dagger_{t,t_0} = \frac{1}{i\hbar}\hat{H}$,

---

old … for $\hat{H}$ independent from time t,

$$\frac{d}{dt}\bar{\mathbf{r}} = \overline{\frac{i}{\hbar}\left[\hat{H},\hat{\mathbf{r}}\right]} = -\frac{1}{ih}\overline{[\hat{H},\hat{\mathbf{r}}]}$$
$$\frac{d}{dt}\bar{\mathbf{p}} = \overline{\frac{i}{\hbar}\left[\hat{H},\hat{\mathbf{p}}\right]} = -\frac{1}{ih}\overline{[\hat{H},\hat{\mathbf{p}}]}$$

Hamiltonian Mechanics

From Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = Q_q$$

q generalized coordinates, $p := \frac{\partial L}{\partial \dot{q}}$ generalized momenta.

Hamiltonian

$$H(p, q, t) = p\dot{q} - L(\dot{q}, q, t)$$

Increment of the Hamiltonian

$$dH = \partial_p H dp + \partial_q H dq + \partial_t H dt$$

$$\begin{aligned} dH &= \dot{q}dp + pd\dot{q} - \partial_{\dot{q}} L d\dot{q} - \partial_q L dq - \partial_t L dt = \\ &= \dot{q}dp - \partial_q L dq - \partial_t L dt = \\ &= \dot{q}dp - \left(\dot{p} + Q_q\right) dq - \partial_t L dt = \end{aligned}$$

$$\begin{cases} \frac{\partial H}{\partial p} = \dot{q} \\ \frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q} = -\dot{p} + Q_q \\ \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \end{cases}$$

Physical quantity $f(q(t), p(t), t)$. Its time derivative reads

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial q}\dot{q} + \frac{\partial f}{\partial p}\dot{p} + \frac{\partial f}{\partial t} = \\ &= \frac{\partial f}{\partial q}\frac{\partial H}{\partial p} + \frac{\partial f}{\partial p}\left[-\frac{\partial H}{\partial q} + Q_q\right] + \frac{\partial f}{\partial t} = \\ &= -\partial_q H \partial_p f + \partial_p H \partial_q f + \partial_t f + Q_q \partial_p f = \\ &= -\{H, f\} + \partial_t f + Q_q \partial_p f \end{aligned}$$

If $Q_q = 0$, the correspondence between quantum mechanics and classical mechanics

$$\begin{aligned} \frac{df}{dt} = -\{H, f\} + \partial_t f \qquad &\leftrightarrow \qquad \frac{d}{dt}\overline{\hat{f}} = -\frac{1}{i\hbar}\overline{[\hat{H}, \hat{f}]} + \overline{\frac{\partial \hat{f}}{\partial t}} \\ \{H, f\} \qquad &\leftrightarrow \qquad \frac{i}{\hbar}[\hat{H}, \hat{f}] \end{aligned}$$

Interaction

14.3.5 Operators

Commuting operators, $0 = [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. Their eigenvalue problems read

$$\hat{A}|a\rangle = a|a\rangle$$
$$\hat{B}|b\rangle = b|b\rangle$$
$$a\hat{B}|a\rangle = \hat{B}\hat{A}|a\rangle = \hat{A}\left(\hat{B}|a\rangle\right)$$

Thus the state $\hat{B}|a\rangle$ is an eigenstate of $\hat{A}$, with the same eigenvalue a. If the eigenvalue is non-degenerate, this eigenstate must be proportional to the original eigenstate, and thus $\hat{B}|a\rangle = b|a\rangle$. If the eigenvalue is degenerate, the state $\hat{B}|a\rangle$ can be a linear combination of all the eigenstates $|a_i\rangle$ with eigenvalue $a_i = a$.

14.3.6 Matrix Mechanics

Attualization of 1925 papers

…to find the canonical commutation relation,

$$[\hat{\mathbf{r}}, \hat{\mathbf{p}}] = i\hbar \mathbb{I}\, \hat{\mathbf{1}} \, .$$

$$\begin{aligned}
[\hat{\mathbf{r}}, \hat{\mathbf{p}}] &= \hat{\mathbf{r}}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{\mathbf{r}} = \\
&= \hat{\mathbf{r}} \int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}| d\mathbf{r} \hat{\mathbf{p}} - \hat{\mathbf{p}} \int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}| d\mathbf{r}\, \hat{\mathbf{r}} \int_{\mathbf{r}'} |\mathbf{r}'\rangle\langle\mathbf{r}'| d\mathbf{r}' = \\
&= -\int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle i\hbar\nabla\langle\mathbf{r}|\, d\mathbf{r} - \hat{\mathbf{p}} \int_{\mathbf{r}} \int_{\mathbf{r}'} |\mathbf{r}\rangle\mathbf{r}' \underbrace{\langle\mathbf{r}|\mathbf{r}'\rangle}_{\delta(\mathbf{r}-\mathbf{r}')} \langle\mathbf{r}'| d\mathbf{r}' = \\
&= -\int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle i\hbar\nabla\langle\mathbf{r}|\, d\mathbf{r} - \hat{\mathbf{p}} \int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle\langle\mathbf{r}| d\mathbf{r} = \\
&= -\int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle i\hbar\nabla\langle\mathbf{r}|\, d\mathbf{r} - \int_{\mathbf{r}'} |\mathbf{r}\rangle\langle\mathbf{r}| d\mathbf{r}\, \hat{\mathbf{p}} \int_{\mathbf{r}'} \mathbf{r}'|\mathbf{r}'\rangle\langle\mathbf{r}'| d\mathbf{r}' = \\
&= -\int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle i\hbar\nabla\langle\mathbf{r}|\, d\mathbf{r} + \int_{\mathbf{r}} |\mathbf{r}\rangle i\hbar\nabla\langle\mathbf{r}| d\mathbf{r} \int_{\mathbf{r}'} \mathbf{r}'|\mathbf{r}'\rangle\langle\mathbf{r}'| d\mathbf{r}' = \ldots \\
[\hat{\mathbf{r}}, \hat{\mathbf{p}}]\, |\Psi\rangle &= -\int_{\mathbf{r}} \mathbf{r}|\mathbf{r}\rangle i\hbar\nabla\Psi(\mathbf{r}, t) + \int_{\mathbf{r}} |\mathbf{r}\rangle i\hbar\nabla\left(\mathbf{r}\Psi(\mathbf{r}, t)\right) = \\
&= -\int_{\mathbf{r}} |\mathbf{r}\rangle i\hbar \left[\mathbf{r}\nabla\Psi(\mathbf{r}, t) + \mathbb{I}\Psi(\mathbf{r}, t) + \mathbf{r}\nabla\Psi(\mathbf{r}, t)\right] = \\
&= i\hbar \underbrace{\int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}| d\mathbf{r}}_{\hat{\mathbf{1}}}\, |\Psi\rangle \, ,
\end{aligned}$$

and since $|\Psi\rangle$ is arbitrary

$$[\hat{\mathbf{r}}, \hat{\mathbf{p}}] = i\hbar\mathbb{I}\hat{\mathbf{1}} \, .$$

$$[\hat{r}_a, \hat{p}_b] = i\hbar\delta_{ab} \, .$$

14.3.7 Heisenberg Uncertainty relation

Uncertainty principle is a relation that holds for "wave descriptions" as it can be proved in the generic framework of Fourier transform, see Fourier transform:Uncertainty relation.

- Heisenberg uncertainty relation is a relation between product of the variance of two physical quantities and their commutator,
- **todo** *relation with measurement process and outcomes. Commutation as a measurement process: first measure B and then A, or first measure A and then B*

$$\sigma_A \sigma_B \geq \frac{1}{2} \left| \overline{[\hat{A}, \hat{B}]} \right| \, .$$

Proof of Heisenberg uncertainty "principle"

$$
\begin{aligned}
\sigma_A^2\sigma_B^2 &= \langle\Psi|\left(\hat{A}-\bar{A}\right)^2|\Psi\rangle\langle\Psi|\left(\hat{B}-\bar{B}\right)^2|\Psi\rangle = \\
&= \langle(\hat{A}-\bar{A})\Psi|(\hat{A}-\bar{A})\Psi\rangle\langle(\hat{B}-\bar{B})\Psi|(\hat{B}-\bar{B})\Psi\rangle = \\
&= \|(\hat{A}-\bar{A})\Psi\|^2\|(\hat{B}-\bar{B})\Psi\|^2 = \\
&\geq \left|\langle(\hat{A}-\bar{A})\Psi|(\hat{B}-\bar{B})\Psi\rangle\right|^2 = \\
&= \left|\langle\Psi|(\hat{A}-\bar{A})(\hat{B}-\bar{B})\Psi\rangle\right|^2 = \\
&= \left|\langle\Psi|\hat{A}\hat{B}-\hat{A}\bar{B}-\bar{A}\hat{B}+\bar{A}\bar{B}|\Psi\rangle\right|^2 = \\
&= \left|\langle\Psi|\hat{A}\hat{B}-\bar{A}\bar{B}|\Psi\rangle\right|^2 \geq \qquad (1) \\
&= \left|\frac{\langle\Psi|\hat{A}\hat{B}-\hat{B}\hat{A}|\Psi\rangle}{2i}\right|^2 = \\
&= \frac{\left|\langle\Psi|[\hat{A},\hat{B}]|\Psi\rangle\right|^2}{4} = \frac{1}{4}\left|\overline{[\hat{A},\hat{B}]}\right|^2
\end{aligned}
$$

having used Cauchy-Schwartz triangle inequality in (1),

$$|z| \geq |\mathrm{im}(z)| = \frac{z-z^*}{2i} \, .$$

Hesienberg uncertainty principles applied to position and momentum reads

$$\sigma_{r_a}\sigma_{p_b} \geq \frac{1}{2}\left|\overline{[\hat{r}_a,\hat{p}_b]}\right| = \frac{\hbar}{2}\delta_{ab} \, .$$

14.4 Many-body problem

Wave function with symmetries: Fermions and Bosons

CHAPTER

FIFTEEN

QUANTUM MECHANICS - NOTES

CHAPTER

SIXTEEN

WAVE QUANTUM MECHANICS

If the system behaves like a wave, it must satisfy a wave function

As the geometrical optics is the short wave-length approximation of the ondulatory optics, classical mechanics may be the short wave-length approximation of ondulatory mechanics

16.1 De Broglie

16.2 Schrodinger

16.2.1 Derivation

From the probability interpretation of the wave function $|\Psi_t\rangle$, and the unitary condition $\langle\Psi_t|\Psi_t\rangle = 1$, for every t, the evolution of the system must be governed by a unitary operator U_{t,t_0},

$$|\Psi_t\rangle = U_{t,t_0}|\Psi_{t_0}\rangle \ ,$$

for the unitary condition to hold at every t,

$$1 = \langle\Psi_t|\Psi_t\rangle = \langle U_{t,t_0}\Psi_{t_0}|U_{t,t_0}\Psi_{t_0}\rangle = \langle\Psi_{t_0}|\underbrace{U^\dagger_{t,t_0}U_{t,t_0}}_{=\hat{\mathbf{1}}}|\Psi_{t_0}\rangle = \langle\Psi_{t_0}|\Psi_{t_0}\rangle \ ,$$

i.e.

$$U^{-1}_{t,t_0} = U^\dagger_{t,t_0} \ .$$

Taking the time derivative w.r.t. t of the evolution relation, it follows

$$|\dot{\Psi}_t\rangle = \partial_t U_{t,t_0}|\Psi_{t_0}\rangle = \underbrace{\partial_t U_{t,t_0}U^\dagger_{t,t_0}}_{\hat{A}(t,t_0)}|\Psi_t\rangle \ .$$

Composition. As $|\Psi_t\rangle = U_{t,t_1}|\Psi_{t_1}\rangle = U_{t,t_1}U_{t_1,t_0}|\Psi_{t_0}\rangle$,

$$U_{t,t_0} = U_{t,t_1}U_{t_1,t_0} \ .$$

Indpendence of $\hat{A}(t,t_0) = \partial_t U_{t,t_0}U_{t,t_0}$ from t_0. The operator $\hat{A}(t,t_0) = \partial_t U_{t,t_0}U_{t,t_0}$ can thus be written as

$$\hat{A}(t,t_0) = \partial_t U_{t,t_0}U^\dagger_{t,t_0} = \partial_t U_{t,t_1}\underbrace{U_{t_1,t_0}U^\dagger_{t_1,t_0}}_{=\hat{\mathbf{1}}}U^\dagger_{t,t_1} = \partial_t U_{t,t_1}U^\dagger_{t,t_1} = \hat{A}(t,t_1) \ ,$$

showing that it's independent from the initial time, $\hat{A}(t,t_0)=\hat{A}(t)$.

$\hat{A}(t)$ **is anti-Hermitian.** As

$$0=\partial_t\underbrace{\left(U_{t,t_0}U_{t,t_0}^\dagger\right)}_{=\hat{\mathbf{1}}}=\underbrace{\partial_t U_{t,t_0}U_{t,t_0}^\dagger}_{=\hat{A}(t)}+\underbrace{U_{t,t_0}\partial_t U_{t,t_0}^\dagger}_{=\hat{A}^\dagger(t)}=\hat{A}(t)+\hat{A}^\dagger(t)\ ,$$

and thus $\hat{A}=-\hat{A}^\dagger$. Thus the operator $\hat{A}$ can be written as $\hat{A}=i\hat{\tilde{H}}$, being $\hat{\tilde{H}}$ and Hermitian operator. For the correspondence principle (classical limit for $\hbar\to 0$) the Hermitian operator is found to be $\hat{\tilde{H}}=-\frac{1}{\hbar}\hat{H}$, being $\hat{H}$ the Hamiltonian operator. **todo** *Add a link to "correspondence principle", either Ehrenfest theorem and/ or equations in Heisenberg picture*

Thus, the wave equation becomes

$$|\dot{\Psi}\rangle=\frac{1}{i\hbar}\hat{H}|\Psi\rangle\qquad\text{or}\qquad i\hbar|\dot{\Psi}\rangle=\hat{H}|\Psi\rangle\ . \tag{16.1}$$

The Hamiltonian can be written as a function of the unitary operator U_{t,t_0} and its time derivative as

$$\hat{H}=i\hbar\,\partial_t U_{t,t_0}U_{t,t_0}^\dagger\ . \tag{16.2}$$

16.2.2 Energy eigen-states

$$\hat{H}|\Psi_k\rangle=E_k|\Psi_k\rangle\ .$$

Let the Hamiltonian be independent from time, thus also the eigenvalues and eigenfunctions are independent from time. Let a state be a superposition (linear combination) of eigenfunctions of the Hamiltonian operator

$$|\Psi_t\rangle=c_k(t)|\Psi_k\rangle\ ,$$

the time evolution immedately follows as

$$i\hbar\dot{c}_k|\Psi_k\rangle=c_k\hat{H}|\Psi_k\rangle=c_kE_k|\Psi_k\rangle\ ,$$

from orthogonality condition $\langle\Psi_j|\Psi_k\rangle=\delta_{jk}$, that gives the dynamical equation for the coefficients c_i,

$$\dot{c}_k=-i\frac{E_k}{\hbar}c_k\ ,$$

whose solution is $c_k(t)=c_{k,0}\exp\left(-i\frac{E_k}{\hbar}t\right)$. The evolution of the state is

$$|\Psi_t\rangle=|\Psi_k\rangle c_{k,0}\exp\left(-i\frac{E_k}{\hbar}t\right)=|\Psi_k\rangle\langle\Psi_k|\Psi_0\rangle\exp\left(-i\frac{E_k}{\hbar}t\right)$$

Properties. If an **isolated system** starts in an eigen-state of the Hamiltonian operator, i.e. $|\Psi_0\rangle=|\Psi_a\rangle$, $|c_{k,0}|^2=\delta_{ka}$, then the evolution of the system reads

$$|\Psi_t\rangle=|\Psi_a\rangle\exp\left(-i\frac{E_a}{\hbar}t\right)\ ,$$

i.e. the probability of measuring the kth energy is

$$p(E=E_k)=|\langle\Psi_k|\Psi_t\rangle|^2=\left|\underbrace{\langle\Psi_k|\Psi_a\rangle}_{\delta_{ka}}\exp\left(-i\frac{E_a}{\hbar}t\right)\right|^2=\delta_{ka}\ ,$$

i.e. there's probability of finding it in the same state as the initial state a is $p(E=E_a)=1$, while the probability of finding in any other state is zero, $p(E=E_k)=0$, $k\neq a$.

16.2.3 Space and momentum operators

Position operator

Definition

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle$$

Wave function in space basis

$$\Psi(\mathbf{r}, t) := \langle \mathbf{r}|\Psi_t\rangle \ .$$

Orthonality, and identity operator

$$\begin{aligned} 1 &= \int_{\mathbf{r}\in\Omega} \Psi^*(\mathbf{r}, t)\Psi(\mathbf{r}, t)d\mathbf{r} = \\ &= \langle\Psi_t| \underbrace{\int_{\mathbf{r}\in\Omega} |\mathbf{r}\rangle\langle\mathbf{r}|d\mathbf{r}}_{=\hat{\mathbf{1}}} |\Psi_t\rangle = \\ &= \langle\Psi_t|\Psi_t\rangle \ . \end{aligned}$$

So that the identity operator in space basis reads $\hat{\mathbf{1}} = \int_{\mathbf{r}\in\Omega} |\mathbf{r}\rangle\langle\mathbf{r}|d\mathbf{r}$. A wave function can be thus written as

$$|\Psi\rangle = \int_{\mathbf{r}'\in\Omega} |\mathbf{r}'\rangle\langle\mathbf{r}'| \, d\mathbf{r}' \, |\Psi\rangle \ . \tag{16.3}$$

Expansion in space basis

$$\langle\mathbf{r}|\Psi\rangle = \Psi(\mathbf{r}, t) = \int_{\mathbf{r}'\in\Omega} \delta(\mathbf{r} - \mathbf{r}')\Psi(\mathbf{r}', t) \, d\mathbf{r}'$$

Comparing the last expression, with the projection of the expression (16.3) over $\langle\mathbf{r}|$,

$$\langle\mathbf{r}|\Psi\rangle = \langle\mathbf{r}| \int_{\mathbf{r}'\in\Omega} |\mathbf{r}'\rangle\langle\mathbf{r}'| \, d\mathbf{r}' \, |\Psi\rangle = \int_{\mathbf{r}'\in\Omega} \langle\mathbf{r}|\mathbf{r}'\rangle\langle\mathbf{r}'|\Psi\rangle \, d\mathbf{r}' = \int_{\mathbf{r}'\in\Omega} \langle\mathbf{r}|\mathbf{r}'\rangle\Psi(\mathbf{r}', t) \, d\mathbf{r}' \ ,$$

if follows the orthogonality condition $\langle\mathbf{r}|\mathbf{r}'\rangle = \delta(\mathbf{r} - \mathbf{r}')$.

The classical Hamiltonian of the system reads

$$H(\mathbf{r}, \mathbf{p}) = \frac{|\mathbf{p}|^2}{2m} - \frac{q^2}{4\pi\varepsilon|\mathbf{r}|} \ ,$$

the promotion to the Hamiltonian operator is thus

$$\hat{H} = \frac{|\hat{\mathbf{p}}|^2}{2m} - \frac{q^2}{4\pi\varepsilon}|\hat{\mathbf{r}}|^{-1} \ .$$

Power of an operator

Positive integer power

$$\hat{A}^n = \underbrace{\hat{A} \dots \hat{A}}_{n \text{ times}}$$

Let the eigenproblem of the operator be $\hat{A}|a\rangle = a|a\rangle$, with $a \in \mathbb{R}$. Then, the eigenproblem for the operator $\hat{A}^n$ reads

$$\hat{A}^n|a\rangle = a^n|a\rangle \ .$$

The inverse of an operator has the eigenproblem

$$\hat{A}^{-1}|a\rangle = \frac{1}{a}|a\rangle \ ,$$

as $|a\rangle = \hat{A}\left(\hat{A}^{-1}|a\rangle\right) = \frac{1}{a}\hat{A}|a\rangle = \frac{1}{a}\, a|a\rangle = |a\rangle$.

Vector operators

Momentum operator, $\hat{\mathbf{p}}$, in space base $\langle \mathbf{r}|\hat{\mathbf{p}} = -i\hbar\nabla_{\mathbf{r}}\langle \mathbf{r}|$. Using Cartesian coordinates

$$\hat{\mathbf{e}}_a \langle \mathbf{r}|\hat{p}_a = -\hat{\mathbf{e}}_a \, i\hbar\partial_a\langle \mathbf{r}|$$

The operator of the square magnitude reads

$$(\widehat{|\mathbf{p}|^2}) = \left(p_x^2 + \widehat{p_y^2} + p_z^2\right) = \widehat{p_x^2} + \widehat{p_y^2} + \widehat{p_z^2} \ ,$$

and in space coordinates

$$\langle \mathbf{r}|(\widehat{|\mathbf{p}|^2}) = -\hbar^2\partial_{aa}\langle \mathbf{r}| = -\hbar^2\nabla^2\langle \mathbf{r}|$$

CHAPTER

SEVENTEEN

ANGULAR MOMENTUM

Before going into details, an introduction is likely to be required.

Spatial (orbital) angular momentum, $\hat{\mathbf{L}}$. Spatial angular motion is the quantum mechanical counterpart of the angular momentum in classical mechanics, $\hat{L} = \mathbf{r} \times \mathbf{p}$, due to the spatial motion of the system w.r.t. a point. The angular momentum operator in quantum mechanics can be defined promoting the classical space and momentum variables to the corresponding operators,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \qquad \rightarrow \qquad \hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} \, ,$$

whose representation using spatial basis reads

$$\langle \mathbf{r} | \hat{\mathbf{L}} = -i\hbar \mathbf{r} \times \nabla_{\mathbf{r}} \langle \mathbf{r}| \, .$$

This operator has several properties that will be investigated in details in the dedicated section. This operator applies to a wave function representing the position, momentum, and physical quantities with classical analogous, here called $|\psi\rangle$.

Spin (intrinsic) angular momentum, $\hat{\mathbf{S}}$. Quantum systems may have intrinsic properties, with no classical counterpart. Spin momentum is one of these variables. Spin angular momentum $\hat{\mathbf{S}}$ is defined in analogy with the spatial orbital momentum, with the same properties and acts on a spin state vector $|s\rangle$.

Thus, the full state of the system belongs to a space state that is a tensor product of a space and a spin state space,

$$\mathcal{H} = \mathcal{H}_{space} \otimes \mathcal{H}_{spin} \, ,$$

defined as the tensor product of a spatial state vector and a spin state vector,

$$|\Psi\rangle = |\psi\rangle \otimes |s\rangle \, .$$

Total angular momentum $\hat{\mathbf{J}}$. Total angular momentum can be defined as the sum of the spatial and the spin angular momentum,

$$\hat{\mathbf{J}} = \left(\hat{\mathbf{L}} \otimes \hat{\mathbf{1}}\right) + \left(\hat{\mathbf{1}} \otimes \hat{\mathbf{S}}\right) \, ,$$

or, remembering that $\hat{\mathbf{L}}$ only acts on $|\psi\rangle$ and $\hat{\mathbf{S}}$ onlt acts on $|s\rangle$, it's usually written as $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$.

17.1 Spatial Angular Momentum

Promoting the classical definition of angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, the angular momentum operator in quantum mechanics reads

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \hat{\mathbf{e}}_a \varepsilon_{abc} \hat{r} \hat{p}_c \, ,$$

and using space basis

$$\langle \mathbf{r} | \hat{\mathbf{L}} = \hat{\mathbf{e}}_a \underbrace{\left(-i \, \hbar \, \varepsilon_{abc} \, r_b \, \partial_c \langle \mathbf{r}| \right)}_{= \langle \mathbf{r} | \hat{L}_a} \, .$$

Properties

Composition of 2 components of the angular momentum operator. Using Cartesian coordinates

$$\hat{L}_a\hat{L}_b = -\hbar^2 r_b\partial_a + \hbar^2\delta_{ab}r_c\partial_c - \hbar^2\varepsilon_{acd}\varepsilon_{bef}r_c r_e\partial_{df} =$$

Details

$$\begin{aligned}\hat{L}_a\hat{L}_b &= -\hbar^2\varepsilon_{acd}r_c\partial_d\left(\varepsilon_{bef}r_e\partial_f\right) = \\ &= -\hbar^2\varepsilon_{acd}\varepsilon_{bef}r_c\left(\delta_{de}\partial_f + r_e\partial_{df}\right) = \\ &= -\hbar^2\varepsilon_{acd}\varepsilon_{bdf}r_c\partial_f - \hbar^2\varepsilon_{acd}\varepsilon_{bdf}r_c r_e\partial_{df} = \\ &= -\hbar^2\left(\delta_{af}\delta_{cb} - \delta_{ab}\delta_{cf}\right)r_c\partial_f - \hbar^2\varepsilon_{acd}\varepsilon_{bef}r_c r_e\partial_{df} = \\ &= -\hbar^2 r_b\partial_a + \hbar^2\delta_{ab}r_c\partial_c - \hbar^2\varepsilon_{acd}\varepsilon_{bef}r_c r_e\partial_{df} =\end{aligned}$$

The last term is symmetric w.r.t. a, b.

Commutators.

$$[\hat{L}_a, \hat{L}_b] = \hat{L}_a\hat{L}_b - \hat{L}_b\hat{L}_a = -\hbar^2(r_b\partial_a - r_a\partial_b)$$

If $a = b$, the obvious result follows, as an operator commutes with itself. If $a \neq b$,

$$[\hat{L}_a, \hat{L}_b] = i\varepsilon_{abc}\hbar\hat{L}_c\ ,$$

or with a compact "vector notation", $\hat{\mathbf{L}} \times \hat{\mathbf{L}} = i\hbar\hat{\mathbf{L}}$.

Magnitude of the angular momentum, $|\hat{L}|^2$. Using Cartesian coordinates,

$$|\hat{L}|^2 = 2\hbar^2 r_a\partial_a - \hbar^2 r_b r_b\partial_{aa} + \hbar^2 r_a r_b\partial_{ab}\ .$$

Details

$$\begin{aligned}|\hat{L}|^2 &= \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \\ &= \hat{L}_a\hat{L}_a = \\ &= -\hbar^2 r_a\partial_a + 3\hbar^2 r_a\partial_a - \hbar^2(\delta_{ce}\delta_{df} - \delta_{cf}\delta_{de})r_c r_e\partial_{df} = \\ &= 2\hbar^2 r_a\partial_a - \hbar^2 r_b r_b\partial_{aa} + \hbar^2 r_a r_b\partial_{ab}\ .\end{aligned}$$

Commutations with magnitude.

$$[\hat{L}^2, \hat{L}_a] = 0$$

Details

The same hold for any integer power of $\hat{L}_a$, like $\hat{L}_a^2$, as it can be easily proved

$$\begin{aligned}\left[\hat{L}^2, \hat{L}_a^2\right] &= \hat{L}^2\hat{L}_a^2 - \hat{L}_a^2\hat{L}^2 = \\ &= \hat{L}^2\hat{L}_a\hat{L}_a - \hat{L}_a^2\hat{L}^2 = \\ &= \hat{L}_a\hat{L}^2\hat{L}_a - \hat{L}_a\hat{L}_a\hat{L}^2 = \\ &= \hat{L}_a[\hat{L}^2, \hat{L}_a] = 0\ .\end{aligned}$$

Analogous relations in **classical mechanics**

$$[L_a, L_b] = i\hbar\varepsilon_{abc}\hat{L}_c \qquad , \qquad \{L_a, L_b\} = \varepsilon_{abc}L_c$$

provides another example of the relation between Poisson brackets in classical mechanics and commutator in quantum mechanics (**Dirac**)

$$\{A, B\} \quad \leftrightarrow \quad \frac{1}{i\hbar}[\hat{A}, \hat{B}]$$

17.1.1 Eigenproblem of angular momentum operators

In this section, the eigenproblems of $\hat{L}^2$ and $\hat{L}_z$ are discussed. Here, all the operators and state functions are written in components using space basis, without explicitly writing the projection $\langle \mathbf{r}|$.

- Using spherical coordinates, the 2π-periodicity of the problem in ϕ implies quantization in the spectral decomposition of $\hat{L}_z$,
-

$\langle \mathbf{r}|\hat{\mathbf{L}}$ using Cartesian coordinates.

$$\ell\psi = \hat{L}_a\psi = -i\hbar\varepsilon_{abc}r_b\partial_c\psi \,,$$

or $\ell\psi = -i\hbar\mathbf{r}\times\nabla\psi$.

…

$\langle \mathbf{r}|\hat{\mathbf{L}}$ using spherical coordinates.

With the set of spherical coordinates $\psi(r, \theta, \phi)$

$$\begin{aligned}\mathbf{r}\times\nabla\psi = \mathbf{r}\times\left[\partial_r\psi\,\hat{\mathbf{r}} + \frac{1}{r}\partial_\theta\psi\,\hat{\theta} + \frac{1}{r\sin\theta}\partial_\phi\psi\hat{\phi}\right] = \\ = \partial_\theta\psi\,\hat{\phi} - \frac{1}{\sin\theta}\partial_\phi\psi\,\hat{\theta}\,.\end{aligned}$$

Thus the z-component, with $\hat{\mathbf{z}}\cdot\hat{\theta} = -\sin\theta$, $\hat{\mathbf{z}}\cdot\hat{\phi} = 0$, can be written as

$$\hat{\mathbf{z}}\cdot(\mathbf{r}\times\nabla\psi) = \partial_\phi\psi$$

Spectral decomposition of the z-component operator, $\hat{L}_z$

$$\ell_z\psi_z(r, \theta, \phi) = -i\hbar\partial_\phi\psi_z(r, \theta, \phi)$$

The eigenfunctions read

$$\psi_{z,\ell}(r, \theta, \phi) = A(r, \theta)\,\exp\left(i\frac{\ell_z}{\hbar}\phi\right) \,.$$

Periodic condition $\psi_z(0) = \psi_z(2\pi)$ implies $\frac{\ell_z}{\hbar} = m_l$, $m_l \in \mathbb{Z}$.

Remark. The constraint $m_l \in \mathbb{Z}$ is just **one of the constraints** on m_l.

In order to keep the notation as intuitive as possible, the eigenvalue ℓ_z and the corresponding eigenstate are denoted as $\{L_z, |L_z\rangle\} = \{\hbar m_l, |m_l\rangle\}$, s.t. $\hat{L}_z|m_l\rangle = \hbar m_l|m_l\rangle$.

Spectral decomposition of commuting operators

Let $\hat{A}|a\rangle = a|a\rangle$, and $\hat{B}|b\rangle = b|b\rangle$. If $0 = [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. Pre-multiplying the first one by $\hat{B}$,

$$a\hat{B}|a\rangle = \hat{B}\hat{A}|a\rangle = \hat{A}\hat{B}|a\rangle \ .$$

So that $\hat{B}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue a.

- If a is an eigenvalue with algebraic multiplicity equal to 1, then $\hat{B}|a\rangle$ must be proportional to the only eigenstate $|a\rangle$ of the operator $\hat{A}$ with eigenvalue a, i.e.

  $$\hat{B}|a\rangle = \mu|a\rangle \ ,$$

  and thus $|a\rangle$ is also an eigenstate of $\hat{B}$.

- If a has algebraic multiplicity larger than 1, then $\hat{B}|a_i\rangle$, can be a linear combination of all the eigenvectors $|a_k\rangle$ with eigenvalues $a_k = a$,

  $$\hat{B}|a_i\rangle = c_{ik}|a_k\rangle \ ,$$

  for $\forall i$.

  …

Relation between $\hat{L}_z$ and $\hat{L}^2$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

As $\hat{L}^2$ commutes with the Cartesian components $\hat{L}_a$, they share the same eigenvalues.The same holds for $\hat{L}_a^2$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \ ,$$

implies

$$\hat{L}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2 \ .$$

Let ψ an eigenstate of $\hat{L}^2$, $\hat{L}_z$ with eigenvalues ℓ^2, ℓ_z, it follows

$$\left(\hat{L}_x^2 + \hat{L}_y^2\right)\psi = \left(\hat{L}^2 - \hat{L}_z^2\right)\psi = (\ell^2 - \ell_z^2)\,\psi$$

As $(\ell^2 - \ell_z^2)|\psi|^2 = \psi^* \hat{L}_x^2 \psi = |\hat{L}_x\psi|^2 \geq 0$, it follows that $\ell^2 - \ell_z^2 \geq 0$, and thus $-\ell \leq \ell_z \leq \ell$.

Combining

$$\left[\hat{L}_y, \hat{L}_z\right] = \hat{L}_y\hat{L}_z - \hat{L}_z\hat{L}_y = i\hbar\hat{L}_x$$
$$\left[\hat{L}_z, \hat{L}_x\right] = \hat{L}_z\hat{L}_x - \hat{L}_x\hat{L}_z = i\hbar\hat{L}_y$$

summing and subtracting $i(1)$ and (2) them

$$\begin{aligned} 0 &= -\hbar\hat{L}_x \mp i\hbar\hat{L}_y - i\hat{L}_y\hat{L}_z + i\hat{L}_z\hat{L}_y \pm \hat{L}_z\hat{L}_x \mp \hat{L}_x\hat{L}_z = \\ &= -(\hat{L}_x \pm i\hat{L}_y)(\hbar \pm \hat{L}_z) + \hat{L}_z(i\hat{L}_y \pm \hat{L}_x) \ , \end{aligned}$$

and thus

$$\hat{L}_z(i\hat{L}_y \pm \hat{L}_x) = (\hat{L}_x \pm i\hat{L}_y)(\hbar \pm \hat{L}_z)$$

or

$$\hat{L}_z(\hat{L}_x \pm i\hat{L}_x) = (\hat{L}_x \pm i\hat{L}_y)(\hat{L}_z \pm \hbar)$$

Applying this operator to the eigenstate ψ

$$\hat{L}_z(\hat{L}_x \pm i\hat{L}_y)\psi = (\hat{L}_x \pm i\hat{L}_y)(\hat{L}_z \pm \hbar)\psi = (\ell_z \pm \hbar)(\hat{L}_x \pm i\hat{L}_y)\psi \tag{17.1}$$

Thus, the state functions $\psi^{\pm} = \left(\hat{L}_x \pm i\hat{L}_y\right)\psi$ are eigenstates of the operator $\hat{L}_z$ with eigenvalues $\ell_z \pm \hbar$. As $\ell_z \in [-\ell, \ell]$, let's call $\ell_{z,min}$ and $\ell_{z,max}$ the minimum and maximum values of ℓ_z. It follows that the operator $\hat{L}^-$ has the smallest eigenvalue equal to $\ell_{z,min} - \hbar$ and $\hat{L}^+$ has the largest eigenvalue equal to $\ell_{max,z} + \hbar$.

Thus, for $\ell_{z,min}$, $\ell_z - \hbar$ can't be an eigenvalue of $\hat{L}_z$, and thus the relation

$$\hat{L}_z\left(\hat{L}_x - i\hat{L}_y\right)\psi = (\ell_{z,min} - \hbar)(\hat{L}_x - i\hat{L}_y)\psi$$

implies that $(\hat{L}_x - i\hat{L}_y)\psi$ can't be an eigenfunction of $\hat{L}_z$, and thus it must be in the kernel or $\hat{L}_z$, i.e.

$$0 = \hat{L}_z(\hat{L}_x - i\hat{L}_y)\psi_{\ell_z,min} = (\hat{L}_x - i\hat{L}_y)(\hat{L}_z + \hbar)\psi_{\ell_z,min} = (\ell_{z,min} + \hbar)(\hat{L}_x - i\hat{L}_y)\psi_{\ell,min}$$

A similar relation holds for the largest eigenvalue

$$0 = \hat{L}_z(\hat{L}_x + i\hat{L}_y)\psi_{\ell_z,max} = (\hat{L}_x + i\hat{L}_y)(\hat{L}_z - \hbar)\psi_{\ell_z,max} = (\ell_{z,max} - \hbar)(\hat{L}_x + i\hat{L}_y)\psi_{\ell,max}$$

Applying $\hat{L}_x + i\hat{L}_y$ to the first equation, the operator becomes

$$(\hat{L}_x + i\hat{L}_y)(\hat{L}_x - i\hat{L}_y) = \hat{L}_x^2 + \hat{L}_y^2 - i[\hat{L}_x, \hat{L}_y] = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z\ .$$

Applying $\hat{L}_x - i\hat{L}_y$ to the second equation, the operator becomes

$$(\hat{L}_x - i\hat{L}_y)(\hat{L}_x + i\hat{L}_y) = \hat{L}_x^2 + \hat{L}_y^2 + i[\hat{L}_x, \hat{L}_y] = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z\ .$$

Thus, the 2 equations are, for a given value of ℓ

$$0 = (\ell_{z,min} + \hbar)(\ell^2 - \ell_{z,min}^2 + \hbar\ell_{z,min})$$
$$0 = (\ell_{z,max} - \hbar)(\ell^2 - \ell_{z,max}^2 - \hbar\ell_{z,max})$$

Either $\ell_{z;min,max} = \mp\hbar$, or

$$0 = \ell^2 - \ell_{z,min}^2 + \hbar\ell_{z,min}$$
$$0 = \ell^2 - \ell_{z,max}^2 - \hbar\ell_{z,max}\ ,$$

and subtracting $0 = (\ell_{z,max} + \ell_{z,min})(\ell_{z,max} - \ell_{z,min} + \hbar)$. As the content of the second bracket is always positive, then $\ell_{z,min} = -\ell_{z,max}$.

Following (17.1), if $|\psi\rangle$ is the eigenvector of $\hat{L}_z$ with eigenvalue ℓ_z, the wave functions $\hat{L}^{\mp}|\psi\rangle$ are eigenvectors of $\hat{L}_z$ with eigenvalues $\ell_z \mp \hbar$. Thus, the set

$$\{\ell_{z,min}, \ell_{z,min} + \hbar, \ell_{z,min} + 2\hbar, \ldots, \ell_{z,max} - \hbar, \ell_{z,max}\} = \{-m_{l,max}\hbar, \ldots, m_{l,max}\hbar\}$$

contains possible values of the eigenvalues of $\hat{L}_z$.

$$\lambda\left(L_z\right) = \hbar m_l \qquad m_l \in$$

Now, for $\ell_{z,min} = -m_{l,max}\hbar$, and $\ell_{z,max} = m_{l,max}\hbar$, the corresponding eigenvalues ℓ^2 are

$$\ell^2 = \hbar^2 m_{l,max}\left(m_{l,max} + 1\right)\ .$$

17.2 Spin

- Properties of spin angular momentum operators
- Pauli matrices
- Pauli equation

17.2.1 Properties of spin angular momentum operators $\hat{S}^2$, $\hat{S}_z$

In analogy with the spatial angular momentum, and using Cartesian coordinates

$$[S_a, S_b] = i\hbar\,\varepsilon_{abc} S_c$$
$$[S_a, S^2] = 0\,.$$

As the operators S_z and S^2 commute, they share the same eigenvectors $|s, m_s\rangle$ (all the eigvecs?) and their eigenproblems read

$$S_z|s, m_s\rangle = \hbar m_s |s, m_s\rangle$$
$$S^2|s, m_s\rangle = \hbar^2 s(s+1)|s, m_s\rangle\,,$$

The value of s depends on the system of interest: for an electron $s = \frac{1}{2}$. The value of the spin projection quantum number m_s belongs to $m_s \in \{-s, -s+1, \dots, s-1, s\}$. For an electron, $m_s \in \{-\frac{1}{2}, \frac{1}{2}\}$.

17.2.2 Pauli matrices

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$
$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Properties

Pauli vectors

Using a Cartesian basis, a Pauli vector is defined as

$$\sigma := \sum_k \sigma_k \hat{\mathbf{x}}_k$$

and the dot product with a vector $\mathbf{a} = \sum_k a_k \hat{\mathbf{x}}_k$ is defined as

$$\mathbf{a}\cdot\sigma = \sum_{l,k} a_l \underbrace{\hat{\mathbf{x}}_l \cdot \hat{\mathbf{x}}_k}_{\delta_{kl}} \sigma_k = \sum_k a_k\,\sigma_k = \begin{bmatrix} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{bmatrix}\,.$$

Properties

17.3 Total Angular Momentum

State function

$$|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle \quad \in \quad \mathcal{H} = \mathcal{H}_{space} \otimes \mathcal{H}_{spin} \, .$$

Total angular momentum operator

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} = \hat{\mathbf{L}} \otimes \hat{\mathbf{1}}_{spin} + \hat{\mathbf{1}}_{space} \otimes \hat{\mathbf{S}}$$

Dot product of $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$,

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = L_x \otimes S_x + L_y \otimes S_y + L_z \otimes S_z \, .$$

Magnitude square of $\hat{\mathbf{J}}$, $\hat{J}^2$,

$$\begin{aligned} \hat{J}^2 &= (\hat{\mathbf{L}} + \hat{\mathbf{S}}) \cdot (\hat{\mathbf{L}} + \hat{\mathbf{S}}) = \\ &= \hat{L}^2 \otimes \hat{\mathbf{1}} + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} + \hat{\mathbf{1}} \otimes \hat{S}^2 \, . \end{aligned}$$

Orbital angular momentum.

$$\begin{aligned} \hat{L}^2|\ell, m_l\rangle &= \hbar^2 \ell(\ell+1)|\ell, m_l\rangle \quad , \quad \ell \in \{0, 1, 2, \dots\} \\ \hat{L}_z|\ell, m_l\rangle &= \hbar m_l |\ell, m_l\rangle \quad , \quad m_l \in \{-\ell, -\ell+1, \dots, \ell-1, \ell\} \end{aligned}$$

Spin angular momentum.

$$\begin{aligned} \hat{S}^2|s, m_s\rangle &= \hbar^2 s(s+1)|s, m_s\rangle \quad , \quad s \in \left\{0, \frac{1}{2}, 1, \dots\right\} \\ \hat{S}_z|s, m_s\rangle &= \hbar m_s |s, m_s\rangle \quad , \quad m_s \in \{-s, -s+1, \dots, s-1, s\} \end{aligned}$$

For an electron $s = \frac{1}{2}$, so that $m_s \in \{-\frac{1}{2}, \frac{1}{2}\}$, and thus, without explicitly writing s in the eigenvectors - being that a given parameter -,

$$\begin{aligned} \hat{S}^2|m_s\rangle &= \frac{3}{4}\hbar^2 |m_s\rangle \\ \hat{S}_z|m_s\rangle &= \hbar m_s |m_s\rangle \quad , \quad m_s \in \left\{-\frac{1}{2}, \frac{1}{2}\right\} \end{aligned}$$

Total angular momentum. The z-component of the total angular momentum, and its magnitude squared are defined as

$$\begin{aligned} \hat{J}^2|j, m_j\rangle &= \hbar^2 j(j+1)|j, m_j\rangle \quad , \quad j \in \left\{0, \frac{1}{2}, 1, \dots\right\} \\ \hat{J}_z|j, m_j\rangle &= \hbar m_j |j, m_j\rangle \quad , \quad m_j \in \{-j, -j+1, \dots, j-1, j\} \end{aligned}$$

17.3.1 Eigenvalue problem of the total angular momentum for an electron

Commutation of operators. The operators $\hat{J}^2$, $\hat{J}_z$, $\hat{L}^2$, $\hat{S}^2$ commute, and thus they share common eigenvectors

$$|j, m_j, \ell, s\rangle \, .$$

The operator $\hat{J}_z$ commutes with $\hat{L}^2$ and $\hat{S}^2$, $\left[\hat{J}_z, \hat{L}^2\right] = 0$, $\left[\hat{J}_z, \hat{S}^2\right] = 0$.

Proof

$$\begin{aligned}\left[\hat{J}_z, \hat{L}^2\right] &= \hat{J}_z\hat{L}^2 - \hat{L}^2\hat{J}_z = \\ &= \hat{L}_z\hat{L}^2 + \hat{S}_z\hat{L}^2 - \hat{L}^2\hat{L}_z - \hat{L}^2\hat{S}_z = \\ &= \underbrace{\left[\hat{L}_z, \hat{L}^2\right]}_{=0} + \underbrace{\left[\hat{S}_z, \hat{L}^2\right]}_{=0} = 0\,.\end{aligned}$$

The last term holds because $\hat{S}_z$ and $\hat{L}^2$ acts on two different sub-spaces, or explicitly for any $|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle$,

$$\begin{aligned}\hat{S}_z\hat{L}^2|\Psi\rangle &= (\hat{\mathbf{1}} \otimes \hat{S}_z)(\hat{L}^2 \otimes \hat{\mathbf{1}})|\psi\rangle \otimes |\sigma\rangle = \\ &= (\hat{\mathbf{1}} \otimes \hat{S}_z)(\hat{L}^2|\psi\rangle \otimes |\sigma\rangle) = \\ &= \hat{L}^2|\psi\rangle \otimes \hat{S}_z|\sigma\rangle\,,\end{aligned}$$

and $\hat{L}^2\hat{S}_z|\Psi\rangle$ provides the same result for any $|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle$.

The operator $\hat{L}_z$ doesn't commute with $\hat{J}^2$, $\left[\hat{L}_z, \hat{J}^2\right] = -2i\hbar\,\hat{\mathbf{z}} \cdot \hat{\mathbf{L}} \times \hat{\mathbf{S}}$.

Proof

$$\begin{aligned}\left[\hat{L}_z, \hat{J}^2\right] &= \hat{L}_z\hat{J}^2 - \hat{J}^2\hat{L}_z = \\ &= \hat{L}_z\left(\hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}\right) - \left(\hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}\right)\hat{L}_z = \\ &= \underbrace{\left[\hat{L}_z, \hat{L}^2\right]}_{=0} + \underbrace{\left[\hat{L}_z, \hat{S}^2\right]}_{=0} + 2[\hat{L}_z, \hat{L}_a]\hat{S}_a = \\ &= 2i\hbar\hat{L}_y\hat{S}_x - 2i\hbar\hat{L}_x\hat{S}_y = \\ &= -2i\hbar\,\hat{\mathbf{z}} \cdot \hat{\mathbf{L}} \times \hat{\mathbf{S}}\,,\end{aligned}$$

as $[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$, $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$, $[\hat{L}_z, \hat{L}_z] = 0$,

z-component operators.

$$\begin{aligned}\hat{J}_z|\Psi_{m_\ell, m_s}\rangle &= (\hat{L}_z + \hat{S}_z)|m_\ell\rangle \otimes |m_s\rangle = \\ &= \hat{L}_z|m_\ell\rangle \otimes |m_s\rangle + \hat{S}_z|m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar m_\ell|m_\ell\rangle \otimes |m_s\rangle + \hbar m_s|m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar\left(m_\ell + m_s\right)|m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar\, m_j\,|m_\ell\rangle \otimes |m_s\rangle\,,\end{aligned}$$

and thus the relation $m_j = m_\ell + m_s$ for every pair of eigenvectors $|m_\ell\rangle$ and $|m_s\rangle$ of the operators $\hat{L}_z$, and $\hat{S}_z$. It also follows, that the extreme values of m_j are $-j$ and $j = \ell + s$.

CHAPTER

EIGHTEEN

DENSITY OPERATOR

Let's start from an example of the preparation of a system of subsystems in pure states $|\psi_i\rangle$, with probability p_i, $\sum_i p_i = 1$. Let M an observable, with the corresponding Hermitian operator $\hat{M}$ with discrete values m_μ and corresponding states $|m_\mu\rangle$.

Now, the probability of measuring the value m_μ for a system in state $|\psi_i\rangle$, i.e. the conditional probability $p(m = m_\mu||\psi\rangle = |\psi_i\rangle)$ reads

$$p(m = m_\mu||\psi\rangle = |\psi_i\rangle) = |\langle m_\mu|\psi_i\rangle|^2 = \langle\psi_i|m_\mu\rangle\langle m_\mu|\psi_i\rangle \ .$$

The probability of measuring m from the ensemble is the marginal probability,

$$\begin{aligned} p(m = m_\mu) &= \sum_i p(m = m_\mu||\psi\rangle = |\psi_i\rangle)\, p(|\psi\rangle = |\psi_i\rangle) = \\ &= \sum_i \langle\psi_i|m_\mu\rangle\langle m_\mu|\psi_i\rangle\, p_i = \\ &= \sum_i \langle\psi_i|\hat{\Pi}_{m_\mu}|\psi_i\rangle\, p_i \ , \end{aligned}$$

having introduced the orthogonal projector over the μth eigenfunction of the operator $\hat{M}$, i.e. $\hat{\Pi}_{m_\mu} = |m_\mu\rangle\langle m_\mu|$.

Let's define the **density operator** as

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i| \ .$$

Using an orthonormal basis $\{|e_k\rangle\}_k$, it's easy to show that

$$\begin{aligned} \mathrm{Tr}\left(\hat{\rho}\hat{\Pi}_{m_\mu}\right) &:= \sum_k \langle e_k|\hat{\rho}\hat{\Pi}_{m_\mu}|e_k\rangle = \\ &= \sum_{k,i} p_i \langle e_k|\psi_i\rangle\langle\psi_i|\hat{\Pi}_{m_\mu}|e_k\rangle = \\ &= \sum_i p_i \langle\psi_i|\hat{\Pi}_{m_\mu} \underbrace{\sum_k |e_k\rangle\langle e_k|}_{=\hat{\mathbf{1}}} \psi_i\rangle = \\ &= \sum_i p_i \langle\psi_i|\hat{\Pi}_{m_\mu}|\psi_i\rangle = \\ &= p(m = m_\mu) \ . \end{aligned}$$

Expected value.

$$\begin{aligned}\mathbb{E}[M] &= \sum_{\mu} m_\mu p(m=m_\mu) = \\ &= \sum_i \sum_\mu m_\mu p_i \langle\psi_i|m_\mu\rangle\langle m_\mu|\psi_i\rangle = \\ &= \sum_i \sum_\mu p_i \langle\psi_i|\hat{M}|m_\mu\rangle\langle m_\mu|\psi_i\rangle = \\ &= \sum_\mu |\langle m_\mu \sum_i p_i|\psi_i\rangle\langle\psi_i|\hat{M}|m_\mu\rangle = \\ &= \sum_\mu \langle m_\mu|\hat{\rho}\hat{M}|m_\mu\rangle = \\ &= \mathrm{Tr}\left(\hat{\rho}\hat{M}\right) \ .\end{aligned}$$

Trace of an operator

Choosing a set of **orthogonal unit vectors** $|\psi_i\rangle$, the trace of an operator $\hat{A}$ can be defined as

$$\mathrm{Tr}\left(\hat{A}\right) = \sum_i \langle\psi_i|\hat{A}|\psi_i\rangle \ .$$

Properties

- $$\mathrm{Tr}\left(\hat{A}|\psi_j\rangle\langle\psi_j|\right) = \sum_i \langle\psi_i|\hat{A}|\psi_j\rangle \underbrace{\langle\psi_j|\psi_i\rangle}_{\delta_{ij}} = \langle\psi_j|\hat{A}|\psi_j\rangle \ .$$

- $$\mathrm{Tr}\left(\hat{A}\right) = \mathrm{Tr}\left(\hat{A}\sum_j |\psi_j\rangle\langle\psi_j|\right) = \sum_{i,j}\langle\psi_i|\hat{A}|\psi_j\rangle \underbrace{\langle\psi_j|\psi_i\rangle}_{\delta_{ij}} = \sum_i \langle\psi_i|\hat{A}|\psi_i\rangle \ .$$

Choosing a **generic basis** $\{|\phi_k\rangle\}_k$, the kth vector of this basis can be written as a linear combination of the vectors of a unit orthogonal basis,

As done in Differential Geometry, a reciprocal basis $\{|\phi^j\rangle\}_j$ exists s.t. $\langle\phi^j|\phi_k\rangle = \delta^j_k$. Defining the components of the metric tensor $g_{ij} = \langle\phi_i|\phi_j\rangle$, $g^{ij} = \langle\phi^i|\phi^j\rangle$ the relations between the original basis and its reciprocal follows

$$|\phi^i\rangle = g^{ij}|\phi_j\rangle \quad , \quad |\phi_i\rangle = g_{ij}|\phi^j\rangle.$$

The identity operator can be written as $\hat{\mathbf{1}} = \sum_j |\phi_j\rangle\langle\phi^j| = \sum_j |\phi^j\rangle\langle\phi_j|$, as

$$|v\rangle = \sum_i v^i|\phi_i\rangle = \sum_{i,j} v^i|\phi_j\rangle \underbrace{\langle\phi^j|\phi_i\rangle}_{=\delta^j_i} = \sum_j |\phi_j\rangle\langle\phi^j| \sum_i v^i|\phi_i\rangle = \hat{\mathbf{1}}|v\rangle \ .$$

The vectors of the basis can be written as a linear combination of the vectors of an orthonormal basis $\{|\psi_k\rangle\}_k$,

$$|\phi_i\rangle = \sum_k T_i{}^k|\psi_k\rangle \ .$$

The dual vectors are written as $|\phi^j\rangle = \sum_l R^{jl}|\psi_l\rangle$. As the dual vectors should be orthogonal to the vectors of the original basis, it follows

$$\delta^j_i = \langle\phi^j|\phi_i\rangle = \sum_{k,l} \left(R^{jl}\right)^* T_i{}^k \underbrace{\langle\psi_l|\psi_k\rangle}_{=\delta_{lk}} = \sum_k \left(R^{jk}\right)^* T_i{}^k \ ,$$

i.e. the transformation matrix R of the vectors of the reciprocal basis is the inverse of the adjoint of the matrix T, i.e.

$$\begin{aligned}\left(\mathsf{T}\right)_{ik} &= T_i{}^k \\ \left(\mathsf{R}\right)_{jk} &= R^{ij} \\ \left(\mathsf{T}^{-1}\right)_{kj} &= \left(R^{jk}\right)^* = \left(\mathsf{R}^H\right)_{kj}\end{aligned}$$

Using matrix formalism, the definition of the inverse matrix gives $\mathsf{I} = \mathsf{T}\mathsf{R}^H = \mathsf{R}^H\mathsf{T}$. (**todo** what happens for infinite dimensional spaces?)

Thus, the relation

$$\sum_i \langle\phi^i|\hat{A}|\phi_i\rangle = \sum_{l,k} \underbrace{\sum_i \left(R^{il}\right)^* T_i{}^k}_{=\left(\mathsf{R}^H\mathsf{T}\right)_{lk}=\delta^{lk}} \langle\psi_l|\hat{A}|\psi_k\rangle = \sum_k \langle\psi_k|\hat{A}|\psi_k\rangle = \mathrm{Tr}\left(\hat{A}\right) \ .$$

todo *Uncomment or delete (more likely)*

Example 18.1 (Difference between mixed states and pure state in superposition)

Pure state in superposition of two orthogonal states $|\psi_1\rangle$, $|\psi_2\rangle$,

$$|\psi\rangle = a_1|\psi_1\rangle + a_2|\psi_2\rangle \ ,$$

with $|a_1|^2 + |a_2|^2 = 1$. A system prepared in this pure state with probability $p = 1$ has density operator

$$\hat{\rho} = |\psi\rangle\langle\psi| = |a_1|^2|\psi_1\rangle\langle\psi_1| + a_1a_2^*|\psi_1\rangle\langle\psi_2| + a_2a_1^*|\psi_2\rangle\langle\psi_1| + |a_2|^2|\psi_2\rangle\langle\psi_2| \ .$$

If $a = b = \frac{1}{\sqrt{2}}$, the components of the density operator in the basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ are

$$\begin{bmatrix} |a_1|^2 & a_1a_2^* \\ a_1^*a_2 & |a_2|^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \ .$$

Mixed state. An ensamble prepared in state $|\psi_1\rangle$ with probability p_1, $p_2 = 1 - p_1$ has density operator

$$\hat{\rho} = p_1|\psi_1\rangle\langle\psi_1| + p_2|\psi_2\rangle\langle\psi_2| \ ,$$

whose components in the $\{|\psi_1\rangle, |\psi_2\rangle\}$ basis are

$$\begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \ .$$

CHAPTER

NINETEEN

DECOHERENCE

19.1 Example: two-dimensional system interacting with a large-dimensional environment

19.1.1 State of the system

A 2-dimensional system is interacting with a large-dimensional environment. The state of the 2-dimensional system is identified by the wave function

$$|\Psi_S\rangle = c_0|0\rangle + c_1|1\rangle \ ,$$

the state of the environment by the wave function $|\Psi_E\rangle$, and the state of the whole system (S+E) by the tensor product of the two,

$$|\Psi_{S+E}\rangle = |\Psi_S\rangle \otimes |\Psi_E\rangle \ .$$

19.1.2 Evolution of the system

The evolution of the system is governed by the Schrodinger equation, with the Hamiltonian of the whole system

$$i\hbar \frac{d}{dt}|\Psi_{S+E}\rangle = \hat{H}_{S+E}|\Psi_{S+E}\rangle \ ,$$

whose evolution can be represented by an unitary operator $\hat{U}_{S+E;t,0}$,

$$|\Psi_{S+E}\rangle_t = U_{S+E;t,0}|\Psi_{S+E}\rangle_0 = \cdots = \alpha|0\rangle \otimes |\Psi_{E;0}\rangle + \beta|1\rangle \otimes |\Psi_{E;1}\rangle$$

Density operator reads

$$\hat{\rho} := |\Psi_{S+E}\rangle\langle\Psi_{S+E}| \ .$$

Tracing out the environment states,

$$\begin{aligned}
\hat{\rho}_E &= \mathrm{Tr}_E\left(\hat{\rho}\right) = \\
&= \sum_k \left(\alpha|0\rangle\langle\phi_k|\Psi_{E;0}\rangle + \beta|1\rangle\langle\phi_k|\Psi_{E;1}\rangle\right)\left(\alpha^*\langle 0|\langle\Psi_{E;0}|\phi_k\rangle + \beta^*\langle 1|\langle\phi_k|\Psi_{E;1}\rangle\right) = \\
&= |\alpha|^2|0\rangle\langle 0|\langle\Psi_{E;0}|\underbrace{\sum_k |\phi_k\rangle\langle\phi_k}_{=\hat{\mathbf{1}}}|\Psi_{E;0}\rangle + \alpha\beta^*|0\rangle\langle 1|\langle\Psi_{E;0}|\underbrace{\sum_k |\phi_k\rangle\langle\phi_k|}_{=\hat{\mathbf{1}}}\Psi_{E;1}\rangle + \\
&\quad + \alpha^*\beta|1\rangle\langle 0|\langle\Psi_{E;1}|\underbrace{\sum_k |\phi_k\rangle\langle\phi_k}_{=\hat{\mathbf{1}}}|\Psi_{E;0}\rangle + |\beta|^2|1\rangle\langle 1|\langle\Psi_{E;1}|\underbrace{\sum_k |\phi_k\rangle\langle\phi_k|}_{=\hat{\mathbf{1}}}\Psi_{E;1}\rangle = \\
&= |\alpha|^2|0\rangle\langle 0| + \alpha\beta^*\langle\Psi_{E;0}|\Psi_{E;1}\rangle|0\rangle\langle 1| + +\alpha^*\beta\langle\Psi_{E;1}|\Psi_{E;0}\rangle|1\rangle\langle 0| + |\beta|^2|1\rangle\langle 1| \ .
\end{aligned}$$

The components of the reduced density operator in the system basis are

$$\begin{bmatrix} |\alpha|^2 & \alpha\beta^* \langle \Psi_{E;0} | \Psi_{E;1} \rangle \\ \alpha^* \beta \langle \Psi_{E;1} | \Psi_{E;0} \rangle & |\beta|^2 \end{bmatrix} .$$

19.1.3 Decoherence time

todo *check and uncomment*

CHAPTER

TWENTY

INTERACTION PICTURE, PERTURBATION THEORY AND EINSTEIN COEFFICIENTS

20.1 Interaction picture

Interaction picture is a common and useful description of a system whose Hamiltonian cna be written as the sum of a term $\hat{H}_0$ that's well known and solvable, and $\hat{H}_1$ that's a perturbation to the system 0, and it's usually either time-dependent and/or hard to solve analytically,

$$\hat{H} = \hat{H}_0 + \hat{H}_1 \, .$$

As $\hat{H}_0$ is time-independent, the corresponding unitary evolution operator is $U^{(0)} = \exp\left[i\frac{\hat{H}_0}{\hbar}t\right]$.

A state vector in interaction picture, $|\Psi_I(t)\rangle$ is defined as

$$|\Psi_I(t)\rangle = U^{(0)\,\dagger}|\Psi_S(t)\rangle \, ,$$

with $|\Psi_S(t)\rangle$ the state vector in Schrodinger picture.

If one requires that the expectation value of operators is the same using different pictures,

$$\begin{aligned}\langle\hat{A}_I\rangle &= \langle\Psi_I|\hat{A}_I|\Psi_I\rangle = \\ &= \langle\Psi_S|\underbrace{U^{(0)}\hat{A}_I U^{(0)\,\dagger}}_{=\hat{A}_S}|\Psi_S\rangle \, ,\end{aligned}$$

it follows that $\hat{A}_I(t) = U^{(0)\,\dagger}(t,t_0)\hat{A}_S U^{(0)}(t,t_0)$. From the commutation of $\hat{H}_0$ with $U^{(0)}$, it follows that it has the same expression in Schrodinger and interaction picture, $H_I^{(0)} = H_S^{(0)}$.

Time evolutions.

Time evolution of a state $|\Psi_I(t)\rangle$

$$i\hbar\frac{d}{dt}|\Psi_I\rangle = \hat{H}_I^{(1)}|\Psi_I\rangle \, .$$

Details

$$
\begin{aligned}
i\hbar\frac{d}{dt}|\Psi_I\rangle &= i\hbar\frac{d}{dt}\left[U^{(0)\,\dagger}(t,t_0)|\Psi_S\rangle\right] = \\
&= i\hbar\left[\partial_t U^{(0)\,\dagger}(t,t_0)|\Psi_S\rangle + U^{(0)\,\dagger}\frac{d}{dt}|\Psi_S\rangle\right] = \\
&= -U^{(0)\,\dagger}\hat{H}_0|\Psi_S\rangle + U^{(0)\,\dagger}\left(\hat{H}^{(0)}+\hat{H}^{(1)}\right)|\Psi_S\rangle = \\
&= \underbrace{U^{(0)\,\dagger}\hat{H}^{(1)}U^{(0)}}_{\hat{H}_I^{(1)}}\ \underbrace{U^{(0)\,\dagger}|\Psi_S\rangle}_{|\Psi_I\rangle} = \\
&= \hat{H}_I^{(1)}|\Psi_I\rangle\ .
\end{aligned}
$$

recalling that $\hat{H}^{(0)} = i\hbar\,\partial_t U^{(0)}U^{(0)\,\dagger} = -i\hbar U^{(0)}\partial_t U^{(0)\,\dagger}$, and that the Hamiltonian operator is Hermitian, $\hat{H} = \hat{H}^\dagger$.

Time evolution of an operator $\hat{A}_I(t)$

Details

20.2 Perturbation theory

For an isolated system with time-independent Hamiltonian operator $\hat{H}_0$, the state of the system can be written as a linear combination of the eigenstates of the Hamiltonian operator

$$|\Psi(t)\rangle = |n\rangle a_n(t)\ ,$$

with $a_n(t) = a_n(0)\exp\left(-i\frac{E_n}{\hbar}t\right) = \langle n|\Psi(0)\rangle\exp\left(-i\frac{E_n}{\hbar}t\right)$.

Time varying perturbation theory. The state of the perturbed system is can be written as a linear combination of the eigenstates of the unperturbed Hamiltonian as well - here explicitly writing a factor $e^{-iE_nt/\hbar}$ in the coefficient,

$$|\Psi(t)\rangle = |n\rangle c_n(t)e^{-iE_nt/\hbar}\ ,$$

so that for unperturbed systems $c_n(t) = a_{n,0}$ constant. Inserting this expression in the Schrodinger equation of the system provides dynamical equations for the coefficients $c_n(t)$, i.e.

$$
\begin{aligned}
0 &= -i\hbar\frac{d}{dt}|\Psi\rangle + \left(\hat{H}_0+\hat{H}_1(t)\right)|\Psi\rangle = \\
&= \sum_n\left[(-i\hbar\dot{c}_n - c_n\,E_n + c_n\,E_n)\,|n\rangle + \hat{H}_1|n\rangle c_n\right]e^{-iE_nt/\hbar} \\
&= \sum_n\left[-i\hbar\dot{c}_n|n\rangle + \hat{H}_1|n\rangle c_n\right]e^{-iE_nt/\hbar}\ .
\end{aligned}
$$

Projecting on $\langle m|$, a system of an infinite number of equations follows

$$\dot{c}_m = -\frac{i}{\hbar}\sum_n\langle m|\hat{H}_1|n\rangle c_n\,e^{i\frac{E_m-E_n}{\hbar}t}\ .$$

20.3 Einstein coefficients

Perturbation theory is applied to an electron around a steady positive nucleus, subject to time-varying electric field. The classical counterpart of the governing equation of a point charge subject to Coulomb potential and a travelling wave in the electric field

$$\mathbf{e}(\mathbf{r},t) = 2\mathbf{E}_0 \cos(\omega t - \mathbf{k}\cdot\mathbf{r}) = q\mathbf{E}_0(e^{i(\omega t - \mathbf{k}\cdot\mathbf{r})} + c.c.) \ ,$$

reads

$$m\ddot{\mathbf{r}} + \frac{q^2}{4\pi\varepsilon}\frac{\mathbf{r}}{|\mathbf{r}|^3} = 2q\mathbf{E}_0 \cos(\omega t - \mathbf{k}\cdot\mathbf{r}) \ .$$

Dipole approximation assumes that the displacement of the charged particle is small compared to the wave-length of the radiation, $\mathbf{k}\cdot\mathbf{r} = \frac{\hat{\mathbf{k}}\cdot\mathbf{r}}{2\pi\lambda} \ll 1$, and thus the dynamic equation becomes

$$m\ddot{\mathbf{r}} + \frac{q^2}{4\pi\varepsilon}\frac{\mathbf{r}}{|\mathbf{r}|^3} = 2q\mathbf{E}_0 \cos(\omega t) \ .$$

The total force is the sum of the Coulomb force and the force due to the incoming electric field. This sum can be written as a gradient of a scalar function, a potential V,

$$\begin{aligned}\mathbf{F} &= -\frac{q^2}{4\pi\varepsilon}\frac{\mathbf{r}}{|\mathbf{r}|^3} + 2q\mathbf{E}_0\cos(2\pi\,\nu\,t) = \\ &= \nabla_{\mathbf{r}}\left[\frac{q^2}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|} + 2q\mathbf{r}\cdot\mathbf{E}_0\cos(2\pi\,\nu\,t)\right] = \\ &= -\nabla_{\mathbf{r}}\left(V_0(\mathbf{r}) + V_1(\mathbf{r},t)\right) \ .\end{aligned}$$

that can be written as the stationary potential V_0 from Coulomb force and the time-varying perturbation $V_1(\mathbf{r},t)$. The vector $q\mathbf{r}$ is usually defined as the **electric dipole**.

Schrodinger equation follows from the promotion of the Hamiltonian function $H = K + V$ to the Hamiltonian operator

$$\hat{H} = \hat{H}_0 + \hat{H}_1(t) = \hat{H}_0 + 2\mathbf{E}_0\cdot q\hat{\mathbf{r}}\,\cos(2\pi\nu t) \ .$$

CHAPTER

TWENTYONE

HISTORICAL APPROACH TO QUANTUM MECHANICS

Contents

- *Timelime* and *topics*
- *Origin of matrix mechanics*
- *Origin of wave mechanics*

References

- JK[0] YouTube channel

21.1 Topics

Atomic models.

- 1891, J.J.Thomson find electrons in cathodic rays. He found out that cathodic rays are made of negatively charged particles[1], deflected by a transverse electromagnetic field. He:
  - evaluated the ratio $\frac{\text{mass}}{\text{el.charge}}$ of these particles. See Scoperta dell'elettrone - J.J.Thomson
  - proposed the pudding atomic model, with negatively charged particles in a positive cloud.
- 1909, Rutherford scattering experiment and Rutherford atomic model: electrons around a compact nucleus
- 1913, Bohr…
- 1916, Sommerfeld
  - relativistic model
  - elliptial orbits
  - introduced another quantum number and degeneracy
  - describe energy level split by magnetic field (Zeeman effect)
- 1925, Heisenberg and Schrodinger first theories about QM; Pauli solution of the H atom system

Spectroscopy.

- 1707, Newton "Optics" described his experiments about diffraction of light with glass prisms

[1] While J.J.Thomson won the Nobel prize in Physics (1906) for having found the first sub-atomic particle, the electron, his son G.P.Thomson won the Noble prize (1937) for its discovery of the diffraction of electrons by crystals in 1927, thus showing wave behavior. Particle-wave duality never ends…

- In the second half of the XIX century, experimental studies on the emission/absorption spectra in chemistry. Each element showed characteristics black lines in otherwise continuous absoprtion spectra (and corresponding bright lines in emission spectra, "hot spectra")
- Empirical laws for the refractive index followed:
  - 1836, Cauchy proposed a relation between the refractive index n of a material and the wave-length of the radiation

    $$n^2 = a + \frac{b}{\lambda^2} + \frac{c}{\lambda^4} + \dots \ ,$$

    with material-dependent parameters a, b, c,...This model proposed $n(\lambda)$ as a continuous function
  - 1870, Christiansen observed anomalous dispersion, with apparently discountinuous refractive index at wave lengths that are characteristics of the element under investigation
  - 1872, Sellmeier proposed that anomalous dispersion should be described in the context of light-matter interaction, in terms of resonance modes of the matter and proposed his law for the refractive index

    $$n^2 - 1 = \sum_k \frac{a_k \lambda_k^2}{\lambda^2 - \lambda_k^2}$$

  - 1875, Helmholtz modified Sellmeier formula, introducing a (wrong, this was not a dampening factor, as the one shown by Drude-Lorentz model) dampening factor.
- *Drude-Lorentz model*. Following the idea of Sellmeier, P.Drude (1900) and then H.Lorentz (1905) proposed a theoretical model for the refractive index of a medium, using a second-order damped oscillator, forced by the electric field of an incoming electromagnetic wave, and evaluated the transmitted wave as the sum of the incident wave and the one generated by the oscillating charges of the matter.
- Generalization of the Drude-Lorentz model for systems with multiple modes became the general semi-empirical classical model for light dispersion

  $$n(\omega) - 1 = \frac{q^2}{2\varepsilon_0 m} \sum_k \frac{N_k}{\omega_k^2 - \omega^2 + i\gamma_k \omega_k}$$

  or

  $$n^2(\omega) - 1 \simeq 2\delta.$$

- Dispersion becomes quantum:
  - 1905, Einstein explanation of photoelectric effect
  - 1906-1908, **Ladenburg** and Loria began experiments about spectral lines, anomalous dispersion, and the microstructure of the light-matter interaction. They found a way to measure the number density of the electrons responsible for spectral lines in absorption spectra in H atom (H_α line only, while they had not enough accuracy for other weaker lines). Only few electrons resonate while the majority appear to be unaffected. They then focused on Na for better resolution. They came back to H few years later (1911), to find $\frac{N_{H_\alpha}}{N_H} \sim \frac{1}{50.000}$, $\frac{N_{H_\beta}}{N_H} \sim \frac{1}{200.000}$.
  - 1913, Bohr model (incompatible with classical mechanics, no e^- fast collapse on the nucleus, due to classical radiation):
    - circular stable orbits with quantized angular momentum nh
    - lines are generated by quantum jumps, with $\nu = \frac{\Delta E}{h}$
    - results compatible with the empirical Rydberg formula (1888)

  - Ladenburg: how do both Bohr model and Drude-Lorentz model work so well in atomic spectra?
  - 1914, H.Kohn: experimental confirmed radiation emitted by incandescent bodies can be treated as thermal radiation. Ladenburg then used thermodynamics to explain thermal radiation.
  - Still Ladenburg
    * 1914, procedure to compute the density of dispersion electrons
    * 1921, quantum counterpart of the classical dispersion, and link with Einstein coefficients (1916)
      · the **physical atom is replaced by a representation of oscillators**: only frequencies and intensities of spectral lines can be observed. They abandoned the idea of building a "physical model of the atom" following the electron along an orbit, and built a model that's just compatible with the phhysical quantities they can measure
  - 1924, **Kramers** (with theoretical derivation from Born, 1924, and Kramers and Heisenberg, 1925):
    * rule for **polarization** with transition frequencies (both to higher and lower energy levels, modeling both emission and absorption)
    * Kramers formula and the idea of virtual oscillators are used by Heisenberg in his formulation of matrix mechanics: this formula set the origin for the non-commutativity of physical quantities and, thus, Heisenberg uncertainty principle
    * tested by Ladenburg, 1928

Radiation and Light-matter interaction.

- 1896, Wien's law
- 1900, Planck described the **continuous spectrum** of black-body radiation, introducing of h, and the idea that light radiation with frequency ν have discrete amount of energy, $h\nu$. He first introduced h as a trick for matching experimental results of the spectrum $\rho(\nu, T)$; a model of statistical mechanics of radiation with discrete levels of energy followed, see *Black-body ratiation using Bose-Einstein statistics*, even though that was the original way Planck first found a theoretical model
- 1905, Einstein explained the photolectric effects using Planck constant h.
- 1916, Einstein
  - derived Planck's law from Bohr's atomic model, using a (quantum) statistical approach to the interaction of atoms (and thier electrons) and electromagnetic radiation in a box. He postulated the existence of three processes, introducing proportionality **Einstein coefficients**, for transition probability (of a e^-) between states: *absorption* (rate: $B_{kj}N_k\rho(\nu)$) *spontaneous emission* (rate: $-A_{jk}N_j$), and **stimulated emission** (rate: $-B_{jk}N_j\rho(\nu)$). Stimulated emissions was found to be necessary in order for this model to be compatible with Planck's law $\rho \propto \frac{1}{e^{h\nu/kT}-1}$ for the system at thermal equilibrium (and not with Wien's law $\rho(\nu) \propto e^{-h\nu/kT}$, the law one would get without stimulated emission)
  - Einstein couldn't calculate his coefficients but he was positive that they could be computed once a theory about the atom had been be established
  - recognized that light quanta have momentum, $p = \hbar k = h\frac{\nu}{c}$, then found by Compton (1923) in experiments about X-rays scattering
  - in 1927, Dirac calculated Einstein coefficients (The Quantum Theory of Emission and Absorption of Radiation)

Origin of quantum mechanics.

- **Matrix mechanics.** (Heisenberg, Born, Jordan), three papers in 1925
- **Ondulatory mechanics.** (Schrodinger)
  - 1923, L.de Broglie: idea of guiding wave for matter particles

  - 1925, E.Schrodinger: wave equation. If matter behaves as a wave, a wave equation should be the governing equation of the system
  - 1923-1927, C.Davisson and L.Germer found a diffraction pattern in an experiment about electron scattering
- **Equivalence.** (Dirac)

21.2 Origins of Matrix and Wave Quantum Mechanics

21.2.1 Origins of Matrix Mechanics

Quantization of radiation energy.

- Planck (1900), Einstein photoelectric effect (1905) and heat capacity of the solids: introduction and evidences of h, $E = h\nu$

$$u(\nu, T) = ... \tag{21.1}$$

Light-matter interaction and light dispersion.

- First experimental evidences: Cauchy; Sellmeier (1872) proposed an empirical law for the refraction index n taking into account resonances of matter; *Lorentz-Drude* (1900-1905) provided a theoretical model of Sellmeier formula, using Maxwell equations for electromagentism,

$$n - 1 = \frac{q^2 N}{2\varepsilon m} \frac{1}{\omega_0^2 - \omega^2 + i\gamma\omega} \approx \frac{q^2 N}{2\varepsilon m} \frac{1}{\omega_0^2 - \omega^2} \ , \tag{21.2}$$

  being m and q the mass and the charge of a particle, N the number density of the charged particles, ω_0 the natural frequency and γ a damping coefficient of the dynamical equation governing the dynamics of the particle, $m\ddot{x} + m\gamma\omega\dot{x} + m\omega_0^2 x = qE(t)$.

- Einstein (1916): quantum theory of the radiation interacting with matter.
  - Statistics of the equilibrium of radiation and matter, with the processes of absorption, spontaneous emission and stimulated emissions. He introduced Einstein coefficiens as proportionality constants
  - E. theory connected Bohr's atomic with Planck's radiation formula (21.1): matching Planck's formula induces relationss between Einstein coefficients,

$$\frac{A_{jk}}{B_{kj}} \frac{g_j}{g_k} = \frac{8\pi\nu^3}{c^3} \qquad , \qquad \frac{B_{jk}}{B_{kj}} \frac{g_j}{g_k} = 1 \ .$$

    Einstein didn't know how to compute A_{ik}, B_{jk}, B_{kj}, but he was confident that it would have been possible to do so, once a theory of quanta was established.

- Ladenburg (1914-24)
  - a material may have more than one resonance. The generalization of the formula (21.2) for the refractive index n from Lorentz-Drude model reads

$$n - 1 = \frac{q^2}{2\varepsilon m} \sum_k \frac{N_k}{\omega_k^2 - \omega^2} \ ,$$

    with N_k the "number density of particles"[1] with natural frequency $f_k = \dfrac{\omega_k}{2\pi}$.

[1] This should be interpreted as a weight representing the contribution of mode k.

– But what's the meaning of N_k? Ladenburg compared the averaged emitted power from ***Larmor's formula*** (21.4) - the classical model - and the emitted power by ***Planck's formula***, or by ***Einstein quantum theory of interaction of radiation and matter*** - the quantum model, for a frequency ν_{ij}. The power emitted by R resonators per unit volume is

…

Dimensional analysis

$$= \frac{\text{power}}{\text{n.parts}}$$
$$[R] = \frac{\text{n.parts}}{\text{length}^3}$$
$$[\langle P \rangle] = \frac{\text{power}}{\text{length}^3}$$
$$\rho(\nu) = ...$$

–>

Particle-wave duality.

- Einstein (1909): radiation has wave and particle behavior

Atomic models.

- Bohr atomic model (1913).

21.3 Timeline

21.4 Origin of matrix mechanics

Experiences:

- Quantization in light-matter interaction:
  - 1900, Planck radiation law: h as a mathematical trick to fit experimental data of black-body radiation; thermodynamics model followed
  - 1905, Einstein work on photoelectric effect: h is not a mathematical trick
- Light dispersion, as a light-matter interaction process:
  - 1900, Lorentz-Drude model for classical dispersion
  - 1913, Bohr atomic model
  - 1917, Einstein derived Planck law from thermodynamic equilibrium between atoms and radiation
  - 1914-24, Ladenburg work on quantization of dispersion law, matching Lorentz-Drude (classical model) and Bohr-Einstein (quantum model) in the classical limit
- Heisenberg matrix mechanics

- …
- *Modern reinterpretation of Heisenberg mechanics*

21.4.1 Light dispersion

First experiences

Classical dispersion

Drude-Lorentz model

…

$$n(\omega) - 1 = \frac{q^2 N}{2\varepsilon_0 m} \frac{1}{\omega_0^2 - \omega^2 + i\gamma\omega}$$

Remark. The refractive index n is a complex number, with

$$\begin{aligned} \mathrm{re}\{n\}(\omega) &= 1 + \Delta n_0 \omega_0^2 \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega)^2 + \gamma^2\omega^2} \\ \mathrm{im}\{n\}(\omega) &= -\Delta n_0 \omega_0^2 \frac{\gamma\omega}{(\omega_0^2 - \omega)^2 + \gamma^2\omega^2} \ \leq 0 \ , \quad \forall \omega \geq 0 \end{aligned}$$

Here the condition $\omega \geq 0$ is required by the condition that the incoming wave travels from left to right.

Thus, a travelling wave

$$\varphi(x,t) = e^{i(\omega t - kx)} = e^{i\omega(t - \frac{x}{c})} = e^{i\omega\left(t - n\frac{x}{c_0}\right)} \ ,$$

thus contains both a oscillating and a dampening contribution by $n = \mathrm{re}\{n\} + i\,\mathrm{im}\{n\}$,

$$\varphi(x,t) = e^{\frac{\omega}{c_0}\mathrm{im}\{n(\omega)\}x} e^{i\omega\left(t - \mathrm{re}\{n(\omega)\}\frac{x}{c_0}\right)}$$

As the imaginary part of the refractive index is negative for all the positive frequencies, than the real exponential represents a damping term. The minimum value of $\mathrm{im}\{n\}(\omega)$ for a slightly damped second-order oscillator occurs approximately at the natural frequency, $\omega_{maxdamp} \simeq \omega_0$. This represents an **absorption of radiation** by the matter.

Remark. Drude-Lorentz model in the form $n - 1 = \delta$ is compatible with empirical formulas in the form $n^2 - 1 = \tilde{\delta}$, as the correction is "small enough" for a linear approximation

$$n^2 = (1 + \delta)^2 \simeq 1 + 2\delta \ .$$

Einstein

Ladenburg

Ladenburg wrote the energy balance at thermodynamical equilibrium, from transitions between each pair of states i, $k > i$, using **Einstein's coefficients** from his *quantum theory of radiation interacting with matter*. The total amount of energy emitted by N_k molecules in the state k transitioning to a lower state i is

$$J = \underbrace{h\nu_{ik}}_{\Delta E_{ik}} N_k \left[A_{ki} + B_{ki} u(\nu_{ik})\right] \ .$$

At thermal equilibrium, this energy is equal to the energy absorbed by N_i molecules in the state i,

$$A = h\nu_{ik} N_i B_{ik} u(\nu_{ik}) \ .$$

Exploiting the relations between Einstein coefficients (21.3), Ladenburg wrote the emitted and absorbed energy as a function of the spontaneous emission foefficient A_{ki} (allowing to relate dispersion with other phenomena involving emission),

$$J = A = h\nu_{ik} u(\nu_{ik}) N_i B_{ik} \ .$$

21.4.2 Light matter interaction

Einstein: quantum theory of radiation interacting with matter

Three mechanisms, whose probabilities are

- Spontaneous emission of radiation, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt}\right)_{spont} = A_{ji} n_j \ ,$$

- Stimulated emission of radiation, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt}\right) = B_{ji} n_j \rho(\nu_{ji}) \ ,$$

  with $\nu_{ji} = \frac{E_j - E_i}{h}$, and $\rho(\nu)$ the density of radiation in the system at frequency ν

- Absorption, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt}\right) = -B_{ij} n_i \rho(\nu_{ij}) \ .$$

The overall rate reads

$$\frac{dn_i}{dt} = \sum_{j, E_j > E_i} \frac{dn_{ij}}{dt} = \sum_{j, E_j > E_i} \left\{ A_{ji} n_j + B_{ji} n_j \rho(\nu_{ij}) - B_{ij} n_i \rho(\nu_{ij}) \right\}$$

At thermodynamic equilibriumm between all the states

$$0 = \frac{dn_{ij}}{dt} = A_{ji} n_j + B_{ji} n_j \rho(\nu_{ij}) - B_{ij} n_i \rho(\nu_{ij})$$

Using Boltzmann distribution

$$\frac{n_i}{n} = \frac{g_i e^{-\frac{E_i}{kT}}}{Z} \ ,$$

the expression of the radiation density follows

$$\begin{aligned}
\rho(\nu_{ij}) = \frac{A_{ji} n_j}{n_i B_{ij} - n_j B_{ji}} = \\
= \frac{A_{ji} g_j}{B_{ij} g_i} \frac{\exp(-E_j/kT)}{\exp(-E_i/kT) - \frac{g_j B_{ji}}{g_i B_{ij}} \exp(-E_j/kT)} = \\
= \frac{A_{ji} g_j}{B_{ij} g_i} \frac{1}{\exp((E_j - E_i)/kT) - \frac{g_j B_{ji}}{g_i B_{ij}}} = \\
= \frac{A_{ji} g_j}{B_{ij} g_i} \frac{1}{\exp(h\nu_{ji}/kT) - \frac{g_j B_{ji}}{g_i B_{ij}}} \ .
\end{aligned}$$

Comparing with Planck's law (with the **radiance**?) $\rho(\nu,T) = \frac{2h\nu^3}{c^3}\frac{1}{\exp(h\nu/kT)-1}$, the relations between Einstein coefficients ($E_j > E_i$) follows

$$\frac{g_j A_{ji}}{g_i B_{ij}} = \frac{2h\nu_{ji}^3}{c^3} \quad , \quad \frac{g_j B_{ji}}{g_i B_{ij}} = 1 \, . \tag{21.3}$$

Planck's law

- Spectral energy density (energy per unit volume, per unit frequency):

$$u_\nu(\nu,T) = \frac{8\pi h\nu^3}{c^3}\frac{1}{\exp(h\nu/kT)-1}$$

  Dimensional analysis:

$$[u] = \frac{[h][\nu]^3}{[c]^3} = \frac{Js\cdot s^{-3}}{m^3 c^{-3}} = \frac{J}{m^3\cdot \text{Hz}} = \frac{\text{energy}}{\text{length}^3\cdot \text{freq.}} \, .$$

- Radiance, $B_\nu(\nu,T) = \frac{c}{4\pi}u_\nu(\nu,T)$ **todo** *Justify this relation. Is this right? Or should the factor be* $\frac{c}{8\pi}$[1]. Thus

$$B_\nu(\nu,T) = \frac{2h\nu^3}{c^2}\frac{1}{\exp(h\nu/kT)-1}$$

Dimensional analysis

$$[B] = \frac{\text{power}}{\text{solid angle}\cdot\text{surface}\cdot\text{Hz}}$$

21.4.3 Radiation of moving charges - Larmor's formula

Green's function solution of wave equations: Liénard-Wiechert potential

Wave equations in electromagmetism. See Classical Electromagnetism: Wave Equations in Electromagnetism. Using **Lorentz gauge**, $\nabla\cdot\mathbf{a} + \frac{1}{c^2}\partial_t\varphi = 0$, the wave equations for the mathbftor potential $\mathbf{a}$ and the scalar potential φ read

$$\frac{1}{c^2}\partial_{tt}\mathbf{a} - \Delta\mathbf{a} = \mu\mathbf{j}$$
$$\frac{1}{c^2}\partial_{tt}\varphi - \Delta\varphi = \frac{\rho}{\varepsilon}$$

Solution of wave equation with Green's function

The solution of a wave equation

$$\frac{1}{c^2}\partial_{tt}u - \Delta u = f \, ,$$

[1] Taltavull, Rudolf Ladenburg and the first quantum interpretation of optical dispersion, Eur. Phys. J. H.

can be written using Math: Green's function method (with assumptions of causality, and no contribution from i.c. and b.c.),

$$u(\mathbf{r},t) = \frac{1}{4\pi}\int_{t'\in T}\int_{\mathbf{r}'\in V}\frac{\delta\left(t'-t+\frac{|\mathbf{r}'-\mathbf{r}|}{c}\right)}{|\mathbf{r}'-\mathbf{r}|}f(\mathbf{r}',t')d\mathbf{r}'dt' = \\ = \frac{1}{4\pi}\int_{\mathbf{r}'\in V}\frac{1}{|\mathbf{r}'-\mathbf{r}|}f\left(\mathbf{r}',t_{ret}(\mathbf{r},t;\mathbf{r}')\right)d\mathbf{r}'$$

with the retarded time $t_{ret}(\mathbf{r},t;\mathbf{r}') = t - \frac{|\mathbf{r}'-\mathbf{r}|}{c}$.

The solution of the wave equations reads

$$\mathbf{a}(\mathbf{r},t) = \frac{\mu}{4\pi}\int_{\mathbf{r}'\in V}\frac{1}{|\mathbf{r}'-\mathbf{r}|}\mathbf{j}\left(\mathbf{r}',t_{ret}(\mathbf{r},t;\mathbf{r}')\right)d\mathbf{r}' \\ \varphi(\mathbf{r},t) = \frac{1}{4\pi\varepsilon}\int_{\mathbf{r}'\in V}\frac{1}{|\mathbf{r}'-\mathbf{r}|}\rho\left(\mathbf{r}',t_{ret}(\mathbf{r},t;\mathbf{r}')\right)d\mathbf{r}'$$

with the retarded time $t_{ret}(\mathbf{r},t;\mathbf{r}') = t - \frac{|\mathbf{r}'-\mathbf{r}|}{c}$.

Point charge. The charge density and the current density of a point charge with electrical charge q, position $\mathbf{r}_s(t)$ and velocity $\mathbf{v}_s = \dot{\mathbf{r}}_s$ are respectively

$$\rho(\mathbf{r},t) = q\,\delta(\mathbf{r}-\mathbf{r}_s(t)) \\ \mathbf{j}(\mathbf{r},t) = q\mathbf{v}_s(t)\,\delta(\mathbf{r}-\mathbf{r}_s(t))$$

Thus, the electromagnetic potentials generated by a point charge are

$$\mathbf{a}(\mathbf{r},t) = \frac{\mu}{4\pi}\frac{1}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}q\mathbf{v}_s(t_{ret}) \\ \varphi(\mathbf{r},t) = \frac{1}{4\pi\varepsilon}\frac{1}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}q$$

Derivatives of the retarded time

$$\partial_t t_{ret} = \frac{1}{1-\hat{\mathbf{r}}\cdot\beta} \\ \nabla t_{ret} = -\frac{1}{c}\frac{\hat{\mathbf{r}}}{1-\hat{\mathbf{r}}\cdot\beta}\,.$$

Details

Time derivative

$$\partial_t t_{ret}(\mathbf{r},t,\mathbf{r}_s(t)) = \partial_t\left(t - \frac{|\mathbf{r}_s(t_{ret}(\mathbf{r},t,\mathbf{r}_s(t_{ret}))-\mathbf{r}|}{c}\right) = \\ = 1 - \frac{1}{c}\frac{d|\mathbf{r}_s-\mathbf{r}|}{dt_{ret}}\partial_t t_{ret} = \\ = 1 + \frac{x_k - x_{s,k}}{|\mathbf{r}_s-\mathbf{r}|}\frac{v_{s,k}}{c}\partial_t t_{ret} = \\ = 1 + \hat{\mathbf{r}}\cdot\beta\,\partial_t t_{ret}\,,$$

and thus

$$\partial_t t_{ret} = \frac{1}{1-\hat{\mathbf{r}}\cdot\beta}\,.$$

Space derivatives

$$\begin{aligned}\partial_k t_{ret}(\mathbf{r},t,\mathbf{r}_s(t)) &= \partial_k\left(t-\frac{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}{c}\right)= \\ &= -\frac{1}{c}\partial_k|\mathbf{r}_s(t_{ret})-\mathbf{r}|= \\ &= \frac{1}{c}\frac{r_i-r_{s,i}(t_{ret})}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}\partial_k(r_{s,i}(t_{ret})-r_i)= \\ &= \frac{1}{c}\frac{r_i-r_{s,i}(t_{ret})}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}\left(v_{s,i}(t_{ret})\partial_k t_{ret}-\delta_{ik}\right)= \\ &= \hat{\mathbf{r}}\cdot\beta\,,\partial_k t_{ret}-\frac{1}{c}\frac{r_k-r_{s,k}(t_{ret})}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}\,,\end{aligned}$$

so that

$$\partial_k t_{ret}=-\frac{1}{c}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}\frac{r_k-r_{s,k}(t_{ret})}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}$$

or, using vector notation,

$$\nabla t_{ret}=-\frac{1}{c}\frac{\hat{\mathbf{r}}}{1-\hat{\mathbf{r}}\cdot\beta}\,.$$

Derivatives of $|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{-1}$

$$\partial_t|\mathbf{r}-\mathbf{r}_s(t_{ret})|^n=-n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-1}\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}c$$

$$\partial_k|\mathbf{r}-\mathbf{r}_s(t_{ret})|^n=n|\mathbf{r}_s(t_{ret})-\mathbf{r}|^{n-1}\left[\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}+1\right]\hat{r}_k$$

Details

$$\begin{aligned}\partial_k|\mathbf{r}-\mathbf{r}_s(t_{ret})|^n &= \partial_k\left[(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot(\mathbf{r}-\mathbf{r}_s(t_{ret}))\right]^{\frac{n}{2}}= \\ &= n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-2}(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot\partial_k(\mathbf{r}-\mathbf{r}_s(t_{ret}))\end{aligned}$$

Now,

$$\begin{aligned}\partial_t(r_i-r_{s,i}(t_{ret})) &= -\partial_t t_{ret}v_{s,i}(t_{ret})= \\ &= -\frac{v_{s,i}(t_{ret})}{1-\hat{\mathbf{r}}\cdot\beta} \\ \partial_k(r_i-r_{s,i}(t_{ret})) &= \delta_{ik}-\partial_k t_{ret}v_{s,i}(t_{ret})= \\ &= \delta_{ik}+\frac{1}{c}\frac{\hat{r}_k}{1-\hat{\mathbf{r}}\cdot\beta}v_{s,i}(t_{ret})\,,\end{aligned}$$

and thus

$$\begin{aligned}\partial_t|\mathbf{r}-\mathbf{r}_s(t_{ret})|^n &= \partial_t\left[(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot(\mathbf{r}-\mathbf{r}_s(t_{ret}))\right]^{\frac{n}{2}}= \\ &= n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-2}(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot\partial_t(\mathbf{r}-\mathbf{r}_s(t_{ret}))= \\ &= -n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-1}\frac{\hat{\mathbf{r}}\cdot\mathbf{v}_s(t_{ret})}{1-\hat{\mathbf{r}}\cdot\beta}\end{aligned}$$

and

$$\begin{aligned}\partial_k|\mathbf{r}-\mathbf{r}_s(t_{ret})|^n &= \partial_k\left[(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot(\mathbf{r}-\mathbf{r}_s(t_{ret}))\right]^{\frac{n}{2}} = \\ &= n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-2}(\mathbf{r}-\mathbf{r}_s(t_{ret}))\cdot\partial_k(\mathbf{r}-\mathbf{r}_s(t_{ret})) = \\ &= n|\mathbf{r}-\mathbf{r}_s(t_{ret})|^{n-1}\left[\frac{1}{c}\frac{\hat{\mathbf{r}}\cdot\mathbf{v}_s(t_{ret})}{1-\hat{\mathbf{r}}\cdot\beta}+1\right]\hat{r}_k\end{aligned}$$

From potentials to the electromagnetic field

$$\mathbf{e} = -\frac{q}{4\pi\varepsilon}\frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}+\frac{q}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|^2}\left(\frac{\hat{\mathbf{r}}-\beta(\hat{\mathbf{r}}\cdot\beta)}{1-\hat{\mathbf{r}}\cdot\beta}\right).$$

$$\mathbf{b} = \frac{q}{4\pi\varepsilon}\frac{1}{c}\left[-\frac{\hat{\mathbf{r}}\times\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}-\frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2}\times\beta\left(\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right].$$

It's easy to realize that $\mathbf{b}=\frac{1}{c}\hat{\mathbf{r}}\times\mathbf{e}$.

Details

$$\begin{aligned}\mathbf{e} &= -\partial_t\mathbf{a}-\nabla\varphi = \\ &= -\partial_t\left(\frac{\mu}{4\pi}\frac{q\mathbf{v}_s(t_{ret})}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}\right)-\nabla\left(\frac{1}{4\pi\varepsilon}\frac{q}{|\mathbf{r}_s(t_{ret})-\mathbf{r}|}\right) = \\ &= -\frac{q\mu}{4\pi}\left[\frac{1}{|\mathbf{r}|}\mathbf{a}_s(t_{ret})\,\partial_t t_{ret}+\mathbf{v}_s|\mathbf{r}|^{-2}\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}c\right]-\frac{q}{4\pi\varepsilon}|\mathbf{r}|^{-2}\left(-1-\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\hat{\mathbf{r}} = \\ &= -\frac{q\mu c^2}{4\pi}\left[\frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}+\frac{\beta}{|\mathbf{r}|^2}\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right]+\frac{q}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|^2}\left(1+\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\hat{\mathbf{r}} = \\ &= -\frac{q}{4\pi\varepsilon}\frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}+\frac{q}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|^2}\left(\frac{-\beta(\hat{\mathbf{r}}\cdot\beta)+\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\beta)}{1-\hat{\mathbf{r}}\cdot\beta}+\hat{\mathbf{r}}\right) = \\ &= -\frac{q}{4\pi\varepsilon}\frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}+\frac{q}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|^2}\left(\frac{\hat{\mathbf{r}}-\beta(\hat{\mathbf{r}}\cdot\beta)}{1-\hat{\mathbf{r}}\cdot\beta}\right).\end{aligned}$$

$$\begin{aligned}\mathbf{b} &= \nabla\times\mathbf{a} = \\ &= \nabla\times\left(\frac{\mu}{4\pi}\frac{q\mathbf{v}_s(t_{ret})}{|\mathbf{r}-\mathbf{r}_s(t_{ret})|}\right) = \\ &= \hat{\mathbf{e}}_k\varepsilon_{klm}\partial_l\left(\frac{\mu}{4\pi}\frac{qv_{s,m}(t_{ret})}{|\mathbf{r}-\mathbf{r}_s(t_{ret})|}\right) = \\ &= \hat{\mathbf{e}}_k\varepsilon_{klm}\frac{\mu q}{4\pi}\partial_l\left(\frac{v_{s,m}(t_{ret})}{|\mathbf{r}-\mathbf{r}_s(t_{ret})|}\right) = \\ &= \hat{\mathbf{e}}_k\varepsilon_{klm}\frac{\mu q}{4\pi}\left[\frac{a_{s,m}(t_{ret})\partial_l t_{ret}}{|\mathbf{r}-\mathbf{r}_s(t_{ret})|}-v_{s,m}(t_{ret})\frac{\hat{r}_l}{|\mathbf{r}|^2}\left(1+\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right] = \\ &= \hat{\mathbf{e}}_k\varepsilon_{klm}\frac{\mu q}{4\pi}\left[-\frac{a_{s,m}(t_{ret})}{|\mathbf{r}-\mathbf{r}_s(t_{ret})|}\frac{1}{c}\frac{\hat{r}_l}{1-\hat{\mathbf{r}}\cdot\beta}-v_{s,m}(t_{ret})\frac{\hat{r}_l}{|\mathbf{r}|^2}\left(1+\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right] = \\ &= \frac{q\mu c^2}{4\pi}\frac{1}{c^2}\left[-\frac{\hat{\mathbf{r}}\times\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}-\frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2}\times\mathbf{v}_s(t_{ret})\left(1+\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right] = \\ &= \frac{q}{4\pi\varepsilon}\frac{1}{c}\left[-\frac{\hat{\mathbf{r}}\times\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}-\frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2}\times\beta\left(1+\frac{\hat{\mathbf{r}}\cdot\beta}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right] = \\ &= \frac{q}{4\pi\varepsilon}\frac{1}{c}\left[-\frac{\hat{\mathbf{r}}\times\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2}\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}-\frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2}\times\beta\left(\frac{1}{1-\hat{\mathbf{r}}\cdot\beta}\right)\right].\end{aligned}$$

Larmor formula

Poynting vector reads

$$\mathbf{s}(\mathbf{r},t) := \frac{\mathbf{e}(\mathbf{r},t)\times\mathbf{b}(\mathbf{r},t)}{\mu}\ ,$$

and thus, using the electromagnetic field of a moving charge,

$$\mathbf{s} = \frac{1}{\mu}\mathbf{e}\times\mathbf{b} = \frac{1}{c\mu}\mathbf{e}\times(\hat{\mathbf{r}}\times\mathbf{e}) = \frac{1}{\mu c}\left[|\mathbf{e}|^2\hat{\mathbf{r}} - (\mathbf{e}\cdot\hat{\mathbf{r}})\mathbf{e}\right]\ .$$

Far-field approximation (also assuming that $\beta\sim\mathbf{0}$. **todo** *slow charge? Non-relativistic limit?*),

$$\mathbf{e}\sim -\frac{q}{4\pi\varepsilon}\frac{\mathbf{a}_{s,ret}}{|\mathbf{r}|c^2}$$

and thus

$$\begin{aligned}\mathbf{s} &= \frac{1}{\mu c}\left(\frac{q}{4\pi\varepsilon|\mathbf{r}|c^2}\right)^2\mathbf{a}\times(\hat{\mathbf{r}}\times\mathbf{a}) = \\ &= \frac{1}{\mu c}\left(\frac{q}{4\pi\varepsilon|\mathbf{r}|c^2}\right)^2(|\mathbf{a}|^2\hat{\mathbf{r}} - (\mathbf{a}\cdot\hat{\mathbf{r}})\mathbf{a}) =\end{aligned}$$

The elementary power flux at $\mathbf{r} = R\hat{\mathbf{r}}$, assuming that the motion is ina region much smaller than R, so that $t_{ret} = t - \frac{R}{c}$, is

$$\hat{\mathbf{r}}\cdot\mathbf{s} = \frac{1}{\mu c}\left(\frac{q}{4\pi\varepsilon Rc^2}\right)^2(|\mathbf{a}|^2 - (\mathbf{a}\cdot\hat{\mathbf{r}})^2) =$$

Let's define the direction of $\mathbf{a}_{ret} = a\hat{\mathbf{t}}$, and sperical coordinates so that $\hat{\mathbf{r}}\cdot\mathbf{t} = \cos\theta$. Integration over the sphere of radius R gives

$$\begin{aligned}\Phi &:= \int_S \hat{\mathbf{r}}\cdot\mathbf{s} = \\ &= \int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi}\frac{1}{\mu c}\left(\frac{q}{4\pi\varepsilon Rc^2}\right)^2(1-\cos^2\theta)^2|\mathbf{a}|^2R^2\sin\theta\,d\theta\,d\phi = \\ &= \frac{1}{\mu c}\left(\frac{q|\mathbf{a}|}{4\pi\varepsilon c^2}\right)^2\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi}(1-\cos^2\theta)\sin\theta\,d\theta\,d\phi = \\ &= \frac{1}{\mu c}\left(\frac{q|\mathbf{a}|}{4\pi\varepsilon c^2}\right)^2\cdot\frac{4}{3}\cdot 2\pi = \\ &= \frac{q^2|\mathbf{a}_{s,ret}|^2}{6\pi\varepsilon c^3}\ .\end{aligned}$$

Average power.

$$\langle\Phi\rangle = \frac{q^2}{6\pi\varepsilon c^3}\langle|\mathbf{a}|^2\rangle$$

Examples:

- **harmonic oscillator** $\mathbf{a}(t) = -A\omega^2\cos(\omega t)\hat{\mathbf{x}}$, $\langle|\mathbf{a}(t)|^2\rangle = \frac{1}{2}A^2\omega^4$
- **circular orbits**, with constant speed. Acceleration is $\mathbf{a}(t) = -R\omega^2\hat{\mathbf{r}}(t)$, and has constan magnitude $|\mathbf{a}(t)| = \omega^2R$

Systems with $V(\mathbf{r}) = \frac{1}{2}m\omega^2|\mathbf{r}|^2$

$$\langle|\mathbf{a}|^2\rangle = \frac{\omega^2}{m}E\,,$$

and thus

$$\langle\Phi\rangle = \frac{q^2\omega^2}{6\pi\varepsilon mc^3}E =: \gamma E\,. \tag{21.4}$$

…**todo** Proof using virial theorem…

Vector identities

$$\varepsilon_{abc}A_b\varepsilon_{cde}B_dC_e = (\delta_{ad}\delta_{be} - \delta_{ae}\delta_{bd})A_bB_dC_e = A_bB_aC_b - A_bB_bC_a = \mathbf{A}\cdot\mathbf{C}\,\mathbf{B} - \mathbf{A}\cdot\mathbf{B}\,\mathbf{C}$$

21.4.4 Spin

- Early 1910s: Experiments about emission spectra of alkali metals (transition between levels obey selection rules) and the Zeeman effect (strong magnetic fields can influence atomic spectra)
- 1924: W.Pauli proposed an additional degree of freedom, the spin, a "two-valuedness without a classical counterpart". This physical quantity is related to angular momentum. This physical quantity was also required to formulated the Pauli exclusion principle, and linking this quantity to symmetry properties of the wave function of systems with many indentical bodies.
- W.Pauli first introduced the new degree of freedom in the non-relativistic Schrodinger equation, later shown to be the non-relativistic limit of the relativistic quantum mehcanics equation by Dirac (1928).
- 1922-1927: Stern-Gerlach experiment. First performed to test Bohr-Sommerfeld atomic model, then used to verify and investigate the properties of spin.

21.4.5 Modern reinterpretation of Heisenberg mechanics

Heisenberg Equations of Motion in the Heisenberg Picture

Time derivative of an operator $\hat{A}_H$ in Heisenberg picture satisfies the relation (14.2),

$$\frac{d\hat{A}_H}{dt} = \frac{1}{i\hbar}\left[\hat{A}_H, \hat{H}_H\right] + \left(\partial_t\hat{A}\right)_H\,.$$

Applying this relation to the position and momentum operators, $\hat{\mathbf{x}}_H$ and $\hat{\mathbf{p}}_H$, Heisenberg found the quantum mechanics counterpart of the equations of motion in classical mechanics,

$$\begin{cases}\dot{\hat{\mathbf{x}}}_H = \dfrac{1}{i\hbar}\left[\hat{\mathbf{x}}_H, \hat{H}_H\right] = \dfrac{\hat{\mathbf{p}}_H}{m} \\ \dot{\hat{\mathbf{p}}}_H = \dfrac{1}{i\hbar}\left[\hat{\mathbf{p}}_H, \hat{H}_H\right] = -\left(\nabla_{\mathbf{r}}V\left(\hat{\mathbf{r}}\right)\right)_H\end{cases} \tag{21.5}$$

Details

If $\hat{H} = \frac{\hat{|\mathbf{p}|}^2}{2m} + V(\hat{\mathbf{r}})$,

$\left[\hat{\mathbf{x}}_H, \hat{H}_H\right] = i\hbar\frac{\hat{\mathbf{p}}_H}{m}$

$$\begin{aligned}\left[\hat{\mathbf{x}}_H, \hat{H}_H\right] &= \mathcal{U}^\dagger_{t,t_0}\left(\left[\hat{\mathbf{x}}, \frac{|\hat{\mathbf{p}}|^2}{2m}\right] + [\hat{\mathbf{x}}, V(\hat{\mathbf{r}})]\right)\mathcal{U}_{t,t_0} = \\ &= \mathcal{U}^\dagger_{t,t_0}\left(i\hbar\frac{\hat{\mathbf{p}}}{m}\right)\mathcal{U}_{t,t_0} = \\ &= i\hbar\frac{\hat{\mathbf{p}}_H}{m}\,.\end{aligned}$$

since

1. $\hat{\mathbf{x}}$ and $V(\hat{\mathbf{r}})$ commute, as it can be proved using power expansion - if required - of $V(\hat{\mathbf{r}})$;
2. the commutator of $\hat{\mathbf{x}}$ and the kinetic contribution of the Hamiltonian operator reads (sum over b repeated index),

$$\begin{aligned}[\hat{x}_a, \hat{p}_b\hat{p}_b] &= \hat{x}_a\hat{p}_b\hat{p}_b - \hat{p}_b\hat{x}_a\hat{p}_b + \hat{p}_b\hat{x}_a\hat{p}_b - \hat{p}_b\hat{x}_b\hat{p}_a = \\ &= [\hat{x}_a, \hat{p}_b]\,\hat{p}_b + \hat{p}_b\,[\hat{x}_a, \hat{p}_b] = \\ &= 2i\hbar\delta_{ab}\hat{p}_b = \\ &= 2i\hbar\hat{p}_a\,.\end{aligned}$$

$\left[\hat{\mathbf{p}}_H, \hat{H}_H\right] = -i\hbar\left(\nabla_{\mathbf{r}}V(\hat{\mathbf{r}})\right)_H$

$$\left[\hat{\mathbf{p}}_H, \hat{H}_H\right] = \mathcal{U}^\dagger_{t,t_0}\left(\left[\hat{\mathbf{p}}, \frac{|\hat{\mathbf{p}}|^2}{2m}\right] + [\hat{\mathbf{p}}, V(\hat{\mathbf{r}})]\right)\mathcal{U}_{t,t_0} =$$

since, the commutator of the momentum and potential energy operator reads $[\hat{\mathbf{p}}, V(\hat{\mathbf{r}})] =$

$$\begin{aligned}\langle\mathbf{r}|\,[\hat{\mathbf{p}}, V(\hat{\mathbf{r}})]\,|\Psi\rangle &= \langle\mathbf{r}|\hat{\mathbf{p}}V(\hat{\mathbf{r}})\,|\Psi\rangle - \langle\mathbf{r}|V(\hat{\mathbf{r}})\,\hat{\mathbf{p}}|\Psi\rangle = \\ &= -i\hbar\nabla_{\mathbf{r}}\langle\mathbf{r}|V(\hat{\mathbf{r}})\,|\Psi\rangle - V(\mathbf{r})\,\langle\mathbf{r}|\hat{\mathbf{p}}|\Psi\rangle = \\ &= -i\hbar\nabla_{\mathbf{r}}\left(V(\mathbf{r})\Psi(\mathbf{r},t)\right) + i\hbar V(\mathbf{r})\nabla_{\mathbf{r}}\Psi(\mathbf{r},t) = \\ &= -i\hbar\nabla_{\mathbf{r}}V(\mathbf{r})\,\Psi(\mathbf{r},t) = \\ &= \langle\mathbf{r}| - i\hbar\nabla_{\mathbf{r}}V(\hat{\mathbf{r}})\,|\Psi\rangle\,.\end{aligned}$$

as $\langle\mathbf{r}|\hat{\mathbf{p}}\rangle = -i\hbar\nabla_{\mathbf{r}}\langle\mathbf{r}|$, and $V(\hat{\mathbf{r}})\,|\mathbf{r}\rangle = V(\mathbf{r})|\mathbf{r}\rangle$, and

$$\begin{aligned}f(x)\Psi(x,t) &= \int_{x'} f(x)\Psi(x',t)\delta(x-x')\,dx' = \\ &= \int_{x'} f(x)\langle x'|\Psi\rangle\langle x|x'\rangle\,dx' = \\ &= \int_{x'}\langle x|f(x)|x'\rangle\langle x'|\Psi\rangle\,dx' = \\ &= \langle x|f(\hat{x})\underbrace{\int_{x'}|x'\rangle\langle x'|dx'}_{=\hat{\mathbf{i}}}|\Psi\rangle = \\ &= \langle x|f(\hat{x})\,|\Psi\rangle\,.\end{aligned}$$

Matrix components in energy basis of the equations of motion

The matrix components in energy basis of the equations of motion (21.5) read

$$\begin{cases} i\omega_{jk} X^H_{jk} = \dfrac{P^H_{jk}}{m} \\ i\omega_{jk} P^H_{jk} = -\left(\nabla V\right)^H_{jk} \end{cases} \tag{21.6}$$

with $\omega_{jk} := \frac{E_j - E_k}{\hbar}$.

Details

If the Hamiltonian is not an explicit function of time, $\mathcal{U}_{t,0} = \exp\left[-i\frac{\hat{H}}{\hbar}t\right]$, and thus

$$\begin{aligned} (\hat{\mathbf{x}}_H) &= (|j\rangle\langle j|\hat{\mathbf{x}}_H|k\rangle\langle k|) = \\ &= \left(|j\rangle\langle j|\mathcal{U}^\dagger_{t,0}\hat{\mathbf{x}}\mathcal{U}_{t,0}|k\rangle\langle k|\right) = \\ &= |j\rangle\langle k| \left[\exp\left(-i\frac{E_k - E_j}{\hbar}t\right)\langle j|\hat{\mathbf{x}}|k\rangle\right] \end{aligned}$$

i.e.

$$X^H_{jk}(t) = X^S_{jk} \exp\left(i\frac{E_j - E_k}{\hbar}t\right) .$$

The time-derivative of the matrix elements of the position operator in Heisenber picture reads

$$\dot{X}^H_{jk} = i\omega_{jk} X^S_{jk} \exp\left(i\omega_{jk}t\right) = i\omega_{jk} X^H_{jk} ,$$

with $\omega_{jk} := \frac{E_j - E_k}{\hbar}$.

Equations of motion as Hamilton's equations

The equations of motion (21.5), or (21.6), can be recast in the form of Hamilton's equations

$$\begin{cases} \dot{\hat{\mathbf{q}}}_H = \dfrac{\partial \mathsf{H}}{\partial \hat{\mathbf{p}}_H} \\ \dot{\hat{\mathbf{p}}}_H = -\dfrac{\partial \mathsf{H}}{\partial \hat{\mathbf{q}}_H} \end{cases}$$

and

$$\begin{cases} \dot{X}^H_{jk} = \dfrac{\partial \mathsf{H}}{\partial P^H_{jk}} \\ \dot{P}^H_{jk} = -\dfrac{\partial \mathsf{H}}{\partial Q^H_{jk}} \end{cases}$$

with $\mathsf{H} = \mathrm{tr}\left(\hat{H}\right) = \sum_a \langle a|\hat{H}|a\rangle$.

Details

CCR and quantization rule

Modern approach. Matrix form of the CCR

Given the CCR

$$[\hat{\mathbf{x}}, \hat{\mathbf{p}}] = i\hbar \mathbb{I} \qquad , \qquad [\hat{x}_a, \hat{p}_b] = i\hbar\delta_{ab} \ ,$$

its matrix components are

$$i\hbar\delta_{ab}\delta_{jk} = \sum_{\ell} \left\{ X^a_{j\ell} P^b_{\ell k} - P^b_{j\ell} X^a_{\ell k} \right\} \ . \tag{21.7}$$

Details

$$\begin{aligned} i\hbar\delta_{ab}\delta_{jk} &= \langle j | i\hbar\delta_{ab} | k \rangle = \\ &= \langle j | \left[\hat{x}_a, \hat{p}_b \right] k \rangle = \\ &= \langle j | \hat{x}_a \hat{p}_b - \hat{p}_b \hat{x}_a | k \rangle = \\ &= \sum_{\ell} \left\{ \langle j | \hat{x}_a | \ell \rangle \langle \ell | \hat{p}_b | k \rangle - \langle j | \hat{p}_b | \ell \rangle \langle \ell | \hat{x}_a | k \rangle \right\} = \\ &= \sum_{\ell} \left\{ X^a_{j\ell} P^b_{\ell k} - P^b_{j\ell} X^a_{\ell k} \right\} \ . \end{aligned}$$

From Bohr-Sommerfeld quantization to CCR

In old quantum mechanics - i.e. quantum mechanics before a quantum mechanics theory - Bohr-Sommerfeld quantization rule was

$$J = nh = \oint p_n dq_n = \oint p_n \dot{q}_n \, dt$$

Classical trajectory. Using Fourier series of a periodic trajectory,

$$q_n(t) = \sum_{\alpha} q_\alpha(n) e^{i\alpha\omega_n t} \ ,$$

the time derivative of $q_n(t)$ reads

$$\dot{q}_n(t) = \sum_{\alpha} i\alpha\omega_n q_\alpha(n) e^{i\alpha\omega_n t} \ ,$$

while momentum can be written as

$$p_n(t) = \sum_{\beta} p_\beta(n) e^{i\beta\omega_n t} \ ,$$

Using these expression in the Bohr-Sommerfeld quantization rule,

$$\begin{aligned} J &= \cdots = \\ &= i\, 2\pi \sum_{\alpha} \alpha p^*_\alpha(n) q_\alpha(n) \ , \end{aligned}$$

so that the derivative w.r.t. J reads

$$1 = i\,2\pi \sum_\alpha \alpha \frac{\partial}{\partial J}\left(p^*_\alpha(n) q_\alpha(n)\right) \ .$$

Quantum reinterpretation. Let a function $\Phi(n,\alpha)$, then

$$\begin{aligned}\Phi(n,\alpha) - \Phi(n-\alpha,\alpha) \sim \alpha \partial_n \Phi(n,\alpha) + o(\alpha) = \\ = \alpha h \partial_J \Phi(n,\alpha) + o(\alpha)\ ,\end{aligned}$$

and the **Born-Kramers correspondence principle follows**,

$$\frac{\Phi(n,\alpha) - \Phi(n-\alpha,\alpha)}{h} \sim \alpha \frac{\partial \Phi}{\partial J}(n,\alpha)\ .$$

Let $p_\alpha(n) = p(n,\alpha) = p_{n+\alpha,n}$, then the reinterpretation of the classical quantization rule gives

$$\begin{aligned}1 = i\,\frac{2\pi}{h} \sum_\alpha \left\{p^*_{n+\alpha,n} q_{n+\alpha,n} - p^*_{n,n-\alpha} q_{n,n-\alpha}\right\} = \\ = i\,\frac{2\pi}{h} \sum_\alpha \left\{p^*_{n+\alpha,n} q_{n+\alpha,n} - \sum_\alpha p^*_{n,n-\alpha} q_{n,n-\alpha}\right\} = \\ = i\,\frac{2\pi}{h} \sum_\alpha \left\{p_{n,n+\alpha} q_{n+\alpha,n} - \sum_\alpha q_{n,n-\alpha} p_{n-\alpha,n}\right\} = \\ = \frac{1}{i\hbar} \sum_\alpha \left\{q_{n,\alpha} p_{\alpha,n} - p_{n,\alpha} q_{\alpha,n}\right\}\end{aligned}$$

These relations are nothing but the diagonal components of the matrix form (21.7) of the CCR. The out-of-diagonal components are identically zero, as P.Jordan proved with the following trick

How P.Jordan proved that out-of-diagonal components of the CCR are identically zero

…

CCR in Energy Basis and Ladenburg Dispersion Formula

Introducing the expression of matrix components of the momentum $\hat{\mathbf{p}}_H = m\dot{\hat{\mathbf{x}}}_H$, $P^a_{jk} = im\omega_{jk} X^a_{jk}$ into the matrix form (21.7) of the CCR relation

$$\begin{aligned}i\hbar\delta_{ab}\delta_{jk} = \sum_\ell \left\{X^a_{j\ell} P^b_{\ell k} - P^b_{j\ell} X^a_{\ell k}\right\} = \\ = \sum_\ell \left\{im\omega_{\ell k} X^a_{j\ell} X^b_{\ell k} - im\omega_{j\ell} X^b_{j\ell} X^a_{\ell k}\right\}\ .\end{aligned}$$

The diagonal components $j = k$ are

$$i\hbar\delta_{ab} = \sum_\ell \left\{im\omega_{\ell k} X^a_{k\ell} X^b_{\ell k} - im\omega_{k\ell} X^b_{k\ell} X^a_{\ell k}\right\} =$$

If $a = b$,

$$\begin{aligned}i\hbar = \sum_\ell \left\{im\omega_{\ell k} X^a_{k\ell} X^a_{\ell k} - im\omega_{k\ell} X^a_{k\ell} X^a_{\ell k}\right\} = \\ = \sum_\ell \left\{im\omega_{\ell k} X^a_{k\ell} X^{a*}_{k\ell} - im\omega_{k\ell} X^a_{k\ell} X^{a*}_{k\ell}\right\} = \\ = i2m \sum_\ell \omega_{\ell k} \left|X^a_{k\ell}\right|^2\ ,\end{aligned}$$

so that

$$\frac{2m}{\hbar}\sum_{\ell}\omega_{\ell k}\left|X^{a}_{k\ell}\right|^{2}=1\,.$$

todo *CHECK if there's a factor* 3*, summing over all the components indexed by* a

Connection to Ladenburg Dispersion Formula. In classical optics/early quantum theory, Ladenburg's quantum dispersion formula for the oscillator strength f_{km} associated with a transition $k \to m$ is defined as:

$$f_{km}=\frac{2m\omega_{km}}{\hbar}|x_{km}|^{2}$$

The diagonal CCR condition directly yields the **Thomas-Reiche-Kuhn (TRK) sum rule**:

$$\sum_{m} f_{km}=1$$

This showed that Heisenberg's matrix mechanics naturally incorporated the empirically validated dispersion theory of Ladenburg, Kramers, and Kronig.

Application: The Linear Harmonic Oscillator

Linear harmonic oscillator

For the harmonic oscillator Hamiltonian $\hat{H}=\frac{\hat{p}^{2}}{2m}+\frac{1}{2}m\omega_{0}^{2}\hat{x}^{2}$:

Equations of Motion.

$$\begin{cases}\dot{\hat{x}}=\dfrac{\hat{p}}{m}\\ \dot{\hat{p}}=-m\omega_{0}^{2}\hat{x}\end{cases}$$

Taking the derivative of the first equation and replacing the second equation for $\dot{\hat{p}}$, a second-order equation for $\hat{x}$ follows

$$\ddot{\hat{x}}+\omega_{0}^{2}\hat{x}=0$$

In matrix components:

$$(-\omega_{mn}^{2}+\omega_{0}^{2})x_{mn}=0$$

Thus, non-zero matrix elements x_{mn} can only exist if $\omega_{mn}=\pm\omega_{0}$, meaning transitions only occur between adjacent levels: $m=n\pm1$.

Matrix Elements & Energy Spectrum. Using the diagonal CCR $\frac{2m}{\hbar}\sum_{m}\omega_{mn}|x_{nm}|^{2}=1$:

$$\frac{2m\omega_{0}}{\hbar}\left(|x_{n,n+1}|^{2}-|x_{n,n-1}|^{2}\right)=1$$

With the boundary condition $x_{0,-1}=0$ for the ground state:

$$|x_{n,n+1}|^{2}=\frac{\hbar}{2m\omega_{0}}(n+1)$$

The energy matrix elements give the discrete eigenvalues:

$$E_{n}=\hbar\omega_{0}\left(n+\frac{1}{2}\right)$$

Transition Probabilities as Dipole Matrix Elements

When an atom interacts with an electromagnetic field, the interaction Hamiltonian is dominated by the electric dipole coupling:

$$\hat{V}(t) = -\hat{\mathbf{d}} \cdot \mathbf{E}(t) = -q\hat{\mathbf{x}} \cdot \mathbf{E}_0 \cos(\omega t)$$

The probability amplitude for a transition from state $|n\rangle$ to state $|m\rangle$ is dictated by the matrix element of the position/dipole operator:

$$x_{mn} = \langle m|\hat{x}|n\rangle$$

The transition rate $W_{n\to m}$ (spontaneous emission / absorption probability) is proportional to the square of the dipole matrix element:

$$W_{n\to m} \propto |\mathbf{d}_{mn}|^2 = q^2 |\langle m|\hat{\mathbf{x}}|n\rangle|^2$$

Perturbation Theory for Transition Probabilities in Weak Electric Fields

Consider a time-dependent perturbation $\hat{V}(t) = \hat{W}\cos(\omega t) = -q\hat{\mathbf{x}} \cdot \mathbf{E}_0 \cos(\omega t)$.

First-Order Amplitude

Writing state expansion $|\psi(t)\rangle = \sum_n c_n(t) e^{-\frac{i}{\hbar}E_n t}|n\rangle$, the equations for $c_m(t)$ are:

$$i\hbar \dot{c}_m(t) = \sum_n \langle m|\hat{V}(t)|n\rangle c_n(t) e^{i\omega_{mn}t}$$

Assuming the system starts in initial state $c_k(0) = 1$ and $c_m(0) = 0$ for $m \neq k$:

$$c_m^{(1)}(t) = -\frac{i}{\hbar}\int_0^t V_{mk}(t') e^{i\omega_{mk}t'}\, dt'$$

Substituting $V_{mk}(t') = -qE_0 x_{mk} \frac{e^{i\omega t'} + e^{-i\omega t'}}{2}$:

$$c_m^{(1)}(t) = \frac{qE_0 x_{mk}}{2\hbar}\left[\frac{1 - e^{i(\omega_{mk}+\omega)t}}{\omega_{mk}+\omega} + \frac{1 - e^{i(\omega_{mk}-\omega)t}}{\omega_{mk}-\omega}\right]$$

Resonant Approximation (Rotating Wave Approximation)

Near resonance ($\omega \approx \omega_{mk}$ with $E_m > E_k$):

The term with denominator $(\omega_{mk} - \omega)$ dominates:

$$|c_m^{(1)}(t)|^2 \approx \frac{q^2 E_0^2 |x_{mk}|^2}{4\hbar^2} \frac{\sin^2\left(\frac{(\omega_{mk}-\omega)t}{2}\right)}{\left(\frac{\omega_{mk}-\omega}{2}\right)^2}$$

Second-Order Expansion for $|c_n(t)|^2$

To compute probability to second order without assuming immediate resonance, we expand $c_k(t)$ and $c_m(t)$:

$$|c_k(t)|^2 = 1 + c_k^{(1)}(t) + c_k^{(1)*}(t) + |c_k^{(1)}(t)|^2 + ...$$

Since $V_{kk} = 0$ for parity-symmetric unperturbed states ($x_{kk} = 0$), the first-order correction to the initial state $c_k^{(1)}(t) = 0$.

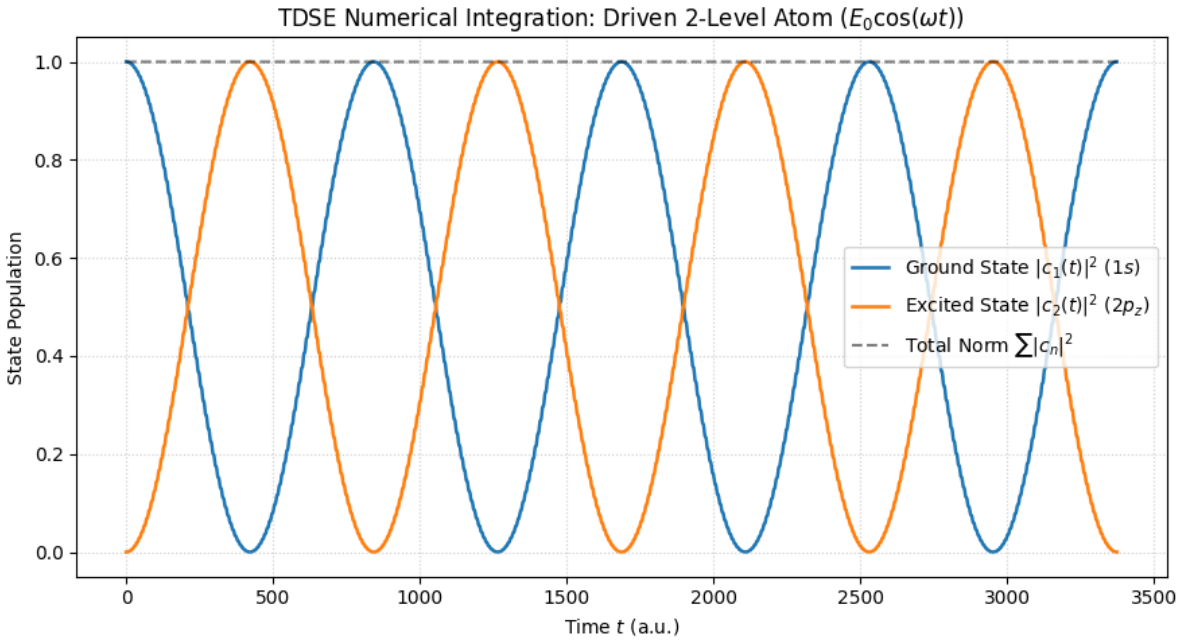
The conservation of total probability yields:

$$|c_k(t)|^2 = 1 - \sum_{m \neq k} |c_m^{(1)}(t)|^2$$

This confirms that weak external monochromatic fields cause probability transitions whose rates are directly dictated by matrix elements x_{mk} of Heisenberg's matrix mechanics.

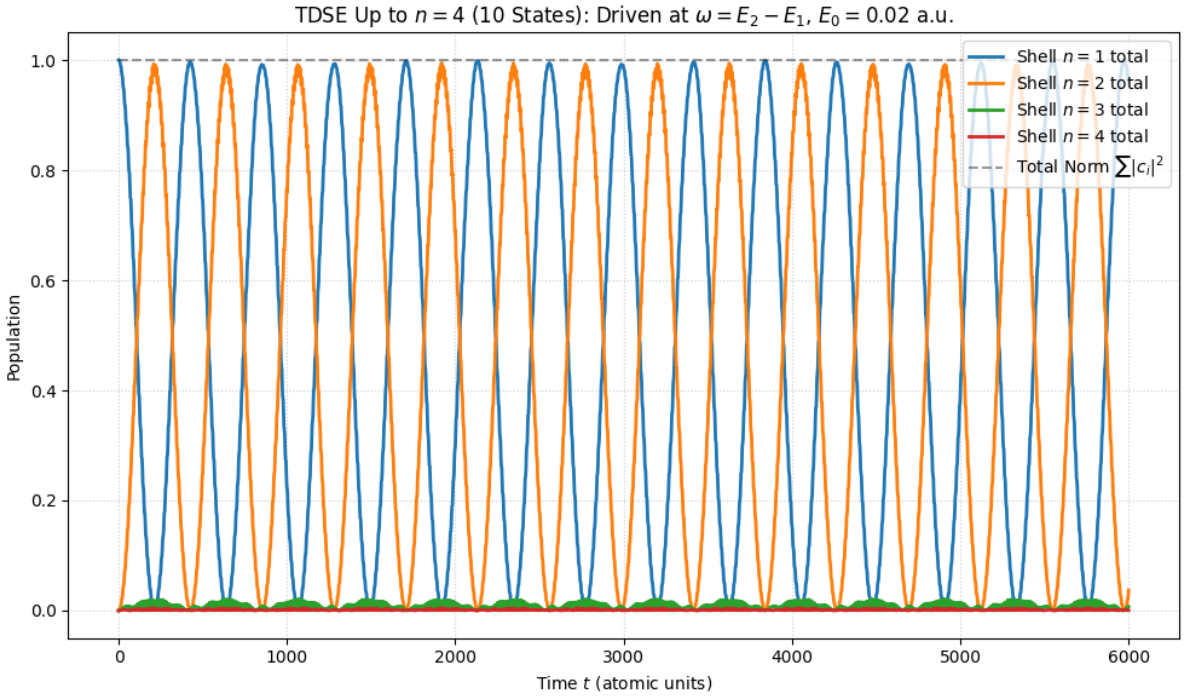
21.4.6 Numerical simulation of the electron pertubation

2-energy level system



4-energy level system

```
Number of m=0 basis states up to n=4: 10
Energy of the stationary states
[-0.5        -0.125      -0.125      -0.05555556 -0.05555556 -0.05555556
 -0.03125    -0.03125    -0.03125    -0.03125    ]
```



Spectra and non-monochromatic radiation

A. Ultrashort / Arbitrary Pulses. For a general pulse (such as a Gaussian envelope centered at t_0 with bandwidth $\Delta\omega$):

$$E(t) = E_0 \exp\left(-\frac{(t-t_0)^2}{2\tau^2}\right) \cos(\omega_0 t + \phi)$$

In the numerical solver, you simply update the field definition in $V_{mn}(t)$:

$$V_{mn}(t) = -e\, D_{mn}\, E(t)$$

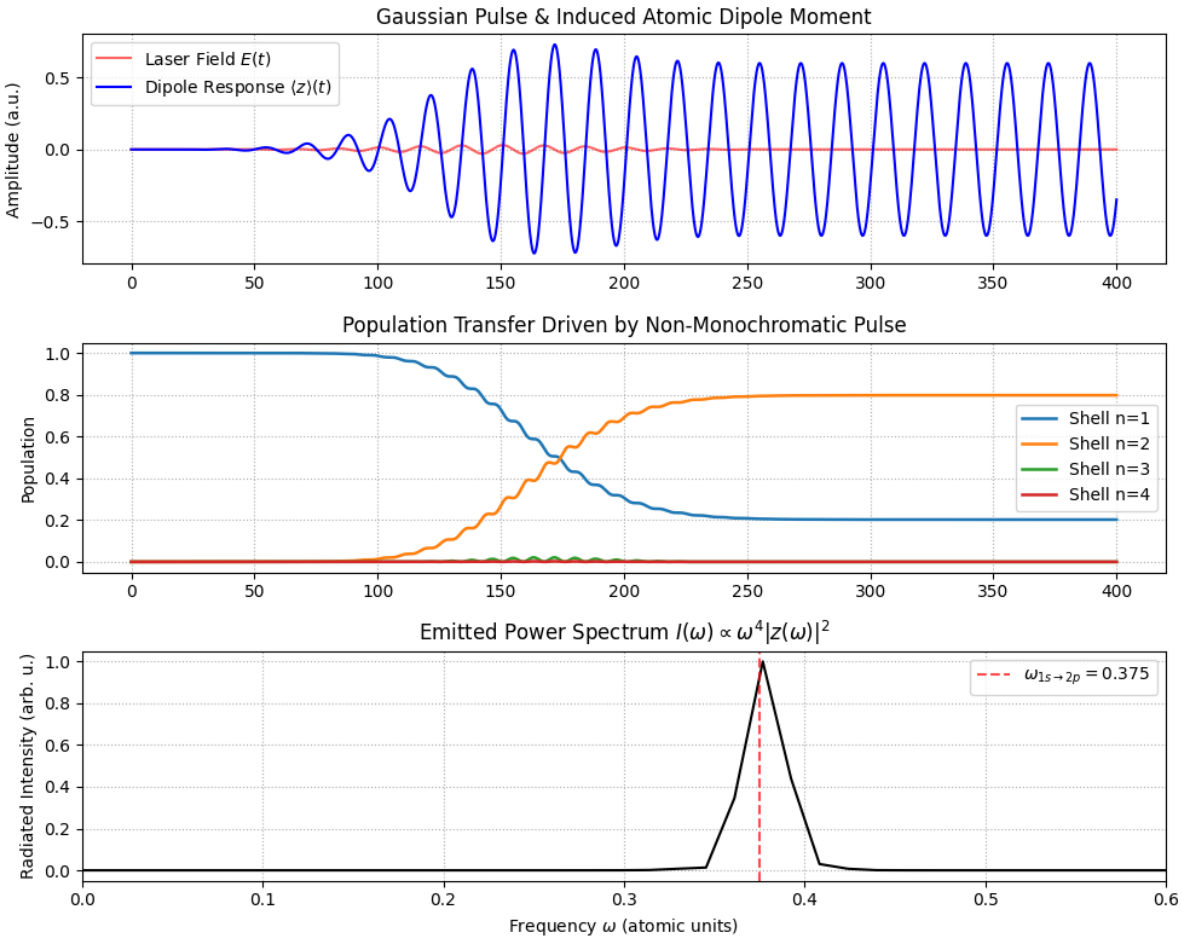
B. Broad-Spectrum / Thermal Light (Fermi's Golden Rule & Rate Equations). If the field is incoherent broadband light characterized by a spectral energy density $\rho(\omega)$ ($\mathrm{J \cdot m^{-3} \cdot Hz^{-1}}$), phase coherence between states drops out on fast timescales.Instead of tracking coherent state amplitudes $c_n(t)$ via the TDSE, the dynamics transition to Einstein Rate Equations for populations $P_n(t) = |c_n(t)|^2$:

$$\frac{dP_b}{dt} = -\frac{dP_a}{dt} = B_{ab}\, \rho(\omega_{ba})\, (P_a - P_b) - A_{ba}\, P_b$$

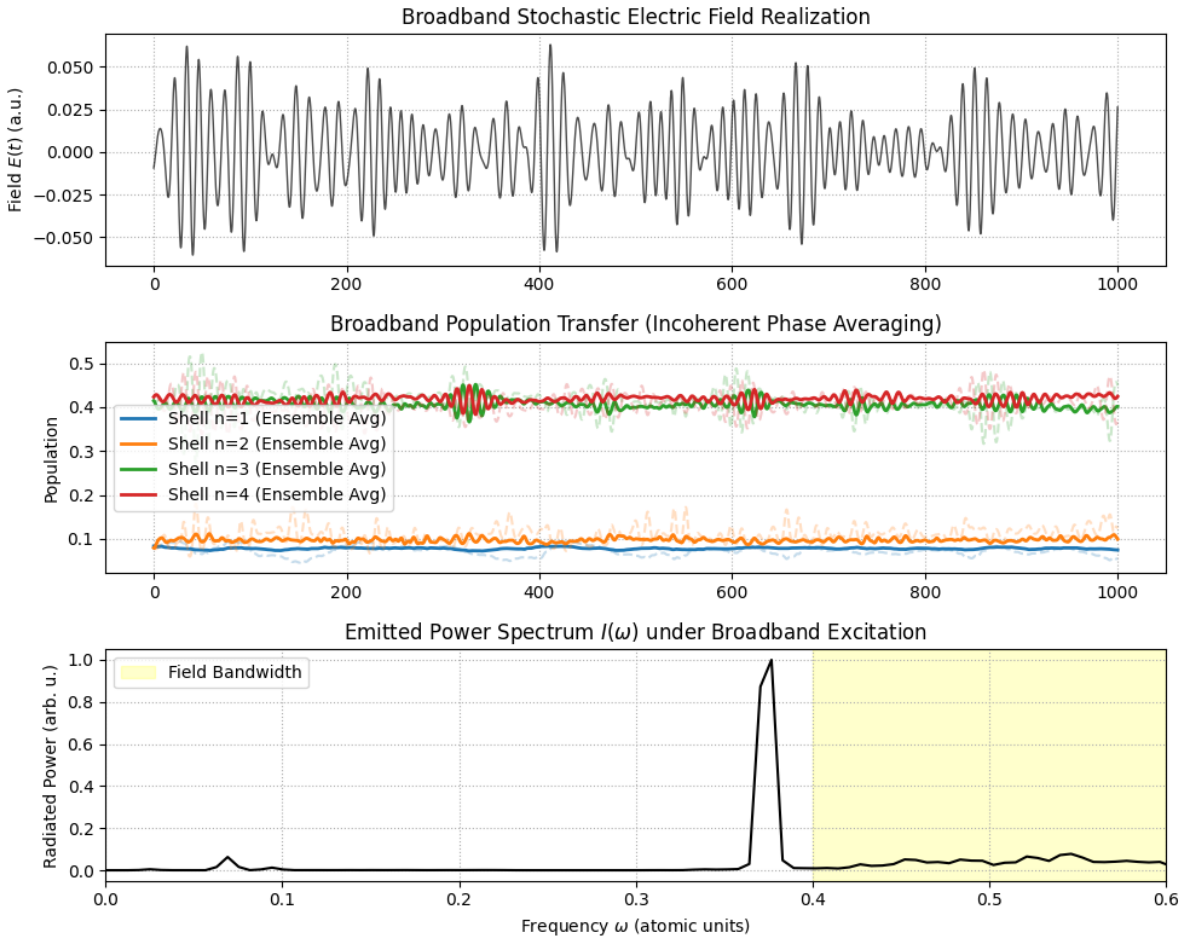
where Fermi's Golden Rule gives the absorption rate $W_{a\to b}$:

$$W_{a\to b} = \frac{2\pi}{\hbar^2} |\langle b|V_0|a\rangle|^2 \delta(E_b - E_a - \hbar\omega) \implies B_{ab} = \frac{\pi e^2}{\epsilon_0 \hbar^2} |D_{ab}|^2$$

Pulse


Gaussian Pulse & Induced Atomic Dipole Moment
Laser Field $E(t)$
Dipole Response $\langle z \rangle(t)$
Amplitude (a.u.)
0.5
0.0
−0.5
0
50
100
150
200
250
300
350
400
Population Transfer Driven by Non-Monochromatic Pulse
Population
1.0
0.8
0.6
0.4
0.2
0.0
Shell n=1
Shell n=2
Shell n=3
Shell n=4
Emitted Power Spectrum $I(\omega) \propto \omega^4 |z(\omega)|^2$
Radiated Intensity (arb. u.)
$\omega_{1s \to 2p} = 0.375$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Frequency ω (atomic units)

Broad spectrum




```
Omega_ij:
[[ 0.          0.375       0.44444444  0.46875   ]
 [-0.375       0.          0.06944444  0.09375   ]
 [-0.44444444 -0.06944444  0.          0.02430556]
 [-0.46875    -0.09375    -0.02430556  0.        ]]
```

21.5 Origin of wave theory of quantum mechanics

Main contributions from:

- *L. de Broglie*, (1922-24): matter waves
- *E. Schrodinger*, (1926) "An Undulatory Theory of the Mechanics of Atoms and Molecules": wave equation. "if a phenomenon behaves like a wave, it must be governed by a wave equation"

Experimental verifications:

- Davisson-Germer experiment (1923-27) confirmed the hypothesis of L. de Broglie: e^- scattered by a surface of a metal crystal showed a diffraction pattern, a characteristic phenomenon of wave processes, consistent with the relation postulated by de Broglie linking the wave vector and the momentum $\mathbf{k} = \frac{\mathbf{p}}{\hbar}$.

Reinterpretation and classical limit:

- …

21.5.1 L. de Broglie, Researches of the Quantum Theory

- Matter waves, with

$$E = hf \qquad , \qquad \mathbf{p} = \hbar \mathbf{k}$$

21.5.2 E. Schrodinger, "An Undulatory Theory of the Mechanics of Atoms and Molecules"

See *wave mechanics*

CHAPTER

TWENTYTWO

ATOMIC MODELS

- *Thomson*: electrons in cathode rays
- *Rutherford*: small nucleus, from scattering
- *Bohr*: atomic model, consistent with the relation $\Delta E = h\nu$ and emission/absorption of light
- *Sommerfeld*: ...
- *Schrodinger*: atomic model from Schrodinger equation $i\hbar|\dot{\Psi}\rangle = \hat{H}|\Psi\rangle$
- *Dirac*: atomic model from Dirac equation $i\hbar|\dot{\Psi}\rangle = \left[c\alpha \cdot \hat{\mathbf{p}} + \beta mc^2 + V(\hat{r})\right]|\Psi\rangle$.

22.1 Thomson model

See J.J. Thomson discovers the electron. Experiments about conduction of electric current in gases, in the High-School book. Italian only, so far: please rely on the automatic translation of your browser.

Topics: motion of a charged particle in an electromagnetic field, and thus subject to Lorentz force.

22.2 Rutherford model

See Geiger-Mardsen scattering experiment, and Rutherford atomic model. Scattering experiment of α-particles, in the High-School book. Italian only, so far: please rely on the automatic translation of your browser.

Topics: motion of a positively charged α-particle in a 2-body problem with positively charged nucleus, and thus subject to repulsive Coulomb force.

22.3 Bohr model

Assumption:

- Electron moves on circular orbits
- All the possible orbits are identified by discrete values of the angular momentum, $L = n\hbar$
- Light emission/absorption occurs when an electron transitions between two orbits

Some mathematical details, using the equations of classical mechanics. The problem of determining the orbit of an electron around the positive nucleus is a two-body problem, subject to a force $\vec{F} \propto \frac{\vec{r}}{|\vec{r}|^3}$ (here the assumption of the center

of mass in the nucleus; otherwise change of coordinates, and equivalent masses…). For an electrically neutral atom with one electron only (H atom?) the dynamical equation governing the motion of the electron

$$m\ddot{\vec{r}} = -\frac{q^2}{4\pi\varepsilon}\frac{\vec{r}}{|\vec{r}|^3} ,$$

being q the charge of the electron.

- Using polar coordinates, $\vec{r} = r\hat{r}$
- Taking time derivative of the position vector $\vec{r}$ identifying the position of the electron w.r.t. the nucleus

$$\frac{d}{dt}(r\hat{r}) = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$$
$$\frac{d^2}{dt^2}(r\hat{r}) = \left(\ddot{r} - r\dot{\theta}^2\right)\hat{r} + \left(2\dot{r}\dot{\theta} + r\ddot{\theta}\right)\hat{\theta}$$

- Assuming circular orbits, $\dot{r} = 0$

the radial and angular components of the dynamical equation read

$$r: \quad -mr\dot{\theta}^2 = -\frac{q^2}{4\pi\varepsilon}\frac{1}{r^2}$$
$$\theta: \quad \ddot{\theta} = 0$$

The angular momentum and the energy of the system are

- Angular momentum,

$$\begin{aligned} n\hbar = L_n = mr_n^2\dot{\theta}_n = & \qquad \text{(radial component)} \\ = mr_n^2\left(\frac{q^2}{4\pi\varepsilon r_n^3 m}\right)^{\frac{1}{2}} = & \\ = \frac{\sqrt{m}q}{\sqrt{4\pi\varepsilon}}r^{\frac{1}{2}} & \\ \rightarrow \quad r_n = 4\pi\varepsilon\frac{n^2\hbar^2}{mq^2} & \end{aligned}$$

- Energy,

$$\begin{aligned} E_n = \frac{1}{2}m|\dot{\vec{r}}_n|^2 - \frac{q^2}{4\pi\varepsilon}\frac{1}{r_n} = & \quad \text{(radial component of the eom)} \\ = \frac{q^2}{8\pi\varepsilon}\frac{1}{r_n} - \frac{q^2}{4\pi\varepsilon}\frac{1}{r_n} = & \\ = -\frac{q^2}{8\pi\varepsilon}\frac{1}{r_n} = & \quad (r_n \text{ from angular momentum}) \\ = -\frac{mq^4}{32\pi^2\varepsilon^2\hbar^2}\frac{1}{n^2} = & \quad (h = 2\pi\hbar) \\ = -\frac{mq^4}{8\varepsilon^2 h^2}\frac{1}{n^2} . & \end{aligned} \tag{22.1}$$

Remark. The value of energy E_n of Bohr model coincides with the value (22.2) of the energy produced by *Schrodinger model*.

From the relation $\Delta E = \nu h$, the relation between the frequency of the emitted (if $E_n > E_m$, n initial state, m final state) or absorbed (if $E_n < E_m$, n initial state, m final state) radiation, follows

$$\nu_{nm} = \frac{E_n - E_m}{h} = -\frac{mq^4}{8\varepsilon^2 h^3}\left(\frac{1}{n^2} - \frac{1}{m^2}\right)$$

22.4 Sommerfeld model

In 1913, Niels Bohr proposed a revolutionary model of the atom that successfully explained the spectral lines of hydrogen. Bohr's model combined classical mechanics with quantum concepts, asserting that electrons revolve around the nucleus in fixed, circular orbits with quantized angular momentum:

$$L = n\hbar = n\frac{h}{2\pi}, \quad n \in \{1, 2, 3, \dots\}$$

While the Bohr model was a monumental step forward, high-resolution spectroscopy soon revealed several limitations:

- **Fine Structure:** Spectral lines that initially appeared singular actually split into closely spaced multiplet lines under high resolution.
- **Complex Atoms:** The model failed to accurately predict the spectra of multi-electron atoms (like helium).
- **Zeeman Effect:** It could not explain the splitting of spectral lines in the presence of a magnetic field.

To resolve these shortcomings, German physicist **Arnold Sommerfeld** extended Bohr's theory in 1916 by:

- relaxing the restriction of purely circular orbits.
- introducing relativistic corrections

22.4.1 Key Postulates and Mathematical Framework

Sommerfeld generalized Bohr's circular orbit condition using the **Wilson-Sommerfeld Quantization Rule** for periodic systems:

$$\oint p_i \, dq^i = n_i h$$

where p_i is the generalized momentum corresponding to coordinate q^i, n_i is an integer quantum number, and h is Planck's constant.

Elliptical Orbits

Sommerfeld proposed that electron orbits are generally **elliptical**, with the nucleus residing at one of the foci of the ellipse. To describe an ellipse in polar coordinates (r, θ), two generalized coordinates are required, leading to two quantization conditions:

1. **Azimuthal Quantization**,

$$\oint p_\theta \, d\theta = n_\theta h \implies L = n_\theta \hbar$$

2. **Radial Quantization**,

$$\oint p_r \, dr = n_r h \, .$$

Here, n_θ is the azimuthal quantum number (also denoted as k) and n_r is the radial quantum number. The total principal quantum number n is defined as:

$$n = n_r + n_\theta$$

The geometry of the ellipse—specifically the ratio of the semi-minor axis b to the semi-major axis a—is governed by these quantum numbers:

$$\frac{b}{a} = \frac{n_\theta}{n} = \frac{n_\theta}{n_r + n_\theta}$$

- When $n_\theta = n$, $b = a$, yielding a circular orbit (Bohr's original case).
- When $n_\theta < n$, $b < a$, producing an elliptical orbit.

Relativistic Correction and Energy Splitting

In non-relativistic mechanics, the energy of an electron in an elliptical orbit depends **only** on the principal quantum number n:

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2 n^2}$$

This means all orbits with the same n would have identical energy (degeneracy).

However, Sommerfeld recognized that an electron moves faster when it passes closer to the nucleus in a highly elliptical orbit. Applying Einstein's special relativity, the electron's mass m varies with velocity v:

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$

When relativistic effects are accounted for, the ellipse undergoes **precession** (perihelion precession), breaking the energy degeneracy. The modified energy expression becomes dependent on both n and n_θ:

$$E_{n,n_\theta} = -\frac{R_y hc Z^2}{n^2}\left[1 + \frac{\alpha^2 Z^2}{n}\left(\frac{1}{n_\theta} - \frac{3}{4n}\right)\right]$$

where:

- R_y is the Rydberg constant,
- Z is the atomic number,
- $\alpha = \dfrac{e^2}{4\pi\epsilon_0 \hbar c} \approx \dfrac{1}{137}$ is the **Fine-Structure Constant**.

22.4.2 Major Achievements

1. **Explanation of Fine Structure:** The dependence of energy levels on n_θ alongside n successfully explained the small energy differences responsible for the fine splitting of hydrogen spectral lines.
2. **Introduction of Spatial Quantization:** Sommerfeld added a third quantum number m (magnetic quantum number) to describe the orientation of orbits in 3D space, providing a framework for the Zeeman effect.
3. **Introduction of α:** It introduced the fine-structure constant α, a fundamental constant in modern physics representing the coupling strength of electromagnetic interactions.

22.4.3 Limitations and Transition to Quantum Mechanics

Despite its empirical success, the Bohr-Sommerfeld model was a **semi-classical (or "Old Quantum Theory")** construction. Its limitations ultimately spurred the transition to modern quantum mechanics:

- **Ad-hoc Assumptions:** It blended classical electrodynamics (planetary orbits) with arbitrary quantization rules without an underlying wave mechanism.
- **Intensity and Selection Rules:** It could not predict the transition probabilities (intensities of spectral lines) accurately.

- **Failure for Multi-Electron Systems:** Like Bohr's model, it could not be extended reliably to helium or larger atoms.

The model was eventually superseded by *the non-relativistic atomic model using* ***Schrödinger's Wave Mechanics*** (1926) and **Dirac's Relativistic Quantum Mechanics** (1928), where n_θ was replaced by the orbital angular momentum quantum number $l = n_\theta - 1$.

22.4.4 Analytical mechanics - short notes

To understand the theoretical justification behind the **Wilson-Sommerfeld Quantization Rule**, we must examine the formal framework of classical analytical mechanics. Old Quantum Theory did not apply quantization conditions to arbitrary coordinates; rather, it relied on the natural canonical coordinates provided by **action-angle variables**.

Lagrangian and Hamiltonian mechanics

Lagrangian mechanics. Consider a classical physical system with f degrees of freedom described by generalized coordinates $\mathbf{q} = (q_1, q_2, \dots, q_f)$ and generalized velocities $\dot{\mathbf{q}} = (\dot{q}^1, \dot{q}^2, \dots, \dot{q}^f)$. The equations of motion can be derived from the principle of stationariety of the action, S,

$$0 = \delta S = \delta \int_{t_0}^{t_1} L\left(\dot{\mathbf{q}}(t), \mathbf{q}(t), t\right)\, dt\ ,$$

with prescribed ends $\delta\mathbf{q}(t_0) = \delta\mathbf{q}(t_1) = \mathbf{0}$, and $L(\dot{\mathbf{q}}(t), \mathbf{q}(t), t)$ the Lagrangian function

$$L(\mathbf{q}, \dot{\mathbf{q}}, t) = T(\mathbf{q}, \dot{\mathbf{q}}) - V(\mathbf{q})\ .$$

The canonical (or conjugate) momentum associated with coordinate q_i is defined as:

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

Hamiltonian mechanics. Via a *Legendre transformation*, we transition from the $(\mathbf{q}, \dot{\mathbf{q}})$ state space to the canonical phase space $(\mathbf{q}, \mathbf{p})$, introducing the **Hamiltonian**, nothing but the mechanical energy of the system as a function of generalized coordinates and momenta,

$$H(\mathbf{q}, \mathbf{p}, t) = \sum_{i=1}^{f} p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t)$$

Taking the differential of H, the equations of motion follows

$$\begin{cases} \dot{q}_i = \dfrac{\partial H}{\partial p_i} \\ \dot{p}_i = -\dfrac{\partial H}{\partial q_i} \end{cases}$$

while $\partial_t H = -\partial_t L$.

Canonical Transformations and Hamilton-Jacobi Theory

Canonical transformations and invariance

Transformation $\mathbf{z} = (\mathbf{q}, \mathbf{p}) \leftrightarrow \mathbf{Z} = (\mathbf{Q}, \mathbf{P})$

$$\begin{cases} \dot{\mathbf{q}} = \partial_{\mathbf{p}} H \\ \dot{\mathbf{p}} = -\partial_{\mathbf{q}} H \end{cases} \quad , \quad \begin{cases} \dot{\mathbf{Q}} = \partial_{\mathbf{P}} K \\ \dot{\mathbf{P}} = -\partial_{\mathbf{Q}} K \end{cases}$$

or

$$\dot{\mathbf{z}} = \mathbf{J} \nabla_{\mathbf{z}} H \quad , \quad \dot{\mathbf{Z}} = \mathbf{J} \nabla_{\mathbf{Z}} K \, ,$$

with

$$\mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix} \, .$$

As $\mathbf{z}(\mathbf{Z}, t)$,

$$\begin{aligned} \dot{z}_i &= \dot{Z}_k \, \partial_{Z_K} z_i + \partial_t z_i \\ \partial_{z_i} f &= \partial_{z_i} Z_k \, \partial_{Z_k} f \end{aligned}$$

for a regular transformation, with non-singular gradient $\partial_{Z_k} z_i \, \partial_{z_i} Z_j = \delta_{kj}$,

$$\begin{aligned} \dot{Z}_j &= \partial_{z_i} Z_j \left(\dot{z}_i - \partial_t z_i \right) = \\ &= \partial_{z_i} Z_j \left(J_{il} \partial_{z_l} H - \partial_t z_i \right) = \\ &= \partial_{z_i} Z_j \, J_{il} \, \partial_{z_l} Z_m \, \partial_{Z_m} H - \partial_{z_i} Z_j \, \partial_t z_i \, , \end{aligned}$$

or with matrix formalism

$$\dot{\mathbf{Z}} = \mathbf{G}^T \mathbf{J} \mathbf{G} \nabla_{\mathbf{Z}} H - \mathbf{G}^T \, \partial_t \mathbf{z} \, .$$

- Stationary change of variables $\mathbf{z}(\mathbf{Z})$, $\partial_t \mathbf{z} = 0$, and thus

  $$\nabla_{\mathbf{Z}} K = \mathbf{G}^T \mathbf{J} \mathbf{G} \nabla_{\mathbf{Z}} H$$

  …
- Non-stationary transformation

  …

…

Properties of canonical transformations

- Inverse of $\mathbf{J}$, $\mathbf{J}^{-1} = -\mathbf{J}$.
- Value of $\mathbf{G}^T \mathbf{J} \mathbf{G}$

  Fix and uncomment
- Invariance of Poisson brackets

  todo

- Determinant of $\mathbf{G}^T\mathbf{J}\mathbf{G} = \mathbf{J}$

$$|\mathbf{G}|^2|\mathbf{J}| = |\mathbf{J}| \quad \rightarrow \quad |\mathbf{G}| = \mp 1 \, .$$

  The orientation-preserving transformation is the one with $|\mathbf{G}| = 1$.

- Elementary volume in the phase space

$$|d\mathbf{q}d\mathbf{p}| = |\mathbf{G}| \, |d\mathbf{Q} \, d\mathbf{P}| = |d\mathbf{Q} \, d\mathbf{P}|$$

A transformation from coordinates $(\mathbf{q}, \mathbf{p})$ to new variables $(\mathbf{Q}, \mathbf{P})$ is **canonical** if it preserves the form of Hamilton's equations under a new Hamiltonian $K(\mathbf{Q}, \mathbf{P}, t)$.

Using two sets of coordinates, the principle of stationary action reads

$$0 = \delta \int_{t_0}^{t_1} L(\dot{\mathbf{q}}, \mathbf{q}, t) \, dt = \delta \int_{t_0}^{t_1} \mathcal{L}(\dot{\mathbf{Q}}, \mathbf{Q}, t) \, dt \, .$$

The two Lagrangian functions can differ by a time-derivative $\dfrac{d}{dt}F_1(\mathbf{q}(t), \mathbf{Q}(t), t)$ at most. Thus, using the relation between the Lagrangian and the Hamiltonian function, it follows

$$\mathbf{p} \cdot \dot{\mathbf{q}} - H(\mathbf{q}, \mathbf{p}, t) = \mathbf{P} \cdot \dot{\mathbf{Q}} - K(\mathbf{Q}, \mathbf{P}, t) + \frac{d}{dt}F_1(\mathbf{q}, \mathbf{Q}, t)$$

Moving $\frac{dF_1}{dt}$ on one side, and expanding the time derivative,

$$\begin{aligned} \frac{dF_1}{dt} &= \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} F_1 + \dot{\mathbf{Q}} \cdot \partial_{\mathbf{Q}} F_1 + \partial_t F_1 = \\ &= \mathbf{p} \cdot \dot{\mathbf{q}} - \mathbf{P} \cdot \dot{\mathbf{Q}} - H(\mathbf{q}, \mathbf{p}, t) + K(\mathbf{Q}, \mathbf{P}, t) \end{aligned}$$

Exploiting the arbitrariness of the functions involved, the following relations hold

$$\mathbf{p} = \partial_{\mathbf{q}} F_1 \qquad , \qquad \mathbf{P} = -\partial_{\mathbf{Q}} F_1 \qquad , \qquad K - H = \partial_t F_1$$

Another *Legendre transformation* introduces a type-2 generating function $F_2(\mathbf{q}, \mathbf{P}, t)$,

$$F_2(\mathbf{q}, \mathbf{P}, t) := F_1(\mathbf{q}, \mathbf{Q}, t) + \mathbf{Q} \cdot \mathbf{P} \, ,$$

that allows to go from the independent pair of variables $(\mathbf{q}, \mathbf{Q})$ to $(\mathbf{q}, \mathbf{P})$, as

$$\begin{aligned} d_t F_2 &= \partial_t F_1 + \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} F_1 + \dot{\mathbf{Q}} \cdot \underbrace{\left(\partial_{\mathbf{Q}} F_1 + \mathbf{P}\right)}_{=\mathbf{0}} + \dot{\mathbf{P}} \cdot \mathbf{Q} = \\ &= \partial_t F_1 + \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} F_1 + \dot{\mathbf{P}} \cdot \mathbf{Q} \, , \end{aligned}$$

so that

$$\partial_{\mathbf{q}} F_2 = \partial_{\mathbf{q}} F_1 = \mathbf{p} \qquad , \qquad \partial_{\mathbf{P}} F_2 = \mathbf{Q} \qquad , \qquad \partial_t F_2 = \partial_t F_1 = K - H$$

The Hamilton-Jacobi Equation

Hamilton-Jacobi equation

Let the canonical transformation so that the Hamiltonian function in $\mathbf{Q}$, $\mathbf{P}$ is identically zero, i.e.

$$K(\mathbf{Q}, \mathbf{P}, t) := 0 \, .$$

Remark. This is not a 1-dimensional constraint between $\mathbf{Q}$, $\mathbf{P}$. This is the expression of the function, like the function $f(x) := 0$ is a real function whose value is 0 for all the x of the domain. Thus, all the derivatives of the function w.r.t. the independent variables are identically zero as well,

$$\partial_{\mathbf{Q}} K = \mathbf{0} \qquad , \qquad \partial_{\mathbf{P}} K = \mathbf{0} \qquad , \qquad \partial_t K = 0 \, .$$

If these relations hold, from the Hamiltonian equations, it follows that $\mathbf{Q}$ and $\mathbf{P}$ are constant in time. Thus,

$$\begin{aligned} 0 = K(\mathbf{Q}, \mathbf{P}, t) = \\ = H(\mathbf{q}, \mathbf{p}, t) + \partial_t F_2(\mathbf{q}, \mathbf{P}, t) = \\ = H(\mathbf{q}, \partial_{\mathbf{q}} F_2(\mathbf{q}, \mathbf{P}, t), t) + \partial_t F_2(\mathbf{q}, \mathbf{P}, t) = \end{aligned}$$

If H is not explicitly dependent on t, and thus the system is conservative as $d_t H = \partial_t H = 0$, it follows $H(\mathbf{q}, \mathbf{p}) = E$, constant. Hamilton-Jacobi equation thus becomes

$$0 = H\left(\mathbf{q}, \partial_{\mathbf{q}} F_2\right) + \partial_t F_2 \, .$$

As H doesn't explicitly depends on time, it's possible to look for a solution of the equation $F_2(\mathbf{q}, \mathbf{P}, t) = S(\mathbf{q}, \mathbf{P}) - T(t)$,

$$0 = H\left(\mathbf{q}, \partial_{\mathbf{q}} S\right) + \dot{T} = 0 \, ,$$

and thus

$$H\left(\mathbf{q}, \partial_{\mathbf{q}} S\right) = -\dot{T} = E \, ,$$

and thus

$$\begin{aligned} T(t) = -Et + T_0 \\ H\left(\mathbf{q}, \partial_{\mathbf{q}} S\right) = E \, . \end{aligned}$$

The ultimate goal of Hamilton-Jacobi theory is to find a canonical transformation to a set canonical variables $(\mathbf{w}, \mathbf{J})$ with **constant** momentum variables $\mathbf{J}$ and making the new Hamiltonian identically zero ($K = 0$). Hamilton equations read

$$\begin{cases} \dot{\mathbf{w}} = \partial_{\mathbf{J}} K \\ \dot{\mathbf{J}} = -\partial_{\mathbf{w}} K \end{cases}$$

If $\mathbf{J}$ is constant, thus $\dot{\mathbf{J}} = \mathbf{0}$ and $\partial_{\mathbf{Q}} K = 0$. Thus $K(\mathbf{J}, t)$. As $K(\mathbf{J}, t) = 0$, and $\mathbf{J}$ constant, thus $0 = d_t K = \partial_t K$. Then it follows that $K(\mathbf{J}) = 0$. As $\mathbf{J}$ are constant, then $\partial_{\mathbf{J}} K$ is constant as well. The first Hamilton equation implies $\mathbf{w}(t) = \partial_{\mathbf{J}} K \cdot t + \mathbf{w}_0$.

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Time derivative of Type-1 generating function, $F_1(\mathbf{q}, \mathbf{Q}, t)$, becomes

$$d_t F_1 = \dot{\mathbf{q}} \cdot \mathbf{p} - \underbrace{\dot{\mathbf{w}}}_{=\partial_{\mathbf{J}} K} \cdot \mathbf{J} + \underbrace{\dot{K}}_{=0} - H(\mathbf{q}, \mathbf{p}, t) =$$

Time derivative of Type-2 generating function, $F_2(\mathbf{q}, \mathbf{P}, t)$, becomes

$$\begin{aligned} d_t F_2 = \partial_t F_1 + \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} F_1 + \underbrace{\dot{\mathbf{J}} \cdot \mathbf{w}}_{=\mathbf{0}} = \\ = -H(\mathbf{q}, \mathbf{p}, t) + \dot{\mathbf{q}} \cdot \mathbf{p} \end{aligned}$$

Choosing $F_2(\mathbf{q}, \alpha, t) = S(\mathbf{q}, \alpha) - Et$, where $S(\mathbf{q}, \alpha)$ is **Hamilton's characteristic function**, Hamilton's principal equation reduces to the time-independent **Hamilton-Jacobi Equation**:

If $H(\mathbf{q}, \mathbf{p}$, then the system is conservative. Exploiting separation of variables, $F_2(\mathbf{q}, \mathbf{J}, t) = S(\mathbf{q}, \mathbf{J}) - Et$

$$H\left(\mathbf{q}, \partial_{\mathbf{q}} S(\mathbf{q}, \mathbf{J})\right) = E \, .$$

As $\dot{\mathbf{J}} = 0$, the time derivative of S reads

$$\dot{S} = \dot{\mathbf{q}} \cdot \partial_{\mathbf{q}} S + \underbrace{\dot{\mathbf{J}} \cdot \partial_{\mathbf{J}} S}_{=\mathbf{0}} = \dot{\mathbf{q}} \cdot \mathbf{p}(\mathbf{q}, \mathbf{J}) \, .$$

If the system is **separable**, $S(\mathbf{q}, \alpha)$ splits into independent single-variable functions:

$$S(\mathbf{q}, \mathbf{J}) = \sum_{i=1}^{f} S_i(q^i, \mathbf{J})$$

so that its time derivative reads

$$\dot{S} = \dot{q}^i \, S_i'(q_i, \mathbf{J}) = \dot{q}^i \, p_i(q^i, \mathbf{J}) \, .$$

Action-Angle Variables

Anction-angle variables

Let the canonical transformation **w**, **J**, so that **J** is constant. From Hamilton equations

$$\begin{cases} \dot{\mathbf{w}} = \partial_{\mathbf{J}} K \\ \dot{\mathbf{J}} = -\partial_{\mathbf{w}} K = \mathbf{0} \end{cases}$$

and thus K is a function of $\mathbf{J}, t$ only, $K(\mathbf{J}, t)$.

…

For bound, periodic systems whose Hamilton-Jacobi equations are separable, the most suitable coordinate system consists of **Action-Angle Variables** $(\mathbf{w}, \mathbf{J})$.

Definition of Action Variables

The **Action Variable** J_i associated with the i-th degree of freedom is defined as the line integral of momentum over one complete cycle of motion in phase space:

$$J_i \equiv \oint p_i \, dq_i = \oint \frac{\partial S_i(q_i, \alpha)}{\partial q_i} \, dq_i$$

Here, the loop $\oint$ denotes:

- A complete libration (back-and-forth oscillation between turning points), or
- A 2π rotation for angular coordinates.

Angle Variables and Frequencies

Because the action variables J_i are constants of motion, we can express the total Hamiltonian purely as a function of the action variables: $H = H(J_1, J_2, \dots, J_f)$.

The canonical conjugates to J_i are the **Angle Variables** w_i, defined via the generating function $S(\mathbf{q}, \mathbf{J})$:

$$w_i = \frac{\partial S(\mathbf{q}, \mathbf{J})}{\partial J_i}$$

Hamilton's equations of motion in action-angle variables simplify to:

$$\dot{J}_i = -\frac{\partial H}{\partial w_i} = 0 \implies J_i = \text{constant}$$

$$\dot{w}_i = \frac{\partial H(J_1, \dots, J_f)}{\partial J_i} \equiv \nu_i(\mathbf{J}) \implies w_i(t) = \nu_i t + w_i(0)$$

Where ν_i is the exact **fundamental frequency** of the classical motion along coordinate q_i.

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4. The Wilson-Sommerfeld Quantization Postulate

The central insight of William Wilson (1915) and Arnold Sommerfeld (1916) was that quantization should **not** be applied arbitrarily to any set of phase space coordinates. Instead, quantization conditions must be invariant under canonical transformations.

The integral invariants of Poincaré show that the total phase space volume element $\sum_i \oint p_i dq_i$ is a canonical invariant. Therefore, quantization must be imposed directly onto the adiabatic invariants of the classical motion: **the Action Variables** J_i.

The Quantization Rule

The Wilson-Sommerfeld quantization rule dictates that each action variable J_i is restricted to integer multiples of Planck's constant h:

$$J_i = \oint p_i \, dq_i = n_i h, \quad n_i \in \mathbb{N}$$

Summary of Connection to Quantum Mechanics

1. **Separability:** The classical system must be separable in coordinates $(q_1, \dots, q_f)$.
2. **Phase Space Loops:** The integral $\oint p_i dq_i$ measures the area enclosed by the periodic trajectory in the (q_i, p_i) 2D projection of phase space.
3. **Discretization:** Quantizing $J_i = n_i h$ slices phase space into discrete cells of volume h^f.
4. **Energy Spectrum:** Inverting $H(J_1, \dots, J_f)$ with $J_i \to n_i h$ yields the quantized energy levels $E(n_1, n_2, \dots, n_f)$.

22.5 Schrodinger model

Contents

- *Mathematical details of the analytical solution*
- *Gallery of orbitals*

- *Auxiliary notes*

Schrodinger equation. Schrodinger equation in space basis for the electron in a H atom reads

$$i\hbar\partial_t\Psi(\mathbf{r},t)=\left[-\frac{\hbar^2}{2m}\nabla^2-\frac{q^2}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|}\right]\Psi(\mathbf{r},t)$$

where the Hamiltonian is the sum of the kinetic energy and a Coulomb potential contribution, $\hat{V}$, so that $\langle\mathbf{r}|\hat{V}|\Psi\rangle=-\frac{q^2}{4\pi\varepsilon|\mathbf{r}|}\Psi(\mathbf{r},t)$

Spectral decomposition of the Hamiltonian operator. The stationary states, and the corresponding energy levels - i.e. the eigenfunctions, and the eigenvalues of the Hamiltonian operator $\hat{H}$ - are the result of the eigenvalue problem, $\hat{H}|\Psi\rangle=E|\Psi\rangle$, or using space basis,

$$\left[-\frac{\hbar^2}{2m}\nabla^2-\frac{q^2}{4\pi\varepsilon}\frac{1}{|\mathbf{r}|}\right]\Psi(\mathbf{r})=E\Psi(\mathbf{r})\ .$$

Quantum numbers. The stationary states depend on **three** - in this model, no spin exists - **quantum numbers**:

- **principal** quantum number $n\in\{1,2,...\}$
- **azimuthal** quantum number $\ell\in\{0,1,...,n-1\}$
- **magnetic** quantum number $m_\ell\in\{-\ell,-\ell+1,...,\ell\}$

Energy levels only depend on the principal quantum number, $E(n)$. Thus, Schrodinger model of the atom has **degenerate** stationary states, i.e. stationary states with the same energy value.

Commutation of $\hat{H}$, $\hat{L}^2$, $\hat{L}_z$. As angular momentum operators $\hat{L}^2$, $\hat{L}_z$ commute with the Hamiltonian operator $\hat{H}$ of the H atom, these three operators share common eigenvectors. These relations are proved in the *note section*.

22.5.1 Schrodinger model - Mathematical details of the analytical solution

Schrodinger equation for the hydrogen atom in space basis reads

$$i\hbar\Psi(\mathbf{r},t)=\left[-\frac{\hbar^2}{2m}\nabla^2-\frac{q^2}{4\pi\varepsilon}\frac{1}{r}\right]\Psi(\mathbf{r},t)\ .$$

The **eigenproblem of the Hermitian operator** (projected onto space basis, as well) reads

$$E_n\Psi_n(\mathbf{r})=-\frac{\hbar^2}{2m}\nabla^2\Psi_n(\mathbf{r})-\frac{q^2}{4\pi\varepsilon r}\Psi_n(\mathbf{r})\ .$$

Given the symmetries of the system, the solution of the eigenproblem is evaluated using spherical coordinates $\{r,\theta,\varphi\}$, and the method of separation of variables. Using **spherical coordinates**,

$$-\frac{\hbar^2}{2m}\left\{\frac{1}{r^2}\partial_r\left(r^2\partial_r\right)+\frac{1}{r^2\sin\theta}\partial_\theta\left(\sin\theta\partial_\theta\right)+\frac{1}{r^2\sin^2\theta}\partial_{\varphi\varphi}\right\}\Psi_n-\frac{q^2}{4\pi\varepsilon_0 r}\Psi_n=E_n\Psi_n$$

Separation of variables, $\Psi_{nlm}(r,\theta,\varphi)=R_n(r)Y_{lm}(\theta,\varphi)$

With the **method of separation of variables**, we look for solutions in product form:

$$\Psi_{nlm}(r,\theta,\varphi)=R_n(r)Y_{lm}(\theta,\varphi)\ ,$$

and substituting into the Schrödinger equation:

$$\left(E_n+\frac{q^2}{4\pi\varepsilon_0 r}\right)R_nY_{lm}=-\frac{\hbar^2}{2m}\left[\frac{1}{r^2}\left(r^2R_n'\right)'Y_{lm}+R_n\left(\frac{1}{r^2\sin\theta}\partial_\theta\left(\sin\theta\partial_\theta Y_{lm}\right)+\frac{1}{r^2\sin^2\theta}\partial_{\varphi\varphi}Y_{lm}\right)\right]$$

Dividing by $R_n Y_{lm}$, and multiplying by $r^2 \frac{2m}{\hbar^2}$,

$$0 = \underbrace{\frac{1}{R_n}(r^2 R_n')' + \frac{2mr^2}{\hbar^2}\left(E_n + \frac{q^2}{4\pi\varepsilon_0 r}\right)}_{f(r)} + \underbrace{\frac{1}{Y_{lm}}\left\{\frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta Y_{lm}) + \frac{1}{\sin^2\theta}\partial_{\varphi\varphi}Y_{lm}\right\}}_{g(\theta,\varphi)}$$

Since $f(r) + g(\theta, \varphi) = 0$, both terms must equal equal-and-opposite constant C for all valid domains $r \in (0, +\infty)$, $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$

$$C = f(r) = -g(\theta, \varphi) \, .$$

Radial equation, and energy levels

The radial component determines the energy level,

$$E_n = -\frac{mq^4}{32\pi^2\varepsilon_0^2\hbar^2}\frac{1}{n^2} \, . \tag{22.2}$$

being $n \in \{1, 2, \dots\}$.

The radial distribution of the energy state functions is

$$R(\rho) = e^{-\rho/2}\rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) \, ,$$

with $\rho = r\frac{\sqrt{-8mE_n}}{\hbar}$ the non dimensional radius, $\ell \in \{0, 1, \dots, n-1\}$, and $L_p^k(x)$ the solution of the associated Laguerre differential equation $xy'' + (k + 1 - x)y' + py = 0$.

Change of variables, and asymptotic behavior

Defining $C = \ell(\ell + 1)$, and dividing the equation $f(r) = C$ by r^2, for the states with $E_n < 0$,

$$\frac{1}{r^2}\frac{d}{dr}(r^2 R') + \left[\frac{2m}{\hbar^2}\left(E + \frac{q^2}{4\pi\varepsilon_0 r}\right) - \frac{\ell(\ell+1)}{r^2}\right]R = 0$$

Defining the **dimensionless radial variable** $\rho = \kappa r$, so $\frac{d}{dr} = \kappa\frac{d}{d\rho}$, and factoring out κ^2

$$0 = \kappa^2\left\{\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{dR}{d\rho}\right) + \left(\frac{2mE}{\hbar^2\kappa^2} + \frac{2mq^2}{\hbar^2\kappa\cdot 4\pi\varepsilon_0\rho} - \frac{\ell(\ell+1)}{\rho^2}\right)\right\}R$$

By choosing $\kappa^2 = -\frac{8mE}{\hbar^2}$ (where $\kappa = \frac{\sqrt{-8mE}}{\hbar}$ for $E < 0$), the first term inside the parentheses becomes -1. Defining the constant parameter ρ_0:

$$\rho_0 = \frac{2mq^2}{\hbar^2\kappa\cdot 4\pi\varepsilon_0} = \frac{2mq^2}{\hbar^2\sqrt{-8mE}\cdot 4\pi\varepsilon_0} = \frac{q^2}{8\pi\varepsilon_0\hbar}\sqrt{\frac{2m}{-E}} \, ,$$

the dimensionless radial equation simplifies to:

$$0 = \frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{dR}{d\rho}\right) + R\left[-\frac{\ell(\ell+1)}{\rho^2} - \frac{1}{4} + \frac{\rho_0}{\rho}\right]$$

Boundary conditions required for regular wavefunctions:

$$R(\rho = 0) < \infty, \quad R(\rho \to +\infty) = 0$$

Asymptotic behavior for $r \to 0$, and $r \to +\infty$

Large Distance Limit ($\rho \to +\infty$). For large ρ, terms proportional to $\frac{1}{\rho}$ and $\frac{1}{\rho^2}$ become negligible:

$$\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2 R'\right) = R'' + \frac{2}{\rho}R' \sim R''$$

The equation reduces to:

$$R'' - \frac{1}{4}R \sim 0$$

General solution:

$$R(\rho) = Ae^{-\rho/2} + Be^{\rho/2}$$

To satisfy the non-divergent boundary condition at infinity, set $B = 0$:

$$R(\rho) \sim e^{-\rho/2} \quad \text{as } \rho \to +\infty$$

Small Distance Limit ($\rho \to 0$). For $\rho \to 0$, the centrifugal term $\frac{\ell(\ell+1)}{\rho^2}$ dominates over the constant and $\frac{1}{\rho}$ terms:

$$R'' + \frac{2}{\rho}R' - \frac{\ell(\ell+1)}{\rho^2}R \sim 0$$

Proposing a power-law ansatz $R(\rho) \sim \rho^s$:

$$s(s-1)\rho^{s-2} + 2s\rho^{s-2} - \ell(\ell+1)\rho^{s-2} = 0$$

$$s^2 + s - \ell(\ell+1) = 0$$

Solving the quadratic characteristic equation for s:

$$s_{1,2} = \frac{-1 \mp \sqrt{1 + 4\ell(\ell+1)}}{2} = \frac{-1 \mp (1+2\ell)}{2}$$

$$s_1 = \ell, \quad s_2 = -(\ell+1)$$

Since $\ell \geq 0$, $s_2 = -(\ell+1)$ leads to divergence at the origin $\rho \to 0$. Retaining only the physically acceptable solution $s = \ell$:

$$R(\rho) \sim \rho^\ell \quad \text{as } \rho \to 0$$

Transformation & Differential Equation for $v(\rho)$. Combining both asymptotic behaviors, we represent the full radial solution as:

$$R(\rho) = \rho^\ell e^{-\rho/2} v(\rho)$$

Writing out derivatives of $R(\rho)$:

$$R' = \left(\ell\rho^{\ell-1} - \frac{1}{2}\rho^\ell\right)e^{-\rho/2}v + \rho^\ell e^{-\rho/2}v'$$

After expanding $\frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2 R'\right) + R\left[-\frac{\ell(\ell+1)}{\rho^2} - \frac{1}{4} + \frac{\rho_0}{\rho}\right] = 0$:

$$= e^{-\rho/2}\left\{\rho^\ell v'' + v'\left[2\ell\rho^{\ell-1} - \frac{1}{2}\rho^\ell + 2\rho^{\ell-1}\right] + v\left[\rho^\ell\ell(\ell+1) - \rho^{\ell+1}(\ell+1) + \frac{1}{4}\rho^{\ell+2} - \ell(\ell+1)\rho^\ell - \frac{1}{4}\rho^{\ell+2} + \rho_0\rho^{\ell+1}\right]\right\} = 0$$

Dividing out non-zero factor $\rho^{\ell-1}e^{-\rho/2}$:

$$\rho v''(\rho) + [2(\ell+1) - \rho]v'(\rho) + (\rho_0 - \ell - 1)v(\rho) = 0$$

Polynomial expansion, energy quantization E_n and radial distribution $R_n(\rho)$

Starting from the transformed radial equation:

$$\rho v''(\rho) + [2(\ell+1) - \rho]v'(\rho) + (\rho_0 - \ell - 1)v(\rho) = 0$$

Looking for a power series solution:

$$v(\rho) = \sum_{k=0}^{\infty} c_k \rho^k$$
$$v'(\rho) = \sum_{k=0}^{\infty} k c_k \rho^{k-1}$$
$$v''(\rho) = \sum_{k=0}^{\infty} k(k-1) c_k \rho^{k-2}$$

Substituting into the differential equation:

$$\sum_{k=0}^{\infty} k(k-1)c_k\rho^{k-1} + \sum_{k=0}^{\infty} 2(\ell+1)kc_k\rho^{k-1} - \sum_{k=0}^{\infty} kc_k\rho^k + (\rho_0 - \ell - 1)\sum_{k=0}^{\infty} c_k\rho^k = 0$$

Shift indices to group terms by power ρ^k:

$$\sum_{k=0}^{\infty} \Big\{ (k+1)(k)c_{k+1} + 2(\ell+1)(k+1)c_{k+1} - kc_k + (\rho_0 - \ell - 1)c_k \Big\} \rho^k = 0$$

Recurrence Relation. Equating coefficients to zero yields the recurrence relation for c_k:

$$c_{k+1} = c_k \frac{k + \ell + 1 - \rho_0}{(k+1)(k+2\ell+2)}$$

Asymptotic Behavior & Series Termination. If the power series does not terminate, as $k \to +\infty$:

$$\frac{c_{k+1}}{c_k} \longrightarrow \frac{1}{k}$$

This asymptotic behavior matches that of the exponential expansion $e^\rho \sim \sum \frac{\rho^k}{k!}$ and thus $\frac{c_{k+1}}{c_k} = \frac{1}{k+1} \sim \frac{1}{k}$. Thus, an infinite series $v(\rho) \sim e^\rho$, would lead to $R(\rho) \sim e^{-\rho/2}e^\rho = e^{\rho/2}$, which diverges as $\rho \to +\infty$. Thus, in order to maintain physical normalizability, the series must terminate at a maximum power $k_{\text{max}} = N$:

$$c_{N+1} = 0 \implies N + \ell + 1 - \rho_0 = 0 \implies \rho_0 = N + \ell + 1 \equiv n$$

where:

- $N \geq 0$ is the radial quantum number.
- $n = N + \ell + 1$ is the **Principal Quantum Number** ($n \geq 1$).
- $\forall n$, the orbital angular momentum quantum number satisfies $0 \leq \ell \leq n - 1$.

Energy Quantization. Recalling the definition of ρ_0, $n = \rho_0 = \sqrt{\frac{2m}{-E}}\frac{q^2}{4\pi\varepsilon_0\hbar}$, and solving for the bound state energies E_n ($E < 0$), the relation between the energy E_n of the nth energy level and the principal quantum number n follows

$$E_n = -\frac{mq^4}{32\pi^2\varepsilon_0^2\hbar^2}\frac{1}{n^2}\,. \tag{22.3}$$

Remark. The value of energy E_n of Schrodinger model coincides with the value (22.1) of the energy produced by *Bohr model*.

Radial solution. Putting everything together, the radial part of the solution reads

$$R(\rho) = e^{-\rho/2}\rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho)\,,$$

being $L_p^k(x)$ the solution of the associated Laguerre differential equation $xy'' + (k+1-x)y' + py = 0$.

Azimuthal and polar equations

Further separation of variables, $Y_{\ell m}(\theta,\varphi)=\Theta(\theta)\Phi(\varphi)$

With $Y(\theta,\varphi)=\Theta(\theta)\Psi(\varphi)$, the equation

$$\frac{1}{Y_{lm}}\left\{\frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta Y_{lm})+\frac{1}{\sin^2\theta}\partial_{\varphi\varphi}Y_{lm}\right\}=-\ell(\ell+1)\ .$$

becomes - after multipyling by $\sin^2\theta$,

$$\underbrace{\frac{\sin\theta}{\Theta}\left(\sin\theta\,\Theta'\right)'+\ell(\ell+1)\,\sin^2\theta}_{F(\theta)}=-\underbrace{\frac{\Phi''}{\Phi}}_{G(\varphi)}\ ,$$

and thus

$$F(\theta)=-G(\varphi)=m_\ell^2\ ,$$

with a non-negative constant m_ℓ^2 in order to match periodic conditions for $\Phi(\varphi=0)=\Phi(\varphi=2\pi)$.

Azimuthal equation

$$\Phi''+m_\ell^2\Phi=0\ ,$$

whose solution is $\Phi(\varphi)\propto\exp\left(\mp im_\ell\varphi\right)$. In order to satisfy periodic conditoins, $m_\ell\in\mathbb{Z}$. This parameter will be furthered constrained later, in the solution of the polar equation, to be $m_\ell\in\{-\ell,-\ell+1,\dots,\ell\}$, with $\ell\in\mathbb{N}$. Thus, selecting only the independent solutions, the azimuthal distribution of the eigenfunctions of the Hamiltonian is

$$\Phi(\varphi)\propto\exp\left(im_\ell\varphi\right)\ ,\quad m_\ell\in\{-\ell,-\ell+1,\dots\ell\}\ .$$

Polar equation

Multiplying the equation $F(\theta)=m_{ell}^2$ by Θ, the polar equation becomes,

$$\sin\theta\left(\sin\theta\,\Theta'\right)'+\left[\ell(\ell+1)\,\sin^2\theta-m_\ell^2\right]\Theta=0\ .$$

Polar equation as an associated Legendre equation

Dividing by $\sin^2\theta=1-\cos^2\theta$, the second term can be immediately recast as the second term in the associated Legendre equation, if $x=\cos\theta$.

As $\frac{dx}{d\theta}=-\sin\theta$, then $\frac{d}{d\theta}=\frac{dx}{d\theta}\frac{d}{dx}=-\sin\theta\frac{d}{dx}$, the first term - divided by $\sin^2\theta$ - becomes

$$-\frac{1}{\sin^2\theta}\sin^2\theta\frac{d}{dx}\left(-\sin^2\theta\frac{d}{dx}\Theta\right)=\frac{d}{dx}\left[(1-x^2)\frac{d}{dx}\Theta\right]$$

Thus the polar equation can be recast as

$$0=\frac{d}{dx}\left[(1-x^2)\frac{d\Theta}{dx}\right]+\left[\ell(\ell+1)-\frac{m_\ell^2}{1-x^2}\right]\Theta\ ,$$

i.e. as the associated Legendre equation whose solution (see below) is $\Theta(x)=P_\ell^{m_\ell}(x)$, for $m_\ell\in\{-\ell,\dots,\ell\}$, $\ell\in\mathbb{N}$.

Legendre differential equation and associated Legendre equation

Legendre differential equations read

$$\begin{aligned} 0 &= (1-x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = \quad , \qquad x \in [-1,1] \\ &= \frac{d}{dx}\left[(1-x^2)P_n'(x)\right] + n(n+1)P_n(x) \, , \end{aligned}$$

with the solution $P_n(x)$ be the Legendre polynomial. The **associated Legendre differential equations** are defined as

$$0 = \frac{d}{dx}\left[(1-x^2)P_n'(x)\right] + \left[n(n+1) - \frac{m^2}{1-x^2}\right] P_n(x) \quad , \qquad x \in [-1,1]$$

and the solution reads $P_n^m(x) = (-1)^m \left(1-x^2\right)^{\frac{m}{2}} \frac{d^m}{dx^m} P_n(x)$, for $m \geq 0$. The soutions for $m < 0$ are defined via symmetry conditions, or Rodrigues' formula, as $P_\ell^{-m} = (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m(x)$, see Wikipedia: associated Lagendre polynomials.

22.5.2 Schrodinger model - Gallery

Stationary states

$$\psi_{n,\ell,m_\ell}(r,\theta,\phi) = R_{n\ell}(r)\, Y_\ell^{m_\ell}(\theta,\phi)$$

Radial Wavefunctions, $R_{n\ell}(r)$. Using the dimensionless variable $\rho = \frac{2r}{na_0}$,

$$R_{n\ell}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]^3}} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) \, ,$$

where $a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}$ is the **Bohr radius**, and $L_p^k(\rho)$ the solution of the associated Laguerre equation.

Spherical Harmonics, $Y_\ell^{m_\ell}(\theta,\phi) = \Theta_{\ell,m_\ell}(\theta)\Phi_{m_\ell}(\phi)$.

$$Y_\ell^{m_\ell}(\theta,\phi) = (-1)^{m_\ell} \sqrt{\frac{(2\ell+1)}{4\pi} \frac{(\ell-m_\ell)!}{(\ell+m_\ell)!}} P_\ell^{m_\ell}(\cos\theta)\, e^{im_\ell\phi}$$

with $P_\ell^{m_\ell}$ the associated Legendre polynomials. The polar functions are real-valued and by definition $\Theta_{\ell,-m_\ell}(\theta) = (-1)^{m_\ell}\Theta_{\ell,m_\ell}(\theta)$.

```
#> Spatial resolution and default parameters
```

```
#> Functions
```

```
#> Evaluate stationary states
```

```
#> Plot functions
```

```
#> Isosurface of probability density of stationary states, colored with phase
```

```
#> Plot a set of stationary states
```

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Real orbitals

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#> Plot a set of real orbitals
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Some details

As $\Theta_{\ell,-m_\ell} = (-1)^{m_\ell}\Theta_{\ell,m_\ell}$, it follows that

$$\begin{cases} Y_\ell^{m_\ell} + (-1)^{m_{ell}} Y_\ell^{-m_\ell} = \Theta_{\ell,m_\ell}(\theta)\left[e^{im_\ell\phi} + e^{-im_\ell\phi}\right] \propto \cos(m_\ell\theta) \\ Y_\ell^{m_\ell} - (-1)^{m_{ell}} Y_\ell^{-m_\ell} = \Theta_{\ell,m_\ell}(\theta)\left[e^{im_\ell\phi} - e^{-im_\ell\phi}\right] \propto i\sin(m_\ell\theta)\ , \end{cases}$$

and thus the real orbitals as the linear combinations

$$\psi_{n,\ell,m_\ell,\text{real}} = \begin{cases} \psi_{n,\ell,0} & \text{for } m_\ell = 0 \\ \frac{1}{\sqrt{2}}\left(\psi_{n,\ell,-|m_\ell|} + (-1)^{m_\ell}\psi_{n,\ell,|m_\ell|}\right) \propto \cos(|m_\ell|\phi) & \text{for real orbital } (x, xy, x^2-y^2, \dots) \\ \frac{i}{\sqrt{2}}\left(\psi_{n,\ell,-|m_\ell|} - (-1)^{m_\ell}\psi_{n,\ell,|m_\ell|}\right) \propto \sin(|m_\ell|\phi) & \text{for real orbital } (y, yz, \dots) \end{cases}$$

Radial and spherical factors

$$\psi_{n,\ell,m_\ell}(r,\theta,\phi) = R_{n\ell}(r)\, Y_\ell^{m_\ell}(\theta,\phi)$$

Radial wavefunctions, $R_{n,\ell}(r)$

Radial Wavefunctions, $R_{n\ell}(r)$. Using the dimensionless variable $\rho = \frac{2r}{na_0}$,

$$R_{n\ell}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]^3}} e^{-\rho/2}\rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho)\ ,$$

where $a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}$ is the **Bohr radius**, and $L_p^k(\rho)$ the solution of the associated Laguerre equation.

Polar and azimuthal wavefunctions, $Y_\ell^{m_\ell}(\theta,\phi) = \Theta_{\ell,m_\ell}(\theta)\Phi_{m_\ell}(\phi)$

$$Y_\ell^{m_\ell}(\theta,\phi) = (-1)^{m_\ell}\sqrt{\frac{(2\ell+1)}{4\pi}\frac{(\ell-m_\ell)!}{(\ell+m_\ell)!}} P_\ell^{m_\ell}(\cos\theta)\, e^{im_\ell\phi}$$

with $P_\ell^{m_\ell}$ the associated Legendre polynomials. The first associated Legendre polynomials are

ℓ	m_ℓ	$P_\ell^{m_\ell}(x)$
0	0	1
1	1	$-(1-x^2)^{\frac{1}{2}}$
	0	x
	-1	$-\frac{1}{2}P_1^1(x)$
2	2	$3(1-x^2)$
	1	$-3x(1-x^2)^{\frac{1}{2}}$
	0	$\frac{1}{2}(3x^2-1)$
	-1	$-\frac{1}{6}P_2^1(x)$
	-2	$\frac{1}{24}P_2^2(x)$

22.5.3 Auxiliary Results on Operators for the Schrödinger Model of the Hydrogen Atom

1. All components of $\hat{\mathbf{L}}$ commute with the kinetic term $\frac{\hat{p}^2}{2m}$ and with $\hat{L}^2$, *proof*

$$[\hat{p}^2, \hat{L}_b] = 0$$

2. All components of $\hat{\mathbf{L}}$ commute with the potential $\hat{V}_{\text{Coulomb}}$, *proof*

$$[\hat{V}_{\text{Coulomb}}, \hat{L}_b] = 0$$

3. Therefore, all components of $\hat{\mathbf{L}}$ commute with the H atom Hamiltonian $\hat{H}_{\text{H-atom}} = \frac{\hat{p}^2}{2m} + \hat{V}_{\text{Coulomb}}$:

$$[\hat{H}_{\text{H-atom}}, \hat{L}_b] = 0$$

4. Simultaneous Commutative Set:

$$\boxed{[\hat{H}_{\text{H-atom}}, \hat{L}^2] = 0, \quad [\hat{H}_{\text{H-atom}}, \hat{L}_a] = 0, \quad [\hat{L}^2, \hat{L}_a] = 0}$$

   i.e. the Hamiltonian $\hat{H}_{\text{H-atom}}$, total angular momentum squared $\hat{L}^2$, and any component $\hat{L}_a$ all mutually commute, and thus share a set of common eigenvectors. For the commutation relation $[\hat{L}^2, \hat{L}_a] = 0$, see the section about *spatial angular momentum*.

Hamiltonian & Basic Definitions

The Hamiltonian of the electron in the Hydrogen atom with Coulomb potential is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}_{Coulomb} \ ,$$

that in space basis becomes

$$\langle \mathbf{r}|\hat{H} = -\frac{\hbar^2}{2m}\nabla^2\langle \mathbf{r}| - \frac{q^2}{4\pi\varepsilon|\mathbf{r}|}\langle \mathbf{r}| \ .$$

Space Basis Representations

$$\langle \mathbf{r}|\Psi\rangle = \Psi(\mathbf{r},t)$$

$$\langle \mathbf{r}|\hat{\mathbf{r}} = \mathbf{r}\langle \mathbf{r}| \quad \Longrightarrow \quad \langle \mathbf{r}|\hat{\mathbf{r}}|\Psi\rangle = \mathbf{r}\,\Psi(\mathbf{r},t)$$

$$\langle \mathbf{r}|\hat{\mathbf{p}} = -i\hbar\nabla_{\mathbf{r}}\langle \mathbf{r}| \quad \Longrightarrow \quad \langle \mathbf{r}|\hat{\mathbf{p}}|\Psi\rangle = -i\hbar\nabla\Psi(\mathbf{r},t)$$

Angular Momentum Operator

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$$

In Levi-Civita component notation, using Cartesian coordinates

$$\hat{\mathbf{L}} = \hat{\mathbf{e}}_a \hat{L}_a = \hat{\mathbf{e}}_a \varepsilon_{abc}\, \hat{r}_b \hat{p}_c$$

In the space (position) basis (Cartesian coordinates):

$$\begin{aligned}\langle \mathbf{r}|\hat{\mathbf{L}}|\Psi\rangle &= \langle \mathbf{r}|\hat{\mathbf{e}}_a \hat{L}_a|\Psi\rangle = \\ &= \hat{\mathbf{e}}_a \varepsilon_{abc}\, r_b(-i\hbar\partial_c\langle \mathbf{r}|\Psi\rangle) = \\ &= -i\hbar\hat{\mathbf{e}}_a \varepsilon_{abc}\, r_b\, \partial_c\Psi(\mathbf{r},t) = \\ &= -i\hbar\mathbf{r}\times\nabla_{\mathbf{r}}\Psi(\mathbf{r},t)\ .\end{aligned}$$

$\langle \mathbf{r}|\hat{\mathbf{p}}|\Psi\rangle$

$$\langle \mathbf{r}|\hat{\mathbf{p}}|\Psi\rangle = \int \langle \mathbf{r}|\hat{\mathbf{p}}|\mathbf{r}'\rangle\langle \mathbf{r}'|\Psi\rangle\, d\mathbf{r}' = \int \delta(\mathbf{r}-\mathbf{r}')\,(-i\hbar\nabla_{\mathbf{r}'}\Psi(\mathbf{r}',t))\, d\mathbf{r}' = -i\hbar\nabla_{\mathbf{r}}\Psi(\mathbf{r},t)$$

$\langle \mathbf{r}|\hat{p}_b\hat{r}_a|\Psi\rangle$

$$\langle \mathbf{r}|\hat{p}_b\hat{r}_a|\Psi\rangle = -i\hbar\partial_b\langle \mathbf{r}|\hat{r}_a|\Psi\rangle = -i\hbar\partial_b\left(r_a\langle \mathbf{r}|\Psi\rangle\right) = -i\hbar\partial_b\left(r_a\Psi(\mathbf{r},t)\right) = -i\hbar(\delta_{ab}\Psi + r_a\partial_b\Psi)$$

Canonical Commutation Relations (CCR)

Using postion basis and Cartesian coordinates, the CCR $[\hat{\mathbf{r}},\hat{\mathbf{P}}] = i\hbar\mathbb{I}$ reads

$$[r_a,p_b]\Psi = r_a(-i\hbar\partial_b\Psi) - (-i\hbar\partial_b(r_a\Psi)) = i\hbar\,\delta_{ab}\Psi\ ,$$

i.e.

$$[r_a,p_b] = i\hbar\,\delta_{ab}$$

Commutators of Angular Momentum with Position and Momentum

Commutator $[r_a,L_b]$

$$r_aL_b\Psi = r_a(-i\hbar\varepsilon_{bcd}\,r_c\partial_d\Psi) = -i\hbar\varepsilon_{bcd}\,r_ar_c\partial_d\Psi$$

$$L_b r_a \Psi = -i\hbar \varepsilon_{bcd}\, r_c \partial_d (r_a \Psi) = -i\hbar \varepsilon_{bcd}\, r_c \delta_{ad} \Psi - i\hbar \varepsilon_{bcd}\, r_c r_a \partial_d \Psi$$

Subtracting the two expressions gives:

$$[r_a, L_b]\Psi = i\hbar \varepsilon_{bcd}\, r_c \delta_{ad} \Psi = i\hbar \varepsilon_{bca}\, r_c \Psi \ ,$$

and thus

$$[r_a, L_b] = -i\hbar \left(\mathbf{r}\times\right)_{ab} \Psi \quad \text{(in general } \neq 0)$$

(Recall: $(\mathbf{a}\times\mathbf{b})_i = \varepsilon_{ijk} a_j b_k = \{\mathbf{a}_\times\}_{ik} \cdot \{\mathbf{b}\}_k$*, i.e.* $\{\mathbf{a}_\times\}_{ik} = \varepsilon_{ijk} a_j$*)*

Commutator $[p_a, L_b]$

$$p_a L_b \Psi = -i\hbar \partial_a \left(-i\hbar \varepsilon_{bcd}\, r_c \partial_d \Psi\right) = -\hbar^2 \varepsilon_{bcd}\, \partial_a (r_c \partial_d \Psi) = -\hbar^2 \varepsilon_{bad}\, \partial_d \Psi - \hbar^2 \varepsilon_{bcd}\, r_c \partial_a \partial_d \Psi$$

$$L_b p_a \Psi = -i\hbar \varepsilon_{bcd}\, r_c \partial_d (-i\hbar \partial_a \Psi) = -\hbar^2 \varepsilon_{bcd}\, r_c \partial_d \partial_a \Psi$$

Subtracting the two expressions gives:

$$[p_a, L_b]\Psi = -\hbar^2 \varepsilon_{bad}\, \partial_d \Psi = \hbar^2 \varepsilon_{abd}\, \partial_d \Psi = i\hbar \varepsilon_{abc}\, p_c \Psi$$

Cartesian coordinates.

- $[p_x, L_x] \equiv 0$
- $[p_x, L_y] = \hbar^2 \varepsilon_{xyz}\, \partial_z \Psi = i\hbar p_z$
- $[p_x, L_z] = \hbar^2 \varepsilon_{xzy}\, \partial_y \Psi = -\hbar^2 \partial_y \Psi = -i\hbar p_y$

Commutator $[\hat{p}^2, L_b]$

$$[p^2, L_b] = \sum_a p_a p_a L_b - L_b \sum_a p_a p_a = \sum_a \Big(p_a [p_a, L_b] + [p_a, L_b] p_a \Big)$$

Using $[p_a, L_b] = i\hbar \varepsilon_{abc} p_c$:

$$\sum_a \left(p_a (i\hbar \varepsilon_{abc} p_c) + (i\hbar \varepsilon_{abc} p_c) p_a \right) = i\hbar \varepsilon_{abc} (p_a p_c + p_c p_a)$$

Since $p_a p_c = p_c p_a$ (symmetric in a, c) and ε_{abc} is antisymmetric in a, c, the sum vanishes identically:

$$[p^2, L_b] = 0 \quad \text{for all components } b$$

Example. For $b = x$:

$$[p^2, L_x] = -i\hbar^3 \Big[1 \cdot \partial_z \partial_y \Psi - 1 \cdot \partial_y \partial_z \Psi \Big] = 0$$

Commutator of Coulomb Potential $\hat{V}$ with Angular Momentum $\hat{L}_b$

In space basis, the Coulomb potential reads:

$$\langle \mathbf{r}|\hat{V}|\Psi\rangle = -\frac{q^2}{4\pi\varepsilon_0}\frac{1}{r}\Psi(\mathbf{r},t) = -k\frac{1}{|\mathbf{r}|}\Psi(\mathbf{r},t)$$

Let's evaluate whether $\hat{V}$ and $\hat{L}_b$ commute:

1. **Action of $\hat{V}\hat{L}_b$:**

$$\langle \mathbf{r}|\hat{V}\hat{L}_b|\Psi\rangle = -k\frac{1}{\sqrt{x_a x_a}}\left(-i\hbar\varepsilon_{bcd}\, r_c\, \partial_d\Psi\right)$$

2. **Action of $\hat{L}_b\hat{V}$, with $|\mathbf{r}| = \sqrt{x_a x_a}$:**

$$\langle \mathbf{r}|\hat{L}_b\hat{V}|\Psi\rangle = -i\hbar\varepsilon_{bcd}\, r_c\, \partial_d\left(-k\frac{\Psi}{\sqrt{x_a x_a}}\right) = i\hbar k\varepsilon_{bcd}\, r_c\left[-\frac{\partial_d|\mathbf{r}|}{|\mathbf{r}|^2}\Psi + \frac{1}{|\mathbf{r}|}\partial_d\Psi\right]$$

   Since $\partial_d|\mathbf{r}| = \partial_d\sqrt{x_a x_a} = \frac{x_d}{|\mathbf{r}|}$:

$$\langle \mathbf{r}|\hat{L}_b\hat{V}|\Psi\rangle = i\hbar k\varepsilon_{bcd}\, r_c\left[-\frac{r_d}{|\mathbf{r}|^3}\Psi + \frac{1}{|\mathbf{r}|}\partial_d\Psi\right]$$

3. **Commutator $[\hat{V},\hat{L}_b]$:**

$$\langle \mathbf{r}|[\hat{V},\hat{L}_b]|\Psi\rangle = -i\hbar k\,\frac{\varepsilon_{bcd}\, r_c r_d}{|\mathbf{r}|^3}\Psi = 0$$

   Why? The product $r_c r_d$ is **symmetric** under swapping $c \leftrightarrow d$, while ε_{bcd} is **antisymmetric** under swapping $c \leftrightarrow d$. Summing over c, d gives zero!

   Example for $b = x$. $\varepsilon_{xcd}r_c r_d = \varepsilon_{xyz}yz + \varepsilon_{xzy}zy = yz - zy = 0$.

22.6 Atomic model using Dirac equation

CHAPTER

TWENTYTHREE

RELATIVISTIC QUANTUM MECHANICS

- Spin - some links in:
  - *historical approach to QM: spin*
  - *angular momentum operator*
  - …
- *Dirac equation*
- *Atomic model with Dirac equation* - in **atomic model** section

23.1 Dirac equation

23.1.1 Klein-Gordon equation for a free particle

The relation between energy and momentum for a massive particle in *special relativity*

$$m^2c^2 = \frac{E^2}{c^2} - |\mathbf{p}|^2 \,, \tag{23.1}$$

can be seamlessly promoted to a operator equation, $\mathbf{p} \to \hat{\mathbf{p}}$, $E \to \hat{H}t$, to only get Klein-Gordon equation

$$\frac{m^2}{c^2}\hbar^2 = \frac{\hat{H}^2}{c^2} - |\hat{\mathbf{p}}|^2 \,,$$

or in position representation, where $\langle \mathbf{r}|\hat{\mathbf{p}} = -i\hbar\nabla\langle \mathbf{r}|$, $\langle \mathbf{r}|\hat{H} = i\hbar\partial_t\langle \mathbf{r}|$, applied to a wave function $|\Psi\rangle$

$$\left\{\Box + \frac{m^2\hbar^2}{c^2}\right\}\Psi(\mathbf{r},t) = 0 \,.$$

23.1.2 Dirac equation for a free particle

Dirac's equation follows from the expression of the Hamiltonian operator as a linear combination of the components of the momentum operator and the mc,

$$\hat{H} = \alpha_0 mc + \sum_{k=1}^{3} \alpha_k \hat{p}_k \,,$$

that satisfies the relation (23.1).

Details of the solution

todo…

The Hamiltonian operator becomes

$$\hat{H} = mc^2\beta + c\alpha \cdot \hat{\mathbf{p}} = mc^2\beta - i\hbar c\alpha \cdot \nabla \,,$$

so that the position representation of Dirac's equation can be written using 3+1 formalism (space+time) as

$$\begin{aligned} i\hbar\partial_t \Psi(\mathbf{r},t) &= \langle \mathbf{r}|\hat{H}|\Psi\rangle = \\ &= -i\hbar c\alpha \cdot \nabla\Psi(\mathbf{r},t) + mc^2\beta\Psi(\mathbf{r},t) \,, \end{aligned}$$

with the set of 4×4 complex matrices $\beta = \gamma^0$, $\gamma^k = \beta\alpha_k$

$$\gamma^0 = \begin{bmatrix} I_2 & \cdot \\ \cdot & -I_2 \end{bmatrix} \quad , \quad \gamma^k = \begin{bmatrix} \cdot & \sigma_k \\ -\sigma_k & \cdot \end{bmatrix} \,,$$

built with 2×2 identity and *Pauli matrices* σ_k. Dirac's equation can be recast in the 4-dimensional formalism of special relativity, after multiplication by β, as

$$\left(i\hbar\gamma^\mu\partial_\mu - mc\right)\Psi = 0 \,,$$

since

$$\begin{aligned} 0 &= i\hbar\left(\beta\partial_t + c\beta\alpha\cdot\nabla\right)\Psi - mc^2\Psi = \\ &= i\hbar c\gamma^\mu\partial_\mu\Psi - mc^2\Psi \,. \end{aligned}$$

Elementary solutions of Dirac equation for a free particle

$$\left\{ i\frac{\hbar}{c}\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}\partial_t + i\hbar\begin{bmatrix} & & & 1 \\ & & 1 & \\ & -1 & & \\ -1 & & & \end{bmatrix}\partial_x + i\hbar\begin{bmatrix} & & & -i \\ & & i & \\ & -i & & \\ i & & & \end{bmatrix}\partial_y + i\hbar\begin{bmatrix} & & 1 & \\ & & & -1 \\ -1 & & & \\ & 1 & & \end{bmatrix}\partial_z - mc \right\}\Psi(t,\mathbf{r}) = \mathbf{0} \,.$$

Dimensional Reduction to 1D and Component Decoupling

To better understand the internal dynamics of the spinor, we can study a simplified scenario by assuming homogeneity along the y and z axes, setting spatial derivatives to zero ($\partial_y = \partial_z = 0$). Dirac's equation reduces to

$$\left\{ i\frac{\hbar}{c}\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & -1 & \\ & & & -1 \end{bmatrix}\partial_t + i\hbar\begin{bmatrix} & & & 1 \\ & & 1 & \\ & -1 & & \\ -1 & & & \end{bmatrix}\partial_x - mc \right\}\Psi(t,x) = \mathbf{0} \,.$$

Expanding the matrix equation for a four-component spinor $\Psi = \begin{pmatrix} \psi_1 & \psi_2 & \psi_3 & \psi_4 \end{pmatrix}^T$, the system splits naturally into **two independent pairs of coupled equations**:

1. A coupling between ψ_1 **and** ψ_4
2. A coupling between ψ_2 **and** ψ_3

For the first pair (ψ_1 and ψ_4), the system reads:

$$\begin{cases} \frac{i\hbar}{c}\partial_t\psi_1 - mc\,\psi_1 + i\hbar\partial_x\psi_4 = 0 \\ -i\hbar\partial_x\psi_1 - \frac{i\hbar}{c}\partial_t\psi_4 - mc\,\psi_4 = 0 \end{cases}$$

Reduction to Klein-Gordon Equation for individual components. By isolating $\partial_x\psi_4$ from the first equation and substituting it into the second, the mixed temporal-spatial derivative terms cancel out, demonstrating that each individual component satisfies the relativistic dispersion relation:

$$\left(\Box + \frac{m^2c^2}{\hbar^2}\right)\psi_1 = 0\,,$$

where $\Box = \frac{1}{c^2}\partial_t^2 - \partial_x^2$ is the d'Alembertian operator.

Details

$$\partial_x\psi_4 = \frac{1}{i\hbar}\left(mc\psi_1 - \frac{i\hbar}{c}\partial_t\psi_1\right)\,,$$

and thus

$$\begin{aligned} 0 &= -i\hbar\partial_{xx}\psi_1 - \left(\frac{i\hbar}{c}\partial_t + mc\right)\partial_x\psi_4 = \\ &= -i\hbar\partial_{xx}\psi_1 + \frac{1}{i\hbar}\left(\frac{i\hbar}{c}\partial_t + mc\right)\left(\frac{i\hbar}{c}\partial_t - mc\right)\psi_1 = \\ &= -i\hbar\partial_{xx}\psi_1 - \frac{i}{\hbar}\left(-\frac{\hbar^2}{c^2}\partial_{tt} - m^2c^2\right)\psi_1\,, \end{aligned}$$

and thus, after dividing by $i\hbar$,

$$\left(\frac{1}{c^2}\partial_{tt} - \partial_{xx} + \frac{m^2c^2}{\hbar^2}\right)\psi_1 = 0\,.$$

Plane Wave Solutions and Component Relations

Assuming a plane wave ansatz for the components of the form $\psi_1 = K_{\mathbf{p}}e^{-\frac{i}{\hbar}(Et - p_x x)}$, the **dispersion relation** $-\frac{E^2}{\hbar^2c^2} + \frac{p_x^2}{\hbar^2} + \frac{m^2c^2}{\hbar^2} = 0$ is found between the energy and the momentum of the free particle, or after multiplying by $\hbar^2$,

$$E^2 = p_x^2c^2 + m^2c^4 = 0 \quad \rightarrow \quad E = \mp c\sqrt{p_x^2 + m^2c^2}\,,$$

i.e. the very same Einstein energy-momentum equation (23.1). Convert the differential coupling into an algebraic relation between the "large" and "small" components, the first equation

$$i\hbar\partial_x\psi_4 = \left(\frac{i\hbar}{c}\partial_t - mc\right)\psi_1$$

becomes

$$i\hbar\cdot i\frac{p_x}{\hbar}\psi_4 = \left(\frac{i\hbar}{c}\left(-\frac{iE}{\hbar}\right) - mc\right)\psi_1$$

Thus, the relation between ψ_1 and ψ_4 is the ratio

$$\psi_4 = \frac{mc^2 - E}{c\,p_x}\psi_1\,.$$

An identical decoupling procedure applies symmetrically to the second pair (ψ_2 and ψ_3).

Physical Insights and Interpretation

1. **Large vs. Small Components:**
   - The upper pair (ψ_1, ψ_2) represents the **large components**, which in the non-relativistic limit ($v \ll c$) reduce to the standard two-component Pauli spinor describing spin.
   - The lower pair (ψ_3, ψ_4) represents the **small components**, suppressed by a factor of v/c, which carry the relativistic corrections responsible for spin-orbit coupling and magnetic moments.
2. **Emergence of Spin:** Unlike non-relativistic quantum mechanics where spin is added ad-hoc via Pauli matrices, spin emerges naturally in the Dirac equation as a direct consequence of the linearization required for Lorentz covariance, combining with orbital angular momentum into a conserved total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$.
3. **Antimatter and Negative Energies:** The dispersion relation permits both positive ($E = +\sqrt{c^2p^2 + m^2c^4}$) and negative ($E = -\sqrt{...}$) energy solutions. In modern quantum field theory, these negative energy states are reinterpreted through second quantization as antiparticles (such as the positron), resolving historical vacuum stability issues.

23.1.3 Dirac equation for a particle in the EM field

Following the **minimal coupling**, the momentum of a free particle is replaced by the momentum of a charged particle in a EM field,[1]

$$\mathbf{P} \to \mathbf{P} + q\mathbf{A} \,,$$

with $\mathbf{P} = m\mathbf{U} = \gamma m(c, \mathbf{v}) = \left(\frac{E}{c}, \mathbf{p}\right)$, and $\mathbf{A} = \left(\frac{\varphi}{c}, \mathbf{a}\right)$, so that

$$\{\mathbf{P} + q\mathbf{A}\}^{\mu} = \left(\frac{E + q\varphi}{c}, \mathbf{p} + q\mathbf{a}\right)$$

$$\{\mathbf{P} + q\mathbf{A}\}_{\mu} = \left(\frac{E + q\varphi}{c}, -\mathbf{p} - q\mathbf{a}\right)$$

and the Hamiltonian in space representation becomes

$$\begin{aligned} & \hat{H} = mc^2\beta - i\hbar c\alpha \cdot \nabla \\ \to \quad & \hat{H} = mc^2\beta + q\varphi + c\alpha \cdot (-i\hbar\nabla + q\mathbf{a}) \end{aligned}$$

$$\begin{aligned} 0 &= i\hbar\beta\partial_t\Psi - \beta\hat{H}\Psi = \\ &= i\hbar\beta\partial_t\Psi - mc^2 - q\beta\varphi\Psi + ic\hbar\beta\alpha_k\partial_k\Psi - cq\beta\alpha_k a_k\Psi = \\ &= i\hbar c\gamma^0\frac{1}{c}\partial_t\Psi + i\hbar c\gamma^k\partial_k\Psi - mc^2\Psi + qc\gamma^0\frac{\varphi}{c}\Psi - qc\gamma^k a_k\Psi = \end{aligned}$$

and thus, after dividing by c,

$$\left(i\hbar\gamma^{\mu}\left(\partial_{\mu} - qA_{\mu}\right) - mc^2\right)\Psi = 0 \,.$$

todo

- check signs, when switching to 3+1 to 4 formalism
- motivation of the introduction of φ in the minimal coupling. Why is it without β? Does it go with E, from the transformation $\frac{E}{c} \to \frac{E+q\varphi}{c}$?
- Invariance in special relativity
- Limit cases

[1] Math: Calculus of Variations: Examples: Lagrange Equations in Special Relativity Using Tensor Formalism

Special Relativity and Invariance

todo…

CHAPTER

TWENTYFOUR

MANY-BODY SYSTEMS

Contents

Indistinguishability of identical systems. Bosons and fermions. Symmetric basis for bosons, anti-symmetric basis for fermions.

Second quantization. Fock space. Asking *which particle is which state?* is meaningless, as particles are indistinguishable. *How many particles are in each state?* could be a reasonable question. Fock space formarlism is introduced.

Second quantization. Bosons. Creation and annihilation operators, number operator, CCR.

Second quantization. Fermions. Creation and annihilation operators, number operator, CCR satisfying **Pauli exclusion principle**.

Quantum fields. From occupation number representation to space/momentum representation. One-body Examples in one-particle, two-particle, and many-particle systems.

Operators in quantum field theory. One-body operators. Two-body operators.

24.1 Identical systems

Let a system be composed of N identical indistinguishable sub-systems. The state of the subsystem k is described by a single-system wave function $|\psi_k\rangle$ belonging to the single-system Hilbert space $\mathcal{H}_1$. Thus, the state of the many-body subsystem is described by a wave function $|\Psi\rangle$ belonging to the N tensor product space

$$\mathcal{H}^{\otimes N} = \underbrace{\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_1}_{n \text{ times}} .$$

Let $\{|\alpha\rangle\}$ be an orthonormal basis for $\mathcal{H}_1$. A basis for $\mathcal{H}^{\otimes N}$ is

$$|\alpha_1, \dots, \alpha_N\rangle = |\alpha_1\rangle \dots |\alpha_N\rangle .$$

Indistinguishability. Applying a permutation operator once

$$P_{ij}\Psi(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots) = \Psi(\dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots)$$

Applying it twice

$$P_{ij}P_{ij}\Psi(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots) = \Psi(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots) ,$$

it follows that $P_{ij}P_{ij} = 1$. Thus, two solutions are possible, corresponding to two different nature of the subsystems

- Bosons, $P_{ij} = 1$,

$$\Psi(\dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots) = \Psi(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots)$$

- Fermions, $P_{ij} = -1$,

$$\Psi(\dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots) = -\Psi(\dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots)$$

24.1.1 Symmetrization and anti-symmetrization

Symmetric basis for bosons

Let $|\alpha_1, \dots, \alpha_N\rangle$

$$|\alpha_1, \dots, \alpha_N\rangle_S := \sqrt{\frac{\prod_k n_k!}{N!}} \sum_{\hat{P} \in S_N} \hat{P} |\alpha_1, \dots, \alpha_N\rangle \, ,$$

with n_k the occupation number of the single-body state k. The factor before the summation is required for normalization of the symmetric state, as it's the square root of the inverse of the number of permutations with repetitions, see Combinatorics: Permutations with Repetitions.

Example 24.1.1 (Low-dimensional example)

Let a 2-particle system be represented in a Hilbert space thats the product $\mathcal{H}^{\otimes 2} = \mathcal{H}_1 \otimes \mathcal{H}_1$. If the two particles are in the same 1-body state $|a\rangle$, the state of the system is $|a, a\rangle$, equal to the symmetric state. If the two particles are in different and orthogonal states $|a\rangle$, $|b\rangle$, the symmetric representation of the state reads

$$|a, b\rangle_S = \frac{1}{\sqrt{2}} \left(|a, b\rangle + |b, a\rangle \right) \, .$$

This state is unit-norm, as

$$\begin{aligned} {}_S\langle a, b | a, b\rangle_S &= \frac{1}{\sqrt{2}} \left(\langle a, b| + \langle b, a| \right) \frac{1}{\sqrt{2}} \left(|a, b\rangle + |b, a\rangle \right) = \\ &= \frac{1}{2} \left(\underbrace{\langle a, b | a, b\rangle}_{=\langle a|a\rangle \langle b|b\rangle = 1} + \underbrace{\langle b, a | a, b\rangle}_{\langle b|a\rangle \langle a|b\rangle = 0} + \underbrace{\langle a, b | b, a\rangle}_{=0} + \underbrace{\langle b, a | b, a\rangle}_{=1} \right) = 1 \, . \end{aligned}$$

Anti-symmetric basis for fermions

Let $|\alpha_1, \dots, \alpha_N\rangle$

$$|\alpha_1, \dots, \alpha_N\rangle_A := \frac{1}{\sqrt{N!}} \begin{vmatrix} |\alpha_1\rangle_1 & \dots & |\alpha_N\rangle_1 \\ \dots & \dots & \dots \\ |\alpha_1\rangle_N & \dots & |\alpha_N\rangle_N \end{vmatrix} \, .$$

If two bodies were in the same state, two rows of the matrix would be equal and thus the determinant be zero. The factor before the determinant is required for normalization, as it's the square root of the inverse of the number of simple pertumations of N objects without repetitions, see Combinatorics: Permutations with Repetitions.

Example 24.1.2 (Low-dimensional example)

Let a 2-particle system be represented in a Hilbert space thats the product $\mathcal{H}^{\otimes 2} = \mathcal{H}_1 \otimes \mathcal{H}_1$. If the two particles are in the same 1-body state $|a\rangle$, the state of the system is $|a, a\rangle$,

$$|a, a\rangle_A = \frac{1}{\sqrt{2}} \begin{vmatrix} |a\rangle_1 & |a\rangle_2 \\ |a\rangle_1 & |a\rangle_2 \end{vmatrix} = 0 \, .$$

If the two particles are in different and orthogonal states $|a\rangle$, $|b\rangle$, the symmetric representation of the state reads

$$|a,b\rangle_A = \frac{1}{\sqrt{2}} \begin{vmatrix} |a\rangle_1 & |b\rangle_1 \\ |a\rangle_2 & |b\rangle_2 \end{vmatrix} = \frac{1}{\sqrt{2}} \left(|a,b\rangle - |b,a\rangle \right) \ .$$

This state is unit-norm, as

$$\begin{aligned} {}_A\langle a,b|a,b\rangle_A &= \frac{1}{\sqrt{2}} \left(\langle a,b| - \langle b,a| \right) \frac{1}{\sqrt{2}} \left(|a,b\rangle - |b,a\rangle \right) = \\ &= \frac{1}{2} \left(\underbrace{\langle a,b|a,b\rangle}_{=\langle a|a\rangle\langle b|b\rangle=1} - \underbrace{\langle b,a|a,b\rangle}_{\langle b|a\rangle\langle a|b\rangle=0} - \underbrace{\langle a,b|b,a\rangle}_{=0} + \underbrace{\langle b,a|b,a\rangle}_{=1} \right) = 1 \ . \end{aligned}$$

24.2 Second quantization. Fock space

Let $\{|\alpha\rangle\}$ a basis of the 1-body system. Let n_α the number of subsystems in the single-body state $|\alpha\rangle$, i.e. the **occupation number** of the state. The sum of the occupation numbers is equal to the number of particles N, i.e. $\sum_\alpha n_\alpha = N$. In the second quantization, the state of the system is represented in the occupation number state

$$|\mathbf{n}\rangle := |n_1, n_2, \dots, n_\alpha, \dots\rangle \ .$$

Ground state. $|\mathbf{0}\rangle = |0_1, 0_2, \dots\rangle$. By definition $\langle 0|0\rangle = 1$.

24.2.1 Fock states

todo *Properties of Fock states*

24.3 Second quantization. Bosons

The state of a many-body bosonic system is symmetric under the echange of 2 particles. This condition is represented in the *CCRs*.

24.3.1 Creation and annihilation operators

Creation operator, $\hat{b}_i^\dagger$. The notation with the dagger is justified below, as the creation operator is the adjoint of the annihilation operator $\hat{b}_i$.

$$\hat{b}_i^\dagger |\dots, n_i, \dots\rangle = \sqrt{n+1} |\dots, (n+1)_i, \dots\rangle \ .$$

A state $|\mathbf{n}\rangle = |n_1, n_2, \dots, n_i, \dots\rangle$ can be obtained from the vacuum equation as

$$|\mathbf{n}\rangle = |n_1, n_2, \dots, n_i, \dots\rangle = \frac{1}{\sqrt{\prod_k n_k!}} \left(\hat{b}_1^\dagger\right)^{n_1} \dots \left(\hat{b}_i^\dagger\right)^{n_i} \dots |\mathbf{0}\rangle \ .$$

Annihilation operator, $\hat{b}_i$.

$$\begin{aligned} &\hat{b}_i |\dots, n_i, \dots\rangle = \sqrt{n} |\dots, (n-1)_i, \dots\rangle \ , \qquad \text{if } n_i > 0 \\ &\hat{b}_i |\dots, 0_i, \dots\rangle = 0 \ . \end{aligned}$$

Tip: Annihilation and creation operators are not Hermitian.

Creation operator is the adjoint of the annihilation operator

The adjoint operator of an operator $\hat{a} : A \to B$ is defined as the operator $\hat{a}^\dagger : B \to A$ so that

$$\begin{aligned}\langle\phi|\hat{a}|\psi\rangle_B = \langle\psi|\hat{a}^\dagger|\phi\rangle_A^* = \\ = \langle\hat{a}^\dagger\phi|\psi\rangle_A \, .\end{aligned}$$

for any $|\phi\rangle \in B$, $|\psi\rangle \in A$. Thus, for the annihilation operator $\hat{b}_i$,

$$\langle\mathbf{m}|\hat{b}_i|\mathbf{n}\rangle = \langle\mathbf{n}|\left(\hat{b}_i\right)^\dagger|\mathbf{m}\rangle^* \, .$$

$$\begin{aligned}\langle\mathbf{n}|\left(\hat{b}_i\right)^\dagger|\mathbf{m}\rangle^* &= \langle\mathbf{m}|\hat{b}_i|\mathbf{n}\rangle = && (1)\\ &= \sqrt{n_i}\langle\mathbf{m}|\mathbf{n}-\mathbf{1}_i\rangle = && (2)\\ &= \sqrt{n_i}\,\delta_{\mathbf{m},\mathbf{n}-\mathbf{1}_i} = && (3)\\ &= \sqrt{m_i+1}\,\delta_{\mathbf{m},\mathbf{n}-\mathbf{1}_i} = && (4)\\ &= \sqrt{m_i+1}\,\delta_{\mathbf{m}+\mathbf{1}_i,\mathbf{n}} = && (5)\\ &= \sqrt{m_i+1}\langle\mathbf{n}|\mathbf{m}+\mathbf{1}_i\rangle = \\ &= \langle\mathbf{n}|\left(\sqrt{m_i+1}|\mathbf{m}+\mathbf{1}_i\rangle\right) = && (6)\\ &= \langle\mathbf{n}|\hat{b}_i^\dagger|\mathbf{m}\rangle \, ,\end{aligned}$$

with the Kronecker's delta acting on all the indices - just as an example, with $\mathbf{a} = (a_1, a_2)$, $\mathbf{b} = (b_1, b_2)$, it follows that $\delta_{\mathbf{a},\mathbf{b}} = \delta_{a_1,b_1}\delta_{a_2,b_2}$ -, and (1) action of the annihilation operator, (2) orthogonality of Fock states, (3-4) properties of Kronecker's delta, (5) orthogonality of Fock states, (6) recognizing the action of the creation operator $\hat{b}_i^\dagger$ on $|\mathbf{m}\rangle$. As the result of the inner product is real, its complex conjugate is the value itself.

Number operator, $\hat{n}_i = \hat{b}_i^\dagger\,\hat{b}_i$. Number operator is self-adjoint. This is consistent with postulates of quantum mechanics: physical observables are represented by self-adjoint operators.

$$\begin{aligned}\hat{n}_i|\mathbf{n}\rangle = \hat{n}_i|\dots, n_i, \dots\rangle = \\ = \hat{b}_i^\dagger\,\hat{b}_i|\dots, n_i, \dots\rangle = \\ = \hat{b}_i^\dagger\,\left(\sqrt{n}|\dots, (n-1)_i, \dots\rangle\right) = \\ = n_i|\dots, n_i, \dots\rangle = \\ = n_i|\mathbf{n}\rangle \, .\end{aligned}$$

24.3.2 Number operator

Total number operator. $\hat{n} = \sum_i \hat{n}_i$

24.3.3 Canonical commutation relations

CCRs.

$$[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij} \, .$$

Proof

$$[\hat{b}_i, \hat{b}_j^\dagger]|\mathbf{n}\rangle = \left(\sqrt{n_j+1}\sqrt{n_i^{(1)}} - \sqrt{n_i}\sqrt{n_j^{(1)}+1}\right)|\dots,(n-1)_i,\dots,(n+1)_j,\dots\rangle$$

with the $n_j^{(1)}$ the state after the application of the first operator. If $i \neq j$, then $n_j^{(1)} = n_j$, and thus the result is zero. If $i = j$, then $n_i^{(1)} = n_i + 1$, $n_j^{(1)} = n_j - 1$, and thus the result reads

$$[\hat{b}_i, \hat{b}_i^\dagger]|\mathbf{n}\rangle = ((n_i+1) - n_i)\,|\mathbf{n}\rangle = 1 \cdot |\mathbf{n}\rangle\ .$$

and thus

$$[\hat{b}_i, \hat{b}_j^\dagger]|\mathbf{n}\rangle = \delta_{ij}|\mathbf{n}\rangle\ .$$

$$[\hat{b}_i, \hat{b}_j] = 0$$
$$[\hat{b}_i^\dagger, \hat{b}_j^\dagger] = 0$$

24.4 Second quantization. Fermions

The state of a many-body fermionic system is anti-symmetric under the exchange of 2 particles. This is the Pauli exclusion principle, that's represented in the *CARs*.

24.4.1 Creation and annihilation operators

$$\hat{c}_k^\dagger|\dots n_k \dots\rangle = \theta_k\sqrt{1-n_k}\,|\dots(1-n)_k\dots\rangle$$
$$\hat{c}_k|\dots n_k \dots\rangle = \theta_k\sqrt{n_k}\,|\dots(1-n)_k\dots\rangle$$

with $\theta_k = (-1)^{\sum_{j=1}^{k-1} n_j}$.

Road to the formalization of creation and annihilation operators

Pauli exclusion principle. Pauli exclusion principle implies that the occupation number of a state can be either 0 or 1. Creation and annihilation operators should satisfy the following conditions

$$\hat{c}_i|0\rangle = 0$$
$$\hat{c}_i|1\rangle = |0\rangle \quad \hat{c}_i^\dagger|0\rangle = |1\rangle$$
$$\hat{c}_i^\dagger|1\rangle = 0$$

A first - but wrong, see **Jordan-Wigner** solution - try of the definition of creation $\hat{c}_i^\dagger$ and annihilation operator $\hat{c}_i$ for a fermionic field could be

$$\hat{c}_k^\dagger|\dots n_k \dots\rangle = \sqrt{1-n_k}|\dots(1-n)_k\dots\rangle$$
$$\hat{c}_k|\dots n_k \dots\rangle = \sqrt{n_k}|\dots(1-n)_k\dots\rangle$$

This definition would provide an inconsistent behavior with the *canonical anticommuation relations*. As an example, with $i < j$

$$\hat{c}_i^\dagger\hat{c}_j^\dagger|0\rangle = -\hat{c}_j^\dagger\hat{c}_i^\dagger|0\rangle\ .$$

Applying the - wrong - expression of the creation and annihilation operators given above, the result would be

$$|1_i, 1_j\rangle = -|1_i, 1_j\rangle \ ,$$

that would imply the state is identically zero.

Jordan-Wigner, "On the Pauli exclusion principle".[1]

$$\hat{c}_k^\dagger |\dots n_k \dots\rangle = \theta_k \sqrt{1-n_k}\, |\dots (1-n)_k \dots\rangle$$
$$\hat{c}_k |\dots n_k \dots\rangle = \theta_k \sqrt{n_k}\, |\dots (1-n)_k \dots\rangle$$

with $\theta_k = (-1)^{\sum_{j=1}^{k-1} n_j}$. With this definition of the creation and annihilation operators the example above, with $i < j$ becomes

$$\hat{c}_i^\dagger \hat{c}_j^\dagger |0\rangle = (-1)^0 \hat{c}_i^\dagger |1_j\rangle = (-1)^0 |1_i, 1_j\rangle = |1_i, 1_j\rangle$$
$$\hat{c}_j^\dagger \hat{c}_i^\dagger |0\rangle = (-1)^0 \hat{c}_j^\dagger |1_i\rangle = (-1)^1 |1_i, 1_j\rangle = -|1_i, 1_j\rangle$$

24.4.2 Number operator

24.4.3 Canonical anti-commutation relations

The commutation relation

$$\{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} = 0$$

implies

- $\hat{c}_i^\dagger \hat{c}_i^\dagger = -\hat{c}_i^\dagger \hat{c}_i^\dagger$, and thus $\left(\hat{c}_i^\dagger\right)^2 = 0$. If $\hat{c}_i^\dagger$ creates the state $|1_i\rangle$ from the vacuum state $|0\rangle$, and a state proportional to $|2_i\rangle$ when applied to state $|1_i\rangle$, i.e. with 2 applications to $|0\rangle$, then this implies that it's not possible to get $|2_i\rangle$, and states with larger occupation numbers. Thus, the occupation number of each state can be either 0 or 1.
- $\hat{c}_i^\dagger \hat{c}_j^\dagger = -\hat{c}_j^\dagger \hat{c}_i^\dagger$

Proof

If $i < j$

$$\left(\hat{c}_i^\dagger \hat{c}_j^\dagger + \hat{c}_j^\dagger \hat{c}_i^\dagger\right) |\dots n_i \dots n_j \dots\rangle =$$
$$= \sqrt{n_j - 1} \cdot (-1)^{\sum_{k=1}^{j-1} n_k} \hat{c}_i^\dagger |\dots n_i \dots (n-1)_j \dots\rangle + \sqrt{n_i - 1}\cot(-1)^{\sum_{l=1}^{i-1} n_l} \hat{c}_j^\dagger |\dots (n-1)_i \dots n_j \dots\rangle =$$
$$= \sqrt{n_j - 1}\sqrt{n_i - 1} \cdot \left\{(-1)^{\sum_{l=1}^{i-1} n_l} \cdot (-1)^{\sum_{k=1}^{j-1} n_k} + (-1)^{\sum_{k=1, l\neq i}^{k-1} n_l} \cdot (-1)^{n_i - 1} \cdot (-1)^{\sum_{l=1}^{i-1} n_l}\right\} |\dots (n-1)_i \dots (n-1)_j \dots\rangle =$$
$$= \sqrt{n_j - 1}\sqrt{n_i - 1} \cdot \left\{\underbrace{((-1)^{n_i} + (-1)^{n_i - 1})}_{=0} (-1)^{\sum_{k=1, l\neq i}^{k-1} n_l} \cdot (-1)^{\sum_{l=1}^{i-1} n_l}\right\} |\dots (n-1)_i \dots (n-1)_j \dots\rangle =$$
$$= 0 \ .$$

If $j = i$, the equality immediately follows from $\left(c_i^\dagger\right)^2 = 0$.

$$\{\hat{c}_i, \hat{c}_j\} = 0$$

[1] Jordan–Wigner (1928), "On the Pauli exclusion principle"

Proof

…

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij} \, .$$

Proof

…

24.5 Quantum fields

In this section, creation and annihilation operators in continuous basis (like space or momentum) are introduced, in terms of creation/annihilation operators in a given (and discrete?) basis $\{\nu_a\}_a$

24.5.1 Field creation and annihilation operators

Creation and annihilation operators in space basis can be written using the basis $\{\nu_a\}$ as

$$\Psi^\dagger(\mathbf{r}) = \sum_a \nu_a^*(\mathbf{r}) \hat{a}_{\nu_a}^\dagger \tag{24.1}$$

$$\Psi(\mathbf{r}) = \sum_a \nu_a(\mathbf{r}) \hat{a}_{\nu_a} \tag{24.2}$$

Creation operator $\Psi^\dagger(\mathbf{r})$ *creates a particle* in state $|\mathbf{r}\rangle$, while annihilation operator $\Psi(\mathbf{r})$ *destroys a particle* in state $|\mathbf{r}\rangle$.

The inverse relations hold

$$\hat{a}_{\nu_a}^\dagger = \int_{\mathbf{r}} \nu_a(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \, d\mathbf{r} \, . \tag{24.3}$$

$$\hat{a}_{\nu_a} = \int_{\mathbf{r}} \nu_a^*(\mathbf{r}) \Psi(\mathbf{r}) \, d\mathbf{r} \, . \tag{24.4}$$

Details. Inverse relations

$$\begin{aligned}
\hat{a}_{\nu_a}^\dagger &= \sum_b \hat{a}_{\nu_b}^\dagger \langle \nu_b | \nu_a \rangle = \\
&= \sum_b \hat{a}_{\nu_b}^\dagger \langle \nu_b | \int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}| \, d\mathbf{r} \, |\nu_a\rangle = \\
&= \int_{\mathbf{r}} \nu_a(\mathbf{r}) \sum_b \nu_b^*(\mathbf{r}) \hat{a}_{\nu_b}^\dagger \, d\mathbf{r} = \\
&= \int_{\mathbf{r}} \nu_a(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \, d\mathbf{r} \, .
\end{aligned}$$

$$\begin{aligned}\hat{a}_{\nu_a} &= \sum_b \hat{a}_{\nu_b}\langle\nu_a|\nu_b\rangle = \\ &= \sum_b \hat{a}_{\nu_b}\langle\nu_a| \int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}|\, d\mathbf{r}\, |\nu_b\rangle = \\ &= \int_{\mathbf{r}} \nu_a^*(\mathbf{r}) \sum_b \nu_b(\mathbf{r})\hat{a}_{\nu_b}\, d\mathbf{r} = \\ &= \int_{\mathbf{r}} \nu_a^*(\mathbf{r})\Psi(\mathbf{r})\, d\mathbf{r}\ .\end{aligned}$$

todo *Need to distinguish between bosons and fermions? Or one can just use the proper set of creation/annihilation operators, without any other major change in the formulas*

CCR

One-particle system

Action of $\hat{\Psi}^\dagger(\mathbf{r})$ on the vacuum state. The action of the operator $\hat{\Psi}^\dagger(\mathbf{r})$ on the vacuum places a particle in the state $|\mathbf{r}\rangle$, i.e.

$$|\mathbf{r}\rangle = \hat{\Psi}^\dagger(\mathbf{r})|0\rangle\ . \tag{24.5}$$

Discrete basis

The state $|\mathbf{r}\rangle$ can be written as a linear combination of the elements of a basis. For a **discrete basis** $\{|\nu_k\rangle\}_k$,

$$|\mathbf{r}\rangle = \sum_k |\nu_k\rangle\langle\nu_k|\mathbf{r}\rangle\ .$$

A vector of the basis of the 1-body system $|\nu_k\rangle$ can be built with the creation operator acting on the ground state, $|\nu_k\rangle = \hat{a}^\dagger_{\nu_k}|0\rangle$, and thus

$$|\mathbf{r}\rangle = \sum_k \langle\nu_k|\mathbf{r}\rangle\, \hat{a}^\dagger_{\nu_k}|0\rangle = \sum_k \nu_k^*(\mathbf{r})\, \hat{a}^\dagger_{\nu_k}|0\rangle\ . \tag{24.6}$$

Comparision of (24.5) and (24.6) gives the expression of the field creation operator

$$\hat{\Psi}^\dagger(\mathbf{r}) = \sum_k \nu_k^*(\mathbf{r})\hat{a}^\dagger_{\nu_k}\ .$$

Equivalent expression of the single-body wave function

The single-body wave function in position representation reads

$$\begin{aligned}\psi(\mathbf{r}) &= \langle\mathbf{r}|\psi\rangle = \\ &= \langle 0|\hat{\Psi}(\mathbf{r})|\psi\rangle = \\ &= \sum_k \sum_j \nu_j(\mathbf{r})\ \underbrace{\langle 0|\hat{a}_{\nu_j}|\nu_k\rangle}_{=\delta_{jk}\langle 0|0\rangle=\delta_{jk}}\ \langle\nu_k|\psi\rangle = \\ &= \sum_k \nu_k(\mathbf{r})\langle\nu_k|\psi\rangle = \\ &= \sum_k \langle\mathbf{r}|\nu_k\rangle\langle\nu_k|\psi\rangle\ .\end{aligned}$$

Continuous basis

Two-particle system

Bosons

Position state can be expressed either with creation operators

$$|\mathbf{r}_1, \mathbf{r}_2\rangle = \hat{\Psi}^\dagger(\mathbf{r}_2)\hat{\Psi}^\dagger(\mathbf{r}_1)|0\rangle \ ,$$

and with as a linear combination of states built with the 1-body basis $\{\nu_k\}$,

$$|\mathbf{r}_1, \mathbf{r}_2\rangle = \sum_{i_1 \leq i_2} |\nu_{i_1}, \nu_{i_2}\rangle\langle\nu_{i_1}, \nu_{i_2}|\mathbf{r}_1, \mathbf{r}_2\rangle \ .$$

Identity operator.

$$\hat{\mathbf{1}}^{(2)} = \sum_{i_1 \leq i_2} |\nu_{i_1}, \nu_{i_2}\rangle\langle\nu_{i_1}, \nu_{i_2}| \ .$$

Proof

$$\begin{aligned}
\hat{\mathbf{1}}^{(2)}|\Psi\rangle = \hat{\mathbf{1}}^{(2)} \sum_{k_1 \leq k_2} c_{k_1,k_2}|\nu_{k_1}\nu_{k_2}\rangle = \\
= \left(\sum_{i_1 \leq i_2} |\nu_{i_1}, \nu_{i_2}\rangle\langle\nu_{i_1}, \nu_{i_2}|\right) \sum_{k_1 \leq k_2} c_{k_1,k_2}|\nu_{k_1}, \nu_{k_2}\rangle = \\
= \sum_{i_1 \leq i_2} \sum_{k_1 \leq k_2} |\nu_{i_1}, \nu_{i_2}\rangle\delta_{i_1 k_1}\delta_{i_2 k_2} c_{k_1,k_2} = \\
= \sum_{i_1 \leq i_2} |\nu_{i_1}, \nu_{i_2}\rangle c_{i_1,i_2} \ .
\end{aligned}$$

Bosons. The state $|\nu_{i_1}, \nu_{i_2}\rangle$ can be built with creation operators from the vacuum state. If $i_1 \neq i_2$, then

$$|\nu_{i_1}, \nu_{i_2}\rangle = \hat{a}^\dagger_{\nu_{i_1}} \hat{a}^\dagger_{\nu_{i_2}}|0\rangle \qquad ' = ' \qquad |\dots \underbrace{1}_{i_1} \dots \underbrace{1}_{i_2} \dots\rangle \ .$$

If $i_2 = i_1$,

$$|\nu_{i_1}, \nu_{i_1}\rangle = \frac{1}{\sqrt{2}} \hat{a}^\dagger_{\nu_{i_1}} \hat{a}^\dagger_{\nu_{i_1}}|0\rangle \qquad ' = ' \qquad |\dots \underbrace{2}_{i_1} \dots\rangle \ ,$$

i.e. in general $|\nu_{i_1}, \nu_{i_2}\rangle = f_{i_1 i_2} \hat{a}^\dagger_{i_1} \hat{a}^\dagger_{i_2}|0\rangle$, with $f_{i_1 i_2} = \frac{1}{\sqrt{1+\delta_{i_1 i_2}}}$. Inserting this expression into the formula for $|\mathbf{r}_1, \mathbf{r}_2\rangle$, the expression for the creator operators follows

$$\begin{aligned}
|\mathbf{r}_1, \mathbf{r}_2\rangle = \hat{\Psi}^\dagger(\mathbf{r}_2)\hat{\Psi}^\dagger(\mathbf{r}_1)|0\rangle = \\
= \sum_{i_1 \leq i_2} f_{i_1 i_2} \nu^*_{i_1}(\mathbf{r}_1)\nu^*_{i_2}(\mathbf{r}_2) \hat{a}^\dagger_{\nu_{i_1}} \hat{a}^\dagger_{\nu_{i_2}}|0\rangle \ ,
\end{aligned}$$

so that

$$\hat{\Psi}^\dagger(\mathbf{r}_2)\hat{\Psi}^\dagger(\mathbf{r}_1) = \sum_{i_1 \leq i_2} f_{i_1 i_2} \nu^*_{i_1}(\mathbf{r}_1)\nu^*_{i_2}(\mathbf{r}_2) \hat{a}^\dagger_{\nu_{i_1}} \hat{a}^\dagger_{\nu_{i_2}}$$

The state function of the system in position representation is

$$\begin{aligned}\Psi(\mathbf{r}_1,\mathbf{r}_2) &= \langle \mathbf{r}_1,\mathbf{r}_2|\Psi\rangle = \\ &= \langle \mathbf{r}_1,\mathbf{r}_2| \sum_{i_1\le i_2} |\nu_{i_1},\nu_{i_2}\rangle\langle\nu_{i_1},\nu_{i_2}|\Psi\rangle = \\ &= \sum_{i_1\le i_2} \nu_{i_1}(\mathbf{r}_1)\,\nu_{i_2}(\mathbf{r}_2)\langle\nu_{i_1},\nu_{i_2}|\Psi\rangle = \\ &= \langle 0|\hat{\Psi}(\mathbf{r}_2)\hat{\Psi}(\mathbf{r}_1)| \sum_{i_1\le i_2} |\nu_{i_1},\nu_{i_2}\rangle\langle\nu_{i_1},\nu_{i_2}|\Psi\rangle = \end{aligned}$$

Fermions.

Many-body systems

Bosons

Identity operator.

- Summing over ordered/independent configurations ("occupation representation")

$$\hat{\mathbf{1}}^{(N)} = \sum_{i_1\le i_2\le\ldots\le i_N} |\nu_{i_1},\nu_{i_2},\ldots,\nu_{i_N}\rangle\langle\nu_{i_1},\nu_{i_2},\ldots,\nu_{i_N}|$$

- Summing over all the indices ("creation operator represetnation")

$$\hat{\mathbf{1}}^{(N)} = \frac{1}{N!}\sum_{i_1,i_2,\ldots,i_N} \left(c^\dagger_{\nu_{i_1}}\ldots c^\dagger_{\nu_{i_N}}|0\rangle\right)\left(\langle 0|c_{\nu_{i_1}}\ldots c_{\nu_{i_N}}\right)$$

Fermions

Identity operator.

- Summing over ordered/independent configurations ("occupation representation")

$$\hat{\mathbf{1}}^{(N)} = \sum_{i_1<i_2<\cdots<i_N} |\nu_{i_1},\nu_{i_2},\ldots,\nu_{i_N}\rangle\langle\nu_{i_1},\nu_{i_2},\ldots,\nu_{i_N}|$$

- Summing over all the indices ("creation operator represetnation")

$$\hat{\mathbf{1}}^{(N)} = \frac{1}{N!}\sum_{i_1,i_2,\ldots,i_N} \left(c^\dagger_{\nu_{i_1}}\ldots c^\dagger_{\nu_{i_N}}|0\rangle\right)\left(\langle 0|c_{\nu_{i_1}}\ldots c_{\nu_{i_N}}\right)$$

24.6 Operators in QFT

One-body operators act on individual particles. Let $\hat{o}$ the operator acting on the single-particle wave function $|\psi\rangle \in \mathcal{H}$, and $o_{ab} = \langle\phi_a|\hat{o}|\phi_b\rangle$ its matrix components w.r.t. the vectors of the basis of $\{\psi_a\}_a$ of $\mathcal{H}$, or in **position representation**

$$o_{ab} = \langle\psi_a|\hat{o}|\psi_b\rangle = \langle\psi_a|\int_{\mathbf{r}} |\mathbf{r}\rangle\langle\mathbf{r}|\hat{o}|\psi_b\rangle\,d\mathbf{r} = \int_{\mathbf{r}} \psi^*_a(\mathbf{r})\,o\psi_b(\mathbf{r})\,d\mathbf{r}\,,$$

being o the linear (tipically differential) operator in space representation, and $o\psi_b(\mathbf{r})$ its action of $|\psi_b\rangle$ in space representation. Just as an example, the space representation of the action of the x-component of the momentum operator $\hat{p}_x$ on vector $|\psi_b\rangle$ reads

$$\langle \mathbf{r}|\hat{p}_x|\psi_b\rangle = -i\hbar\partial_x\langle \mathbf{r}|\psi_b\rangle = -i\hbar\partial_x\psi_b(\mathbf{r}) \ ,$$

so that the linear differential operator in space representation acting on the space representation of the wave vector reads $p_x = -i\hbar\partial_x$, and $p_x\psi_b(\mathbf{r}) = -i\hbar\partial_x\psi_b(\mathbf{r})$.

24.6.1 One-body operators

First quantization. The operator $\hat{O}_1$ is the sum of the 1-particle operators $\hat{o}$, each one acting on the corresponding particle,

$$\hat{O}_1 = \sum_{i=1}^{N} \hat{o}(q^i, p_i) \ . \tag{24.7}$$

Let the state of the system $|\Psi\rangle$ be represented in the basis $\{|\psi_{i_1}, \psi_{i_2}, \dots\rangle\}$, with the vectors $|\psi_i\rangle$ being the eigenvectors of the operator $\hat{o}$, with eigenvalues o_k. Let the system be in state $|\Psi\rangle = |\psi_{i_1}, \psi_{i_2}, \dots\rangle$. Then the action of the one-body operator on the state of the system reads

$$\hat{O}_1|\Psi\rangle = \sum_{k=1}^{N} \hat{o}^{(k)}|\Psi\rangle = \sum_{k=1}^{N} o_{i_k}|\Psi\rangle \ , \tag{24.8}$$

i.e. the state $|\psi_{i_1}, \psi_{i_2}, \dots\rangle$ is an eigenstate of the one-body operator $\hat{O}_1$, with eigenvalue $\sum_{k=1}^{N} o_{i_k}$.

Examples for a 2-particle system.

Let the state a two-particle system be represented by the wave vector $|\Psi\rangle = |\psi_1, \psi_2\rangle$. Its space representation reads

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \langle \mathbf{r}_1, \mathbf{r}_2|\Psi\rangle = \psi_1(\mathbf{r}_1)\psi_2(\mathbf{r}_2) \ .$$

todo *Discuss statistical independence…*

Some examples of one-body operators acting on a two-particle system are:

- **Momentum** (1-dimensional domain), $\hat{P}_x = \hat{p}_x^{(1)} + \hat{p}_x^{(2)}$, or $= \hat{p}_x \otimes \hat{1} + \hat{1} \otimes \hat{p}_x$

  $$\hat{P}|\Psi\rangle = \hat{P}|\Psi\rangle = |\hat{p}\psi_1, \psi_2\rangle + |\psi_1, \hat{p}\psi_2\rangle \ ,$$

  or in space representation

  $$\langle \mathbf{r}_1, \mathbf{r}_2|\hat{P}|\Psi\rangle = -i\hbar\partial_{x_1}\psi_1(\mathbf{r}_1)\psi_2(\mathbf{r}_2) - i\hbar\psi_1(\mathbf{r}_1)\partial_{x_2}\psi_2(\mathbf{r}_2) \ .$$

- **Kinetic energy**, $\hat{T} = \hat{t}^{(1)} + \hat{t}^{(2)}$

  …

- **External potential energy**, $\hat{V} = \hat{v}^{(1)} + \hat{v}^{(2)}$

  …

- **Total energy** (kinetic + external potential energy), $\hat{H} = \hat{h}^{(1)} + \hat{h}^{(2)}$. If the state of the system $|\psi_1, \psi_2\rangle$ is in $|\psi_a, \psi_b\rangle$, with $|\psi_a\rangle$ a stationary state of the 1-particle system, with energy E_a, then it follows that the action of the total energy operator is

$$\begin{aligned}\hat{H}|\Psi\rangle &= \left(\hat{h}^{(1)} + \hat{h}^{(2)}\right)|\psi_1, \psi_2\rangle = \\ &= \left(\hat{h}^{(1)} + \hat{h}^{(2)}\right)|\psi_1, \psi_2\rangle = \\ &= \left(\hat{h}^{(1)} + \hat{h}^{(2)}\right)|\psi_a, \psi_b\rangle = \\ &= |\hat{h}\psi_a, \psi_b\rangle + |\psi_a, \hat{h}\psi_b\rangle = \\ &= (E_a + E_b)\,|\psi_a, \psi_b\rangle = (E_a + E_b)\,|\Psi\rangle\ .\end{aligned}$$

Second quantization.

$$\hat{O}_1 = \sum_{\alpha,\beta} o_{\alpha\beta}\hat{a}^\dagger_\alpha \hat{a}_\beta\ , \tag{24.9}$$

with $\hat{a}^\dagger_\alpha$, $\hat{a}_\beta$ the creation and annihilation operators for one particle in states $|\psi_a\rangle$, and $|\psi_b\rangle$ respectively, and $o_{\alpha\beta} = \langle\psi_a|\hat{o}|\psi_b\rangle$, with $\{|\psi_a\rangle\}$ vectors of a basis of the 1-particle space. The summations in α, and β here occur on all the 1-particle eigenstates.

If the operator is built using the 1-particle eigenstates of the operator $\hat{o}$, then the matrix $o_{\alpha\beta}$ is diagonal, and the operator can be written as

$$\hat{O}_1 = \sum_\alpha o_\alpha \hat{a}^\dagger_\alpha \hat{a}_\alpha = \sum_\alpha o_\alpha \hat{n}_\alpha\ ,$$

with o_α the eigenvalues of the 1-particle operator $\hat{o}$ and $\hat{n}_\alpha = \hat{a}^\dagger_\alpha \hat{a}_\alpha$ the number operator of the state α. The action of the operator $\hat{O}_1$ on the state of the system, represented in the occupation number basis $|\Psi\rangle = |n_{o_1}, n_{o_2}, \dots\rangle$ built on the 1-particle eigenstates of the operator $\hat{o}$,

$$\hat{O}_1|\Psi\rangle = \left(\sum_\alpha o_{\alpha\beta}\hat{a}^\dagger_\alpha \hat{a}_\alpha\right)|\Psi\rangle = \left(\sum_\alpha o_{\alpha\beta}\hat{n}_\alpha\right)|\Psi\rangle\ .$$

If the system is in an eigenstate of the operator $\hat{O}$, represented by only one vector $|\Psi\rangle = |\mathbf{n}\rangle = |n_1, n_2, \dots\rangle$, then

$$\begin{aligned}\hat{O}_1|\mathbf{n}\rangle &= \left(\sum_\alpha o_\alpha \hat{n}_\alpha\right)|\mathbf{n}\rangle = \\ &= \left(\sum_\alpha o_\alpha \hat{n}_\alpha\right)|\dots n_\alpha \dots\rangle = \\ &= \left(\sum_\alpha o_\alpha\, n_\alpha\right)|\dots n_\alpha \dots\rangle = \\ &= \underbrace{\left(\sum_\alpha n_\alpha o_\alpha\right)}_{=:o_{\mathbf{n}}}|\mathbf{n}\rangle\ ,\end{aligned} \tag{24.10}$$

i.e. the state $|\mathbf{n}\rangle$ of the occupation numbers of the eigenstates of $\hat{o}$ is an eigenvector of the operator $\hat{O}$ with eigenvalue $o_{\mathbf{n}} := \sum_\alpha o_\alpha n_\alpha$.

Operators in first and second quantization

The action of the operator $\hat{O}_1$ can be expressed using second quantization formalism with (24.10) or using the analogous first quantization formalism with (24.8): while summation in second quantization runs over the eigenstates, the summation

runs over "particles" in first quantization. These two different points of view provide the same result, with the eigenvalue of the eigenstate $|\mathbf{n}\rangle = |\psi_{i_1}, \psi_{i_2}, \dots\rangle$ (with n_1 of the ψ_{i_k} in eigenstate ψ_1,…) equal to

$$o_{\mathbf{n}} = \sum_{\alpha} n_a o_a = \sum_k o_{i_k} \, .$$

This analogy also provides a justification for the expressions (24.7) and (24.9) of the one-body operator $\hat{O}_1$.

Space representation

Using the relations (24.3) and (24.4),

$$\hat{a}_a^\dagger = \int_{\mathbf{r}} \psi_a(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \, d\mathbf{r} \, .$$

$$\hat{a}_a = \int_{\mathbf{r}} \psi_a^*(\mathbf{r}) \Psi(\mathbf{r}) \, d\mathbf{r} \, .$$

it's possible to find another equivalent expression of the expression (24.9) of the $\hat{O}_1$ operator,

todo *Review these steps. There's too much free-style in swapping operators…*

$$\begin{aligned}
\hat{O}_1 &= \sum_{\alpha,\beta} o_{\alpha\beta} \hat{a}_\alpha^\dagger \hat{a}_\beta = \\
&= \sum_{\alpha,\beta} o_{\alpha\beta} \left(\int_{\mathbf{r}} \psi_a(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \, d\mathbf{r} \right) \left(\int_{\mathbf{r}'} \psi_\beta^*(\mathbf{r}') \Psi(\mathbf{r}') \, d\mathbf{r}' \right) = \\
&= \sum_{\alpha,\beta} \int_{\mathbf{r}} \int_{\mathbf{r}'} o_{\alpha\beta} \psi_a(\mathbf{r}) \Psi^\dagger(\mathbf{r}) \psi_\beta^*(\mathbf{r}') \Psi(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' = \\
&= \sum_{\alpha,\beta} \int_{\mathbf{r}} \int_{\mathbf{r}'} \psi_a(\mathbf{r}) o_{\alpha\beta} \psi_\beta^*(\mathbf{r}') \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' = \qquad (1) \\
&= \int_{\mathbf{r}} \int_{\mathbf{r}'} \sum_{\alpha,\beta} \langle \mathbf{r} | \psi_a \rangle \langle \psi_a | \hat{o} | \psi_\beta \rangle \langle \psi_\beta | \mathbf{r}' \rangle \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' = \\
&= \int_{\mathbf{r}} \int_{\mathbf{r}'} \sum_{\alpha,\beta} \langle \mathbf{r} | \hat{o} | \mathbf{r}' \rangle \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' = \qquad (2) \\
&= \int_{\mathbf{r}} \int_{\mathbf{r}'} \sum_{\alpha,\beta} o_{\mathbf{r};\partial_{\mathbf{r}}} \underbrace{\langle \mathbf{r} | \mathbf{r}' \rangle}_{\delta(\mathbf{r}-\mathbf{r}')} \Psi^\dagger(\mathbf{r}) \Psi(\mathbf{r}') \, d\mathbf{r} \, d\mathbf{r}' = \\
&= \int_{\mathbf{r}} \sum_{\alpha,\beta} \Psi^\dagger(\mathbf{r}) o_{\mathbf{r};\partial_{\mathbf{r}}} \Psi(\mathbf{r}') \, d\mathbf{r} \, ,
\end{aligned}$$

Example 24.6.1 (Kinetic energy)

$$\begin{aligned}
\hat{T} &= \int_{\mathbf{r}} \hat{\Psi}^\dagger(\mathbf{r}) \langle \mathbf{r} | \hat{t} \hat{\Psi}(\mathbf{r}) = \\
&= \dots = \\
&= \int_{\mathbf{r}} \hat{\Psi}^\dagger(\mathbf{r}) \left(-\frac{\hbar^2}{2m} \nabla^2 \right) \hat{\Psi}(\mathbf{r}) = \\
&= \int_{\mathbf{r}} \frac{\hbar^2}{2m} \nabla \hat{\Psi}^\dagger(\mathbf{r}) \cdot \nabla \hat{\Psi}(\mathbf{r}) \, ,
\end{aligned}$$

with integration by parts and vanishing operators for $|\mathbf{r}| \to +\infty$.

24.6.2 Two-body operators

CHAPTER

TWENTYFIVE

QUANTIZATION OF THE ELECTROMAGNETIC FIELD

In non-relativistic quantum mechanics, Coulomb gauge $\nabla \cdot \mathbf{a} = 0$ is usually used.

CHAPTER

TWENTYSIX

INTRODUCTION TO QUANTUM FIELD THEORY

Harmonic oscillator.

Field quantization (Bosons). Modes of a field governed by a wave equation, and quantization of a classical field.

Photons. EM field quantization Quantization of the EM field. Introduction: Fermi ideal cavity, EM in free space. Classical limit of the quantum field: incoherent and coherent states (Glauber, 1963 *Photon Correlations*)

Multi-body quantum mechanics and QFT. System with many identical bodies. Bosons: creation and annihilation operatos; quantization of the EM field. Fermions: Pauli exclusion principle; creation and annihilation operators; second quantization of the electron field.

Interacting fields. Perturbation theory and Fermi goldern rule. Interaction between electron and EM field.

Summary

1. **Fock space** is built from commutators (bosons) or anticommutators (fermions) of creation/annihilation operators; Pauli exclusion is the algebraic fact $(c^\dagger)^2 = 0$, and one-body operators are $\sum t_{ij} a_i^\dagger a_j$ for both statistics.
2. **Symmetries** act by unitary operators $e^{-i\theta T}$ generated by one-body operators; their eigenvalues are additive over occupied modes. Number conservation of electrons (global $U(1)$) versus the absence of such a charge for the photon explains why photons are created singly.
3. **Local $U(1)$ gauge invariance** fixes the electron–photon coupling $\mathbf{p} \to \mathbf{p} + e\mathbf{A}$, producing both $H_1 \propto \mathbf{A}\cdot\mathbf{p}$ and the seagull $H_2 \propto \mathbf{A}^2$ with locked coefficients.
4. **The photon field** is a set of oscillators with $[a, a^\dagger] = 1$; the classical limit is the coherent state with $\Delta E/E = 1/(2\sqrt{\bar{n}})$, and radiation by a classical current produces exactly that state.
5. **The electron field** obeys the Schrödinger equation as an operator equation. Free and bound electrons are two different bases of one and the same field, related by a unitary transformation.
6. **Emission and absorption**: matrix elements $\propto \sqrt{n}$ and $\sqrt{n+1}$ give stimulated and spontaneous emission; in the dipole approximation $\Gamma = \omega^3 |\mathbf{d}|^2/(3\pi\varepsilon_0\hbar c^3)$, which reproduces the measured hydrogen $2p$ lifetime of 1.6 ns, equals (oscillator strength) × (classical radiation-damping rate), and enforces Einstein's thermal balance.
7. **Free electrons cannot absorb a single photon**; the lowest-order free-electron process is Compton scattering. In the non-relativistic theory only the seagull contributes for an electron at rest, giving the Thomson cross-section $\sigma_{\mathrm{T}} = \frac{8\pi}{3} r_e^2$ and the Compton shift to leading order in $\hbar\omega/mc^2$; the relativistic Klein–Nishina result requires the Dirac field.

26.1 Harmonic Oscillator

Harmonic oscillator is used as the fundamental element in *field quantization*.

26.1.1 Classical model

The Hamiltonian function of a harmonic oscillator is

$$H = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2} \, .$$

26.1.2 From classical to quantum

Promoting generalized coordinate and momentum to operators, the Hamiltonian operator of the quantum harmonic oscillator reads

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2} \, .$$

Stationary states

The **stationary states** are defined as the eigenstates of the Hamiltonian operators, whose energy level is the corresponding eigenvalue

$$E_n|\psi_n\rangle = \left(\frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2} \right) |\psi_n\rangle \, ,$$

or in position representation, with decaying wave function at infinity,

$$\begin{cases} E_n \psi_n = -\dfrac{\hbar^2}{2m}\psi_n'' + \dfrac{m\omega^2 x^2}{2}\psi_n \\ \psi\big|_{\pm\infty} \to 0 \, . \end{cases}$$

A solution exists only for a set of **discrete values** of E_n, while for all the other values the b.c. are not satisfied. The feasible solutions of the eigenproblem are

$$\psi_n(x) = c_n H_n(x) e^{-\frac{1}{2}x^2}, \qquad E_n = \left(\tfrac{1}{2} + n\right)\hbar\omega$$

n	$\psi_n(x)$	E_n
0	$\psi_0 = c_0 e^{-x^2/2}$	$\frac{1}{2}\hbar\omega$
1	$\psi_1 = c_1 x\, e^{-x^2/2}$	$\frac{3}{2}\hbar\omega$
2	$\psi_2 = c_2(2x^2-1)e^{-x^2/2}$	$\frac{5}{2}\hbar\omega$
…	…	…

Important: The lowest energy is $\neq 0$ because of Heisenberg uncertainty, i.e. a particle can't be exactly in $x = 0$ with $p = 0$!

Details of the spectral decomposition of the Hamiltonian operator

Let $\xi = ax$. Then, as $\frac{d}{dx} = \frac{d\xi}{dx}\frac{d}{d\xi} = a\frac{d}{d\xi}$ the equation can be recast as

$$E_n\psi_n(\xi) = -a^2\frac{\hbar^2}{2m}\psi_n''(\xi) + \frac{m\omega^2\xi^2}{2a^2}\psi_n(\xi) \,.$$

Defining $a = \sqrt{\frac{m\omega}{\hbar}}$, $\lambda_n = \frac{2E_n}{\hbar\omega}$, the equation becomes

$$\psi_n''(\xi) + (\lambda_n - \xi^2)\psi_n(\xi) = 0 \,.$$

Asymptotic behavior

For $|\xi| \to +\infty$, the elastic term dominates the eigenvalue term, and thus

$$0 \simeq -\psi'' + \xi^2\psi \,.$$

The asymptotic behavior reads $\psi_{n,\infty}(\xi) = e^{-\xi^2/2}$. This can be checked evaluating the first two derivatives

$$\frac{d}{d\xi}e^{-\xi^2/2} = -\xi e^{-\xi^2/2}$$
$$\frac{d^2}{d\xi^2}e^{-\xi^2/2} = \left(-1+\xi^2\right)e^{-\xi^2/2} \,.$$

and comparing the leading order contributions $O(\xi^2 e^{-\xi^2/2})$.

Full solution

Let the solution be $\psi(\xi) = h(\xi)e^{-\xi^2/2}$, with unknown $h(\xi)$. Its derivatives read

$$\psi' = \left(h' - \xi h\right)e^{-\xi^2/2}$$
$$\psi'' = \left(h'' - h - 2\xi h' + \xi^2 h\right)e^{-\xi^2/2}$$

Introducing this expression in the differential equation, and removing the term $e^{-\xi^2/2} \neq 0$,

$$\begin{aligned} 0 &= (\lambda - \xi^2)h(\xi) + (h''(\xi) - h(\xi) - 2\xi h'(\xi) + \xi^2 h(\xi)) = \\ &= \lambda h(\xi) + h''(\xi) - h(\xi) - 2\xi h'(\xi) \,. \end{aligned}$$

Let the power expansion of $h(\xi)$ be

$$h(\xi) = \sum_{k=0}^{+\infty} c_k \xi^k \,,$$

thus the equation becomes

$$\begin{aligned} 0 &= \lambda\sum_{k=0}^{+\infty} c_k\xi^k + \sum_{k=0}^{+\infty} k(k-1)c_k\xi^{k-2} - \sum_{k=0}^{+\infty} c_k\xi^k - 2\xi\sum_{k=0}^{+\infty} k\,c_k\xi^{k-1} = \\ &= \lambda\sum_{k=0}^{+\infty} c_k\xi^k + \sum_{k=0}^{+\infty}(k+2)(k+1)c_{k+2}\xi^k - \sum_{k=0}^{+\infty} c_k\xi^k - 2\sum_{k=0}^{+\infty} k\,c_k\xi^k = \\ &= \sum_{k=0}^{+\infty}\left\{-\left(-\lambda+1+2k\right)c_k + (k+2)\,(k+1)\,c_{k+2}\right\}\xi^k \,. \end{aligned}$$

This equation implies the a recursion relation between the coefficients of the power expansion

$$c_{k+2} = \frac{2k+1-\lambda}{(k+2)(k+1)}c_k \,. \tag{26.1}$$

Symmetry of the solutions

The recurrence relation (26.1) involves coefficients with the same parity, c_k and c_{k+2}. Thus it's possible to select:

- even solutions, by setting $c_1 = 0$, that implies $c_3 = c_5 = \cdots = 0$
- odd solutions, by setting $c_0 = 0$, that implies $c_2 = c_4 = \cdots = 0$

Asymptotic behavior of the power series

Let's discuss even solution here. The discussion for odd solutions is identical.

Even solutions. Let $c_1 = 0$. Then the power series of $h(\xi)$ only contains even powers

$$h(\xi) = \sum_{m=0}^{+\infty} c_{2m} \xi^{2m} \, .$$

For "large" m, the recurrence relation becomes

$$c_{2m+2} \sim \frac{2}{2m} c_{2m} = \frac{c_{2m}}{m} \sim \frac{c_{2m}}{m+1} \, ,$$

so that the asymptotic behavior of the **infinite power series** becomes

$$h(\xi) = \cdots + \sum_{m=1}^{+\infty} c_{2m} \xi^{2m} = \cdots + c_0 \sum_{m=0}^{+\infty} \frac{\xi^{2m}}{m!} = \cdots + e^{\xi^2} \, .$$

Thus, the asymptotic behavior of the full solution, $\psi(\xi) = e^{-\xi^2/2} h(\xi)$, is $\psi(\xi) \sim e^{-\xi^2/2} e^{\xi^2} = e^{\xi^2/2}$, for $\xi \to \mp + \infty$. As the boundary conditions of the original problem implies $\psi(\xi \to \mp\infty) \to 0$, this asymptotic behavior is not allowed, and thus the power series can't have an infinite number of terms. Thus the recurrence relation provides a **stopping criterion**.

If $c_{n+2} = 0$ is the first zero coefficient with that parity (i.e. $c_{n+1} = 0$), then the last non-zero term is c_n. The recurrence relation provides the condition

$$\lambda_n = 2n + 1 \, .$$

The following table contains the first solutions (not normalized, not orthogonal **!**)

k	λ_k	$E_k = \frac{\hbar\omega}{2}\lambda_k = \hbar\omega\left(k + \frac{1}{2}\right)$	$\psi_k(\xi)$
0	1	$\frac{1}{2}\hbar\omega$	$c_0 e^{-\xi^2/2}$
1	3	$\frac{3}{2}\hbar\omega$	$c_1 x e^{-\xi^2/2}$
2	5	$\frac{5}{2}\hbar\omega$	$c_0\left(1 - x^2\right) e^{-\xi^2/2}$
3	7	$\frac{7}{2}\hbar\omega$	$c_1\left(x - \frac{2}{3}x^3\right) e^{-\xi^2/2}$

Creator and annihilation operators

Let's call $\psi_n(x) = |n\rangle$. It's possible to prove (**todo** *prove it!*) that there are two operators

- $\hat{a}^\dagger |n\rangle \to |n+1\rangle$, **creation operator**, that moves the state $|n\rangle$ to the state $|n+1\rangle$
- $\hat{a} |n\rangle \to |n-1\rangle$, **annihilation operator**, that moves the state $|n\rangle$ to the state $|n-1\rangle$

These operators are defined with the following proportionality constants - whose reason appears in the definition of the *number operator* -

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1}\, |n+1\rangle$$
$$\hat{a} |n\rangle = \sqrt{n}\, |n-1\rangle \, ,$$

with the condition

$$\hat{a}|0\rangle = 0 \,.$$

With $c = \hbar = 1$, creation/destruction in terms of $\hat{x}, \hat{p}$:

$$\hat{a}^\dagger = \sqrt{\tfrac{m\omega}{2\hbar}}\left(\hat{x} - \tfrac{i\hat{p}}{m\omega}\right), \qquad \hat{a} = \sqrt{\tfrac{m\omega}{2\hbar}}\left(\hat{x} + \tfrac{i\hat{p}}{m\omega}\right)$$

Creation and annihilation operators, and the Hamiltonian

$$\begin{aligned}
\hat{H} &= \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} = \\
&= \hbar\omega \underbrace{\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - i\frac{\hat{p}}{m\omega}\right)}_{=\hat{a}^\dagger} \underbrace{\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + i\frac{\hat{p}}{m\omega}\right)}_{=\hat{a}} - i\hbar\omega\frac{m\omega}{2\hbar}\underbrace{\left(\hat{x}\frac{\hat{p}}{m\omega} - \frac{\hat{p}}{m\omega}\hat{x}\right)}_{\frac{1}{m\omega}[\hat{x},\hat{p}]=\frac{i\hbar}{m\omega}} = \\
&= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) \,.
\end{aligned}$$

Commutation of annihilation and creation operator

$$\begin{aligned}
[\hat{a}, \hat{a}^\dagger] &= \frac{m\omega}{2\hbar}\left[\hat{x} + i\frac{\hat{p}}{m\omega}, \hat{x} - i\frac{\hat{p}}{m\omega}\right] = \\
&= \frac{m\omega}{2\hbar}\left\{-\frac{i}{m\omega}[\hat{x},\hat{p}] + \frac{i}{m\omega}[\hat{p},\hat{x}]\right\} = \\
&= -i\frac{1}{2\hbar}\left\{[\hat{x},\hat{p}] + [\hat{x},\hat{p}]\right\} = \\
&= -i\frac{1}{2\hbar}\cdot 2i\hbar = \\
&= 1 \,.
\end{aligned}$$

Details about the action of creation and annihilation operators on eigenstates in position basis

The n mode in space basis reads

$$\psi_n(x) = e^{-x^2/2}h_n(x) = e^{-x^2/2}\sum_{k=0}^{n} c_k x^k \,,$$

with the coefficients of the power series linked by the recurrence relation (26.1). The application of the position operator gives

$$x\psi_n(\xi) = \sqrt{\frac{\hbar}{m\omega}}\,\xi\psi_n(\xi) \,,$$

while the application of the momentum operator (on an even function, with $c_1 = 0$) gives, recalling that $\xi = x\sqrt{\frac{m\omega}{\hbar}}$, and thus $\partial_x = \xi'(x)\partial_\xi = \sqrt{\frac{m\omega}{\hbar}}\partial_\xi$,

$$\begin{aligned}
\langle x|\hat{p}|\psi_n\rangle &= -i\hbar\partial_x\psi_n(x) = \\
&= -i\hbar\sqrt{\frac{m\omega}{\hbar}}\left\{-\xi\sum_{k=0}^{n}c_k\xi^k + \sum_{k=0}^{n}kc_k\xi^{k-1}\right\}e^{-\xi^2/2} = \\
&= -i\hbar\sqrt{\frac{m\omega}{\hbar}}\left\{-\sum_{k=0}^{n}c_k\xi^{k+1} + \underbrace{c_1}_{=0} + \sum_{k=0}^{n}(k+2)c_{k+2}\xi^{k+1}\right\}e^{-\xi^2/2} = \\
&= -i\hbar\sqrt{\frac{m\omega}{\hbar}}\left\{-\sum_{k=0}^{n}\left(c_k - (k+2)c_{k+2}\right)\xi^{k+1}\right\}e^{-\xi^2/2}\,.
\end{aligned}$$

Creation operator gives

$$\begin{aligned}
&\sqrt{\frac{m\omega}{2\hbar}}\left\langle x\left|\hat{x} - i\frac{\hat{p}}{m\omega}\right|\psi_n\right\rangle = \\
&= \sqrt{\frac{m\omega}{2\hbar}}\left\{\sqrt{\frac{\hbar}{m\omega}}e^{-\xi^2/2}\sum_{k=0}^{n}c_k\xi^{k+1} - \frac{i}{m\omega}\cdot(i\hbar)\cdot\sqrt{\frac{m\omega}{\hbar}}\sum_{k=0}^{n}\left(c_k - (k+2)c_{k+2}\right)\xi^{k+1}\right\} = \\
&= \frac{1}{\sqrt{2}}e^{-\xi^2/2}\left\{\sum_{k=0}^{n}c_k\xi^{k+1} + \sum_{k=0}^{n}\left(c_k - (k+2)c_{k+2}\right)\xi^{k+1}\right\} = \\
&= \frac{1}{\sqrt{2}}e^{-\xi^2/2}\sum_{k=0}^{n}\left(2c_k - (k+2)c_{k+2}\right)\xi^{k+1} = \\
&= \frac{1}{\sqrt{2}}e^{-\xi^2/2}\sum_{k=0}^{n}c_k\left(2 - \frac{2k+1-\lambda_n}{k+1}\right)\xi^{k+1} = \\
&= \frac{1}{\sqrt{2}}e^{-\xi^2/2}\sum_{k=0}^{n}\underbrace{c_k\left(\frac{2n+2}{k+1}\right)}_{=:d_{k+1}}\xi^{k+1}\,,
\end{aligned}$$

with $\lambda_n = 2n+1$

$$d_{k+1} = c_k\frac{2n+2}{k+1}\,,$$

and thus

$$\begin{aligned}
d_{k+3} &= c_{k+2}\frac{2n+2}{k+3} = \\
&= -2c_k\frac{n-k}{(k+1)(k+2)}\frac{2n+2}{k+3} = \\
&= -2\frac{n-k}{(k+2)(k+3)}\left(c_k\frac{2n+2}{k+1}\right) = \\
&= -2\frac{(n+1)-(k+1)}{(k+2)(k+3)}d_{k+1} = \\
&= -2\frac{(n+1)-(k+1)}{(k+2)(k+3)}d_{k+1}\,,
\end{aligned}$$

and, recognizing the degree of the expansion $n+1$, it follows

$$d_{k+3}^{(n+1)} = -2\frac{(n+1)-(k+1)}{(k+2)(k+3)}d_{k+1}^{(n+1)}\,,$$

compatible with the recurrence relation

$$c_{k+2}^{(n)} = \frac{2k+1-\lambda_n}{(k+1)(k+2)} c_k^{(n)} = -2\frac{n-k}{(k+1)(k+2)} c_k^{(n)} \ .$$

Annihilation operator. Algebra is left as an exercise for readers or contributors…

Number operator

The number operator is defined as $\hat{a}^\dagger \hat{a} =: \hat{n}$, with $\hat{n}|n\rangle = n|n\rangle$.

$$\hat{H} = \hbar\omega \left(\tfrac{1}{2} + \hat{a}^\dagger \hat{a}\right), \qquad [\hat{a}, \hat{a}^\dagger] = 1$$

All these results apply to a single particle, in a 1-dim domain, with a single oscillation frequency.

TODO: generalization to more complex systems — is it possible?

26.1.3 Classical limit

Coherent states

- Definition
- Properties:
  - time evolution of a coherent state
  - classical evolution of expected values
  - minimum uncertainty

Definition

Through the action of the displacement operator on the vacuum state.

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle = \exp()\,|0\rangle$$

Displacement operator

Eigen-state of the annihilation operator. A coherent state $|\alpha\rangle$ is the eigenstate of the annihilation operator $\hat{a}$.

Schrodinger picture. In Schrodinger picture, the operator is independent of time,

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle \ .$$

Coherent state as a power series

$$|\alpha\rangle = \sum_{n=0}^{+\infty} b_n |\phi_n\rangle \ ,$$

with

$$b_n = \frac{\alpha^n}{\sqrt{n!}}\, b_0 = \frac{\alpha^n}{\sqrt{n!}} e^{-\frac{|\alpha|^2}{2}} e^{i\varphi} \ .$$

Details

$$\alpha \sum_{n=0}^{+\infty} b_n |n\rangle = \sum_{n=0}^{+\infty} \sqrt{n+1} b_{n+1} |n\rangle \ ,$$

and thus

$$\sqrt{n+1}\, b_{n+1} = \alpha b_n \ ,$$

and

$$b_n = \frac{\alpha^n}{\sqrt{n!}} b_0$$

Normalization gives

$$1 = \langle \alpha | \alpha \rangle = \sum_{m,n=0}^{+\infty} b_m^* b_n \langle m|n\rangle = |b_0|^2 \sum_{n=0}^{+\infty} \frac{\alpha^{2n}}{n!} = |b_0|^2 \cdot e^{|\alpha|^2} \ ,$$

i.e.

$$b_0 = e^{i\varphi} e^{-\frac{|\alpha|^2}{2}} \ .$$

Poisson distribution of probability coefficients in energy basis. The probabilities $|c_n|^2$ of being in energy states $|\phi_n\rangle$ follows a **Poisson distribution**,

$$|c_n|^2 = \frac{|\alpha|^{2n} e^{-|\alpha|^2}}{n!} = p^{Poisson}(n; |\alpha|^2) \ ,$$

and thus it has both **expected value** and **variance** equal to $|\alpha|^2$.

Time evolution of the state of a system, starting in a coherent state $|\Psi\rangle_0 = |\alpha\rangle$. The evolution of a system governed by the Sschrodinger equation

$$i\hbar \frac{d}{dt} |\Psi\rangle_t = \hat{H} |\Psi\rangle_t = \left[\frac{\hat{p}^2}{2m} + \frac{m\omega^2 \hat{x}^2}{2} \right] |\Psi\rangle_t \ ,$$

can be written as

$$\begin{aligned} |\Psi\rangle_t &= \exp\left[-i \frac{\hat{H}}{\hbar} t \right] |\Psi\rangle_0 = \\ &= \exp\left[-i \frac{\hat{H}}{\hbar} t \right] \sum_{n=1}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-\frac{|\alpha|^2}{2}} e^{i\varphi} |\phi_n\rangle = \\ &= e^{i\varphi} e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{+\infty} \frac{\alpha^n}{\sqrt{n!}} e^{-i\frac{E_n}{\hbar} t} |\phi_n\rangle \ . \end{aligned}$$

For a harmonic oscillator $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$,

$$|\Psi\rangle_t = e^{-i\frac{\omega}{2}t} \left[e^{i\varphi} e^{-\frac{|\alpha|^2}{2}} \sum_{n=1}^{+\infty} \frac{\left(\alpha e^{-i\omega t}\right)^n}{\sqrt{n!}} |\phi_n\rangle \right] \ .$$

In space representation

The annihilation operator reads

$$\begin{aligned}\langle x|\hat{a} &= \sqrt{\frac{m\omega}{2\hbar}}\langle x|\left(\hat{x}+\frac{i\hat{p}}{m\omega}\right) = \\ &= \sqrt{\frac{m\omega}{2\hbar}}\left(x+\frac{\hbar}{m\omega}\partial_x\right)\end{aligned}$$

and its spectral decomposition

$$\begin{aligned}\langle x|\hat{a}|\alpha\rangle &= \alpha\langle x|\alpha\rangle \\ \sqrt{\frac{m\omega}{2\hbar}}\left(x+\frac{\hbar}{m\omega}\partial_x\right)\psi_\alpha(x) &= \alpha\psi_\alpha(x)\end{aligned}$$

whose solution is

$$\psi_\alpha(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{m\omega}{2\hbar}\left(x-\sqrt{\frac{2\hbar}{m\omega}}\mathrm{Re}\{\alpha\}\right)^2 + i\sqrt{\frac{m\omega}{2\hbar}}\mathrm{Im}\{\alpha\}x + i\,\mathrm{Im}\{C\}\right] .$$

Details

Solution of the equation

$$xy(x) + ay'(x) = by(x)$$

with separation of variables,

$$\begin{aligned}ay'(x) &= (x-b)y(x) \\ \int\frac{dy}{y} &= \int\frac{1}{a}(b-x)\,dx \\ \ln|y| &= \frac{1}{a}\left(-\frac{x^2}{2}+bx+C\right) \\ \rightarrow\quad y(x) &= \exp\left[-\frac{x^2}{2a}+\frac{b}{a}x+C\right]\end{aligned}$$

The factor $K = \exp(C)$ is determined with the normalization condition, up to an irrelevant phase factor. In the problem $a \in \mathbb{R}$, $b \in \mathbb{C}$, and thus the squared magnitude is

$$\begin{aligned}|y(x)|^2 &= \exp\left(-\frac{x^2}{a}\right)\cdot\exp\left(\frac{b+b^*}{a}x\right)\cdot\exp\left(C+C^*\right) = \\ &= \exp\left(-\frac{x^2}{a}+2\frac{\mathrm{Re}\{b\}}{a}x+2\mathrm{Re}\{C\}\right) = \\ &= \exp\left[-\frac{1}{a}\left(x-\mathrm{Re}\{b\}\right)^2+\frac{\mathrm{Re}\{b\}^2}{a}+2\mathrm{Re}\{C\}\right] =\end{aligned}$$

and its integral

$$1 = \int_{-\infty}^{+\infty}|y(x)|^2dx = \int_{-\infty}^{+\infty}\exp\left(-\frac{1}{a}\left(x-\mathrm{Re}\{b\}\right)^2\right)dx\cdot\exp\left[\dots\right] = \sqrt{\pi a}\cdot\exp[\dots],$$

so that the normalization factor reads $\exp[\dots] = \frac{1}{\sqrt{2a}}$, having exploited

$$I^2 = \left[\int_{x=-\infty}^{+\infty}e^{-\frac{x^2}{a}}\right]^2 = 2\pi\cdot\frac{a}{2}\cdot\int_{r=0}^{+\infty}e^{-\frac{r^2}{a}}d\frac{r^2}{a} = \pi a\,.$$

As $\exp\left[\frac{\mathrm{Re}\{b\}^2}{a}+2\mathrm{Re}\{C\}\right]=\frac{1}{\sqrt{2a}}$, the normalized solution becomes

$$\begin{aligned}y(x)&=\exp\left[-\frac{x^2}{2a}+\frac{\mathrm{Re}\{b\}}{a}x+i\frac{\mathrm{Im}\{b\}}{a}x+\mathrm{Re}\{C\}+i\mathrm{Im}\{C\}\right]=\\&=\exp\left[-\frac{x^2}{2a}+\frac{\mathrm{Re}\{b\}}{a}x+i\frac{\mathrm{Im}\{b\}}{a}x+i\mathrm{Im}\{C\}\right]\cdot\exp\left[-\frac{\mathrm{Re}\{b\}^2}{2a}\right]\cdot\frac{1}{(\pi a)^{1/4}}=\\&=\exp\left[-\frac{1}{2a}\left(x-\mathrm{Re}\{b\}\right)^2+i\frac{\mathrm{Im}\{b\}}{a}x+i\mathrm{Im}\{C\}\right]\cdot\frac{1}{(\pi a)^{1/4}}\,.\end{aligned}$$

Thus the solution of the eigenproblem of the annihilation operator is readily found with $a=\frac{\hbar}{m\omega}$, $b=\alpha\sqrt{\frac{2\hbar}{m\omega}}$, and thus $\frac{b}{a}=\alpha\sqrt{\frac{m\omega}{2\hbar}}$,

$$\psi_\alpha(x)=\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\exp\left[-\frac{m\omega}{2\hbar}\left(x-\sqrt{\frac{2\hbar}{m\omega}}\mathrm{Re}\{\alpha\}\right)^2+i\sqrt{\frac{m\omega}{2\hbar}}\mathrm{Im}\{\alpha\}x+i\,\mathrm{Im}\{C\}\right]\,.$$

Heisenberg picture. Creation operator in Heisenberg picture satisfies Heisenberg dynamical equation,

$$i\hbar\frac{d\hat{a}_H}{dt}=[\hat{a},\hat{H}]_H=\cdots=\hbar\omega\hat{a}_H\,,$$

and thus its evolution is $\hat{a}_H(t)=\hat{a}_H(0)\,e^{-i\omega t}$. Using the relation $\{\}$ between Heisenberg and Schrodinger pictures of the same operator, $\hat{A}_H=\mathcal{U}^\dagger_{t,t_0}\hat{A}_S\mathcal{U}_{t,t_0}$, with $\mathcal{U}_{t_0,t_0}=\hat{\mathbf{1}}$, it follows that $\hat{a}_H(0)=\hat{a}$, and $\hat{a}_H(t)=\hat{a}\,e^{-i\omega t}$, without explicitly writing the index S for Schrodinger picture.

Thus the spectral decomposition of the annihilation operator in Heisenberg picture, $\hat{a}_H|\alpha\rangle_H=\alpha_H|\alpha\rangle_H$, becomes

$$e^{-i\omega t}\hat{a}|\alpha\rangle_H=\alpha_H|\alpha\rangle_H$$

$$\hat{a}|\alpha\rangle_H=e^{i\omega t}\alpha_H|\alpha\rangle_H$$

and thus

$$|\alpha\rangle_H=$$

…

Classical limit

The expected value of the position and the momentum of the system starting in a coherent state is

$$\langle\hat{x}\rangle=\sqrt{\frac{\hbar}{2m\omega}}\left(\alpha^*e^{i\omega t}+\alpha e^{-i\omega t}\right)=\sqrt{\frac{2\hbar}{m\omega}}\left(\mathrm{Re}\{\alpha\}\cos(\omega t)+\mathrm{Im}\{\alpha\}\sin(\omega t)\right)$$

$$\langle\hat{p}\rangle=\sqrt{\frac{m\hbar\omega}{2}}\left(i\alpha^*e^{i\omega t}-i\alpha e^{-i\omega t}\right)=\sqrt{2m\hbar\omega}\left(-\mathrm{Re}\{\alpha\}\sin(\omega t)+\mathrm{Im}\{\alpha\}\cos(\omega t)\right)=m\frac{d}{dt}\langle\hat{x}\rangle$$

Details

Using Heisenberg picture, the expected value of the position reads

$$\begin{aligned}\langle\hat{x}\rangle&=\langle\alpha|\hat{x}_H|\alpha\rangle=\\&=\langle\alpha|\sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}^\dagger_H+\hat{a}_H\right)|\alpha\rangle=\\&=\sqrt{\frac{\hbar}{2m\omega}}\langle\alpha|\left(\hat{a}^\dagger e^{i\omega t}+\hat{a}e^{-i\omega t}\right)|\alpha\rangle=\\&=\sqrt{\frac{\hbar}{2m\omega}}\left(\alpha^*e^{i\omega t}+\alpha e^{-i\omega t}\right)\,,\end{aligned}$$

since $\langle\alpha|\hat{a}|\alpha\rangle = \alpha$, $\langle\alpha|\hat{a}^\dagger|\alpha\rangle = \alpha^*$ (see the section *One-mode coherent state, for the EM field*). The expected value of the momentum, $\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}\,(\hat{a}^\dagger - \hat{a})$, reads

$$\begin{aligned}
\langle\hat{p}\rangle &= \langle\alpha|\hat{p}_H|\alpha\rangle = \\
&= \langle\alpha|i\sqrt{\frac{m\hbar\omega}{2}}\left(\hat{a}_H^\dagger - \hat{a}_H\right)|\alpha\rangle = \\
&= \sqrt{\frac{m\hbar\omega}{2}}\langle\alpha|i\left(\hat{a}^\dagger e^{i\omega t} - \hat{a}e^{-i\omega t}\right)|\alpha\rangle = \\
&= \sqrt{\frac{m\hbar\omega}{2}}\left(i\alpha^* e^{i\omega t} - i\alpha e^{-i\omega t}\right) = \\
&= \sqrt{\frac{m\hbar\omega}{2}}\cdot 2\cdot \mathrm{Re}\{i\alpha^* e^{i\omega t}\} = \\
&= \sqrt{2m\hbar\omega}\left(\mathrm{Im}\{\alpha\}\cos(\omega t) - \mathrm{Re}\{\alpha\}\sin(\omega t)\right)\ ,
\end{aligned}$$

Non-spreading wave-packets

Minimal uncertainty

---

old version

Creation and destruction

Motivation

- EM energy is quantized. These quanta (photons) have $E = \hbar\omega$, $\vec{p} = \hbar\vec{k}$.
- Photons can be created/destroyed. Example, γ emission:
  1. destruction of a high-energy e^-,
  2. creation of a lower-energy e^- + γ.
- e^- can be represented by wavefunctions.
- 1917, A. Einstein: re-derived Planck's law; spontaneous and stimulated emission.

Models for creation and destruction

Single-frequency oscillators at frequency ν

- creation → 1 more photon → $\Delta E = h\nu$
- destruction → 1 photon less → $\Delta E = -h\nu$

Classical oscillator

$$H = \frac{p^2}{2m} + \frac{m\omega^2 x^2}{2} \quad \longrightarrow \quad \text{Quantum: } \hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}$$

Eigenproblem $E_n|\psi_n\rangle = \left(\frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}\right)|\psi_n\rangle$. In position representation:

$$E_n\psi_n = -\frac{\hbar^2}{2m}\psi_n'' + \frac{m\omega^2 x^2}{2}\psi_n \quad \text{+ b.c. s.t. } \psi\big|_{\pm\infty} \to 0$$

A solution exists only for a set of discrete values of E_n; for all the other values the b.c. are not satisfied.

$$\psi_n(x) = c_n H_n(x) e^{-\frac{1}{2}x^2}, \qquad E_n = \left(\tfrac{1}{2} + n\right)\hbar\omega$$

n	$\psi_n(x)$	E_n
0	$\psi_0 = c_0 e^{-x^2/2}$	$\frac{1}{2}\hbar\omega$
1	$\psi_1 = c_1 x\, e^{-x^2/2}$	$\frac{3}{2}\hbar\omega$
2	$\psi_2 = c_2(2x^2-1)e^{-x^2/2}$	$\frac{5}{2}\hbar\omega$

Important: The lowest energy is $\neq 0$ because of Heisenberg uncertainty, i.e. a particle can't be exactly in $x = 0$ with $p = 0$!

Creation / destruction operators

Let's call $\psi_n(x) = |n\rangle$.

- $\hat{a}^\dagger|n\rangle \to |n+1\rangle$: moves from the state $|n\rangle$ to the state $|n+1\rangle$
- $\hat{a}|n\rangle \to |n-1\rangle$

The right proportionality constants give:

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad \hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle, \qquad \hat{a}|0\rangle = 0$$

With $c = \hbar = 1$, creation/destruction in terms of $\hat{x}, \hat{p}$:

$$\hat{a}^\dagger = \sqrt{\tfrac{\omega}{2}}\left(\hat{x} + \tfrac{i\hat{p}}{\omega}\right), \qquad \hat{a} = \sqrt{\tfrac{\omega}{2}}\left(\hat{x} - \tfrac{i\hat{p}}{\omega}\right)$$

[check] the usual convention has the signs the other way round ($\hat{a} \propto \hat{x} + i\hat{p}/\omega$).

Number operator: $\hat{a}^\dagger\hat{a} =: \hat{n}$, with $\hat{n}|n\rangle = n|n\rangle$.

$$\hat{H} = \hbar\omega\left(\tfrac{1}{2} + \hat{a}^\dagger\hat{a}\right), \qquad [\hat{a}, \hat{a}^\dagger] = 1$$

All these results apply to a single particle, in a 1-dim domain, with a single oscillation frequency.

TODO: generalization to more complex systems — is it possible?

26.2 Field quantization

First, a *classical system* governed by a wave equation is discussed, along with its mode decomposition: a general solution can be written as the superposition of modes $u(x,t) = \sum_k q_k(t) n_k(x)$ whose contribution is modulated by coefficients $q_k(t)$ satisfying the ODE of a *"single-frequency" harmonic oscillator*. Thus, quantization of the system is performed in the modal basis using the same procedure used for the "single-frequency" harmonic oscillator.

26.2.1 Classical system

Let a classical field $u(x,t)$ be governed by the wave equation

$$\partial_t^2 u - c^2 \partial_{xx} u = f \ ,$$

supplied with proper initial and boundary conditions. This equation is usually the continuous model (exploiting **continuum hypotesis**, or some equivalent principle) of a macroscopic system, made of a huge number - but **finite** number - of sub-systems at the microscopic level. This is quite a relevant remark, as continuous models have an infinite number of modes, while discrete models with a finite number of sub-systems have a finite number of modes.

The wave equation is a PDE that can be interpreted as the continuous counter-part of a set of ODE for the sub-systems of a discrete model, e.g.

$$\ddot{u}_i - c^2 \frac{u_{i-1} - 2u_i + u_{i+1}}{\Delta x^2} = f_i \ ,$$

having used centered finite difference for the second-order space derivative. As done in mechanics of elastic structures, the solution of the continuous equation can be written in the modal basis, that is computed as all the possible standing waves of the homogeneous problem

$$\partial_t^2 u - c^2 \partial_{xx} u = 0$$

supplied with homogeneous boundary conditions $u(0) = u(L) = 0$, using the methods of separation of variables. Modes are the set of functions $n_k(x) = \sin\left(\frac{k\pi}{L} x\right)$, with $k \in \mathbb{N}^+$. The general solution can be written as a linear combination of the modes

$$u(x,t) = \sum_k q_k(t) \sin\left(\frac{k\pi}{L} x\right) \ ,$$

with time-dependent coefficients $q_k(t)$ satisfying the harmonic oscillator uncoupled ODEs

$$\ddot{q}_k + \underbrace{\frac{c^2 k^2 \pi^2}{L^2}}_{\omega_k^2} q_k = 0 \ . \tag{26.2}$$

An **energy density** of the system can be defined as

$$e(x,t) := \frac{|\partial_t u|^2}{2} + c^2 \frac{|\partial_x u|^2}{2} \ .$$

and the energy in the whole domain is

$$\begin{aligned}
E &= \int_{x=0}^{L} e(x,t)\, dx = \\
&= \int_{x=0}^{L} \left\{ \frac{1}{2} \sum_{k,k'} \dot{q}_k \dot{q}_{k'} \sin\left(\frac{k\pi x}{L}\right) \sin\left(\frac{k'\pi x}{L}\right) + \frac{1}{2} \sum_{k,k'} c^2 q_k q_{k'} \frac{k k' \pi^2}{L^2} \cos\left(\frac{k\pi x}{L}\right) \cos\left(\frac{k'\pi x}{L}\right) \right\} dx = \\
&= \frac{L}{2} \left\{ \frac{1}{2} \sum_k \dot{q}_k^2 + \frac{1}{2} \sum_k c^2 \frac{k^2 \pi^2}{L^2} q_k^2 \right\} = \\
&= \frac{L}{2} \sum_k \left\{ \frac{1}{2} \dot{q}_k^2 + \frac{1}{2} \omega_k^2 q_k^2 \right\} = \\
&= \sum_k E_k(q^k, \dot{q}^k) \ ,
\end{aligned}$$

having exploited orthogonality of the modes. The latter expression shows that the total energy can be written as the sum of energy contributions from the different modes. As the amplitudes of the modes satisfy the harmonic oscillator ODE, then they can be written as

$$q_k(t) = Q_k e^{i(\omega_k t+\varphi_k)} + c.c. = 2Q_k \cos(\omega_k t + \varphi_k)$$
$$\dot{q}_k(t) = i\omega_k Q_k e^{i(\omega_k t+\varphi_k)} + c.c. = -2\omega_k Q_k \sin(\omega_k t + \varphi_k)$$

and the energy in the domain becomes

$$E = \frac{L}{4}\sum_k \left\{4\omega_k^2 Q_k^2 \sin^2(\omega_k t + \varphi_k) + \omega_k^2 4Q_k^2 \cos^2(\omega_k t + \varphi_k)\right\} =$$
$$= L\sum_k \omega_k^2 Q_k^2 \,,$$

so that the elementary contribution of the kth mode to the energy of the system is $E_k = L\omega_k^2 Q_k^2$.

26.2.2 From classical to quantum

The amplitude $q_k(t)$ of mode k is governed by the dynamical equation (26.2), compatible with the Hamiltonian

$$H_k(q^k, p_k) = \frac{p_k^2}{2} + \frac{\omega_k^2 (q^k)^2}{2} \,.$$

Wavefunction of the mode k:

$$\psi_{n_k}(q^k) \propto H_{n_k}(\cdot)\, e^{-\frac{1}{2}(q^k)^2}, \qquad E_k = \left(n_k + \tfrac{1}{2}\right)\hbar\omega_k$$

Wavefunction of the system of oscillators:

$$\Psi(q^1, q^2, \dots) = \psi_1(q^1)\,\psi_2(q^2)\cdots$$

or, using the occupation-number representation: $|n_1, n_2, \dots\rangle = |n_1\rangle|n_2\rangle\cdots$

- creation/destruction and number operators for each individual mode;
- for independent modes: energy $E = \sum_k E_k$, with $E_k = \hbar\omega_k\left(n_k + \frac{1}{2}\right)$.

Infinite energy

Even if $n_k = 0\ \forall k$, then $E = \sum_{k=1}^{\infty} \frac{1}{2}\hbar\omega_k = \infty$.

But physical systems are made of a **finite number** (even if huge) of particles N, and thus the system has only N modes: so the zero-point energy for mechanical systems is non-∞.

But for continuous fields, say $\vec{e}(\vec{r}, t)$, we don't have this argument: as it's usually possible to measure only energy differences, zero-point energy "can be declared by fiat", i.e. "the vacuum state has zero energy".

26.3 Photons - EM field quantization

Photons are the quanta of EM fields. After reviewing classical electromagnetism, two different introductions to EM field quantization are presented

- *Quantization of the EM field in an ideal cavity (Fermi, 1931)*
- *Quantization of the EM field in free space*

Then, the *classical limit* is discussed, comparing *a field with incoherent states* and *a field with coherent states*

26.3.1 Classcal Electromagnetism

Link to Classical Electromagnetism.

26.3.2 EM field quantization

Fermi 1931, Introduction to QFT lessons

EM Field in free space

Field in 1D free space

Travelling wave along z:

$$\vec{A}(\vec{r},t) = \hat{u}_x\, b\cos(kz-\omega t+\theta)$$

$$b\cos(kz-\omega t+\theta) = b\cos(kz)\cos(\omega t-\theta) + b\sin(kz)\sin(\omega t-\theta)$$

i.e. a linear combination of 2 standing waves.

Definitions

$$q(t) = b\cos(\omega t-\theta)\ \text{(gen. coord.)}, \qquad \dot{q}(t) = -b\omega\sin(\omega t-\theta) =: p\ \text{(gen. mom.)}, \qquad \ddot{q} = -\omega^2 q$$

$$\vec{A} = \hat{u}_x\left[q\cos(kz) - \frac{p}{\omega}\sin(kz)\right]$$

In Coulomb gauge:

$$\vec{E} = -\partial_t\vec{A} = -\hat{u}_x\left[p\cos kz + \omega q\sin kz\right], \qquad \vec{B} = \nabla\times\vec{A} = -\hat{u}_y\left[kq\sin(kz) + \frac{k}{\omega}p\cos(kz)\right]$$

[check] the $\vec{E}$ expression above is re-derived from $\dot{q}=p,\ \dot{p}=-\omega^2 q$; the handwriting is hard to read.

with $\dot{q}=p, \dot{p}=-\omega^2 q$.

Energy

$$H = \tfrac{1}{2}\left(|\vec{E}|^2+|\vec{B}|^2\right) = \tfrac{1}{2}\left\{2p^2\cos^2(kz) + 2\omega^2 q^2\sin^2(kz) + 4\omega pq\sin(kz)\cos(kz)\right\}$$

(using $k^2/\omega^2 = 1$ as normalization). Averaging over a domain with $L = n\lambda = n\frac{2\pi}{k}$, $n\in\mathbb{N}$:

$$\langle H\rangle_L = \tfrac{1}{2}\left(p^2+\omega^2 q^2\right)$$

Quantization: from q, p to $\hat{q}, \hat{p}$:

$$\hat{H} = \tfrac{1}{2}\left(\hat{p}^2+\omega^2\hat{q}^2\right), \qquad \hat{q} = \sqrt{\tfrac{1}{2\omega}}\left(\hat{a}+\hat{a}^\dagger\right), \qquad \hat{p} = -i\sqrt{\tfrac{\omega}{2}}\left(\hat{a}-\hat{a}^\dagger\right)$$

(**[check]** sign of $\hat{p}$, chosen to be consistent with the final result below.)

$$\hat{A} = \hat{u}_x\left[\hat{q}\cos(kz) - \frac{\hat{p}}{\omega}\sin(kz)\right] = \hat{u}_x\sqrt{\tfrac{1}{2\omega}}\Big\{\cos(kz)(\hat{a}+\hat{a}^\dagger) + i\sin(kz)(\hat{a}-\hat{a}^\dagger)\Big\} = \hat{u}_x\sqrt{\tfrac{1}{2\omega}}\Big\{\hat{a}^\dagger e^{-ikz} + \hat{a}\,e^{ikz}\Big\}$$

In a 3D domain

$$\hat{A} = \sum_{\vec{k},\mu} \vec{e}_{\vec{k}\mu} \frac{1}{\sqrt{2\omega_k}} \left\{ \hat{a}^\dagger_{\vec{k},\mu} e^{-i\vec{k}\cdot\vec{r}} + \hat{a}_{\vec{k},\mu} e^{i\vec{k}\cdot\vec{r}} \right\}$$

$$\hat{H} = \sum_{\vec{k},\mu} \left(\hat{N}_{\vec{k},\mu} + \tfrac{1}{2} \right) \hbar\omega_k, \qquad |\psi\rangle = |\, \ldots n_{\vec{k},\mu} \ldots \rangle$$

26.3.3 Quantum description of the EM field — classical limit

Example: 1D domain, 1 propagating wave.

$$A(z,t) = u_x\, b \cos(kz - \omega t + \theta) \xrightarrow{\text{quant}} \hat{A} = u_x \sqrt{\tfrac{1}{2\omega}} \left(\hat{a} e^{ikz} + \hat{a}^\dagger e^{-ikz} \right)$$

How to recover the classical limit?

Try 1 — Incoherent states

Let $|\psi\rangle(t)$ be the state with n photons, $|n\rangle$. Time-dependent Schrödinger: $i\hbar|\dot{\psi}\rangle = \hat{H}|\psi\rangle$.

$$|\psi\rangle(t) = c_n(t)|n\rangle, \qquad i\hbar\dot{c}_n = E_n c_n \;\rightarrow\; c_n(t) = c_{n0} \exp\!\left(-\frac{iE_n}{\hbar} t \right)$$

with $E_n = (n + \frac{1}{2})\hbar\omega$. For the system in state $|n\rangle$, with $|\psi\rangle_0 \equiv |n\rangle \rightarrow c_{n0} = 1$: $|\psi\rangle_t = \exp(-iE_n t/\hbar)|n\rangle$.

The expected value of the operator $\hat{A}$ reads

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle = \langle n | \sqrt{\tfrac{1}{2\omega}} \left(\hat{a} e^{ikz} + \hat{a}^\dagger e^{-ikz} \right) | n \rangle = \frac{e^{ikz}}{\sqrt{2\omega}} \sqrt{n}\, \underbrace{\langle n | n-1 \rangle}_{=0} + \frac{e^{-ikz}}{\sqrt{2\omega}} \sqrt{n+1}\, \underbrace{\langle n | n+1 \rangle}_{=0}$$

$$\Rightarrow \quad \langle \hat{A} \rangle \equiv 0$$

- At a given time and place, the expected value of $\hat{A}$ is zero (this is the case of **incoherent light**, random phase).
- The expected value is zero, but the field can be $\neq 0$!!

Try 2 — Coherent states

Particular superposition of energy states [Glauber '63, "Photon Correlations"].

Definition: $\hat{a}|\psi\rangle = c|\psi\rangle$, with $|\psi\rangle = \sum_{n=0}^{\infty} b_n |n\rangle$. It follows

$$\hat{a}|\psi\rangle = \sum_{n=1}^{\infty} b_n \sqrt{n}\, |n-1\rangle = \sum_{n=0}^{\infty} b_{n+1} \sqrt{n+1}\, |n\rangle = c|\psi\rangle$$

Exploiting orthogonality of $\{|n\rangle\}$: $c\, b_n = b_{n+1}\sqrt{n+1}$.

Recursive relation for the coefficients b_n:

$$\boxed{b_{n+1} = c\, \frac{b_n}{\sqrt{n+1}}} \qquad (\circledast)$$

If $b_0 \neq 0$: $b_1 = c\,b_0/\sqrt{1},\ b_2 = c\,b_1/\sqrt{2} = c^2 b_0/\sqrt{1\cdot 2},\ \ldots\ b_n = c^n b_0/\sqrt{n!}$.
Normalization $1 = \langle\psi|\psi\rangle = \sum_n |b_n|^2$ gives, using $e^x = \sum_n x^n/n!$:

$$1 = |b_0|^2 \sum_{n=0}^{\infty} \frac{|c|^{2n}}{n!} = |b_0|^2 e^{|c|^2} \ \Rightarrow\ b_0 = e^{-|c|^2/2} \ \text{(if } b_0 \text{ is real)}$$

Thus, for a given value of c:

$$|\psi\rangle = \sum_{n=0}^{+\infty} e^{-\frac{|c|^2}{2}} \frac{c^n}{\sqrt{n!}} |n\rangle$$

Number of photons. Using the probability $|b_n|^2$ of the system being in state $|n\rangle$:

- expected value of $\hat{n}$ (~ *signal*):

$$\langle n\rangle = \sum_{n=0}^{\infty} n|b_n|^2 = \sum_{n=0}^{\infty} n\, e^{-|c|^2} \frac{|c|^{2n}}{n!} = |c|^2 \sum_{n=1}^{\infty} e^{-|c|^2} \frac{|c|^{2(n-1)}}{(n-1)!} = |c|^2$$

- standard deviation / variance σ_n^2 (~ *quantum noise*):

$$\sigma_n = \sqrt{\langle (n - \langle n\rangle)^2\rangle} = \cdots = \sqrt{\langle n\rangle} = |c|$$

(The distribution $|b_n|^2$ is a bell-shaped curve centred at $|c|^2$.)

Expected values of the fields

Expected value of $\hat{a}$:

$$\langle\psi|\hat{a}|\psi\rangle = \sum_{m,n} b_{m0}^* b_{n0} \langle m|\hat{a}|n\rangle e^{-i(n-m)\omega t} = \sum_{m,n} b_{m0}^* b_{n0}\sqrt{n}\ \underbrace{\langle m|n-1\rangle}_{\delta_{m,n-1}} e^{-i\omega t} = \sum_m b_{m0}^* b_{m+1,0}\sqrt{m+1}\, e^{-i\omega t}$$

and using the recursion ($\circledast$):

$$= \sum_n c\,|b_{n0}|^2 e^{-i\omega t} = c\, e^{-i\omega t}$$

Expected value of $\hat{a}^\dagger$: $\langle\psi|\hat{a}^\dagger|\psi\rangle = \cdots = \sum_n c^*|b_{n0}|^2 e^{+i\omega t} = c^* e^{i\omega t}$.

Thus the **expected value of** $\hat{A}$:

$$\langle\psi|\hat{A}|\psi\rangle = u_x\sqrt{\tfrac{1}{2\omega}}\,\langle \hat{a}e^{ikz} + \hat{a}^\dagger e^{-ikz}\rangle = u_x\sqrt{\tfrac{1}{2\omega}}\,(c\,e^{-i\omega t}e^{ikz} + c^* e^{i\omega t}e^{-ikz}) = u_x\sqrt{\tfrac{1}{2\omega}}\ 2|c|\cos(kz - \omega t + \theta)$$

with $c = |c|e^{i\theta} \approx$ **classical electromagnetic wave**.

Obs. Time evolution of the state: $|\psi\rangle = \sum_n b_n(t)|n\rangle$. Substituting in $i\hbar|\dot{\psi}\rangle = \hat{H}|\psi\rangle$ and using orthogonality: $i\hbar\dot{b}_n = E_n b_n \to b_n(t) = b_{n0}\exp(-iE_n t/\hbar)$, i.e. $|\psi\rangle = \sum_n \exp(-i\omega_n t)\, b_{n0}|n\rangle$.

Obs. With $E_n = \hbar\omega(n + \frac{1}{2})$: $\omega_m - \omega_n = \omega(m-n)$ — **without the ½ factor!!**

Summary

- The expected value of $\hat{A}$ ~ classical wave.
- The field is a random field whose variance reads:

$$\sigma_A = \sqrt{\langle\psi|\hat{A}^2|\psi\rangle - \langle\psi|\hat{A}|\psi\rangle^2} = \sqrt{\tfrac{1}{2\omega}}$$

- *Obs.* The variance is independent of $\vec{r}, t$ and independent of the number of photons ($|c|^2$). Thus the quantum noise is constant while the signal grows with the field amplitude: **classical limit** — quantum noise → 0 (relative to the signal) as the signal power increases.

- But as the field amplitude goes to zero ($|c| \to 0$, # photons $\to 0$), only **quantum fluctuations** remain.

Heisenberg vs Schrödinger pictures

- *Heisenberg*: time dependence appears in the operator $\hat{A}(\vec{r},t)$ **— quite convenient!!**
- *Schrödinger*: only space dependence appears in $\hat{A}$, while t appears in $|\psi\rangle$.

26.4 Photons - EM field quantization with Coulomb gauge

26.4.1 Classical fields

In free space without charge and current densities, the vector potential satisfies the homogeneous wave equation $\Box\vec{a} = \vec{0}$, whose solution can be written as a linear combination of Fourier modes. For finite (or periodic) domains

$$\vec{a}(\vec{r},t) = \sum_{\vec{k}} \vec{a}(\vec{k},t)e^{i\vec{k}\cdot\vec{r}} \ ,$$

with $k_a L_a = n_a 2\pi$, $n_a \in \mathbb{Z}$. For infinite domains

$$\vec{a}(\vec{r},t) = \frac{1}{(2\pi)^{3/2}} \int_{\vec{k}} \vec{a}(\vec{k},t)e^{i\vec{k}\cdot\vec{r}}\, d\vec{k} \ .$$

Coulomb gauge. Exploiting the orthogonality of the base functions $\frac{e^{i\vec{k}\cdot\vec{r}}}{(2\pi)^{3/2}}$, Coulomb gauge $\nabla\cdot\vec{a} = 0$ implies

$$i\vec{k}\cdot\vec{a}(\vec{k},t) = 0 \quad , \quad \forall\vec{k} \ .$$

Thus, as each coefficient $\vec{a}(\vec{k},t)$ is orthogonal to its corresponding wave vector $\vec{k}$, it only has 2 independent components (corresponding to **polarization**). Let $\{\vec{\epsilon}_{\vec{k},\mu}\}_{\mu\in\{1,2\}}$ an orthogonal basis - with $\vec{\epsilon}_{\vec{k},\mu}\cdot\vec{k} = 0$ - for the coefficient $\vec{a}(\vec{k},t)$, then this coefficient can be written as

$$\vec{a}(\vec{k},t) = \sum_{\mu} \vec{\epsilon}_{\vec{k},\mu} a(\vec{k},\mu,t) \ .$$

Real-valued field, $\vec{a}(\vec{r},t)$. Since $\vec{a}(\vec{r},t)$ is a real-valued field, then

$$\vec{a}(\vec{k},t) = \vec{a}^*(-\vec{k},t) \ . \tag{26.3}$$

Solution of the homogeneous wave equation. Exploiting the orthogonality of the base functions, and the linearity of the wave equation, each coefficient of the Fourier expansion satisfies a linear second-order equation (simple harmonic oscillator),

$$\frac{1}{c^2}\partial_{tt}\vec{a}(\vec{k},t) + |\vec{k}|^2\vec{a}(\vec{k},t) = \vec{0} \ ,$$

whose solution reads

$$\vec{a}(\vec{k},t) = \vec{a}_-(\vec{k})e^{i\omega_{\vec{k}}t} + \vec{a}_+(\vec{k})e^{-i\omega_{\vec{k}}t} \ ,$$

with $\omega_{\vec{k}} = c|\vec{k}|$, and the coefficients to be determined with the initial conditions for the potential governed by the wave equation. The condition (26.3) about real-valued field $\vec{a}(\vec{r},t)$ implies

$$\vec{a}_+(\vec{k},t) = \vec{a}_-^*(-\vec{k},t) \quad , \quad \vec{a}_-(\vec{k},t) = \vec{a}_+^*(-\vec{k},t) \ .$$

Details

$$\begin{aligned}\vec{a}(\vec{k},t) &= \vec{a}_-(\vec{k})e^{i\omega_{\vec{k}}t} + \vec{a}_+(\vec{k})e^{-i\omega_{\vec{k}}t} \\ \vec{a}^*(-\vec{k},t) &= \left(\vec{a}_-(-\vec{k})e^{i\omega_{-\vec{k}}t} + \vec{a}_+(-\vec{k})e^{-i\omega_{-\vec{k}}t}\right)^* = \\ &= \vec{a}^*_-(-\vec{k})e^{-i\omega_{\vec{k}}t} + \vec{a}^*_+(-\vec{k})e^{i\omega_{\vec{k}}t}\end{aligned}$$

Solution of the wave equation. Putting everything together, the solution of the wave equation for the vector potential can be written - here in a periodic domain - as

$$\begin{aligned}\vec{a}(\vec{r},t) &= \sum_{\vec{k}} \vec{a}(\vec{k},t)e^{i\vec{k}\cdot\vec{r}} = \\ &= \sum_{\vec{k}} \left[\vec{a}_-(\vec{k})e^{i\omega_{\vec{k}}t} + \vec{a}_+(\vec{k})e^{-i\omega_{\vec{k}}t}\right] e^{i\vec{k}\cdot\vec{r}} = \qquad \text{(see re-arrangement)} \\ &= \sum_{\vec{k}} \left[\vec{a}_-(-\vec{k})e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})} + \vec{a}_+(\vec{k})e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\right] .\end{aligned} \tag{26.4}$$

Re-arrangement of the summation

For each $\vec{k}$ (except for $\vec{k}=\vec{0}$, to avoid summing twice the same contributions), summing over $\vec{k}$ and $-\vec{k}$ produces 4 terms

$$\begin{aligned}&\vec{a}(\vec{k},t) + \vec{a}(-\vec{k},t) = \\ &= \vec{a}_-(\vec{k})e^{i(\omega_{\vec{k}}t+\vec{k}\cdot\vec{r})} + \vec{a}_+(\vec{k})e^{i(-\omega_{\vec{k}}t+\vec{k}\cdot\vec{r})} + \vec{a}_-(-\vec{k})e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})} + \vec{a}_+(-\vec{k})e^{i(-\omega_{\vec{k}}t-i\vec{k}\cdot\vec{r})}\end{aligned}$$

that can be re-grouped as

$$= \underbrace{\vec{a}_-(\vec{k})e^{i(\omega_{\vec{k}}t+\vec{k}\cdot\vec{r})} + \vec{a}_+(-\vec{k})e^{-i(\omega_{\vec{k}}t+i\vec{k}\cdot\vec{r})}}_{} + \underbrace{\vec{a}_+(\vec{k})e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})} + \vec{a}_-(-\vec{k})e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}}_{} ,$$

s.t. each pair of terms gives a real contribution. Here the sum of the first pair fives a travelling wave in $-\hat{k}$ direction, while the second pair gives a travelling wave in $+\hat{k}$ direction.

EOMs and solutions as a function of generalized coordinates and momenta.

…

Electro-magnetic field.

…

26.4.2 Quantization

…**todo** *discuss promotion. Link to classical limit, other quantization procedures,…*

Promotion of the expression (26.4) to operator. In Heisenberg picture

$$\hat{\vec{a}}(\vec{r},t) = \sum_{\vec{k},\mu} \left[e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\hat{a}^\dagger_{\vec{k},\mu} + e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\hat{a}_{\vec{k},\mu}\right]$$

…

with Heisenberg picture for annihilation and creation operators being $\hat{a}^\dagger_H(t) = \hat{a}^\dagger e^{i\omega t}$ and $\hat{a}_H(t) = \hat{a}e^{-i\omega t}$ respectively.

todo *Discuss both Schrodinger and Heisenberg picture.*

Schrodinger and Heisenberg picture of annihilation/creation operators. Single simple harmonic oscillator

The Hamiltonian operator of a single simple harmonic oscillator reads

$$\hat{H} = \frac{\hat{p}^2}{2} + \frac{\omega^2 \hat{q}^2}{2} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) .$$

Annihilation operator. Heisenberg equations of motion for the annihilation operator reads

$$\begin{aligned} i\hbar \frac{d\hat{a}_H}{dt} &= \left[\hat{a}, \hat{H}\right]_H + 0(\partial_t \hat{a})_H = \\ &= \left[\hat{a}, \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right)\hbar\omega\right]_H + 0\left(\partial_t \hat{A}\right)_H = \\ &= \left\{(\hat{a}\hat{a}^\dagger \hat{a} - \hat{a}^\dagger \hat{a}\hat{a})\right\}_H \hbar\omega = \\ &= \left\{[\hat{a}, \hat{a}^\dagger]\, \hat{a}\right\}_H \hbar\omega = \\ &= \hbar\omega \hat{a}_H , \end{aligned}$$

whose solution provides the relation between the Heisenberg and Schrodinger picture of the annihilation operator,

$$\hat{a}_H(t) = \hat{a}_H(0) e^{-i\omega t} = \hat{a} e^{-i\omega t} .$$

Creation operator. Heisenberg equations of motion for the creation operator reads

$$\begin{aligned} i\hbar \frac{d\hat{a}^\dagger_H}{dt} &= \left[\hat{a}^\dagger, \hat{H}\right]_H + 0(\partial_t \hat{a}^\dagger)_H = \\ &= \left[\hat{a}^\dagger, \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right)\hbar\omega\right]_H + 0\left(\partial_t \hat{A}\right)_H = \\ &= \left\{(\hat{a}^\dagger \hat{a}^\dagger \hat{a} - \hat{a}^\dagger \hat{a}\hat{a}^\dagger)\right\}_H \hbar\omega = \\ &= \left\{\hat{a}^\dagger\, [\hat{a}^\dagger, \hat{a}]\right\}_H \hbar\omega = \\ &= -\hbar\omega \hat{a}^\dagger_H , \end{aligned}$$

whose solution provides the relation between the Heisenberg and Schdoringer picture of the creation operator,

$$\hat{a}^\dagger_H(t) = \hat{a}^\dagger_H(0) e^{i\omega t} = \hat{a}^\dagger e^{i\omega t} .$$

Energy

Momentum

...

26.4.3 Classical limit

One-mode coherent state

The state of an electromagnetic field can be written as a linear combination of states in the Fock space, with the occupation numbers of each possible modes $|\vec{k}, \mu\rangle$. A one-mode coherent state is defined as the $|\alpha_{\vec{k},\mu}\rangle$ that's an eigenstate of the annihilation operator $\hat{a}_{\vec{k},\mu}$ for the mode $|\vec{k}, \mu\rangle$,

$$\hat{a}_{\vec{k},\mu} |\alpha_{\vec{k},\mu}\rangle = c_{\vec{k},\mu} |\alpha_{\vec{k},\mu}\rangle .$$

The one-mode coherent state is a linear combination of states in the Fock space

$$|\alpha_{\vec{k},\mu}\rangle = \sum_n b_n^{\vec{k},\mu} |\dots n_{\vec{k},\mu} \dots\rangle \ ,$$

with zero occupation number of any state different from $|\vec{k},\mu\rangle$, or dropping the indices,

$$|\alpha\rangle = \sum_{n=0}^{+\infty} b_n |n\rangle \ ,$$

with $b_n = b_0 \frac{c^n}{\sqrt{n!}}$ from the definition and coefficient matching, and $b_0 = e^{-\frac{|c|^2}{2}}$ from the normalization condition $\langle\alpha|\alpha\rangle = 1$.

Annihilation operator

Expected value when the system is in state $|\alpha\rangle$,

$$\langle\alpha|\hat{a}|\alpha\rangle = c$$

Details

$$\begin{aligned}
\langle\alpha|\hat{a}|\alpha\rangle &= \sum_{n,m=0}^{+\infty} b_m^* b_n \langle m|\hat{a}|n\rangle = \\
&= \sum_{m=0}^{+\infty} \sum_{n=1}^{+\infty} b_m^* b_n \sqrt{n} \langle m|n-1\rangle = \\
&= \sum_{m=0}^{+\infty} \sum_{n=0}^{+\infty} b_m^* b_{n+1} \sqrt{n+1} \underbrace{\langle m|n\rangle}_{=\delta_{mn}} = \\
&= \sum_{n=0}^{+\infty} c \cdot |b_n|^2 = \\
&= c \cdot \sum_{n=0}^{+\infty} \frac{|c|^{2n}}{n!} = \\
&= c \cdot \underbrace{|b_0|^2 \cdot e^{|c|^2}}_{=1} = c \ ,
\end{aligned}$$

since $\sqrt{n+1} b_{n+1} = c \cdot b_n$.

Creation operator

Expected value when the system is in state $|\alpha\rangle$,

$$\langle\alpha|\hat{a}^\dagger|\alpha\rangle = c^*$$

Details

$$\begin{aligned}
\langle\alpha|\hat{a}^\dagger|\alpha\rangle &= \left(\langle\alpha|\hat{a}|\alpha\rangle\right)^* = \\
&= \left(\sum_{n,m=0}^{+\infty} b_m^* b_n \langle m|\hat{a}|n\rangle\right)^* = \\
&= \sum_{m=0}^{+\infty}\sum_{n=1}^{+\infty} b_m b_n^* \sqrt{n}\langle m|n-1\rangle = \\
&= \sum_{m=0}^{+\infty}\sum_{n=0}^{+\infty} b_m b_{n+1}^* \sqrt{n+1}\underbrace{\langle m|n\rangle}_{=\delta_{mn}} = \\
&= \sum_{n=0}^{+\infty} c^* \cdot |b_n|^2 = \\
&= c^* \cdot \sum_{n=0}^{+\infty} \frac{|c|^{2n}}{n!} = \\
&= c^* \cdot \underbrace{|b_0|^2 \cdot e^{|c|^2}}_{=1} = c^* \,.
\end{aligned}$$

Number operator

Expected value of the number operator

$$\langle\alpha|\hat{n}|\alpha\rangle = |c|^2$$

Details

With $\hat{n} = \hat{a}^\dagger\hat{a}$,

$$\begin{aligned}
\langle\alpha|\hat{a}^\dagger\hat{a}|\alpha\rangle &= \sum_{m,n=0}^{+\infty} b_m^* b_n \langle m|\hat{a}^\dagger\hat{a}|n\rangle = \\
&= \sum_{m,n=0}^{+\infty} b_m^* b_n \sqrt{n}\langle m|\hat{a}^\dagger|n-1\rangle = \\
&= \sum_{m,n=0}^{+\infty} b_m^* b_n \sqrt{n}\sqrt{n}\underbrace{\langle m|n\rangle}_{=\delta_{mn}} = \\
&= \sum_{n=0}^{+\infty} |b_n|^2 \cdot n = \\
&= |b_0|^2 \sum_{n=1}^{+\infty} n\frac{|c|^{2n}}{n!} = \\
&= |c|^2 \cdot |b_0|^2 \cdot \underbrace{\sum_{n=0}^{+\infty} \frac{|c|^{2n}}{n!}}_{=e^{|c|^2}=|b_0|^{-2}} = \\
&= |c|^2 \,.
\end{aligned}$$

Potential vector

Expected value of the potential vector field operator

$$\langle\alpha|\hat{\vec{a}}(\vec{r},t)|\alpha\rangle = e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\, c^*_{\vec{k},\mu} + e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\, c_{\vec{k},\mu}\ .$$

Details

$$\begin{aligned}\langle\alpha|\hat{\vec{a}}(\vec{r},t)|\alpha\rangle &= \langle\alpha|\sum_{\vec{k}',\mu'}\left[e^{i(\omega_{\vec{k}'}t-\vec{k}'\cdot\vec{r})}\vec{\epsilon}_{\vec{k}',\mu'}\hat{a}^\dagger_{\vec{k}',\mu'} + e^{-i(\omega_{\vec{k}'}t-\vec{k}'\cdot\vec{r})}\vec{\epsilon}_{\vec{k}',\mu'}\hat{a}_{\vec{k}',\mu'}\right]|\alpha\rangle = \\ &= \langle\alpha|\left[e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\hat{a}^\dagger_{\vec{k},\mu} + e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\hat{a}_{\vec{k},\mu}\right]|\alpha\rangle = \\ &= e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\langle\alpha|\hat{a}^\dagger_{\vec{k},\mu}|\alpha\rangle + e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\langle\alpha|\hat{a}_{\vec{k},\mu}|\alpha\rangle = \\ &= e^{i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\, c^* + e^{-i(\omega_{\vec{k}}t-\vec{k}\cdot\vec{r})}\vec{\epsilon}_{\vec{k},\mu}\, c\ .\end{aligned}$$

Comparison with (26.4) identifies the coefficients of the Fourier expansion with the eigenvalue of the annihilation operator corresponding to the coherent state,

$$\vec{a}_+(\vec{k},\mu) \sim c_{\vec{k},\mu} \quad , \quad \vec{a}_-(-\vec{k},\mu) \sim c^*_{\vec{k},\mu}\ .$$

26.5 Multi-body quantum mechanics and QFT

In multi-body quantum systems the number of identical particles may change in time, and the interactions may occur as non-smooth dynamics involving creation and destruction of particles. First, the limits of the quantum systems represented with so called first quantization process are discussed: these systems can't represent systems with a variable number of particles and non smooth interactions. Then, the second quantization of the electron field is introduced, along with the Pauli exclusion principle. Eventually, the properties and the formalism for the second quantization of boson and fermion fields are recalled.

26.5.1 First quantization

First quantization moves the description of a particle to a wave process, promoting generalized coordinates and momenta x, p to operators $\hat{q}, \hat{p}$. The state of a system is described by a wave function $|\Psi\rangle$ belonging to a Hilbert space $\mathcal{H}$, satisfying the Schrodinger equation of the system

$$i\hbar|\dot{\Psi}\rangle = \hat{H}|\Psi\rangle\ .$$

The wavefunction $|\Psi\rangle$ satisfies the normalization condition $\langle\Psi|\Psi\rangle = 1$. If the Hamiltonian operator $\hat{H}$ is independent from time, its spectral decomposition provides an orthogonal basis for $\mathcal{H}$ of stationary states $\{|\psi_n\rangle\}$ and their energy levels $\{E_n\}$. The wavefunction of the system $|\Psi\rangle$ can be represented in the orthonormal basis as

$$|\Psi(t)\rangle = \sum_n |\psi_n\rangle c_n(t)\ ,$$

with the coefficients c_n that can be linked to the probability $|c_n|^2$ of finding the system whose state is represented by $|\Psi\rangle$ in the state $|\psi_n\rangle$, as from normalization condition it follows

$$1 = \langle\Psi|\Psi\rangle = \sum_{n,m} c^*_n c_m \underbrace{\langle\psi_n|\psi_m\rangle}_{\delta_{nm}} = \sum_n |c_n|^2\ .$$

The Schrödinger equation makes $c_n(t)$ vary continuously. In energy basis, the coefficients of teh expansion are governed by the ODEs

$$i\hbar \dot{c}_n = E_n c_n \ ,$$

and thus they evolve as $c_n(t) = c_{n,0} \exp\left(-i\frac{E_n}{\hbar}t\right)$.

But Interactions between systems may occur as non-smooth dynamics. As an example, EM radiation absorption or emission by an atom occurs in **quantum jumps**. As an example, spontaneous emission of EM radiation from an atom involves:

1. the destruction of a high-energy e^-,
2. the creation of a lower-energy e^- + a γ.

26.5.2 Second quantization of the electron field

First quantization. The wave function describing an electron can be written as a linear combination of the energy eigenstates. In space representation

$$\psi(\mathbf{r}, t) = \sum_k b_k(t)\psi_k(\mathbf{r}) = \sum_k b_{k0}\psi_k(\mathbf{r})e^{-i\omega_k t} \ .$$

with $\omega_k = \frac{E_k}{\hbar}$. The expected value of the energy is

$$\langle\psi|\hat{H}|\psi\rangle = \sum_{m,n}\langle m|\hat{H}|n\rangle \, b^*_{m0}b_{n0} \, e^{+i\omega_m t}e^{-i\omega_n t} = \sum_n E_n|b_{n0}|^2 \ .$$

Second quantization: $b_{k0} \to \hat{b}_k$ (destruction operator). In analogy with multi-body QM:

$$\hat{H} = \sum_k E_k\hat{n}_k = \sum_k E_k\hat{b}^\dagger_k\hat{b}_k$$

26.5.3 Hamiltonian and state of the non-interacting e^- and radiation fields

Electron field. The state can be represented in the occupation number basis as a linear combination of the basis of the Fock space, namely

$$\{|\mathbf{n}\rangle\} = \{|\mathbf{0}\rangle, |1,0,0,...\rangle, |0,1,0,...\rangle, ..., |2,0,0,...\rangle, |1,1,0,...\rangle, ...\} \ .$$

$$e^- : \ |\phi_e\rangle = |n_{e1}, n_{e2}, ...\rangle, \quad \hat{H}_e = \sum_k E_k\hat{b}^\dagger_k\hat{b}_k$$

Radiation field

$$\text{rad} : \ |\psi_r\rangle = |n_{k\alpha} ...\rangle, \quad \hat{H}_{rad} = \sum_{k,\alpha}\hbar\omega_k\hat{a}^\dagger_{k\alpha}\hat{a}_{k\alpha}$$

These are **non-interacting fields**.

An **interaction is required** to get quantum jumps of e^- associated with creation/annihilation of a γ.

26.5.4 Bosons and fermions

(This page of the notes is written upside-down; the content is transcribed in condensed form.)

Bosons — occupation-number formalism:

- $\hat{a}^\dagger$ acts on a state by raising it, $|0\rangle \to |1\rangle \to |2\rangle \to \cdots$, and $\hat{a}$ lowers it.
- Number operator: $\hat{n}_k = \hat{a}_k^\dagger \hat{a}_k$, s.t. $\hat{n}_k |\dots n_k \dots\rangle = n_k |\dots n_k \dots\rangle$.

$$\hat{a}_k^\dagger |\dots n_k \dots\rangle = \sqrt{n_k+1}\,|\dots (n_k+1) \dots\rangle, \qquad \hat{a}_k |\dots n_k \dots\rangle = \sqrt{n_k}\,|\dots (n_k-1) \dots\rangle$$

CCR (canonical commutation relations):

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = \hat{a}_k \hat{a}_{k'}^\dagger - \hat{a}_{k'}^\dagger \hat{a}_k = \delta_{kk'}$$

Fermions — with occupation-number formalism (Pauli exclusion principle):

- $\hat{b}^\dagger \hat{b}^\dagger |0\rangle = 0$, and $\hat{b}\,\hat{b}|1\rangle = 0$ (a second particle cannot be added to an occupied orbital);
- $\hat{b}|0\rangle = 0$, $\hat{b}^\dagger|1\rangle = 0$;
- thus $\{\hat{b}, \hat{b}'\}$ and $\{\hat{b}^\dagger, \hat{b}'^\dagger\}$ vanish: **anticommutators**, $\{\hat{b}, \hat{b}^\dagger\} = 1$ (CCR/anticommutation);
- number operator $\hat{n} = \hat{b}^\dagger \hat{b}$ satisfies $\hat{n}^2 = \hat{b}^\dagger \hat{b} \hat{b}^\dagger \hat{b} = \hat{b}^\dagger (1 - \hat{b}^\dagger \hat{b}) \hat{b} = \hat{n} \Rightarrow n = 0$ or 1.

[check] Fermion/boson page: written upside-down and partly hard to read; verify against the original.

26.6 Interacting Fields

Contents

- *Perturbation theory*
- *Interaction between electron and EM fields*

A model for interacting electron and electomagnetic fields requires an interaction contribution in the Hamiltonian of the system, namely

$$\hat{H} = \hat{H}_{atom} + \hat{H}_{rad} + \hat{H}_{inter}$$

being

$$\hat{H}_{atom} = \sum E_k \hat{b}_k^\dagger \hat{b}_k \quad , \quad \hat{H}_{rad} = \sum E_{k,\alpha} \hat{a}_{k,\alpha}^\dagger \hat{a}_{k,\alpha}$$

the Hamiltonian of the electron bound to the atom, and the Hamiltonian of the EM field. Interaction comes from the coupling of the electron dynamics and the EM field: electron may act as a source of the EM field, and its state is influenced by the EM field, as shown…

References

todo *Add references to*

- *classical electromagnetism*
- *electromagnetism in special relativity*
- *notes about calculus of variations and Lagrangian approach*

As an example, minimal coupling for the motion of acharged particle in an EM field is readily obtained replacing the momentum $\mathbf{P}$ with the sum $\mathbf{P} + q\mathbf{A}$, or in classical mechanics $\vec{p} \to \vec{p} + q\vec{a}$. In classical mechanics the Hamiltonian reads

$$\mathcal{H} = \frac{1}{2}\frac{|\vec{p} - q\vec{a}|^2}{m} + q\varphi \ ,$$

producing two coupling terms, one linear and one quadratic in $\vec{a}$.

- The linear term produces the dominant interaction processes - for EM with low intensity - in the Hamiltonian terms with three creation/annihilation operators: a pair of creation/annihilation operators of the electron field and one creation or annihilation operator of the photon field, represent absorption and emission of light from the atom. As an example
  - **Absorption** process: $\hat{b}_n^\dagger \hat{b}_m \hat{a}_{k,\alpha}$ — atom $|\psi_m\rangle \to |\psi_n\rangle$, photon $|k, \alpha\rangle$ absorbed (destroyed from the EM field).
  - **Emission** process: $\hat{b}_m^\dagger \hat{a}_{k,\alpha}^\dagger \hat{b}_n$ — atom $|\psi_n\rangle \to |\psi_m\rangle$, photon $|k, \alpha\rangle$ emitted (created in the EM field).
- The quadratic term is responsible for scattering…

26.6.1 Perturbation theory

Other sections dealing with perturbation theory:

- *Perturbation theory for transition probabilities in weak electric fields*
- *Interaction picture and perturbation theory*

todo *Need for cleaning*

Perturbation theory applies to systems governed by a Hamiltonian that can be written as the sum

$$\hat{H} = \hat{H}_0 + \hat{H}_{int} \ ,$$

with $\hat{H}_0$ an easy-to-solve (unperturbed) system, whose spectral decomposition is usually known, and $\hat{H}_{int}$ a "small" perturbation, either time independent or time dependent. Let $|\phi_n\rangle$, and E_n the eigenstates and the eigenvalues of the unperturbed Hamiltonian $\hat{H}_0$, i.e.

$$\hat{H}_0|\phi_n\rangle = E_n|\phi_n\rangle \ .$$

Now, the perturbation Hamiltonian is explicitly written as the product of a "small" parameter λ and a scaled perturbation Hamiltonian $\hat{H}_1$, with the same order of the unperturbed Hamiltonian $\hat{H}_0$, i.e.

$$\hat{H} = \hat{H}_0 + \lambda\hat{H}_1 \ ,$$

and a generic state $|\psi\rangle$ of the system written as a linear combination of the eigenstates $\{|\phi_n\rangle\}_n$ of the unperturbed system,

$$|\psi\rangle = \sum_n c_n(t) e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle \ .$$

Inserting this expression in the Schrodinger equation $i\hbar|\dot{\psi}\rangle = (\hat{H}_0 + \hat{H}_{int})|\psi\rangle$,

$$i\hbar \sum_n \left(\dot{c}_n - i\frac{E_n}{\hbar}c_n\right) e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle = (\hat{H}_0 + \hat{H}_{int}) \sum_n c_n e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle$$
$$\sum_n i\hbar\dot{c}_n e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle = \hat{H}_{int} \sum_n c_n e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle \ ,$$

since $E_n|\phi_n\rangle = \hat{H}_0|\phi_n\rangle$. Multiplying by $\langle\phi_m|$, and exploiting orthogonality $\langle\phi_m|\phi_n\rangle = \delta_{mn}$,

$$i\hbar\dot{c}_m e^{-i\omega_m t} = \sum_n c_n e^{-i\omega_n t}\langle\phi_m|\hat{H}_{int}|\phi_n\rangle$$

i.e. the infinite system of equations - $m \in \{0, 1, 2, ...\}$ - follows

$$\dot{c}_m = \frac{1}{i\hbar}\sum_n c_n e^{i(\omega_m - \omega_n)t}\langle\phi_m|\hat{H}_{int}|\phi_n\rangle \tag{26.5}$$

Perturbation theory looks for a solution writing the coefficients $c_m(t)$ as power expansions in the "small" parameter λ,

$$c_m(t) = c_m^{(0)}(t) + \lambda c_m^{(1)}(t) + \lambda^2 c_m^{(2)}(t) + ...$$

Inserting this expression in (**??**),

$$\dot{c}_m^{(0)} + \lambda\dot{c}_m^{(1)} + \lambda^2\dot{c}_m^{(2)} + \cdots = \frac{1}{i\hbar}\sum_n \left(c_n^{(0)} + \lambda c_n^{(1)} + \lambda^2 c_n^{(2)} + ...\right)\lambda H_{mn}^{(1)} e^{i(\omega_m - \omega_n)t} \ ,$$

an (approximate) solution can be found by coefficient matching:

$$\begin{cases} (0): & \dot{c}_m^{(0)} = 0 \\ (k): & \dot{c}_m^{(k)} = \dfrac{1}{i\hbar}\displaystyle\sum_n c_n^{(k-1)} H_{mn}^{(1)} e^{i(\omega_m - \omega_n)t} \end{cases}$$

0th-order solution

System starting in $|n\rangle$, $c_n(0) = 1$, $c_{m\neq n}(0) = 0$. As the 0th-order equations read $\dot{c}_m^{(0)} = 0$, then 0th-order coefficients are constant in time.

1st-order solution

If $c_n^{(0)}(t) = 1$, $c_{m\neq n}(t) = 0$, then infinite set of 1st order equations is

$$\dot{c}_m^{(1)}(t) = \frac{1}{i\hbar}e^{i\omega_{mn}t}H_{mn}^{(1)}$$

- coefficient $c_n^{(1)}$
  - if $H_{nn}^{(1)} = 0$, then the dynamical equation reads $\dot{c}_n^{(1)} = 0$, and the constant solution reads $c_n^{(1)} = 0$, as the i.c. for the non-zero order terms are identically zero
  - if $H_{nn}^{(1)} \neq 0$, then the dynamical equation reads $\dot{c}_n^{(1)} = \frac{1}{i\hbar}H_{nn}^{(1)}$, the solution reads $c_n^{(1)} = \frac{H_{nn}^{(1)}}{i\hbar}t$
- coefficients $c_m^{(1)}$, for $m \neq n$,
  - if $H_{mn}^{(1)} = 0$, the dynamical eqaution reads $\dot{c}_m^{(1)} = 0$, and the constant solution reads $c_m^{(1)} = 0$

- else
  - if $\omega_{mn} := \omega_m - \omega_n \neq 0$, the dynamical equation reads $\dot{c}_m^{(1)} = \frac{e^{i\omega_{mn}t}}{i\hbar} H_{mn}^{(1)}$, then the solution reads

$$c_m^{(1)}(t) = \left[-\frac{H_{mn}^{(1)} e^{i\omega_{mn}\tau}}{\hbar\omega_{mn}} \right]\Bigg|_{\tau=0}^{t} = \frac{1 - e^{i\omega_{mn}t}}{\hbar\omega_{mn}} H_{mn}^{(1)} \tag{26.6}$$

  - if $\omega_{mn} := \omega_m - \omega_n = 0$, the dynamical equation reads $\dot{c}_m^{(1)} = \frac{1}{i\hbar} H_{mn}^{(1)}$, and the solution is $c_m^{(1)} = \frac{1}{i\hbar} H_{mn}^{(1)}$.

Probabilities. For a system starting in state $|n\rangle$, the power expansion of the coefficients $c_m(t)$ reads

$$c_m(t) = \delta_{mn} + \sum_{k=1}^{+\infty} c_m^{(k)} \lambda^k \ .$$

Thus,

- the probability $|c_n(t)|^2$ contains zero and first order terms,

$$\begin{aligned} |c_n|^2 &= \left(\sum_{k=0}^{+\infty} c_n^{(k)} \lambda^k \right)^* \left(\sum_{j=0}^{+\infty} c_n^{(j)} \lambda^j \right) = \\ &= \sum_{k=0}^{+\infty} \sum_{j=0}^{+\infty} {c_n^{(k)}}^* c_n^{(j)} \lambda^{k+j} = \\ &= \sum_{\ell=0}^{+\infty} \sum_{j=0}^{\ell} {c_n^{(\ell-j)}}^* c_n^{(j)} \lambda^{\ell} \ , \end{aligned}$$

  or a bit more explicitly

$$\begin{aligned} |c_n|^2 &= |c_n^{(0)}|^2 + \lambda \left({c_n^{(0)}}^* c_n^{(1)} + {c_n^{(1)}}^* c_n^{(0)} \right) + \lambda^2 \left({c_n^{(0)}}^* c_n^{(2)} + |c_n^{(1)}|^2 + {c_n^{(2)}}^* c_n^{(0)} \right) + \cdots = \\ &= 1 + \lambda \left(c_n^{(1)} + {c_n^{(1)}}^* \right) + \lambda^2 \left(c_n^{(2)} + |c_n^{(1)}|^2 + {c_n^{(2)}}^* \right) + \ldots \ , \end{aligned}$$

- the probabilities $|c_m(t)|^2$ with $m \neq n$ only contains powers larger than or equal to 2,

$$\begin{aligned} |c_m|^2 &= \left(\sum_{k=1}^{+\infty} c_m^{(k)} \lambda^k \right)^* \left(\sum_{j=1}^{+\infty} c_m^{(j)} \lambda^j \right) = \\ &= \sum_{k=1}^{+\infty} \sum_{j=1}^{+\infty} {c_m^{(k)}}^* c_m^{(j)} \lambda^{k+j} = \\ &= \sum_{\ell=2}^{+\infty} \sum_{j=1}^{\ell-1} {c_m^{(\ell-j)}}^* c_m^{(j)} \lambda^{\ell} \ , \end{aligned}$$

  or a bit more explicitly

$$|c_m|^2 = \lambda^2 |c_m^{(1)}|^2 + \lambda^3 \left({c_m^{(2)}}^* c_m^{(1)} + {c_m^{(1)}}^* c_m^{(2)} \right) + \ldots$$

Here the parameter λ is assumed to be real.

If $H_{mn}^{(1)} \neq 0$, and $\omega_{mn} \neq 0$, the coefficient $c_m^{(1)}$ is given by (**??**), and its square modulus reads

$$\begin{aligned} |c_m^{(1)}|^2 &= \left(\frac{1 - e^{i\omega_{mn}t}}{\hbar\omega_{mn}} H_{mn}^{(1)} \right)^* \left(\frac{1 - e^{i\omega_{mn}t}}{\hbar\omega_{mn}} H_{mn}^{(1)} \right) = \\ &= \left| H_{mn}^{(1)} \right|^2 \frac{4 \sin^2 \left(\frac{\omega_{mn}t}{2} \right)}{\hbar^2 \omega_{mn}^2} \ , \end{aligned}$$

as $(1-e^{ia})^*(1-e^{ia}) = 2-e^{ia}-e^{-ia} = 2-2\cos(a) = 4\sin^2\left(\frac{a}{2}\right)$. Thus the leading order of the probability $|c_n(t)|^2$ - this is the probability of finding the state of the system in $|\phi_n\rangle$ at time t, as this is the coefficient of $|\phi_n\rangle$ of the state $|\psi\rangle$ except for an irrelevant phase factor $e^{i\omega_n t}$ - is

$$|c_n(t)|^2 \sim \lambda^2 |H^{(1)}_{mn}|^2 \frac{4\sin^2\left(\frac{\omega_{mn}t}{2}\right)}{\hbar^2\omega^2_{mn}}$$

As $t \to +\infty$ this function goes to **Dirac's delta**:

$$\int_{\omega=-\infty}^{+\infty} \left\{\frac{\sin(\omega t/2)}{\omega}\right\}^2 d\omega = \frac{\pi}{2}t$$

so that the transition rate probability between the initial state m and the final state n at the first order is

$$\frac{|c_n|^2}{t} \longrightarrow \frac{2\pi}{\hbar^2}\,|H^{int}_{mn}|^2\,\delta(\omega_m - \omega_n)\,.$$

todo *See, update and cross-link* Math: Dirac delta.

Fermi golden rule

Definition 26.6.1 (Fermi golden rule - infinite time)

For $t \to +\infty$, for a system with unperturbed Hamiltonian $\hat{H}_0$, with stationary states $|\phi_n\rangle_t = |\phi_n\rangle e^{-i\omega_n t}$, an additive "small" perturbation Hamiltonian $\hat{H}_{int}$ causes the initial state $|i\rangle_t = |\phi_i\rangle e^{-i\omega_i t}$ to transition to $|f\rangle = |\phi_f\rangle e^{-i\omega_f t}$ at the rate

$$2\pi\,|H^{int}_{mn}|^2\,\delta(\omega_m - \omega_n) \qquad \text{(to the 1st order)}$$

Definition 26.6.2 (Fermi golden rule - finite time)

For a system with unperturbed Hamiltonian $\hat{H}_0$, with stationary states $|\phi_n\rangle_t = |\phi_n\rangle e^{-i\omega_n t}$, an additive "small" perturbation Hamiltonian $\hat{H}_{int}$ causes the initial state $|i\rangle_t = |\phi_i\rangle e^{-i\omega_i t}$ to transition to $|f\rangle = |\phi_f\rangle e^{-i\omega_f t}$ at the rate

$$|c_n(t)|^2 \sim \lambda^2 |H^{(1)}_{mn}|^2 \frac{4\sin^2\left(\frac{\omega_{mn}t}{2}\right)}{\hbar^2\omega^2_{mn}}$$

Higher orders

Check it, and provide details about the mutliple interactions involving intermediate states.

$$\dot{c}^{(p+1)}_f(t) = \frac{1}{i\hbar}\sum_n c^{(p)}_n(t)\langle f|\hat{H}^{int}|n\rangle e^{i(\omega_f-\omega_n)t}$$

E.g. 2nd order: $c^{(1)}_n \propto \langle n|\hat{H}^{int}|m\rangle \times \frac{1-e^{i\omega_{mn}t}}{\hbar\omega_{mn}}$, so

$$\dot{c}^{(2)}_f(t) = \frac{1}{i\hbar}\sum_n \langle f|\hat{H}^{int}|n\rangle\langle n|\hat{H}^{int}|m\rangle \times \frac{1-e^{-i\omega_{mn}t}}{\hbar\omega_{mn}}\,e^{i\omega_{fn}t}$$

(Diagram in the notes: states $|1\rangle, |2\rangle, |i\rangle, \ldots, |I_1\rangle, \ldots, |I_2\rangle, \ldots, |f\rangle$.)

- 1st order: direct jump $|i\rangle \to |f\rangle$.
- 2nd order: $|i\rangle \to |I\rangle \to |f\rangle$ — need to sum over all possible intermediate states.
- 3rd order: $|i\rangle \to |I_1\rangle \to |I_2\rangle \to |f\rangle$ — need to sum over all possible combinations of intermediate states.

26.6.2 Interaction of the electron and electromagnetic field

The behavior of the interactions between electron and electromagnetic fields strongly depends on the characteristic of the electron field:

- *Bound electrons - Atom spectra*
- *Free electrons - Compton scattering*

Non-relativistic model of the electron field. Schrodinger equation

For an electron in a potential $V(\mathbf{x})$ we impose $\{\psi_\sigma(\mathbf{x}), \psi^\dagger_{\sigma'}(\mathbf{y})\} = \delta_{\sigma\sigma'}\delta^3(\mathbf{x}-\mathbf{y})$ and use the one-body Hamiltonian:

$$H_{\text{el}} = \sum_\sigma \int d^3x\; \psi^\dagger_\sigma(\mathbf{x}) \Big[-\frac{\hbar^2\nabla^2}{2m} + V(\mathbf{x}) \Big] \psi_\sigma(\mathbf{x}).$$

The Heisenberg equation follows from $[A, BC] = \{A,B\}C - B\{A,C\}$:

$$[\psi_\sigma(\mathbf{x}),\, \psi^\dagger_{\sigma'}(\mathbf{y})\, h\, \psi_{\sigma'}(\mathbf{y})] = \delta(\mathbf{x}-\mathbf{y})\, h\, \psi_\sigma(\mathbf{y}) - \psi^\dagger(\mathbf{y})\, h\, \underbrace{\{\psi_\sigma(\mathbf{x}), \psi_{\sigma'}(\mathbf{y})\}}_{=0},$$

so, integrating over $\mathbf{y}$,

$$i\hbar\, \partial_t \psi_\sigma(\mathbf{x},t) = [\psi_\sigma, H_{\text{el}}] = \Big[-\frac{\hbar^2\nabla^2}{2m} + V \Big] \psi_\sigma(\mathbf{x},t).$$

The *operator* ψ obeys the *Schrödinger equation*, and the one-particle wave function is recovered as $\Phi(\mathbf{x},\sigma) = \langle 0|\psi_\sigma(\mathbf{x})|\Phi\rangle$. "Second quantisation" is therefore not a second quantisation of the same object: what is quantised is a field whose classical equation of motion happens to be the Schrödinger equation.

Free electrons

For $V = 0$ in the periodic box use plane waves $\phi_{\mathbf{k}\sigma}(\mathbf{x}) = V^{-1/2} e^{i\mathbf{k}\cdot\mathbf{x}} \chi_\sigma$:

$$\psi_\sigma(\mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} c_{\mathbf{k}\sigma}, \qquad H_{\text{el}} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m}\, c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}, \qquad \mathbf{P} = \sum_{\mathbf{k}\sigma} \hbar\mathbf{k}\, c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}.$$

A single free electron is $c^\dagger_{\mathbf{k}\sigma}|0\rangle$, with energy $\hbar^2k^2/2m$ and momentum $\hbar\mathbf{k}$. The ground state of N non-interacting electrons fills the N lowest $(\mathbf{k},\sigma)$ levels, the Fermi sea (a consequence of $(c^\dagger)^2 = 0$). The spin label is conserved by H_{el}.

Bound electrons

For a hydrogen-like atom, $V = -Ze^2/4\pi\varepsilon_0 r$, use the orbitals $\phi_{nlm}(\mathbf{x})\chi_\sigma$ with $E_n = -Z^2\,\text{Ry}/n^2$:

$$\psi_\sigma(\mathbf{x}) = \sum_{nlm} \phi_{nlm}(\mathbf{x})\, c_{nlm\sigma} + \int d^3k\; \phi^{(+)}_{\mathbf{k}}(\mathbf{x})\, c_{\mathbf{k}\sigma}, \qquad H_{\text{el}} = \sum_{nlm\sigma} E_n\, c^\dagger_{nlm\sigma} c_{nlm\sigma} + \int d^3k\, \frac{\hbar^2k^2}{2m} c^\dagger_{\mathbf{k}\sigma} c_{\mathbf{k}\sigma}.$$

The second term, the **Coulomb scattering states** $\phi^{(+)}_{\mathbf{k}}$, completes the basis: bound and continuum states together span $\mathcal{H}$. By §1.6 these operators are related to the plane-wave operators of §4.2 by a unitary transformation, $c^{\text{plane}}_{\mathbf{p}} = \sum_b \langle \mathbf{p}|b\rangle c_b + \int \langle \mathbf{p}|\mathbf{k}^{(+)}\rangle c_{\mathbf{k}}$, with the bound-state momentum distribution $\langle \mathbf{p}|b\rangle$ spread over $|\mathbf{p}| \sim \hbar/a_0$. **The Fock space is the same; free and bound electrons differ only by which single-particle basis diagonalises H_{el}.** A transition $c^\dagger_{\mathbf{k}} c_b$ is a photoionisation; $c^\dagger_{b'} c_b$ is a bound–bound transition; both are matrix elements of the *same* operator. For an atom with

several electrons one adds the two-body Coulomb term of §1.5 and, in the single-active-electron approximation, treats the outermost electron in an effective potential; the light–matter formalism below is unchanged.

Spin. Besides the orbital coupling, the electron has a Zeeman-type term $H_{\text{spin}} = (e\hbar/2m)\,\sigma\cdot\mathbf{B}$ ($g = 2$). Compared with $H_1 = (e/m)\mathbf{A}\cdot\mathbf{p}$, using $p \sim \hbar/a_0$ and $B \sim kA$, the ratio is $\sim ka_0 \sim 10^{-3}$ for optical and UV light, so we neglect it; it produces magnetic-dipole transitions. The fully consistent treatment of spin and its relativistic corrections is given by the Dirac field (§5.6).

Old - from Via Science - to be cleaned

Photon field (Heisenberg picture). The vector potential field operator and the Hamiltonian read

$$\hat{\vec{a}}(\vec{r},t) = \sum_{\vec{k},\alpha} \frac{\vec{\epsilon}_{\vec{k},\alpha}}{\sqrt{2\omega_k}} \left(\hat{a}_{\vec{k},\alpha} e^{-i\omega_k t} e^{i\vec{k}\cdot\vec{r}} + \hat{a}^\dagger_{\vec{k},\alpha} e^{i\omega_k t} e^{-i\vec{k}\cdot\vec{r}} \right)$$

$$\hat{H}_r = \sum_{\vec{k},\alpha} \hat{n}_{\vec{k},\alpha}\, \hbar\omega_k \,,$$

after **energy renormalization**, i.e. zero-point energy set $\equiv 0$.

Expected value of the enerdy of the EM field

$$\langle \hat{H}_r \rangle = \langle \dots n_{\vec{k},\alpha} \dots | \hat{H}_r | \dots n_{\vec{k},\alpha} \dots \rangle = \sum_{\vec{k},\alpha} n_{\vec{k},\alpha} \hbar\omega_k$$

Electron field. Annihilation and creation field operators for the electron read

$$\hat{\psi}(\vec{r},t) = \sum_k \hat{b}_k e^{-i\omega_k t} \varphi_k(\vec{r}) \quad , \qquad \hat{\psi}^\dagger(\vec{r},t) = \sum_k \hat{b}^\dagger_k e^{i\omega_k t} \varphi^*_k(\vec{r})$$

being φ_k the "orbitals" of the atom of interest, solution of the eigenproblem for the Hamiltonian of the system, $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$. The Hamiltonian for one electron (assuming no interaction between electrons?) is

$$\hat{H} = \frac{1}{2m_e}\hat{p}^2 + V(\hat{x}) \,.$$

Expected value of $\hat{H}$

$$\begin{aligned}
\langle \hat{H} \rangle &= \langle \psi | \hat{H} | \psi \rangle = \\
&= \sum_{m,n} c^*_m c_n \langle m | \hat{H} | n \rangle = \\
&= \sum_{m,n} c^*_m c_n \, E_n \, \underbrace{\langle m | n \rangle}_{\delta_{mn}} = \\
&= \sum_n |c_n|^2 \, E_n \,.
\end{aligned}$$

QFT form of the Hamiltonian

…

todo *Check and uncomment*

Interaction Hamiltonian

Minimal coupling. The Hamiltonian of a particle with mass m and electric charge q in a prescribed EM field is

$$\mathcal{H} = \frac{1}{2} m \left|\dot{\vec{r}}\right|^2 + q\varphi = \frac{|\vec{p} - q\vec{a}|^2}{2m} + q\varphi \, .$$

If the EM radiation field can be written with the vector EM potential field $\vec{a}$, while the scalar EM potential field fully represents the EM field of the nucleus, the Hamiltonian operator for a electron bound to an atom in an external EM field is obtained by the promotion to operators of the electron position and momentum, and the vector EM field,

$$\hat{H} = \frac{|\hat{\mathbf{p}}|^2}{2m_e} \underbrace{- \frac{q}{m_e} \hat{\mathbf{p}} \cdot \hat{\mathbf{a}} + \frac{q^2}{2m_e} |\hat{\mathbf{a}}|^2}_{\hat{H}_{int}} + V(\hat{\mathbf{r}}) =$$

Do $\hat{p}$ and $\hat{a}$ commute? Likely not! Maybe yes, if $\hat{p}$ only acts on e^- while $\hat{a}$ acts on $\vec{k}, \alpha$.

To get the interaction Hamiltonian, put the 2 extra terms (due to the e.m. field) into the expectation formula:

$$\hat{H} = \int_x \sum_m \hat{b}_m^\dagger \psi_m^*(x) \, \hat{H}_{int} \sum_n \hat{b}_n \psi_n(x) \, dx$$

One-body operator for a bosonic field

Let $\hat{o}$, with $\hat{a}^\dagger$ and $\hat{a}$ creation and annihilation operators in the $\{|\psi_k\rangle\}_k$ basis. Let $o_{\alpha\beta} = \langle \psi_\alpha | \hat{o} | \psi_\beta \rangle$.

$$\hat{O}_1 = \sum_{\alpha,\beta} o_{\alpha\beta} \hat{a}_\alpha^\dagger \hat{a}_\beta$$

or in space representation

$$\hat{O}_1 =$$

One-body operator for a fermionic field

(1) $\vec{p} \cdot \vec{a}$ term

$$\hat{H}^{(1)} = \int_x \sum_m \ldots \frac{q}{m_e} \left(\hat{A}_x \hat{p}_x + \hat{A}_y \hat{p}_y + \hat{A}_z \hat{p}_z \right) \sum_n \ldots \, dx$$

with

$$\hat{A}_x = \sum_{\vec{k},\alpha} \frac{\{\vec{e}_{k\alpha}\}_x}{\sqrt{2\omega_k}} \left(\hat{a}_{\vec{k}\alpha} e^{i\vec{k}\cdot\vec{x}} + \hat{a}_{\vec{k}\alpha}^\dagger e^{-i\vec{k}\cdot\vec{x}} \right)$$

- first part: **absorption** (destruction of a γ),
- second part: **emission** (creation of a γ),

and $\hat{p}_x = -i\hbar\partial_x$. For the absorption part along x:

$$\hat{H}^{(1)}_{abs,x} = -i\hbar \int_x \sum_m \hat{b}^\dagger_m \psi^*_m(x) \sum_{\vec{k},\alpha} \frac{\{\vec{e}_{k\alpha}\}_x}{\sqrt{2\omega_k}} \hat{a}_{\vec{k}\alpha} e^{i\vec{k}\cdot\vec{x}}\, \partial_x \sum_n \hat{b}_n \psi_n(x)\, dx$$

$$= -i\hbar \sum_m \sum_n \sum_{\vec{k},\alpha} \left[\int_x \psi^*_m(x) \frac{\{\vec{e}_{k\alpha}\}_x}{\sqrt{2\omega_k}} e^{i\vec{k}\cdot\vec{x}} \partial_x \psi_n(x) dx \right] \hat{b}^\dagger_m \hat{a}_{\vec{k}\alpha} \hat{b}_n = \sum_m \sum_n \sum_{\vec{k},\alpha} M_{abs,x}(m,n,\vec{k},\alpha)\, \hat{b}^\dagger_m \hat{b}_n \hat{a}_{\vec{k},\alpha}$$

Collecting all the terms for x, y, z, and doing so for the emission part:

$$\hat{H}^{(1)}_{abs} = \sum_{m,n,\vec{k},\alpha} M_{abs}(m,n,\vec{k},\alpha)\, \hat{b}^\dagger_m \hat{b}_n \hat{a}_{\vec{k},\alpha}$$

$$\hat{H}^{(1)}_{em} = \sum_{m,n,\vec{k},\alpha} M_{em}(m,n,\vec{k},\alpha)\, \hat{b}^\dagger_m \hat{b}_n \hat{a}^\dagger_{\vec{k},\alpha}$$

$$\hat{H}^{(1)} = \hat{H}^{(1)}_{abs} + \hat{H}^{(1)}_{em}$$

These coefficients determine the probability of the events $\hat{b}^\dagger_m \hat{b}_n \hat{a}^{(\dagger)}_{\vec{k},\alpha}$ **— quantum jumps**.

Energy conservation. Time-dependent factors have been absorbed in the creation/annihilation operators. It follows that:

- ABS: $\omega_m - \omega_n - \omega_k = 0$
- EM: $\omega_m - \omega_n + \omega_k = 0$

give the "relevant processes", i.e. those with constant time dependence, as $e^{-i(\dots)t} = 1$:

$$\text{ABS: } \omega_k + \omega_n = \omega_m, \qquad \text{EM: } \omega_k = \omega_n - \omega_m \qquad \text{(cons. of energy)}$$

(2) a^2 terms

$$\hat{H}^{(2)} = \sum_{m,n} \int_x \hat{b}^\dagger_m \psi^*_m(x)\, \frac{q^2}{2m_e} \left(\hat{a}^2_x + \hat{a}^2_y + \hat{a}^2_z\right) \sum_n \hat{b}_n \psi_n(x)\, dx$$

with $\hat{a}_x = \sum_{\vec{k},\alpha} \frac{\{\vec{e}_{k\alpha}\}_x}{\sqrt{2\omega_k}} \left(\hat{a}_{\vec{k}\alpha} e^{i\vec{k}\cdot\vec{x}} + \hat{a}^\dagger_{\vec{k}\alpha} e^{-i\vec{k}\cdot\vec{x}}\right)$, hence

$$\hat{a}^2_x = \sum_{\vec{k},\alpha} \sum_{\vec{l},\beta} \frac{\{\vec{e}_{k\alpha}\}_x \{\vec{e}_{l\beta}\}_x}{\sqrt{2\omega_k}\sqrt{2\omega_l}} \left(\hat{a}_{k\alpha} e^{i\vec{k}\cdot\vec{x}} + \hat{a}^\dagger_{k\alpha} e^{-i\vec{k}\cdot\vec{x}}\right) \left(\hat{a}_{l\beta} e^{i\vec{l}\cdot\vec{x}} + \hat{a}^\dagger_{l\beta} e^{-i\vec{l}\cdot\vec{x}}\right)$$

4 types of terms (3 types):

- 2 destructions
- 2 creations
- 1 destruction & 1 creation (two orderings, counted as one type)

→ **2-photon processes**, much weaker than the 1-photon processes. (Space–time diagrams with two photon lines are drawn in the notes.)

* Conservation of energy just provides **one** constraint → there's a *continuum* of possible interactions.

$$\hat{H}_{int} = \hat{H}^{(1)}_{int} + \hat{H}^{(2)}_{int} = \hat{H}^{(1)}_{abs} + \hat{H}^{(1)}_{em} + \hat{H}^{(2)}_{int}$$

Dipole approximation

Usually the size of the atom (e.g. H_2) orbitals $\propto 10^{-9}$ m, while $\lambda_{EM,\,jump} \propto 10^{-7}$ m. Thus $\vec{k}\cdot\vec{x}$ is small within the atom:

$$e^{i\vec{k}\cdot\vec{x}} = 1 + i\vec{k}\cdot\vec{x} + \cdots \sim 1 \qquad \Rightarrow \qquad \textbf{DIPOLE APPROXIMATION}$$

In the dipole approximation the ABS and EM coefficients are identical.

Selection rules

Interesting physical interpretation using angular momentum

Quantum numbers n, ℓ, m_ℓ, m_s.

- $E \propto \frac{1}{n^2}$ (energy)
- $L = \sqrt{\ell(\ell+1)}\,\hbar$: $|\vec{L}|$ is known, but not the direction in space.
  - $\ell = 0 \to L = 0$ (s orbital)
  - $\ell = 1 \to L = \sqrt{2}\,\hbar$ (p orbital)
- $m_\ell \to z$-component of orbital angular momentum
- $m_s \to z$-component of spin

Photon angular momentum: $\sqrt{2}\,\hbar$.

Angular momentum conservation (feasible jumps):

$$\gamma + s \to p, \qquad p \to \gamma + s$$

Linear momentum

Smells like Compton effect.

Before: atom at rest; photon momentum

$$p = \hbar k = \hbar\omega\frac{k}{\omega} = \frac{E}{c}$$

After: atom moving with V_{atom}:

$$p_\gamma = p_{atom} \simeq m\,V_{atom} \;\; (\sim m_{proton}V_{atom}) \;\; \to \;\; V_{atom} \simeq \frac{E}{m_p c} \;\; (\sim 3 \text{ m/s for a } 1s \to 2p \text{ transition})$$

The **nucleus** absorbs the γ linear momentum.

Example 1: $1s + \gamma \to 2p$ (absorption)

$$|i\rangle = |1, 0, \dots\rangle_a \, |0 \dots n_{\vec{k}\alpha} \dots 0\rangle_r = |1_1\rangle_a |n_{\vec{k}\alpha}\rangle_r$$

$$|f\rangle = |0, 0, 1, 0, \dots\rangle_a \, |0 \dots (n-1)_{\vec{k}\alpha} \dots 0\rangle_r = |1_3\rangle_a |(n-1)_{\vec{k}\alpha}\rangle_r$$

1-photon interaction: absorption of $\gamma \propto \hat{b}_3^\dagger \hat{b}_1 \hat{a}_{\vec{k}\alpha}$, modulated by $M(3, 1, \vec{k}, \alpha)$.

$$|M_{fi}|^2 = |\langle f|\hat{H}_{int}|i\rangle|^2 = \left|\langle 1_3|_a \langle (n-1)_{\vec{k}\alpha}|_r \, M(3,1,\vec{k},\alpha) \, \hat{b}_3^\dagger \hat{b}_1 \hat{a}_{\vec{k}\alpha} \, |1_1\rangle_a |n_{\vec{k}\alpha}\rangle_r \right|^2$$

$$= \left|\sqrt{n_{\vec{k},\alpha}} \, M(3,1,\vec{k},\alpha)\right|^2 = n_{\vec{k},\alpha} \, |M(3,1,\vec{k},\alpha)|^2$$

(here $\hat{a}_{\vec{k}\alpha}|n\rangle = \sqrt{n}|n-1\rangle$ and $\hat{b}_3^\dagger \hat{b}_1 |1_1\rangle = |1_3\rangle$.)

Important: The transition rate $\propto$ the **number of photons in the state** $\vec{k}, \alpha$.

This can be used in the Fermi golden rule.

Example 2: emission process $2p \to 1s + \gamma$

$$|i\rangle = |1_3\rangle_a |n_{\vec{k}\alpha}\rangle_r, \qquad |f\rangle = |1_1\rangle_a |(n+1)_{\vec{k}\alpha}\rangle_r$$

$$|M_{fi}|^2 = |\langle f|\hat{H}_{int}|i\rangle|^2 = \left|\langle f|M_e(1,3,\vec{k},\alpha) \, \hat{b}_1^\dagger \hat{b}_3 \hat{a}_{\vec{k}\alpha}^\dagger |i\rangle\right|^2 = \underbrace{(n_{\vec{k},\alpha} + 1)}_{\substack{n\text{: stimulated emission} \\ 1\text{: spontaneous emission}}} \underbrace{|M(1,3,\vec{k},\alpha)|^2}_{\text{symmetry properties}}$$

Revisiting Einstein coefficients (1917)

…

Photon–e^- scattering

A free e^- can't absorb a γ (conditions about conservation of E and $\vec{p}$ cannot be met simultaneously).

→ no $\hat{H}^{(1)}$ processes → $\hat{H}^{(2)}$ is the **dominant interaction**!

→ a free e^- with given momentum is completely **not-localized** → Feynman diagrams are not to be taken literally.

→ Conditions about **conservation of momentum** in scattering! → Conditions about **conservation of energy**. → Alignment of the photon polarization.

Transition rate …

$$2\pi \left(\frac{q^2}{2m_e}\right)^2 \frac{\left(\vec{e}_{k\alpha} \cdot \vec{e}_{l\beta}\right)^2}{\omega_k \omega_l} \, M_{e,p} \left(n_{\vec{k}\alpha} \ , \ +1\right)$$

Annotations in the notes: $n_{\vec{k}\alpha}$ = "photon in initial state"; the $(+1)$ term = "spontaneous + stimulated scattering"; $(\vec{e}_{k\alpha} \cdot \vec{e}_{l\beta})^2$ = "alignment of the photon polarization".

[check] Final formula: the exact placement of the factors $n_{\vec{k}\alpha}$ and $(n_{\vec{l}\beta} + 1)$, and the $M_{e,p}$ term, are hard to read; verify against the original.

(In the notes, a small diagram shows the initial state $|\vec{p}\rangle$ + photon $|\vec{k}, \alpha\rangle$ → final state $|\vec{p}'\rangle$ + photon $|\vec{l}, \beta\rangle$ in a space–time plot.)

Bound electrons - Atom spectra

…

todo

- Schrodinger, Heisenberg and interaction pictures
- Time dependence of the interaction Hamiltonian, in order to make energy conservation applear
- …

Bound electron

An electron bound to an isolated atom is governed by the Schrodinger equation - here neglecting spin for simplicity; use Pauli equation for non-relativistic approximation, or Dirac equation -

$i\hbar|\dot{\Psi}\rangle = \left(\frac{|\hat{\mathbf{p}}|^2}{2m} + V(\hat{\mathbf{r}})\right)|\Psi\rangle$,$

being $V(\hat{\mathbf{r}})$ the potential energy operator, due to Coulomb force from the nucleus.

The Hamiltonian

The full Hamiltonian is $\hat{H} = \hat{H}_{\text{el}} + \hat{H}_{\text{rad}} + \hat{H}_1 + \hat{H}_2$ with, from §2.3 (second-quantised via §1.5),

$$\hat{H}_1(t) = \frac{q}{m}\int d^3\mathbf{r}\hat{\Psi}^\dagger(\mathbf{r})\mathbf{A}(\mathbf{r},t)\cdot(-i\hbar\nabla)\hat{\Psi}(\mathbf{r}) \quad , \quad \hat{H}_2(t) = \frac{q^2}{2m}\int d^3\mathbf{r}\hat{\Psi}^\dagger(\mathbf{r})|\mathbf{A}(\mathbf{r},t)|^2\hat{\Psi}(\mathbf{r}) \ .$$

Now,

- expanding $\mathbf{A}(\mathbf{r},t)$ with

$$\hat{\mathbf{A}}(\mathbf{r},t) = \sum_{\mathbf{k},\mu}\sqrt{\frac{\hbar}{2\varepsilon_0\omega_{\mathbf{k}}V}}\epsilon_{\mathbf{k},\mu}\left(\hat{a}_{\mathbf{k},\mu}e^{-i(\omega_{\mathbf{k}}t-\mathbf{k}\cdot\mathbf{r})} + \hat{a}^\dagger_{\mathbf{k},\mu}e^{i(\omega_{\mathbf{k}}t-\mathbf{k}\cdot\mathbf{r})}\right) \ ,$$

- writing the electron fields as linear combinations of creation an annihilation operators $\hat{c}^\dagger$, $\hat{c}$ in any basis $\{|\alpha\rangle\}$ as

$$\begin{aligned}\hat{\Psi}^\dagger(\mathbf{r}) &= \sum_\alpha \alpha^*(\mathbf{r})\hat{c}^\dagger_\alpha \quad &= \sum_\alpha \langle\alpha|\mathbf{r}\rangle\hat{c}^\dagger_\alpha \\ \hat{\Psi}(\mathbf{r}) &= \sum_\beta \beta(\mathbf{r})\hat{c}_\beta \quad &= \sum_\beta \langle\mathbf{r}|\beta\rangle\hat{c}_\beta\end{aligned}$$

the $\hat{H}_1$ interaction Hamiltonian becomes

$$\hat{H}_1(t) = \frac{e}{m}\sum_{\alpha\beta}\sum_{\mathbf{k}\mu}N_k\Big[\langle\alpha|e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}\,\epsilon_{\mathbf{k}\mu}\cdot\hat{\mathbf{p}}|\beta\rangle\, e^{-i\omega_{\mathbf{k}}t}\, c^\dagger_\alpha c_\beta\, a_{\mathbf{k}\lambda} + \langle\alpha|e^{-i\mathbf{k}\cdot\hat{\mathbf{r}}}\,\epsilon_{\mathbf{k}\mu}\cdot\hat{\mathbf{p}}|\beta\rangle\, e^{i\omega_{\mathbf{k}}t}\, c^\dagger_\alpha c_\beta\, a^\dagger_{\mathbf{k}\lambda}\Big], \tag{26.7}$$

(the orderings of $e^{\pm i\mathbf{k}\cdot\mathbf{x}}$ and $\varepsilon\cdot\mathbf{p}$ do not matter since $\varepsilon\cdot\mathbf{k} = 0$).

Details

$$\hat{H}_1(t) = \frac{q}{m}\int \hat{\Psi}^\dagger(\mathbf{r})\mathbf{A}(\mathbf{r},t)\cdot(-i\hbar\nabla)\hat{\Psi}(\mathbf{r})d^3\mathbf{r} =$$

$$= \frac{q}{m}\int \left(\sum_\alpha \langle\alpha|\mathbf{r}\rangle\hat{c}^\dagger_\alpha\right)\left(\sum_{\mathbf{k},\mu} N_\mathbf{k}\epsilon_{\mathbf{k},\mu}\left(\hat{a}_{\mathbf{k},\mu}e^{-i(\omega_\mathbf{k}t-\mathbf{k}\cdot\mathbf{r})}+\hat{a}^\dagger_{\mathbf{k},\mu}e^{i(\omega_\mathbf{k}t-\mathbf{k}\cdot\mathbf{r})}\right)\right)\cdot\left(-i\hbar\nabla\sum_\beta\langle\mathbf{r}|\beta\rangle\hat{c}_\beta\right)d^3\mathbf{r} = \quad (1)$$

$$= \frac{q}{m}\sum_{\alpha,\beta}\sum_{\mathbf{k},\mu}N_k\int\langle\alpha|\mathbf{r}\rangle\langle\mathbf{r}|\epsilon_{\mathbf{k},\mu}\cdot\hat{\mathbf{p}}|\beta\rangle d^3\mathbf{r}\,\hat{c}^\dagger_\alpha\hat{c}_\beta\left(\hat{a}_{\mathbf{k},\mu}e^{-i(\omega_\mathbf{k}t-\mathbf{k}\cdot\mathbf{r})}+\hat{a}^\dagger_{\mathbf{k},\mu}e^{i(\omega_\mathbf{k}t-\mathbf{k}\cdot\mathbf{r})}\right) = \quad (2)$$

$$= \frac{q}{m}\sum_{\alpha,\beta}\sum_{\mathbf{k},\mu}N_k\langle\alpha|e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}\,\epsilon_{\mathbf{k},\mu}\cdot\hat{\mathbf{p}}|\beta\rangle\,\hat{c}^\dagger_\alpha\hat{c}_\beta\left(\hat{a}_{\mathbf{k},\mu}e^{-i\omega_\mathbf{k}t}+\hat{a}^\dagger_{\mathbf{k},\mu}e^{i\omega_\mathbf{k}t}\right)\,,$$

since (1) $\langle\mathbf{r}|\hat{\mathbf{p}}|\beta\rangle = -i\hbar\nabla\langle\mathbf{r}|\beta\rangle$, and (2)

$$\int_\mathbf{r}\langle\alpha|\mathbf{r}\rangle e^{i\mathbf{k}\cdot\mathbf{r}}\langle\mathbf{r}|\epsilon_{\mathbf{k},\mu}\cdot\hat{\mathbf{p}}|\beta\rangle d^3\mathbf{r} = \int_\mathbf{r}\langle\alpha|e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}|\mathbf{r}\rangle\langle\mathbf{r}|\epsilon_{\mathbf{k},\mu}\cdot\hat{\mathbf{p}}|\beta\rangle d^3\mathbf{r} =$$
$$= \langle\alpha|e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}\,\epsilon_{\mathbf{k},\mu}\cdot\hat{\mathbf{p}}|\beta\rangle d^3\mathbf{r}\,,$$

since $e^{i\mathbf{k}\cdot\mathbf{r}}|\mathbf{r}\rangle = e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}|\mathbf{r}\rangle$, following from $\hat{x}^n|\mathbf{r}\rangle = x^n|\mathbf{r}\rangle$, and $e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{j=0}^{+\infty}\frac{(i\mathbf{k}\cdot\mathbf{r})^n}{n!}$, and $e^{i\mathbf{k}\cdot\hat{\mathbf{r}}} = \sum_{j=0}^{+\infty}\frac{(i\mathbf{k}\cdot\hat{\mathbf{r}})^n}{n!}$, and $\int_\mathbf{r}|\mathbf{r}\rangle\langle\mathbf{r}|\,d^2\mathbf{r} = \hat{\mathbf{1}}$.

Reading the structure:

- $c^\dagger_\alpha c_\beta\, a$: the electron goes from β to α **absorbing** a photon; $c^\dagger_\alpha c_\beta\, a^\dagger$: it goes from β to α **emitting** one. Electron number is conserved (§2.2), photon number changes by ±1.
- H_2 contains $a^\dagger a$ (scattering, photon number unchanged) and aa, $a^\dagger a^\dagger$ (two photons absorbed or emitted), each with a factor $c^\dagger_\alpha c_\beta$.
- Total momentum $\mathbf{P}_\text{el}+\mathbf{P}_\text{rad}$ is conserved when $V=0$, and the total angular momentum and parity in general (§2).

Transition rates

Perturbation theory allows to evaluate the transition probability between an initial state $|i\rangle$ and a final state $|f\rangle$ of the unperturbed system, with Hamiltonian $\hat{H}_0$.

For $\hat{H}=\hat{H}_0+\hat{H}_I$ with the Hamiltonian $\hat{H}_0=\hat{H}_\text{el}+\hat{H}_\text{rad}$ the Hamiltonian of the non-interacting fields. Let the initial and final states be eigenvectors of the unperturbed Hamiltonian, i.e.

$$\hat{H}_0|i\rangle = \left(\hat{H}_\text{el}+\hat{H}_\text{rad}\right)|\mathbf{n}^\text{el}_i\rangle_\text{el}\otimes|\mathbf{n}^\text{rad}_i\rangle_{\mathbf{k}\mu} = \left(\sum_a n^\text{el}_{a,i}E^\text{el}_a+\sum_{\mathbf{k}\mu}n^\text{rad}_{\mathbf{k}\mu,i}E_{\mathbf{k}\mu}\right)|i\rangle = E_i|i\rangle$$

$$\hat{H}_0|f\rangle = \left(\hat{H}_\text{el}+\hat{H}_\text{rad}\right)|\mathbf{n}^\text{el}_f\rangle_\text{el}\otimes|\mathbf{n}^\text{rad}_f\rangle_{\mathbf{k}\mu} = \left(\sum_a n^\text{el}_{a,f}E^\text{el}_a+\sum_{\mathbf{k}\mu}n^\text{rad}_{\mathbf{k}\mu,f}E_{\mathbf{k}\mu}\right)|f\rangle = E_f|f\rangle$$

Fermi golden rule provides the ratio of transition probabilities from $|i\rangle$ to $|f\rangle$ as

$$\Gamma_{i\to f} = \frac{2\pi}{\hbar}|T_{fi}|^2\,\delta(E_f-E_i), \qquad T_{fi} = \langle f|H_I|i\rangle + \sum_m\frac{\langle f|H_I|m\rangle\langle m|H_I|i\rangle}{E_i-E_m+i0}+\ldots$$

where m runs over *all* intermediate states of H_0 (including states with extra photons). The sum over final photon modes is $\sum_\mathbf{k}\to V\int d^3k/(2\pi)^3 = V\int\omega^2\,d\omega\,d\Omega\big/((2\pi)^3c^3)$, using $k=\omega/c$.

Absorbption and emission are first-order processes and the transition probabilities are dominated by the term $\langle f|\hat{H}^{int}|\rangle\sim\langle f|\hat{H}_1|i\rangle$, i.e. using the expression (**??**)

- Absorption, **assuming** an existing electron in $|\beta\rangle$ and non existing electron in $|\alpha\rangle$ (to avoid unnecessary complications with the notation, at least here),

$$\begin{aligned}
&\langle f|\hat{H}_1^{abs}|i\rangle = \\
&= \langle \mathbf{n}_f^{\text{el}}| \otimes \langle \mathbf{n}_f^{\text{rad}}| \left(\sum_{\alpha\beta} \sum_{\mathbf{k}\mu} M^{\text{abs}}_{\alpha\beta\mathbf{k}\mu} c_\alpha^\dagger c_\beta \, a_{\mathbf{k}\lambda} \right) |\mathbf{n}_i^{\text{el}}\rangle \otimes |\mathbf{n}_i^{\text{rad}}\rangle = \\
&= \langle \mathbf{n}_f^{\text{el}}| \otimes \langle \mathbf{n}_f^{\text{rad}}| \sum_{\alpha\beta\mathbf{k}\mu} M^{\text{abs}}_{\alpha\beta\mathbf{k}\mu} \sqrt{n_{\mathbf{k}\mu}} |\mathbf{n}^{\text{el}}_{i+\alpha-\beta}\rangle \otimes |\mathbf{n}_i^{\text{rad}} - \mathbf{1}_{\mathbf{k}\mu}\rangle
\end{aligned}$$

  Thus in a final state with non-zero transition probability

  - there's one electron more in $|\alpha\rangle$, and one electron less in $|\beta\rangle$ w.r.t. the initial state, and thus $E_f^{\text{el}} = E_i^{\text{el}} + E_a - E_b$,
  - a photon with $\mathbf{k}\mu$ is annihilated (absorbed), so that the radiation energies of the initial and final states are related by $E_f^{\text{rad}} - E_i^{\text{rad}} = -\hbar\omega_{\mathbf{k}}$.

  As the delta condition implies $E_f = E_i$, then $E_a - E_b = \hbar\omega_{\mathbf{k}}$ (energy conservation), the transition probability is proportional to

$$|\langle f|\hat{H}_1^{\text{abs}}|i\rangle|^2 = n_{\mathbf{k}\mu} |M^{\text{abs}}_{\alpha\beta\mathbf{k}\mu}|^2 \, .$$

- Emission, **assuming** an existing electron in $|\beta\rangle$ and non existing electron in $|\alpha\rangle$ (to avoid unnecessary complications with the notation, at least here),

$$\begin{aligned}
&\langle f|\hat{H}_1^{\text{em}}|i\rangle = \\
&= \langle \mathbf{n}_f^{\text{el}}| \otimes \langle \mathbf{n}_f^{\text{rad}}| \left(\sum_{\alpha\beta} \sum_{\mathbf{k}\mu} M^{\text{em}}_{\alpha\beta\mathbf{k}\mu} c_\alpha^\dagger c_\beta \, a^\dagger_{\mathbf{k}\lambda} \right) |\mathbf{n}_i^{\text{el}}\rangle \otimes |\mathbf{n}_i^{\text{rad}}\rangle = \\
&= \langle \mathbf{n}_f^{\text{el}}| \otimes \langle \mathbf{n}_f^{\text{rad}}| \sum_{\alpha\beta\mathbf{k}\mu} M^{\text{em}}_{\alpha\beta\mathbf{k}\mu} \sqrt{n_{\mathbf{k}\mu} + 1} |\mathbf{n}^{\text{el}}_{i+\alpha-\beta}\rangle \otimes |\mathbf{n}_i^{\text{rad}} - \mathbf{1}_{\mathbf{k}\mu}\rangle
\end{aligned}$$

  Thus in a final state with non-zero transition probability

  - there's one electron more in $|\alpha\rangle$, and one electron less in $|\beta\rangle$ w.r.t. the initial state, and thus $E_f^{\text{el}} = E_i^{\text{el}} + E_a - E_b$,
  - a photon with $\mathbf{k}\mu$ is created (emitted), so that the radiation energies of the initial and final states are related by $E_f^{\text{rad}} - E_i^{\text{rad}} = \hbar\omega_{\mathbf{k}}$.

  As the delta condition implies $E_f = E_i$, then $E_a = E_b - \hbar\omega_{\mathbf{k}}$ (energy conservation), the transition probability is proportional to

$$|\langle f|\hat{H}_1^{\text{em}}|i\rangle|^2 = (n_{\mathbf{k}\mu} + 1) |M^{\text{em}}_{\alpha\beta\mathbf{k}\mu}|^2 \, .$$

Absorption, and stimulated and spontanteous emission

Absorption is proportional to $n_{\mathbf{k}\mu}$, while emission is proportional to $n_{\mathbf{k}\mu} + 1$, as the sum of the stimulated and the spontaneous emission transition probability.

Symmetry of absorption and stimulated emission factors

$$M^{\text{abs}}_{\alpha\beta\mathbf{k}\mu} = \left(M^{\text{abs}}_{\beta\alpha\mathbf{k}\mu} \right)^* \, .$$

Proof

$$M^{\text{em}}_{\alpha\beta\mathbf{k}\mu} = \langle\alpha|e^{i\mathbf{k}\cdot\hat{\mathbf{r}}}\epsilon_{\mathbf{k}\mu}\cdot\hat{\mathbf{p}}|\beta\rangle = \ldots$$

Dipole approximation and Einstein coefficients

…

Absorption and emission of light by an atom

Time dependence and energy conservation

$$\begin{aligned}\hat{a}^\dagger_{\mathbf{k},\mu} &\sim e^{i\frac{E_{\mathbf{k},\mu}}{\hbar}t}\\ \hat{a}_{\mathbf{k},\mu} &\sim e^{-i\frac{E_{\mathbf{k},\mu}}{\hbar}t}\\ \hat{c}^\dagger_\alpha &\sim e^{i\frac{E_\alpha}{\hbar}t}\\ \hat{c}_\beta &\sim e^{-i\frac{E_\beta}{\hbar}t}\end{aligned}$$

$$\hat{c}^\dagger_\alpha\hat{c}_\beta\hat{a}_{\mathbf{k},\mu}|n_\alpha n_\beta n_{\mathbf{k},\mu}\rangle = |(n+1)_\alpha(n-1)_\beta(n-1)_{\mathbf{k},\mu}\rangle$$

Matrix elements. Let the atom go from level a to b. With n photons in the mode $(\mathbf{k},\lambda)$:

$$\text{absorption } (E_b>E_a):\ \langle b;n{-}1|H_1|a;n\rangle = \frac{e}{m}N_k\sqrt{n}\,\langle b|e^{i\mathbf{k}\cdot\mathbf{x}}\varepsilon\cdot\mathbf{p}|a\rangle,$$

$$\text{emission } (E_b<E_a):\ \langle b;n{+}1|H_1|a;n\rangle = \frac{e}{m}N_k\sqrt{n+1}\,\langle b|e^{-i\mathbf{k}\cdot\mathbf{x}}\varepsilon\cdot\mathbf{p}|a\rangle,$$

using $a|n\rangle = \sqrt{n}|n-1\rangle$ and $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$. The rates are therefore proportional to n for absorption and to $(n+1)$ for emission: the part $\propto n$ is **stimulated emission** (the mirror image of absorption), and the part $\propto 1$ is **spontaneous emission**, which survives in the vacuum ($n=0$) and is caused by the zero-point field of §3.3. A classical field acting on quantised matter gives no spontaneous emission at all (cf. Chapter 1, §2.4).

Dipole approximation. Over the size of the atom, $\mathbf{k}\cdot\mathbf{x}\sim ka_0$. For the hydrogen Lyman-α line, $\omega a_0/c = \frac{3}{8}\alpha = 2.7\times10^{-3}$, so $e^{\pm i\mathbf{k}\cdot\mathbf{x}} = 1+O(10^{-3})$ and we replace it by 1. The remaining matrix element is converted using $[\mathbf{x},H_0]=i\hbar\mathbf{p}/m$:

$$\langle b|[\mathbf{x},H_0]|a\rangle = (E_a-E_b)\langle b|\mathbf{x}|a\rangle = \frac{i\hbar}{m}\langle b|\mathbf{p}|a\rangle \implies \langle b|\mathbf{p}|a\rangle = -\,i\,m\,\omega\,\langle b|\mathbf{x}|a\rangle \quad (\hbar\omega = E_a-E_b),$$

(for absorption $E_a<E_b$ the same identity holds with a sign change, which drops out of $|\cdot|^2$). With the dipole matrix element $\mathbf{d}_{ba} = -e\langle b|\mathbf{x}|a\rangle$ the emission matrix element has modulus

$$|M|^2 = \left(\frac{e}{m}\right)^2 N_k^2\,(n+1)\,m^2\omega^2|\varepsilon\cdot\langle b|\mathbf{x}|a\rangle|^2 = \frac{\hbar\omega}{2\varepsilon_0 V}\,(n+1)\,|\varepsilon\cdot\mathbf{d}_{ba}|^2.$$

Spontaneous emission rate ($n=0$). Apply the golden rule, with $\delta(E_a-E_b-\hbar\omega)=\delta(\omega-\omega_{ab})/\hbar$ and the sum of §5.2:

$$\Gamma = \frac{2\pi}{\hbar}\cdot\frac{1}{\hbar}\cdot\frac{V\omega^2}{(2\pi)^3c^3}\cdot\frac{\hbar\omega}{2\varepsilon_0 V}\int d\Omega\sum_\lambda|\varepsilon_\lambda\cdot\mathbf{d}_{ba}|^2 = \frac{\omega^3}{8\pi^2\varepsilon_0\hbar c^3}\int d\Omega\sum_\lambda|\varepsilon_\lambda\cdot\mathbf{d}_{ba}|^2.$$

By polarisation completeness, $\sum_\lambda |\varepsilon_\lambda\cdot\mathbf{d}|^2 = |\mathbf{d}|^2 - |\hat{\mathbf{k}}\cdot\mathbf{d}|^2$, and with $\int d\Omega\,\hat{k}_i\hat{k}_j = \frac{4\pi}{3}\delta_{ij}$ the angular integral is $4\pi|\mathbf{d}|^2 - \frac{4\pi}{3}|\mathbf{d}|^2 = \frac{8\pi}{3}|\mathbf{d}|^2$. The result is

$$\boxed{\Gamma_{a\to b} = \frac{\omega^3\,|\mathbf{d}_{ba}|^2}{3\pi\varepsilon_0\hbar c^3}} \qquad \Big(\text{angular distribution } \propto \sin^2\theta \text{ from } \mathbf{d}\Big),$$

the Einstein A coefficient derived from first principles (Dirac, 1927). It is the number quoted in Chapter 1, §2.4.

Example 26.6.1 (Worked example: hydrogen $2p \to 1s$ (Lyman-α).)

The photon energy is $\hbar\omega = \frac{3}{4}\,\text{Ry} = 10.204$ eV, i.e. $\lambda = 121.5$ nm. With $R_{10} = 2e^{-r/a_0}a_0^{-3/2}$, $R_{21} = \frac{1}{2\sqrt{6}}(r/a_0)\,e^{-r/2a_0}a_0^{-3/2}$, the radial integral is $\int R_{10}\,r\,R_{21}\,r^2dr = 1.2903\,a_0$ and the angular factor $\langle 00|\cos\theta|10\rangle = 1/\sqrt{3}$, giving

$$\langle 1s|z|2p, m{=}0\rangle = \frac{2^{15/2}}{3^5}\,a_0 = 0.7449\,a_0.$$

(The rate is the same for $m = 0, \pm 1$ by rotation symmetry.) Then $|\mathbf{d}_{ba}|^2 = e^2(0.7449\,a_0)^2$ and

$$\Gamma = 6.27\times10^8\ \text{s}^{-1}, \qquad \tau = 1/\Gamma = 1.60\ \text{ns}, \qquad \Delta\nu_{\text{nat}} = \Gamma/2\pi \simeq 100\ \text{MHz}.$$

The measured $2p$ lifetime is 1.60 ns. By contrast, the $2s$ level cannot decay to $1s$ by a one-photon E1 transition (no parity change, $\Delta l = 0$, §2.4); in isolated hydrogen it decays by emitting *two* photons at a rate of only $\approx 8\ \text{s}^{-1}$ (lifetime ≈ 0.12 s), about eight orders of magnitude slower than the allowed $2p$ decay.

The classical limit of the rate. A classical electron oscillator (Lorentz model) of angular frequency ω radiates, by Larmor's formula $P = e^2\omega^4x_0^2/(12\pi\varepsilon_0c^3)$, with energy $\frac{1}{2}m\omega^2x_0^2$, so its energy decays at rate $\gamma_{\text{cl}} = e^2\omega^2/(6\pi\varepsilon_0mc^3)$. Define the oscillator strength $f_{ba} = \frac{2m\omega}{3\hbar g_b}\sum_{ab}|\langle a|\mathbf{r}|b\rangle|^2$ (g = degeneracy; the sum runs over all sublevels). Substituting into Γ gives

$$\Gamma_{a\to b} = 3\,\frac{g_b}{g_a}\,f_{ba}\,\gamma_{\text{cl}}.$$

For Lyman-α, $f = 0.416$, $g_{1s}/g_{2p} = 1/3$ (orbital degeneracies) and $\gamma_{\text{cl}} = 1.51\times10^9\ \text{s}^{-1}$, so $\Gamma = f\gamma_{\text{cl}} = 6.27\times10^8\ \text{s}^{-1}$ again: **the quantum emission rate is the classical radiation-damping rate times the oscillator strength**, the dimensionless measure of "how much of a classical oscillator" the transition is. The factor $\hbar$ has dropped out of the structure, as it must in a correspondence-principle statement.

Thermal equilibrium and Einstein's relations. In blackbody radiation each mode has mean occupation $\bar{n} = 1/(e^{\hbar\omega/k_BT}-1)$. Summing the above over modes, the total emission rate from a level a is $A(\bar{n}+1)$ and the absorption rate of an atom in b is $(g_a/g_b)A\bar{n}$ (because the double sum $\sum_{ab}|\mathbf{d}_{ab}|^2$ is common). Balancing the populations, $N_a\,A(\bar{n}+1) = N_b\,(g_a/g_b)A\bar{n}$, gives

$$\frac{N_a}{N_b} = \frac{g_a}{g_b}\frac{\bar{n}}{\bar{n}+1} = \frac{g_a}{g_b}\,e^{-\hbar\omega/k_BT},$$

exactly the Boltzmann distribution. This is Einstein's 1917 argument (Chapter 1, §2.4), where A, B_{12}, B_{21} were related by consistency, now obtained as a *theorem* of the quantised theory: the "+1" is spontaneous emission.

Beyond the dipole approximation. Expanding $e^{i\mathbf{k}\cdot\mathbf{x}} = 1 + i\mathbf{k}\cdot\mathbf{x} + \ldots$ gives electric-quadrupole and magnetic-dipole transitions, suppressed by $(ka_0)^2 \sim 10^{-5}$ in rate: these account for "forbidden" lines.

Free electrons - Compton scattering

Setup. Take $V = 0$ and an electron initially at rest with spin σ:

$$|i\rangle = c^\dagger_{\mathbf{0}\sigma}\, a^\dagger_{\mathbf{k}\lambda}|0\rangle, \qquad |f\rangle = c^\dagger_{\mathbf{p}'\sigma'}\, a^\dagger_{\mathbf{k}'\lambda'}|0\rangle, \qquad \mathbf{p}' = \hbar(\mathbf{k}-\mathbf{k}') \ \text{(momentum conservation)}.$$

There are three contributions at order e^2, the time-ordered diagrams: (a) the seagull H_2 acting once; (b) H_1 twice, absorbing $\mathbf{k}$ first then emitting $\mathbf{k}'$; (c) H_1 twice, emitting $\mathbf{k}'$ first then absorbing $\mathbf{k}$.

The H_1H_1 terms vanish. In (b) the first vertex is $\langle \hbar\mathbf{k}|e^{i\mathbf{k}\cdot\mathbf{x}}\varepsilon\cdot\mathbf{p}|\mathbf{0}\rangle \propto \varepsilon\cdot\mathbf{p}_i = 0$ because $\mathbf{p}$ acts on the initial state, which has $\mathbf{p}_i = 0$. In (c) the first vertex is $\propto \varepsilon'\cdot\mathbf{p}_i = 0$ for the same reason. (This is exact in Coulomb gauge: $\varepsilon\cdot\mathbf{p}$ acts *before* the intermediate momentum appears.) For a moving electron these terms are of relative order p/mc, producing the Doppler broadening of the Compton line.

The seagull term. The matrix element of $\mathbf{A}^2$ between $|\mathbf{k}\lambda\rangle$ and $|\mathbf{k}'\lambda'\rangle$ has two contributions ($q = \mathbf{k}, q' = \mathbf{k}'$ and the swap), so

$$\langle f|H_2|i\rangle = \frac{e^2}{2m}\cdot 2N_kN_{k'}\,(\varepsilon\cdot\varepsilon')\,\underbrace{\frac{1}{V}\int d^3x\, e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{x}}\, e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}_{\delta_{\mathbf{p}',\hbar(\mathbf{k}-\mathbf{k}')}} = \frac{e^2\hbar}{2m\varepsilon_0 V\sqrt{\omega\omega'}}\,(\varepsilon\cdot\varepsilon')\,\delta_{\mathbf{p}',\hbar(\mathbf{k}-\mathbf{k}')},$$

(using $N_kN_{k'} = \hbar/(2\varepsilon_0 V\sqrt{\omega\omega'})$; the electron plane waves $V^{-1/2}e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ supply the $1/V$, and $\mathbf{p}_i = 0$).

Cross-section. The golden rule with the final photon density of states $V\omega'^2 d\Omega/((2\pi)^3c^3\hbar)$ (we neglect the electron recoil in the Jacobian $dE_f/d\omega' = \hbar(1 + O(\hbar\omega/mc^2))$) gives

$$\frac{d\Gamma}{d\Omega} = \frac{2\pi}{\hbar}\Big(\frac{e^2\hbar}{2m\varepsilon_0 V\sqrt{\omega\omega'}}\Big)^2 |\varepsilon\cdot\varepsilon'|^2 \frac{V\omega'^2}{8\pi^3c^3\hbar} = \frac{e^4}{16\pi^2m^2\varepsilon_0^2c^3V}\,\frac{\omega'}{\omega}\,|\varepsilon\cdot\varepsilon'|^2.$$

The incident flux is one photon per volume V at speed c, i.e. c/V. Dividing,

$$\boxed{\frac{d\sigma}{d\Omega} = r_e^2\,\frac{\omega'}{\omega}\,|\varepsilon\cdot\varepsilon'|^2, \qquad r_e = \frac{e^2}{4\pi\varepsilon_0 mc^2} = 2.818\ \text{fm}}$$

(using $e^4/(16\pi^2m^2\varepsilon_0^2c^4) = r_e^2$). This is **Thomson scattering**, derived here from the quantised theory, with a polarisation dependence $|\varepsilon\cdot\varepsilon'|^2$ (the scattered photon keeps the projected polarisation of the incident one).

Unpolarised light. Summing over the final polarisations with completeness, $\sum_{\lambda'}|\varepsilon\cdot\varepsilon'_{\lambda'}|^2 = 1 - (\varepsilon\cdot\hat{\mathbf{k}}')^2$, and averaging over the two incident polarisations, $\frac{1}{2}\sum_\lambda(\varepsilon_\lambda\cdot\hat{\mathbf{k}}')^2 = \frac{1}{2}\sin^2\theta$ (the two polarisations transverse to $\hat{\mathbf{k}}$ together project onto $\hat{\mathbf{k}}'$ with weight $1-\cos^2\theta$). The polarisation-averaged factor is thus $1 - \frac{1}{2}\sin^2\theta = \frac{1}{2}(1+\cos^2\theta)$. Hence

$$\frac{d\sigma}{d\Omega}\Big|_{\text{unpol}} = \frac{r_e^2}{2}\,\frac{\omega'}{\omega}\,(1+\cos^2\theta), \qquad \sigma_{\text{T}} = \frac{8\pi}{3}r_e^2 = 6.652\times10^{-29}\ \text{m}^2 = 0.665\ \text{b},$$

using $\int(1+\cos^2\theta)\,d\Omega = 16\pi/3$ and $\omega'/\omega \to 1$. The cross-section is independent of the frequency and of $\hbar$: it is the classical result of an electron driven by the wave and radiating as a Hertzian dipole, recovered as the leading term of a quantum calculation.

Two physical consequences follow from the factor $|\varepsilon\cdot\varepsilon'|^2$:

- The component of the incident polarisation perpendicular to the scattering plane scatters with weight 1, the in-plane component with weight $\cos^2\theta$. At $\theta = 90^\circ$ the scattered light is therefore **100% linearly polarised** (perpendicular to the plane), with degree of polarisation $(1-\cos^2\theta)/(1+\cos^2\theta)$ in general. This is why scattered light, in the sky and in the cosmic microwave background, is polarised.
- Where electrons are free, mean free paths are set by $\ell = 1/(n_e\sigma_{\text{T}})$: in the solar core ($n_e \approx 6\times10^{31}\ \text{m}^{-3}$) this is ≈ 0.25 mm.

Kinematics: the Compton shift from energy conservation. The delta function enforces $\hbar\omega = \hbar\omega' + \hbar^2|\mathbf{k}-\mathbf{k}'|^2/2m$. Write $|\mathbf{k}-\mathbf{k}'|^2 = (k-k')^2 + 2kk'(1-\cos\theta)$. The first term is second order in the small shift, so to leading order in $\hbar\omega/mc^2$,

$$c\,(k-k') = \frac{\hbar k^2(1-\cos\theta)}{m} \quad\Longrightarrow\quad \frac{1}{k'} - \frac{1}{k} = \frac{k-k'}{kk'} \simeq \frac{\hbar(1-\cos\theta)}{mc},$$

and with $\lambda = 2\pi/k$,

$$\lambda' - \lambda = \frac{h}{mc}\,(1-\cos\theta),$$

the Compton formula of Chapter 1, now as the energy conservation of the Fock-space matrix element above. (The exact *relativistic* kinematics gives the same formula without approximation.)

Where the non-relativistic theory fails. We can quantify the neglected corrections by comparing the scattered-photon energy at 90° from exact relativistic kinematics, $E' = E/(1+E/mc^2)$, with that from the non-relativistic energy balance used above (computed numerically):

Incident E	Exact E' (90°)	Non-relativistic E'	Error
17.44 keV (Mo Kα)	16.8644 keV	16.8641 keV	2×10^{-5}
100 keV	83.63 keV	83.41 keV	0.27%
661.66 keV (^{137}Cs)	288.3 keV	195.8 keV	32%

(For the Mo line, the wavelength shift is 2.4263 pm exactly, against 2.4277 pm from the non-relativistic balance.) The relativistic theory (quoted: **Klein–Nishina**, 1929, requiring the Dirac field) gives for unpolarised light

$$\frac{d\sigma}{d\Omega} = \frac{r_e^2}{2}\Big(\frac{\omega'}{\omega}\Big)^2\Big[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\Big],$$

which reduces to our result when $\omega' \to \omega$: $\tfrac{1}{2}(2-\sin^2\theta) = \tfrac{1}{2}(1+\cos^2\theta)$. Integrated over angles, the total cross-section relative to Thomson is $\sigma_{\rm KN}/\sigma_{\rm T} = 0.937$ at 17.44 keV, 0.741 at 100 keV, 0.385 at 662 keV and 0.125 at 5 MeV: the forward-peaked, energy-dependent suppression that makes high-energy photons penetrating.

Gauge invariance and the seagull. In the calculation above only H_2 contributed. Had we dropped it as "a second-order nuisance", the amplitude for a free electron at rest would vanish and scattering would be absent, which contradicts the classical (and experimental) Thomson result. The coefficient of H_2 is tied to that of H_1 by gauge invariance (§2.3), and the amplitude at $\omega \to 0$, $-r_e\,\varepsilon\cdot\varepsilon'$, is fixed by the charge and mass alone (a *low-energy theorem*).

Scattering by a bound electron (quoted). For an atomic electron initially in $|a\rangle$ and finally in $|b\rangle$, with the same ingredients (the three diagrams and the dipole approximation), the amplitude is

$$T_{ba} \propto \frac{e^2}{m}\Big[\varepsilon\cdot\varepsilon'\,\delta_{ba} + \frac{1}{m}\sum_m\Big(\frac{\langle b|\varepsilon'\cdot\mathbf{p}|m\rangle\langle m|\varepsilon\cdot\mathbf{p}|a\rangle}{E_a+\hbar\omega-E_m} + \frac{\langle b|\varepsilon\cdot\mathbf{p}|m\rangle\langle m|\varepsilon'\cdot\mathbf{p}|a\rangle}{E_a-\hbar\omega'-E_m}\Big)\Big],$$

the **Kramers–Heisenberg formula** (1925), with the intermediate states m of the atom (including the continuum). The three regimes are all contained in it: for $\hbar\omega \gg$ binding energies the denominators are large, the second term is negligible and one recovers free-electron **Thomson** scattering; for $\hbar\omega \ll E_m - E_a$ the ω-independent terms cancel against the seagull by the Thomas–Reiche–Kuhn sum rule $\sum_m (E_m-E_a)|\langle m|x|a\rangle|^2 = \hbar^2/2m$, leaving the **Rayleigh** law $\sigma \propto \omega^4$ (blue sky); and for $\hbar\omega \to E_m - E_a$ the denominators vanish and the amplitude has a **resonance** (resonance fluorescence), regulated by the natural linewidth Γ of §5.3.

CHAPTER

TWENTYSEVEN

ANALTYCAL MECHANICS - NOTES FOR QUANTUM MECHANICS

Contents

- *Lagrange and Hamiltonian mechanics*
- *Canonical transformations* Canonical transformations are pure kinematics (kinematics+inertia, as they involve generalized momentum), independent from the physics of the system of interest, i.e. independent from the specific form of the Lagrangian function or of the Hamiltonian function.
- *Hamilton-Jacobi formulation of mechanics*
- *Angle-Action variables*

Topics and tools

- Calculus of variations
- Legendre transformation, in the definition of functions with different independent variables.

Examples

  - From Lagrangian function to Hamiltonian function $\mathcal{H}(\mathbf{q}, \mathbf{p}, t) := \mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t)$
  - From one generating function to another, e.g. $F_2(\mathbf{q}, \mathbf{P}, t) := \mathbf{P} \cdot \mathbf{Q} + F_1(\mathbf{q}, \mathbf{Q}, t)$
  - …

- Properties of canonical transformations can be derived with a proper application of rules of derivation of **composite functions**, **paying attention to the set of independent variables** of these functions.

27.1 Lagrange and Hamiltonian Mechanics

Lagrangian mechanics

Equations of motion follow the principle of stationary action functional

$$0 = \delta S[\mathbf{q}(t)] = \int_{t_0}^{t_1} \mathcal{L}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t)\, dt\,, \tag{27.1}$$

with prescribed extreme values, so that $\delta\mathbf{q}(t_0) = \delta\mathbf{q}(t_1) = \mathbf{0}$. Here $\mathcal{L}$ is the Lagrangian function, with $\mathbf{q}(t)$ the vector of generalized coordinates, and t the time. Equations of motion are the Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{0}\,. \tag{27.2}$$

The generalized momentum is defined as $\mathbf{p} := \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}}$. If the Lagrangian function doesn't explicitly depend on the generalized coordinate $q^k(t)$, the generalized momentum $p_k(t) = \frac{\partial \mathcal{L}}{\partial q^k}$ is constant.

Hamiltonian mechanics.

Hamiltonian function is defined as $\mathcal{H}(\mathbf{q}, \mathbf{p}, t) := \mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t)$, with $\dot{\mathbf{q}}(\mathbf{q}, \mathbf{p}, t)$. It's differential reads

$$\begin{aligned} d\mathcal{H} &= \dot{\mathbf{q}} \cdot d\mathbf{p} + \mathbf{p} \cdot d\dot{\mathbf{q}} - \partial_{\dot{\mathbf{q}}}\mathcal{L} \cdot d\dot{\mathbf{q}} - \partial_{\mathbf{q}}\mathcal{L} \cdot d\mathbf{q} - \partial_t \mathcal{L} dt = \\ &= \dot{\mathbf{q}} \cdot d\mathbf{p} - \partial_{\mathbf{q}}\mathcal{L} \cdot d\mathbf{q} - \partial_t \mathcal{L} dt \ , \end{aligned} \tag{27.3}$$

by the definition of the generalized momentum, $\mathbf{p} = \partial_{\dot{\mathbf{q}}}\mathcal{L}$. Hamilton equations immediately follows

$$\begin{cases} \dot{\mathbf{q}}(t) = \partial_{\mathbf{p}}\mathcal{H}(\mathbf{q}(t), \mathbf{p}(t), t) \\ \dot{\mathbf{p}}(t) = -\partial_{\mathbf{q}}\mathcal{H}(\mathbf{q}(t), \mathbf{p}(t), t) \end{cases}$$

with the relation

$$\partial_t \mathcal{H}|_{\mathbf{q},\mathbf{p}}(\mathbf{q}(t), \mathbf{p}(t), t) = -\partial_t \mathcal{L}|_{\mathbf{q},\dot{\mathbf{q}}}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) \ .$$

Using Hamilton's equations, it's immediately proved that $d_t\mathcal{H} = \partial_t \mathcal{H}$. The Hamiltonian function is thus conserved - an integral of motion - if the Lagrangian function doesn't explicitly depend on t.

27.2 Canonical transformations

Definition

A canonical transformation is defined as a change of coordinates from $(\mathbf{q}, \mathbf{p})$ to $(\mathbf{Q}, \mathbf{P})$ so that the equations of motion have the same expression

$$\begin{cases} \dot{\mathbf{q}} = \partial_{\mathbf{p}}\mathcal{H} \\ \dot{\mathbf{p}} = -\partial_{\mathbf{q}}\mathcal{H} \end{cases} \quad , \quad \begin{cases} \dot{\mathbf{Q}} = \partial_{\mathbf{P}}\mathcal{H} \\ \dot{\mathbf{P}} = -\partial_{\mathbf{Q}}\mathcal{H} \end{cases} \tag{27.4}$$

Simplectic structure

Let's define

$$\eta = \begin{pmatrix} \mathbf{q} \\ \mathbf{p} \end{pmatrix} \quad , \quad \zeta = \begin{pmatrix} \mathbf{Q} \\ \mathbf{P} \end{pmatrix}$$

whose relation is assumed to be invertible for each time t,

$$\zeta(\eta, t) \quad , \quad \eta(\zeta, t) \ . \tag{27.5}$$

Hamilton equations. Hamilton equations (**??**) can be recast as

$$\dot{\eta} = \mathbf{J}\nabla_{\eta}\mathcal{H} \quad , \quad \dot{\zeta} = \mathbf{J}\nabla_{\zeta}\mathcal{H} \ , \tag{27.6}$$

with the matrix $\mathbf{J} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{pmatrix}$.

Recasting Hamilton equations in new coordinates, using derivatives of composite functions. As the the vector of the new coordinates $\zeta(t)$ can be written as a function of the old coordinates $\eta(t)$ through the function (**??**)

$$\begin{aligned}\frac{d}{dt}\zeta\left(\eta(t),t\right) &= \dot{\eta}\cdot\nabla_\eta\zeta|_t + \partial_t\zeta|_\eta = \\ &= \mathbf{M}^T(\eta(t),t)\dot{\eta} + \partial_t\zeta|_\eta = \\ &= \mathbf{M}^T\,\mathbf{J}\,\nabla_\eta\mathcal{H} + \partial_t\zeta|_\eta\,,\end{aligned}$$

and changing from (ζ,t) to (η,t) variables

$$\dot{\zeta} = \mathbf{M}^T\mathbf{J}\mathbf{M}\,\nabla_\zeta\,\mathcal{H}\big|_t + \partial_t\zeta|_\eta\,, \tag{27.7}$$

with $\{\mathbf{M}\}_{ij} = \frac{\partial\zeta_j}{\partial\eta_i}(\eta,t)$, and the rule of transformation of the gradient of a scalar function,

$$f = f(\eta,t) = f(\eta(\zeta,t),t) = f(\zeta,t) = f(\zeta(\eta,t),t)\,,$$

$$\nabla_\eta f(\eta(\zeta(t),t),t) = \nabla_\eta\zeta\big|_t\cdot\nabla_\zeta f\big|_t = \mathbf{M}\nabla_\zeta f\,.$$

The equation (**??**) must be compared with the Hamilton equations $\dot{\zeta} = \mathbf{J}\nabla_\zeta\mathcal{H}$ from (**??**).

Comparison between different forms of Hamilton equations - canonical transformations are pure kinematics, independent from the physics of the system of interest

From the comparison

$$\begin{cases}\dot{\zeta} = \mathbf{M}^T\mathbf{J}\mathbf{M}\,\nabla_\zeta\,\mathcal{H}\big|_t + \partial_t\zeta|_\eta \\ \dot{\zeta} = \mathbf{J}\nabla_\zeta\mathcal{H}\end{cases}$$

it follows

$$\mathbf{M}^T\mathbf{J}\mathbf{M}\,\nabla_\zeta\,\mathcal{H}\big|_t + \partial_t\zeta|_\eta = \mathbf{J}\nabla_\zeta\mathcal{H}$$

System 0. Let the system zero be a physical system whose Hamiltonian $\mathcal{H}_0(\mathbf{q},\mathbf{p},t) := 0$. **Remark** This is the functional definition of the Hamiltonian, not 1-dimensional constraint between variables. It follows that

$$\partial_t\zeta|_\eta = \mathbf{J}\nabla_\zeta\mathcal{H}_0\,.$$

Generic system. Exploiting the latter relation, for a generic system

$$\mathbf{M}^T\mathbf{J}\mathbf{M}\,\nabla_\zeta\,\mathcal{H}\big|_t = \mathbf{J}\nabla_\zeta\left(\mathcal{H}-\mathcal{H}_0\right)\,.$$

As this relation must hold for any system, two relations follow:

- the simplectic condition

$$\mathbf{M}^T\mathbf{J}\mathbf{M} = \mathbf{J}\,.$$

- a relation between the Hamiltonian functions

$$\mathcal{H}-\mathcal{H}+\mathcal{H}_0 = f(t)$$

A comparison with the partial derivatives (**??**) of of **type-2** generating function $F_2(\mathbf{q},\mathbf{P},t)$, it follows that[1]

$$\partial_t F_2\big|_{\mathbf{q},\mathbf{P}} = \mathcal{H}-\mathcal{H} = \mathcal{H}_0 - f(t)\,.$$

[1] The link between simplectic structure and generating functions can be derived with a proper use of derivatives of composit functions.

27.3 Generating functions

Generating functions

The pair of Hamilton equations (**??**) can be derived from the same variational principle of Lagrangian mechanics for two different choices of the generalized variables,

$$0 = \delta \int_{t_0}^{t_1} \mathcal{L}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t)\, dt$$
$$0 = \delta \int_{t_0}^{t_1} \mathcal{L}(\mathbf{Q}(t), \dot{\mathbf{Q}}(t), t)\, dt$$

In order to get the same equations of motion, the two Lagrangian functions may differ by a time derivative $\frac{dF_1}{dt}(\mathbf{q}(t), \mathbf{Q}(t), t)$, whose integral reads $F_1(\dots,)$ evaluated in the extremes of integration, and whose variation is thus zero. Thus

$$\mathcal{L}(\mathbf{q}(t), \dot{\mathbf{q}}(t), t) = \mathcal{L}(\mathbf{Q}(t), \dot{\mathbf{Q}}(t), t) + \frac{dF_1}{dt}(\mathbf{q}(t), \mathbf{Q}(t), t)$$

or as a function of the Hamiltonian functions,

$$\mathbf{p} \cdot \dot{\mathbf{q}} - \mathcal{H}(\mathbf{q}, \mathbf{p}, t) = \mathbf{P} \cdot \dot{\mathbf{Q}} - \mathcal{H}(\mathbf{Q}, \mathbf{P}, t) + \frac{dF_1}{dt}(\mathbf{q}(t), \mathbf{Q}(t), t)\ ,$$

and thus

$$\frac{dF_1}{dt}(\mathbf{q}(t), \mathbf{Q}(t), t) = \mathbf{p} \cdot \dot{\mathbf{q}} - \mathbf{P} \cdot \dot{\mathbf{Q}} - \mathcal{H}(\mathbf{q}, \mathbf{p}, t) + \mathcal{H}(\mathbf{Q}(t), \mathbf{P}(t), t) =$$
$$= \partial_{\mathbf{q}} F_1 \cdot \dot{\mathbf{q}} + \partial_{\mathbf{Q}} F_1 \cdot \dot{\mathbf{Q}} + \partial_t F_1$$

so that - using the arbitariness of the result on the choice of coordinates and the physics of the system - its differential reads

$$dF_1 = \mathbf{p} \cdot d\mathbf{q} - \mathbf{P} \cdot d\mathbf{Q} + (\mathcal{H} - \mathcal{H})dt\ .$$

and its partial derivatives

$$\partial_{\mathbf{q}} F_1\big|_{\mathbf{q},t} = \mathbf{p} \qquad , \qquad \partial_{\mathbf{Q}} F_1\big|_{\mathbf{Q},t} = -\mathbf{P} \qquad , \qquad \partial_t F_1\big|_{\mathbf{q},\mathbf{Q}} = \mathcal{H} - \mathcal{H}\ . \tag{27.8}$$

This is **Type-1** generating function. **Type-2** generating function is defined as

$$F_2(\mathbf{q}, \mathbf{P}, t) = \mathbf{P} \cdot \mathbf{Q} + F_1(\mathbf{q}, \mathbf{P}, t)\ ,$$

and its differential gives

$$dF_2 = \mathbf{p} \cdot d\mathbf{q} + \mathbf{Q} \cdot d\mathbf{P} + (\mathcal{H} - \mathcal{H})dt\ ,$$

and its partial derivatives

$$\partial_{\mathbf{q}} F_2\big|_{\mathbf{P},t} = \mathbf{p} \qquad , \qquad \partial_{\mathbf{P}} F_2\big|_{\mathbf{q},t} = \mathbf{Q} \qquad , \qquad \partial_t F_2\big|_{\mathbf{q},\mathbf{P}} = \partial_t F_1\big|_{\mathbf{q},\mathbf{Q}} = \mathcal{H} - \mathcal{H}\ . \tag{27.9}$$

Type-3.

Type-4.

27.4 Hamilton-Jacobi formulation of mechanics

Let a set of canonical variables $(\mathbf{Q}, \mathbf{P})$ exist so that $\mathcal{H}(\mathbf{Q}, \mathbf{P}, t) = 0$.[2] **Remark** This is the definition of the function, not a 1-dimensional constraint. For this set of canonical variables, Hamilton's equations implies that $\mathbf{Q}$, and $\mathbf{P}$ are integrals of motion

$$\begin{cases} \dot{\mathbf{Q}} = \nabla_{\mathbf{P}} \mathcal{H} & \rightarrow \quad \mathbf{Q}(t) = \overline{\mathbf{Q}} \\ \dot{\mathbf{P}} = -\nabla_{\mathbf{Q}} \mathcal{H} & \rightarrow \quad \mathbf{P}(t) = \overline{\mathbf{P}} \end{cases}$$

Using type-2 generating function, and its partial derivatives (**??**), $\mathbf{p} = \partial_{\mathbf{q}} F_2(\mathbf{q}, \mathbf{P}, t)$,

$$0 = \partial_t F_2 \left(\mathbf{q}, \mathbf{P}, t\right) - \underbrace{\mathcal{H}(\mathbf{Q}, \mathbf{P}, t)}_{=0} + \mathcal{H}(\mathbf{q}, \mathbf{p}, t) \ ,$$

or, calling $F_2(\mathbf{q}(t), \overline{\mathbf{P}}, t) := S(\mathbf{q}(t), t; \overline{\mathbf{P}})$ - as it can be seen as the action functional as a function of the ending time t and generalized coordinates $\mathbf{q}(t)$, see *below* - the common form of the Hamilton-Jacobi equation follows

$$0 = \partial_t S \left(\mathbf{q}(t), t; \overline{\mathbf{P}}\right) + \mathcal{H} \left(\mathbf{q}(t), \partial_{\mathbf{q}} S(\mathbf{q}(t), t; \overline{\mathbf{P}}), t\right) \ . \tag{27.10}$$

Stationary problems. For problems whose Hamiltonian is not explicitly function of time, $\mathcal{H}(\mathbf{q}, \mathbf{p})$, the Hamiltonian is constant and equal to the energy of the system, $\mathcal{H}(\mathbf{q}, \mathbf{p}) = E$. Integration in t gives

$$S \left(\mathbf{q}(t), t; \overline{\mathbf{P}}\right) = -Et + W \left(\mathbf{q}(t); \overline{\mathbf{P}}\right) \ .$$

The constant energy E is a function of the integrals of motion $\overline{\mathbf{P}}$ only.

Introducing this expression in the Hamilton-Jacobi equation (**??**) gives an equation for $W \left(\mathbf{q}, \overline{\mathbf{P}}\right)$,

$$E = \mathcal{H} \left(\mathbf{q}, \partial_{\mathbf{q}} W \left(\mathbf{q}, \overline{\mathbf{P}}\right)\right) \ .$$

27.4.1 Action functional and type-2 generating function

27.5 Angle-action variables

Let the system of interest be a **conservative system** using the original set of variables $(\mathbf{q}, \mathbf{p})$, so that the Hamiltonian doesn't explicitly depend on time t, and it's constant $\mathcal{H}(\mathbf{q}, \mathbf{p}) = E$. Let a set of canonical variables $(\mathbf{Q}, \mathbf{P}) =: (\mathbf{w}, \mathbf{J})$ exist so that $\mathbf{J}(t) = \overline{\mathbf{J}}$ are integrals of motion, and $\mathcal{H}(\mathbf{w}, \overline{\mathbf{J}}, t) = \mathcal{H}(\mathbf{q}, \mathbf{p}) = E$, *trajectory-wise*.

$$\begin{cases} \dot{\mathbf{w}} = \nabla_{\mathbf{J}} \mathcal{H} \\ \dot{\mathbf{J}} = -\nabla_{\mathbf{w}} \mathcal{H} = \mathbf{0} & \rightarrow \quad \mathcal{H}(\overline{\mathbf{J}}, t) \end{cases}$$

Partial derivative of $F_1(\mathbf{q}, \mathbf{Q}, t)$ w.r.t. time t is therefore zero

$$\partial_t F_1|_{\mathbf{q},\mathbf{w}} = \mathcal{H} - \mathcal{H} = 0 \qquad \rightarrow \qquad F_1(\mathbf{q}, \mathbf{w}) \ .$$

and alos the hamiltonian $\mathcal{H}$ can't be a function of time, $\mathcal{H}(\mathbf{q}, \mathbf{p}) = \mathcal{H}(\overline{\mathbf{J}}) = E(\overline{\mathbf{J}})$, *trajectory-wise*. Thus the gradient of the Hamiltonian $\mathcal{H}$ w.r.t. $\mathbf{J}$ is constant as well $\nabla_{\mathbf{J}} \mathcal{H} =: \overline{\omega}$, and the evolution of the generalized coordinate is linear (including periodic orbits, if the coordinates are cycical),

$$\mathbf{w}(t) = \mathbf{w}_0 + \overline{\omega}\, t \ .$$

[2] Does this set of variables exist? For short-time? For long-time?

Differentials of the generating functions. With this choice of conjugated variables, the differential of the type-1 generating function becomes

$$dF_1(\mathbf{q}, \mathbf{w}, t) = \mathbf{p} \cdot d\mathbf{q} - \overline{\mathbf{J}} \cdot d\mathbf{w} \,,$$

and the integration along a closed trajectory γ in the space $(\mathbf{q}, \mathbf{w})$ gives, if the function F_1 is regular in the region where integration occurs, so that $\oint dF_1 = 0$.

$$\begin{aligned}\oint_{\gamma_k} dF_1 &= \oint_{\gamma_k} \mathbf{p} \cdot d\mathbf{q} - \oint_{\gamma_k} \overline{\mathbf{J}} \cdot \overline{\omega} dt = \\ &= \oint_{\gamma_k} \mathbf{p} \cdot d\mathbf{q} - \overline{\mathbf{J}} \cdot \overline{\omega}\, T_k = \\ &= \oint_{\gamma_k} \mathbf{p} \cdot d\mathbf{q} - \frac{1}{2\pi} \sum_i \overline{J}_i \frac{T_k}{T_i} \,,\end{aligned}$$

with T_γ the period of the trajectory γ. The differential of the type-2 generating function reads

$$\begin{aligned}dF_2 &= \mathbf{J} \cdot d\mathbf{w} + \mathbf{w} \cdot d\mathbf{J} + dF_1 = \\ &= \mathbf{p} \cdot d\mathbf{q} + \mathbf{w} \cdot d\mathbf{J}\end{aligned}$$

and its integration over a closed trajectory in the $(\mathbf{q}, \mathbf{J})$ space, with $\mathbf{J} = \overline{\mathbf{J}}$, $\mathbf{w} = \mathbf{w}_0 + \overline{\omega}\, t$, gives

$$\oint dF_2 = \oint dF_1 + \oint \overline{\mathbf{J}} \cdot \overline{\omega}\, dt = \oint \mathbf{p} \cdot d\mathbf{q} \,.$$

Remark. todo Discussion about the regularity of F_2 and the domain of integration. Integration of an exact differential over a closed path different from zero implies that something "strange" is happening.

todo

- Separable systems
- definition of action variables in separable systems

Part V

Quantum Field Theory

CHAPTER

TWENTYEIGHT

INTRODUCTION TO QFT

Part VI

Solid State Physics

CHAPTER

TWENTYNINE

INTRODUCTION TO SOLID STATE PHYSICS

Contents.

- *Solid State Physics*
- *Semiconductors*

References

- D.Tong, University of Cambridge, Solid State Physics
  - Particles in a Magnetic Field
  - Band Structure: lattices, Bloch's theorem, tight-binding and nearly-free electron models,…
  - Electron dynamics in Solids
  - Phonons
- C.Kittel, Introduction to Solid State Physics
  - Semiconductor Crystals (Ch.8, p.185)
  - Fermi Surfaces and Metals (Ch.9, p.221)
  - Surface and Interface Physics (Ch.17, p.487)
    - …, p-n junctions, semiconductor lasers, LED,…
- D.A.Neamen, Semiconductor Physics and Devices
  - Part I, Semiconductor Material Properties
    - Introduction to the QM of Solids (Ch.3, p.58)
    - The Semiconductors in Equilibrium (Ch.4, p.106)
    - Carrier Transport Phenomena (Ch.5, p.156)
    - Non-equilibrium Excess Carriers in Semiconductors (Ch.6, p.192)
  - Part II, Fundamental Semiconductor Devices
    - p-n Junction (Ch.7, p.241)
    - p-n junction diode (Ch.8, p.276)
    - MOSFET (Ch.10,11)
    - …
  - Part III, Specialized Semiconductor Devices

    - Optical Devices (Ch.14, p.618)
        - Solar cells
        - Photodetectors
        - Photoluminiscence and electroluminescence
        - LED
        - Laser diodes
- LibreTexts Engineering, Material Science, Supplemental Modules: Semiconductors

CHAPTER

THIRTY

NOTES ABOUT SOLID STATE PHYSICS

30.1 Crystals

Primary lattice vectors, $\mathbf{a}_j$, defines periodicity (and symmetry) of the lattice,…

$\$f(\mathbf{r}) = f\left(\mathbf{r} + \sum_j n_j \mathbf{a}_j\right)$,

with $n_j \in \mathbb{Z}$.

30.1.1 Bloch's theorem

…

CHAPTER

THIRTYONE

INTRODUCTION TO SEMICONDUCTORS

Following mainly *D.A.Naemen, Semiconductor Physics and Devices*

- *Introduction to the QM of Solids*
- *Semiconductors in Equilibrium*
- *Carrier Transport Phenomena*
- *Non-equilibrium Excess Carriers in Semiconductors*
- *p-n Junction*
- *p-n Junction Diode*

31.1 Introduction to the QM of Solids

31.1.1 Energy bands

First, the energy level splitting is discussed. Then, two models (Kronig-Penny, and nearly free electron models) are introduced to describe the concept of allowed and forbidden energy bands, starting from Schrodinger equation.

Energy level splitting

This section deal with energy level splitting in a two-element system.

Non-interacting systems

Let $|L\rangle$, and $|R\rangle$ two localized eigen-states of non-interacting identical subsystems, with the same energy E_0,

$$\hat{H}_0|L\rangle = E_0|L\rangle$$
$$\hat{H}_0|R\rangle = E_0|R\rangle$$

As these states are localized in space, they're (approximately?) orthogonal, $\langle L|R\rangle = 0$.

Interacting subsystems - Perturbation potential

When the two systems are brought close together, they start interacting weakly, so that the Hamiltonian becomes

$$\hat{H} = \hat{H}_0 + \hat{V} \, .$$

The matrix elements of the Hamiltonian in the base $\{|L\rangle, |R\rangle\}$ (where are all the other eigen-functions? We're not interested in them, right now, but only on the pair of eigenfunctions with the same energy when the syb-systems are not interacting) read

$$\begin{aligned}
\langle L|\hat{H}|L\rangle &=: E_0' \\
\langle R|\hat{H}|R\rangle &=: E_0' \\
\langle L|\hat{H}|R\rangle = \langle R|\hat{H}|L\rangle^* &=: -t
\end{aligned}$$

Energy levels - Eigenvalues of the Hamiltonian operator

If the new eigen-functions of the interacting system can be written as a linear combination of the non-interacting functions, $|\Psi_{1,2}\rangle = \ell_{1,2}|L\rangle + r_{1,2}|R\rangle$, the eigen-problem becomes

$$\hat{H}|\Psi_{1,2}\rangle = E_{1,2}|\Psi_{1,2}\rangle \, .$$

Projecting onto $\langle L|$, and $\langle R|$,

$$\begin{cases} \ell E_0' - rt = E\ell \\ -\ell t + rE_0' = Er \end{cases} \quad \rightarrow \quad \begin{cases} \ell(E_0' - E) - rt = 0 \\ -\ell t + r(E_0' - E) = 0 \end{cases}$$

This system has non trivial solutions, if the determinant of the linear system is zero,

$$0 = \begin{vmatrix} E_0' - E & -t \\ -t & E_0' - E \end{vmatrix} = (E_0' - E) - t^2 \, .$$

The energy levels of the interacting states are slightly shifted w.r.t. the non-interacting level,

$$E = E_0' \mp t \, .$$

Eigen-states

The eigen-states are

- the **bonding state**, for $E = E_0' - t$, $\ell = -r$, and with normalization

$$|\Psi\rangle_- = \frac{1}{\sqrt{2}} \left(|L\rangle + |R\rangle \right)$$

- the **anti-bonding state**, for $E = E_0' + t$, $\ell = r$, and with normalization

$$|\Psi\rangle_+ = \frac{1}{\sqrt{2}} \left(|L\rangle - |R\rangle \right)$$

Kronig-Penny model

Kronig-Penny potential

Kronig-Penny model is a 1-dimensional model with periodic square wave potential,

$$V(x) = \begin{cases} V_0 & , \quad x \in \text{regions II} = [na + (n-1)b, n(a+b)] \\ 0 & , \quad x \in \text{regions I } = [(n-1)(a+b), na + (n-1)b] \end{cases}$$

General expression of Bloch states

Following *Bloch's theorem*, the eigen-states of an electron (in position base) in a infinite periodic lattice can be written as

$$\Psi(\mathbf{r}) = u(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}} ,$$

with $u(\mathbf{r}) = u\left(\mathbf{r} + \sum_j n_j \mathbf{a}_i\right)$ with the same periodicity as the lattice, $n_j \in \mathbb{Z}$, or in 1-dimensional problems,

$$\Psi(x) = u(x) e^{ikx} .$$

Eigenvalue problem for the Hamiltonian operator

The eigenvalue problem for the Hamiltonian operator reads

$$E\Psi = \hat{H}\Psi = \left[\frac{\hat{p}^2}{2m} + V(x)\right] \Psi ,$$

with $\hat{p}$ the momentum operator, whose expression in position base is given by (14.1), and in 1-dimensional problems $\hat{p}\Psi = -i\hbar\partial_x \Psi$. The operator acting twice gives $\hat{p}^2\Psi = -\hbar^2 \partial_{xx}\Psi$.

Introducing the expression of the eigenfunctions from Bloch's theorem[1], it follows

$$Eu(x) = -\frac{\hbar^2}{2m}\left(u''(x) + 2iku'(x) - k^2 u(x)\right) + V(x)u(x) ,$$

or

$$\begin{aligned} 0 &= u'' + 2iku'(x) - \left(k^2 - \frac{2mE}{\hbar^2}\right) u(x) && \text{regions I} \\ 0 &= u'' + 2iku'(x) - \left(k^2 - \frac{2m(E-V_0)}{\hbar^2}\right) u(x) && \text{regions II} \end{aligned}$$

Solution of the ODEs.

Continuity of $\Psi(x)$, and $\partial_x \Psi(x)$. for matching solutions.**

Energy levels.

Nearly free electron model

[1] The derivatives read $\Psi' = (u' + iku)\, e^{ikx}$, and $\Psi'' = \left(u'' + 2iku' - k^2 u\right) e^{ikx}$.

31.2 Semiconductors in Equilibrium

31.2.1 Structure of a semi-semiconductor

Symbol	Concentration of
p	Holes in the valence bands of Si-atom lattice
n	Free electrons in the conduction band
N_A	Acceptor atoms, usually class III
N_D	Donor atoms, usually class V
N_A^-	Ionized acceptor atoms, having collected e^- (usually from neighboring Si atoms)
N_D^+	Ionized donor atoms, having released e^-

Usually, N_A, N_D, N_A^-, N_D^+ only depends on the space coordinate, as a consequencen of doping. The concentration of free electrons and holes may be time dependent instead (…).

Intrinsic semiconductor

Lattice of Si atoms. Some free electrons have left the valence bands of the atoms and entered the conduction band. For a new free electron, a new hole is left behind in the lattice. If the medium is **electrically neutral** the numbers of free electrons and holes are locally equal, and thus their concentrations

$$n_i(\mathbf{r}) = p_i(\mathbf{r}) \ .$$

Doped semiconductors

p-type

The semiconductor is doped with atoms of class III, replacing some of the Si atoms in the lattice. These atoms introduce a hole in the lattice and are prone to collect e^- from neighboring Si atoms.

Under *full-ionization condition*, all the doping atoms are ionized, and thus $N_A^- = N_A$. In a **electrically netural** region with no donor atoms, the concentration of the holes is equal to the sum of the concentration of the ionized acceptor atoms and the free electrons,

$$p = n + N_A^- \ .$$

In a standard p-type semiconductor, $N_A >> n_i$, and thus $p \sim N_A^-$.

n-type

The semiconductor is doped with atoms of class V, replacing some of the Si atoms in the lattice. These atoms introduce a “extra” loosely bound electron in the lattice and are prone to release it in the conduction band of the semiconductor.

Under *full-ionization condition*, all the doping atoms are ionized, and thus $N_D^+ = N_D$. In a **electrically netural** region with no acceptor atoms, the concentration of the free electrons is equal to the sum of the concentration of the ionized donor atoms and holes,

$$n = p + N_D^+ \ .$$

In a standard n-type semiconductor, $N_D >> p_i$, and thus $n \sim N_D^+$.

Full ionization

todo

31.2.2 Electric charge density

$$\rho(\mathbf{r},t) = -qn(\mathbf{r},t) + qp(\mathbf{r},t) + qN_D^+(\mathbf{r},t) - qN_A^-(\mathbf{r},t)\ , \tag{31.1}$$

being

- q the elementary charge, the charge of the electron (here the opposite to get a positive numerical value of q in Coulomb)
- $n(\mathbf{r},t)$ the number volume density of the mobile electrons
- $p(\mathbf{r},t)$ the number volume density of the holes in the valence bands of the lattice
- $N_D^+(\mathbf{r},t)$ the number volume density of the positive donor ions
- $N_A^-(\mathbf{r},t)$ the number volume density of the negative acceptor ions

Usually, $N_D^+(\mathbf{r})$, $N_A^-(\mathbf{r})$, corresponding to the density of the donor and acceptor atoms in the lattice, that have fixed positions.

Remark. todo *Discuss the values of N_D^+ and N_D in terms of energy levels at different temperatures.* Add this discussion to *full-ionization section* or in another section and then point to it?

31.3 Carrier Transport Phenomena

In this section:

- *Currents in semiconductors*:hole and electron number current, and electric current density
- *Einstein relation*
- *Balance equations*: number and charge density dynamical equations

31.3.1 Currents in semiconductors

Two main processes:

1. **diffusion** due to non-uniform density
2. **drift** due to an electric field $\mathbf{e}(\mathbf{r})$

The overall current has the contribution of the motion of both electrons (n, for negative charges) and holes (p, for positive "charges"),

$$\mathbf{j}(\mathbf{r},t) = \mathbf{j}_n(\mathbf{r},t) + \mathbf{j}_p(\mathbf{r},t)\ ,$$

and both these contributions have their diffusion and drift part,

$$\mathbf{j}_n = \mathbf{j}_{n,diff} + \mathbf{j}_{n,drift}$$
$$\mathbf{j}_p = \mathbf{j}_{p,diff} + \mathbf{j}_{p,drift}$$

For the contribution of the electrons, the diffusion current follows a Fick's law for the number density n (multiplied by the constant charge $-q$ of the elementary charge), while the drift coefficient is written as the product of the negative charge density and their drift velocity,

$$\begin{aligned}\mathbf{j}_n &= \mathbf{j}_{n,diff} + \mathbf{j}_{n,drift} = \\ &= -D_n(-q)\nabla n + \underbrace{(-q)n}_{\rho_n}\mathbf{v}_n = \\ &= D_n q\nabla n + q\mu_n n\mathbf{e}\,,\end{aligned} \tag{31.2}$$

being the average drift velocity, $\mathbf{v}_n = -\mu_n\mathbf{e}$, a function on the local electric field through the electron mobility μ_n. Here the minus sign follows the definition of the electron mobility with positive value, and from the opposite direction of the local electric field $\mathbf{e}$ and the drift velocity of negative charges.

For the contribution of the holes,

$$\mathbf{j}_p = -D_p q\nabla p + q\mu_p p\mathbf{e}\,. \tag{31.3}$$

	$\mu_n\,[\dots]$	$\mu_p\,[\dots]$
Silicon	1350	480
Gallium Arsenide	8500	400
Germanium	3900	1900

Remark. While charges in free-space have acceleration proportional to the electric field, in solids they have drift velocity proportional to the electric field, due to collisions with the lattice. This mechanism can be described in terms of conductivity σ (or its inverse, resistivity ρ_R) of the medium, similarly to Ohm's law

$$\mathbf{j} = \sigma\mathbf{e} \quad , \quad \mathbf{e} = \rho_R\mathbf{j}\,.$$

Using the expression of the drift current, as the sum of the n and p contributions in (**??**) and (**??**) respectively,

$$\mathbf{j}_{drift} = q(\mu_n n + \mu_p p)\mathbf{e}\,,$$

it immediatley follows the formula for the resistivity (and the conductivity)

$$\rho_R = \frac{1}{\sigma} = \frac{1}{q(\mu_n n + \mu_p p)}\,. \tag{31.4}$$

todo *See discussion about this relation in 5.1.4. Velocity Saturation*

31.3.2 Einstein relation

…

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q}\,. \tag{31.5}$$

31.3.3 Balance equations

The density of free electrons and holes are governed by the following PDEs

$$\begin{aligned} \partial_t n - \nabla \cdot \left(\frac{\mathbf{j}_n}{q} \right) &= (G_n - R_n) \\ \partial_t p + \nabla \cdot \left(\frac{\mathbf{j}_p}{q} \right) &= (G_p - R_p) \end{aligned} \tag{31.6}$$

where the expression of current densities $\mathbf{j}_n$, $\mathbf{j}_p$ are given in (**??**), (**??**) respectively, and the terms $G_{n,p}$, $R_{n,p}$ represent source and sink terms representing **generation** or **recombination** of free charges and holes.

As for a new free electron there's a new hole, then $G_n = G_p$. As recombination occurs between the same number of free electrons and holes, $R_n = R_g$.

Multiplying the first and the second equation in (**??**) by $-q$ and q respectively, the balance equation for the *free* charge density $\rho_f = -qn + qp$,

$$\partial_t \rho_f + \nabla \cdot \mathbf{j} = 0 \, .$$

31.4 Non-equilibrium Excess Carriers in Semiconductors

31.5 p-n Junctions

31.5.1 Introduction to the pn Junction and its Operational Regimes

When a p-type and an n-type semiconductor are brought into intimate contact, the immense carrier concentration gradients at the interface drive a transient diffusion process. Mobile electrons and holes annihilate each other at the metallurgical interface, leaving behind uncompensated, fixed dopant ions. This region, stripped of mobile carriers, is known as the **depletion region**. The fixed charges generate an internal electric field that opposes further diffusion, establishing a balance.

Depending on the external voltage V_a applied across the device, the junction operates in one of three primary regimes:

- **Zero Bias (Thermal Equilibrium, $V_a = 0$):** No external voltage is applied. The internal electric field perfectly balances the carrier diffusion gradients. The net current density for both electrons and holes is identically zero ($J_n = 0, J_p = 0$).
- **Forward Bias ($V_a > 0$):** A positive potential is applied to the p-side relative to the n-side. This external potential opposes the built-in field, lowering the electrostatic barrier and narrowing the depletion width. Diffusion forces win over drift, resulting in an exponential injection of minority carriers across the junction.
- **Reverse Bias ($V_a < 0$):** A negative potential is applied to the p-side relative to the n-side. The external voltage reinforces the built-in field, widening the depletion region and increasing the potential barrier. Diffusion drops to zero, and the current is limited to a minute reverse saturation current driven by thermal generation.

31.5.2 The Depletion Approximation and Shockley Transport

To derive closed-form analytical expressions for the electrostatics and current-voltage relations, we employ the **Depletion Approximation** (also called the *Abrupt Junction Approximation*). This framework assumes that the transition between doping zones is perfectly sharp and that the depletion region is completely devoid of mobile carriers ($\rho =$ constant inside the zone, $\rho = 0$ outside).

Concentration	p-bulk	p-interface	n-interface	n-bulk
N_A^-	$\sim N_A$	$\sim N_A$		
N_D^+			$\sim N_D$	$\sim N_D$
p	$\sim N_A^-$			
n				$\sim N_D^+$
ρ		$-N_A^- q$	$N_D^+ q$	

Zero Bias (Thermal Equilibrium)

By applying the depletion approximation, the net space-charge density $\rho(x)$ is treated as piecewise constant:

$$\rho(x) = \begin{cases} 0 x x_p \\ -qN_A x_p \leq x < 0 \\ qN_D 0 \leq x \leq x_n \\ 0 x > x_n \, . \end{cases}$$

Integrating Gauss' law, $\partial_x e(x) = \frac{\rho(x)}{\varepsilon}$, with the assumption of electrical neutrality, $0 = q(N_A x_p + N_D x_n)$, and zero field outside the interface,

$$e(x) = \begin{cases} 0 x x_p \\ -\dfrac{qN_A}{\varepsilon}\left(x - x_p\right) x_p \leq x < 0 \\ \dfrac{qN_D}{\varepsilon}\left(x - x_n\right) 0 \leq x \leq x_n \\ 0 x > x_n \, . \end{cases}$$

Integrating the relation between the electric field and the electric potential, $e(x) = -\partial_x \phi(x)$, with the reference $\phi(x) = 0$ in the p-bulk, $x < x_p$,

$$\phi(x) = \begin{cases} 0 x x_p \\ \dfrac{qN_A}{2\varepsilon}\left(x - x_p\right)^2 x_p \leq x < 0 \\ \dfrac{qN_D}{2\varepsilon}\left(x - x_n\right)^2 + \dfrac{qN_A}{2\varepsilon} x_p^2 - \dfrac{qN_D}{2\varepsilon} x_n^2 0 \leq x \leq x_n \\ \dfrac{q}{2\varepsilon}\left(N_A x_p^2 - N_D x_n^2\right) =: V_{bi} x > x_n \, . \end{cases}$$

Global charge neutrality requires $-N_A x_p = N_D x_n$, yielding the equilibrium depletion width W_0:

$$W_0 = x_p + x_n = \sqrt{\frac{2\varepsilon_s V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

Details

For the continuity of the potential, for $0 \leq x \leq x_n$,

$$\begin{aligned}\phi(x) &= -\frac{qN_D}{\varepsilon}\left[\frac{x^2}{2} - x_n x\right] + \phi(0) = \\ &= -\frac{qN_D}{\varepsilon}\left[\frac{x^2}{2} - x_n x\right] + \frac{qN_A}{2\varepsilon}x_p^2 = \\ &= -\frac{qN_D}{2\varepsilon}\left(x - x_n\right)^2 + \frac{qN_A}{2\varepsilon}x_p^2 + \frac{qN_D}{2\varepsilon}x_n^2\end{aligned}$$

Thus the potential difference across the depletion region V_{bi} reads,

$$\phi(x_n) = \frac{q}{2\varepsilon}\left(N_A x_p^2 + N_D x_n^2\right) =: V_{bi}\ .$$

The width of the depletion region be w can be written as

$$\begin{aligned}w &= x_n - x_p = \\ &= x_n\left(1 + \frac{N_D}{N_A}\right) = x_n N_D\left(\frac{1}{N_D} + \frac{1}{N_A}\right) = \\ &= -x_p\left(\frac{N_A}{N_D} + 1\right) = -x_p N_A\left(\frac{1}{N_D} + \frac{1}{N_A}\right)\ ,\end{aligned}$$

then the potential can be written as

$$\begin{aligned}V_{bi} &= \frac{q}{2\varepsilon}\left(\frac{N_A^2 x_p^2}{N_A} + \frac{N_D^2 x_n^2}{N_D}\right) = \\ &= \frac{q}{2\varepsilon}N_A^2 x_p^2\left(\frac{1}{N_A} + \frac{1}{N_D}\right) = \\ &= \frac{q}{2\varepsilon}w^2\left(\frac{1}{N_A} + \frac{1}{N_D}\right)^{-1}\ .\end{aligned}$$

Thus the width of the depletion region can be written as a function of the built-in potential as

$$w = \sqrt{\frac{2\varepsilon V_{bi}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}\ .$$

---

todo *Deal with some statistical mechanics to show this*

The total voltage drop across the region is the **built-in potential** V_{bi}, determined by the bulk doping concentrations and the Law of Mass Action ($n_0 p_0 = n_i^2$):

$$V_{bi} = V_t \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

where $V_t = \frac{k_B T}{q}$ is the thermal voltage.

Forward Bias Under Shockley Assumptions

To find the explicit I-V relationship, we introduce the **Shockley Ideal Conditions**:

1. Low-level injection ($n \ll N_A$ in the p-bulk; $p \ll N_D$ in the n-bulk).
2. No recombination or generation occurs inside the depletion region.
3. The electric field in the neutral bulk regions is negligible; transport there is purely diffusion-driven.

Width of the depletion region. The applied forward voltage V_a lowers the barrier, modifying the depletion width to:

$$W(V_a) = \sqrt{\frac{2\varepsilon_s(V_{bi} - V_a)}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

Minority charge injection. Lowering the barrier causes minority charge injection at the boundaries of the depletion region. Compared to the equilibrium condition with no external voltage,

$$\Delta n(x_p) = n_{p0}\left[\exp\left(\frac{V_a}{V_t}\right) - 1\right]$$
$$\Delta p(x_n) = p_{n0}\left[\exp\left(\frac{V_a}{V_t}\right) - 1\right]$$

Charge transport dynamics. If Shockley conditions hold, no recombination in the depletion region occurs; recombination occurs in the bulk of p- and n- sections: in bulk regions, the electric field $\vec{e}$ is negligible, and thus the current is driven by diffusion only.

Governing equation of the holes in the n-bulk immediately follows from equation (**??**), with the assumption of no drift current in (**??**),

$$\begin{aligned}\partial_t \Delta p_n &= -\nabla \cdot \left(\frac{j_{p,diff}}{q}\right) + \Delta(G - R) = \\ &= -\nabla \cdot (-D\Delta p_n \nabla n) + \Delta(G - R)\ ,\end{aligned}$$

or in 1-dimensional problems with constant coefficients

$$\partial_t \Delta p_n = D\partial_{xx}\Delta p_n + \Delta(G - R)$$

Assuming *steady-state diffusion equation* (**todo** *justify this limit*) governs the diffusion of the minority charge carriers in the bulk, and using **the law of mass action** to write the source term (see box below), the governing equation becomes

$$D_p \Delta p_n'' = \frac{1}{\tau_p}\Delta p_n\ ,$$

with $\tau_p = \frac{1}{k_r n_{n0}}$.

Law of mass action for $G - R$

In the n-bulk region,

- $n_{n0} \gg p_{n0}$,
- $\Delta n = \Delta p$ if the electrical charge remains zero
- $\Delta p_n \gg p_{n0}$
- $\Delta n_n \ll n_{n0}$

Thus the source term becomes

$$\begin{aligned} R - G = k_r pn - k_r p_0 n_0 = \\ = k_r (p_{n0} + \Delta p_n)(n_{n0} + \Delta n_n) - k_r p_{n0} n_{n0} = \\ = k_r (p_{n0} \Delta n_n + n_{n0} \Delta p_n + \Delta n_n \Delta p_n) = \\ \simeq k_r n_{n0} \Delta p_n \end{aligned}$$

as:

- $G \sim G_0 = R_0$ at equilibrium. **todo** *Find some time/space to justify all these sentences about equilibrium*
- the conditions at the beginning of the box give:
  - $n_{n0} \Delta p_n \gg p_{n0} \Delta n_n$ (because the deltas are equal, and $n_{n0} \gg p_{n0}$)
  - $n_{n0} \Delta p_n \gg \Delta p_n \Delta n_n$ (because the deltas are equal, and $n_{n0} \gg \Delta n_n$

The differential porblem supplied with proper boundary conditions becomes

$$\begin{cases} \dfrac{\Delta p_n(x)}{\tau_p} = D_p \Delta p_n''(x) \\ p(x_n) = p_{n0} \left[\exp\left(\dfrac{V_a}{V_t} \right) - 1 \right] \\ p(x \to +\infty) = 0 \end{cases}$$

and its solution reads

$$\Delta p_n(x) = p_{n0} \left[\exp\left(\frac{V_a}{V_t} \right) - 1 \right] \exp\left(-\frac{x - x_n}{L_p} \right) ,$$

with $L_p = \sqrt{\tau_p D_p}$ for the holes in the n-bulk, $x > x_n$

$$\Delta n_p(x) = n_{p0} \left[\exp\left(\frac{V_a}{V_t} \right) - 1 \right] \exp\left(\frac{x - x_p}{L_n} \right) ,$$

with $L_n = \sqrt{\tau_n D_n}$ for the electrons in the p-bulk, $x < x_p$.[1][2]

Current under steady conditions.

- Depletion region $x \in [x_n, x_p]$. Under steady conditions, no charge accumulation occurs along the p-n junction and thus the current must be constant. There's no recombination or generation inside the depletion region $x \in [x_p, x_n]$, thus the following conditions hold for holes and free electrons

  $$\begin{aligned} \text{const} = AJ_p(x) = AJ_p(x_p) = AJ_p(x_n) \quad \forall x \in [x_p, x_n] \\ \text{const} = AJ_n(x) = AJ_n(x_p) = AJ_n(x_n) , \end{aligned}$$

  and thus

  $$\text{const} = AJ(x) \quad \forall x \in [x_p, x_n] .$$

  Evaluating the current at the boundary of the depletion region (in the model, just ouside that, so that the electric field is zero and the current is driven only by diffusion)

  $$\begin{aligned} I = AJ(x) = AJ_n(x) + AJ_p(x) = \\ = AJ_n(x_p) + AJ_p(x_n) = \\ = -A(-q) D_n n_p'(x_p) - AqD_p p_n'(x_n) = \\ = AqD_n \frac{1}{L_n} n_{p0} \left[\exp\left(\frac{V_a}{V_t} \right) - 1 \right] + AqD_p \frac{1}{L_p} p_{n0} \left[\exp\left(\frac{V_a}{V_t} \right) - 1 \right] = \\ = qA \left(\frac{D_n n_{p0}}{L_n} + \frac{D_p p_{n0}}{L_p} \right) \left[\exp\left(\frac{V_a}{V_t} \right) - 1 \right] . \end{aligned}$$

[1] The boundary conditions of the diffusion of n in the p-bulk are $n(x \to -\infty) = 0$, $n(x_p) = n_{p0}[\dots]$.

[2] Usually $n_0 \ll \Delta n \ll N_A$.

This equation readily gives the $V(I)$ relation for an ideal p-n junction,

$$I(V) = I_0 \left[\exp\left(\frac{V}{V_t}\right) - 1 \right] , \tag{31.7}$$

with $I_0 = qA\,(\dots) = qAn_i^2 \left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right)$.

- n-bulk, $x \in [x_n, +\infty)$. In this region, the electric field is assumed to be negligible, and thus only diffusion drives the current of the charge carriers. For the holes,

$$\begin{aligned} J_p(x) &= -D_p p_n'(x) = \\ &= \frac{D_p}{L_p} p_{n0} \exp\left(\frac{V_a}{V_t}\right) \exp\left(-\frac{x - x_n}{L_p}\right) = \\ &= J_p(x_n) \exp\left(-\frac{x - x_n}{L_p}\right) . \end{aligned}$$

  and thus $J_n(x) = J - J_p(x)$.
- p-bulk, $x \in (-\infty, x_p]$…

31.5.3 Advanced Description: Quasi-Fermi Levels and Non-Equilibrium Transport

todo *Uncomment*

31.6 p-n Diodes

PROOF INDEX

Adjoint Operator

`Adjoint Operator` (*ch/quantum-mechanics/intro*), 107

Operator

`Operator` (*ch/quantum-mechanics/intro*), 107

Self-Adjoint Operator

`Self-Adjoint Operator` (*ch/quantum-mechanics/intro*), 107

example-0

`example-0` (*ch/statistical-mechanics/notes*), 75

example-1

`example-1` (*ch/statistical-mechanics/notes*), 78

example-2

`example-2` (*ch/relativity-special/lorentz*), **??**

example-3

`example-3` (*ch/relativity-special/lorentz*), **??**

qft:fermi-golden-rule

`qft:fermi-golden-rule` (*ch/quantum-mechanics/qft/interactions-perturbation*), **??**

qft:fermi-golden-rule-finite

`qft:fermi-golden-rule-finite` (*ch/quantum-mechanics/qft/interactions-perturbation*), **??**

theorem-0

`theorem-0` (*ch/statistical-mechanics/liouville-boltzmann*), 67